

Topology, Uncertainty, and Precision in Passive Potential-Flow Optimization

Abstract

In passive potential-flow networks, such as gas and water networks without active elements, each nomination and choice of edge-law coefficients determines a unique physical state. Optimizing a pressure difference or arc flow of this state over uncertain nominations and coefficients is a nonconvex problem arising in robust validation and design. We determine how its complexity in the Turing bit model depends on topology, observable, uncertainty model, and required output. At fixed maximum cycle rank per biconnected block, the largest prescribed potential difference over a balanced nomination box and independent resistance intervals is computable to additive accuracy ε in time polynomial in the input length and $\log(1/\varepsilon)$, with an exactly admissible rational scenario; arc-flow extrema are computable exactly, so robust capacity validation is exact. On cacti, this answers the additive version of an open question recorded in a recent survey, while exact pressure comparison contains Square-Root Sum and, for one source and sink, is equivalent to it. For quadratic laws and fixed nominations, interval hulls of independent resistance sets preserve all extrema precisely on trees for weighted potentials, cacti for potential differences and linear flows, and series-parallel graphs for arc flows. Over nomination boxes, weighted potentials with growing support are NP-hard on trees, yet admit accuracy-bit algorithms in three structured settings with fixed nomination dimension or support. On cacti with fixed nominations, exact capacity recovery survives global coefficient correlations, which nevertheless make total flow and minimum dissipation strongly NP-hard; maximum dissipation over a resistance polytope is tractable on every connected graph. Rational energy, Bregman, and curvature certificates turn numerical solutions into exactly checkable guarantees.

1 Introduction

In a passive network, conservation and the potential drops across edges determine the flow. Each scenario of nominations (injections and withdrawals)

and physical coefficients therefore has a unique state, the minimizer of a strictly convex energy. When nominations or coefficients are uncertain, however, finding a scenario with the largest pressure difference or arc flow is a nonconvex global optimization problem with direct operational meaning: a worst-case arc flow tests a capacity, a potential difference measures a required pressure range, and a weighted potential objective measures several locations together.

Potential-based models of this kind describe passive gas and water transport and nonlinear resistive circuits; the survey of Pfetsch et al. [44] describes their common structure and physical interpretations. In gas networks the questions above arise in robust booking and nomination validation [5, 33, 49]. Network design adds discrete or continuous investment decisions to the same physical model [32, 47], and robust design contains worst-case state optimization as a subproblem [50]. We work with the abstract passive equations, principally with the quadratic law $\pi_u - \pi_v = \beta_e x_e |x_e|$. The potential may already be a transformed physical quantity, such as squared pressure. Our results concern this mathematical model; they do not replace a system’s complete engineering model, controls, or admissible operating conditions.

Four choices govern the complexity: how cycles interact inside blocks; whether the objective inspects one arc, compares two potentials, or sums measurements; whether uncertain coefficients are independent intervals, finite sets, or correlated; and whether the output is an exact threshold decision or an additive interval with an admissible near-optimal scenario. The last choice is easy to overlook. Two parallel quadratic paths already produce a square root, and a chain of independent cycles accumulates many roots whose sum is easy to enclose but has no known polynomial-time exact comparison. We therefore require running time polynomial in $\log(1/\varepsilon)$ (accuracy-bit polynomial), not in $1/\varepsilon$, and require returned rational parameters to satisfy the original scenario constraints exactly, although their physical state may be irrational.

1.1 Main results

Let $r_{\max}$ be the largest cycle rank of a biconnected block. Cacti are the graphs with $r_{\max} \leq 1$; series-parallel graphs are those without a K_4 minor. Unless stated otherwise, laws are quadratic and scenario domains carry no additional operating constraints.

1. *Block rank localizes nominations* (section 3). A law-independent face reduction confines the free nominations of a potential-difference maximizer to one block. With fixed-dimensional real algebraic computation

this gives, for each fixed bound on $r_{\max}$, accuracy-bit optimization of a prescribed potential difference over a nomination box and independent resistance intervals, and exact algebraic arc extrema, hence exact robust capacity validation, also for dense polynomial laws (theorem 3.1). Block rank cannot be replaced by treewidth two, series-parallel structure, or feedback vertex number one unless $P = NP$ (proposition C.12).

2. *Exact comparison is an arithmetic problem* (theorem 3.2). On cacti with fixed resistances, exact single-source/sink comparison of the maximum potential difference (MPD) is polynomial-time equivalent to Square-Root Sum; general exact quadratic MPD lies in $\exists\mathbb{R} \subseteq PSPACE$. Fixed fractional power laws with exponents above one keep the additive algorithms at fixed block rank, but exact arc comparison is Square-Root-Sum hard on one cycle for every reduced exponent above one with even denominator, such as the water head-loss exponent $1.852 = 463/250$.
3. *The observable determines the topology boundary* (section 4). For quadratic laws and fixed nominations, the graphs on which independent resistance sets can always be replaced by their interval hulls are precisely the trees for weighted potentials, the cacti for potential differences and linear flows, and the series-parallel graphs for individual arcs (theorem 4.1). The attainable flow region is convex for every nomination and resistance box precisely on cacti; potential regions (with one grounded vertex) and joint flow-potential regions, precisely on trees. For fixed nominations on series-parallel graphs, choosing an extreme allowed law on each edge by its electrical sign gives polynomial additive arc optimization with no rank bound (theorem E.6). With independent finite resistance choices, weighted potentials on one cycle, potential differences and linear flows on a rank-two block, and arc flows on a rank-three block are already NP-hard (table 2).
4. *Weighted objectives need more structure* (section 5). With growing support and a nomination box, weighted-potential threshold attainment is NP-complete on trees (theorem F.2). Accuracy-bit algorithms hold on cacti with fixed nomination dimension or fixed support, for bounded-rank blocks along one nomination line, and for several nomination parameters when the total rank of noncactus blocks is fixed (theorem 5.1). The remaining case, several shared parameters and many higher-rank blocks, is isolated in equation (87).
5. *Correlations and the direction of optimization decide design complexity*

(section 6). For fixed nominations on a cactus, parameter polytopes within individual cycles give exact rational parameter recovery for arc extrema and capacities, additive minimization of dense convex polynomial flow objectives, and additive convex maximization through a fixed number of linear measurements (theorem 6.1). Capacity feasibility, robust capacity tests, and exact arc extrema survive even a global parameter polytope. Maximizing a convex function that couples all flows is strongly NP-hard even with independent intervals, and global coefficient correlations make total-flow maximization and minimum dissipation strongly NP-hard on cacti. In contrast, with fixed nominations on any connected graph, maximum dissipation over a positive rational resistance polytope admits a polynomial additive algorithm (theorem 6.2).

6. *Rational certificates connect numerical and exact results* (section 7). Rational energy, Bregman, support, and curvature witnesses certify flows, potentials, linear goals, and original uncertainty scenarios without trusting a solver’s status (theorem 7.1); the curvature bound is sharp for its quadratic error set.

The primary sources we inspected do not supply the combined fixed-block-rank nomination and coefficient guarantees of theorem 3.1, the weighted extensions of theorem 5.1, or the objective-dependent classifications and output boundaries; within the limits of the comparison below, these scoped results and their network-specific proofs are the contributions of this paper. The energy principle, electrical adjoints, circuit confluence, parametric linear optimization, quantifier elimination, analytic-panel approximation, and zonotope enumeration are classical ingredients, credited where used. We claim no novelty for these ingredients or for general principles of nonlinear network sensitivity, circuit monotonicity, or analytic approximation, and no priority for endpoint and envelope identities (we could not inspect Hasler and Wang [30] in full).

1.2 Relation to earlier work

Passive states and circuit structure. The primitive-energy characterization of passive states goes back to nonlinear network theory: Duffin [22], Birkhoff and Diaz [10], and Minty [40] establish variational and monotone foundations under their respective law hypotheses. We give a self-contained existence proof for continuous strictly increasing laws vanishing at zero, including bounded laws; surjectivity is not needed. The grounded Laplacian and

adjoint calculation have direct predecessors: Misra et al. [41, Section IV-A] identify potentials and the inverse weighted Laplacian as the gradient and Hessian of the conjugate energy. For linear two-terminal networks, effective resistance is concave in the edge resistances [48], and the electrical cut/cycle projection is classical [37, Chapter 2, Section 4]. Our maximum-dissipation theorem uses the cubic analogue of the first fact, and the conservation-aware certificates use the second.

The topology-based sign mechanism is classical circuit confluence [23]. Chauffoureaux and Hasler [18, Theorem 3 and conclusion] use a linear-circuit criterion to study nonlinear parameter monotonicity and extremal evaluation. The undated Wang–Hasler report studies convexity of scalar transfer characteristics as source values vary, including structural sufficient conditions and ladder networks [54, Theorems 1, 4, and 5]. Our convexity statements concern a different object: the vector of attainable flows or potentials under independent resistance variation. The universal nondegeneracy characterization of cacti by Brandenberg and Stursberg [15] concerns linear differential-flow polytopes, quantified over all positive elasticity vectors and all nomination and capacity bounds; this property differs from our resistance-attainable nonlinear state regions.

Robust feasibility and network complexity. The joint product of a balanced nomination box and positive resistance intervals is established in Aßmann [4, Section 3.3], and the circulation sign-region method and cycle-capacity linearization appear in Aßmann et al. [5]. González Grandón, Heitsch, and Henrion combine random loads with robust roughness uncertainty on trees [27]; we treat deterministic worst-case scenarios on graphs with cycles. Labbé et al. [33, Section 6] leave booking feasibility on a single nonlinear cycle open; Labbé et al. [34, Section 6] later give a polynomial Turing-time algorithm for single-cycle booking validation under rational-data hypotheses. Our block-rank algorithms allow unboundedly many blocks and cycles and handle the arithmetic of their combined objective. Thürauf’s nomination hardness construction [49] supplies an important boundary: its main block has unbounded rank although the graph is series-parallel. We credit that construction and derive the required bit-gap and arc-probe consequences.

Groß et al. [29, Lemma 4.2 and Theorem 4.3] give homogeneous series-parallel reductions and discuss approximate root arithmetic in the Turing model. Our exact comparison results separate the cost of performing those reductions from the cost of ordering the resulting radical expression. The January 2026 overview still asks for the complexity of nonlinear cactus MPD

[44, pp. 9–10]. We resolve its additive version and classify the exact single-source/sink subclass, but not exact MPD over arbitrary cactus nomination boxes.

Local formulas, approximation, and design. Gotzes, Heitsch, Henrion, and Schultz derive quadratic-radical circulation formulas on affine nomination rays and discuss node-disjoint cycles with attached trees [28, Theorem 6, pp. 18 and 24 of the author manuscript]. Vigneron combines algebraic functions through common arrangements and approximate summation; his bit-model schemes have polynomial dependence on $1/\varepsilon$ [52, Theorem 6 and Sections 2.3, 3.2]. Our weighted results require polynomial dependence on accuracy bits, express all surrogates in the same original nomination coordinates, and recover rational original parameters. Appendix J compares the scalar construction with analytic enclosure, rational algebraic approximation, and semialgebraic parametrization results [9, 13, 43, 55]; the difference lies in the interface.

The exact cycle-root recovery refines a classical LP-bisection mechanism [1, 38] with dense-degree separation and parameter output bounds. Few-measurement convex maximization uses zonotope methods [26, 42]. In network design, Robinius et al. [47, Theorem 4.7] treat hardness of minimum-cost discrete arc sizing subject to potential feasibility, already on trees, and credit earlier water-network hardness of Yates, Templeman, and Boffey; our finite resistance choices are adversarial scenarios rather than investment decisions. Klimm et al. [32] choose installed arcs and conductances under costs and a potential range (convex without fixed costs, Corollary 4; an FPTAS on series-parallel networks with strictly positive variable costs and finite conductance bounds, Theorem 11); these investment objectives and relative-error guarantees differ from our scenario optimization. Thürauf et al. [50] reduce fixed-design robust checking to polynomially many nonlinear worst-case subproblems and use an adversarial design algorithm; a polynomial number of subproblems does not by itself make each of them polynomial.

Section 2 fixes the model and output conventions; each later main-text result names the appendix statements it combines, and Appendices A–K contain all proofs in dependency order.

2 Model, graph parameters, and output contracts

Let $G = (V, E)$ be a finite connected loopless graph with a chosen edge orientation. Parallel edges are permitted except in the universal graph characterizations, where graphs are simple. Its incidence matrix A has $+1$

at each tail and -1 at each head. A balanced nomination $b \in \mathbb{R}^V$, $\mathbf{1}^\top b = 0$, specifies injections and withdrawals. The signed flow x and potential π satisfy

$$Ax = b, \quad A^\top \pi = g(x). \quad (1)$$

A general passive law g_e is continuous, strictly increasing, and vanishes at zero. The flow is unique and the potentials are unique up to an additive constant: the primitives $F_e(t) = \int_0^t g_e(u) du$ are strictly convex and coercive, so $\sum_e F_e(x_e)$ has a unique minimizer on $Ax = b$, and stationarity says $g(x) \in \text{im } A^\top$. Theorem A.1 gives the full argument, including bounded laws and the one-vertex case.

Unless stated otherwise, algorithmic results use symmetric quadratic laws $g_e(t) = \beta_e t|t|$, with positive coefficients. A directional quadratic law instead uses $\beta_e^+ t^2$ for $t \geq 0$ and $-\beta_e^- t^2$ for $t \leq 0$. Reversing an edge swaps its two coefficients. An output state has one potential grounded to zero. Every nonzero physical flow points down a strictly decreasing potential, so there is no directed circulation, and a source-to-sink path decomposition bounds $|x_e|$ by the total positive nomination. This bound is the uniform compactness estimate behind the perturbation and rounding arguments.

2.1 Scenario domains

A balanced nomination box is

$$\mathcal{B} = \{b \in \mathbb{R}^V : \ell \leq b \leq u, \mathbf{1}^\top b = 0\}, \quad \ell, u \in \mathbb{Q}^V, \quad \ell \leq u. \quad (2)$$

It is nonempty exactly when $\sum_v \ell_v \leq 0 \leq \sum_v u_v$. Shifted intervals, sign-changing nominations, and singleton intervals are allowed. Another input is an affine balanced family $b(z) = b^0 + Mz$ on a nonempty compact rational polytope $P \subseteq \mathbb{R}^d$, supplied by linear inequalities and equalities. Fixed dimension means that d is independent of graph size.

Independent resistance uncertainty means $\mathcal{R} = \prod_e [L_e, U_e]$, with rational $0 < L_e \leq U_e$. Finite uncertainty means a product of explicitly listed nonempty finite subsets of $\mathbb{Q}_{>0}$. A global polytope may couple coefficients from different edges and blocks. A cycle-local polytope couples any number of law parameters within one cycle, while different cycle polytopes are independent; the attainable regions of these two models can differ even on two cycles. Directional and symmetric intervals are also distinct models: directional intervals are independent positive intervals for β_e^+ and β_e^- , whereas a symmetric interval constrains one common coefficient for both directions.

A scenario is an admissible original parameter vector, such as (b, β) ; its state is determined by (1). Most nomination theorems use the full *unfiltered*

scenario domain. Robust capacity validation asks whether the arc flow of every such state satisfies given signed bounds. Capacity *feasibility* and constrained design instead ask whether some scenario satisfies the bounds, and optimize only over those scenarios. Such a filter changes the parameter domain and can remove all rational points even for rational input. Where filters are allowed, the theorem states their form and the resulting output type.

2.2 Observables and precision

We study a prescribed terminal difference $F_{s,t} = \pi_s - \pi_t$, a weighted potential $W = c^\top \pi$ with $c \in \mathbb{Q}^V$ and $\mathbf{1}^\top c = 0$, an individual arc flow x_a , and a linear flow objective $d^\top x$. Since $\mathbf{1}^\top c = 0$, W does not depend on the gauge; let $p = |\text{supp } c|$. Maximum potential difference (MPD) means maximizing $F_{s,t}$ for specified ordered terminals, not a search over terminal pairs.

For a maximization objective F , write $\text{OPT} = \max_{\theta \in \Theta} F(\theta)$. An exact threshold question is $\text{OPT} \geq \gamma$ for rational γ ; equality belongs to the yes case. The additive contract, for rational $\varepsilon > 0$, is

$$\underline{v} \leq \text{OPT} \leq \bar{v}, \quad \bar{v} - \underline{v} \leq \varepsilon, \quad \hat{\theta} \in \Theta \cap \mathbb{Q}^k, \quad F(\hat{\theta}) \geq \text{OPT} - \varepsilon. \quad (3)$$

The output interval is rational and the parameter scenario is exactly admissible; minimization reverses the inequality. Theorems that promise only a parameter or an exact algebraic value say so. Algebraic values are encoded by an integer polynomial and an isolating interval, or an equivalent real univariate representation. A list of local algebraic values is an expression for their sum, not necessarily a polynomial-size algebraic encoding of that sum.

All complexity bounds use the Turing model. Let N denote total binary input length, including rational tolerances, and put $N_\varepsilon = N + \max\{0, \lceil \log_2(1/\varepsilon) \rceil\} + 1$. An accuracy-bit polynomial algorithm is polynomial in N_ε , so each additional bit of accuracy costs only polynomially more; an algorithm polynomial in $1/\varepsilon$ need not have this property. A parameter-dependent exponent, such as $N_\varepsilon^{O(r)}$, gives a polynomial for each fixed r ; this is an XP-type bound, not fixed-parameter tractability, which would require $f(r)N_\varepsilon^C$ with an absolute exponent C .

An additive interval decides a threshold when it lies on one side of it, but repeated refinement alone does not settle equality. Moreover, rational scenarios need not have rational states. For two unloaded parallel branches of total resistances 1 and 2, a unit transfer splits as $\sqrt{2}/(1+\sqrt{2})$ and $1/(1+\sqrt{2})$, with common drop $2/(1+\sqrt{2})^2$. All inputs are rational, while the flow and

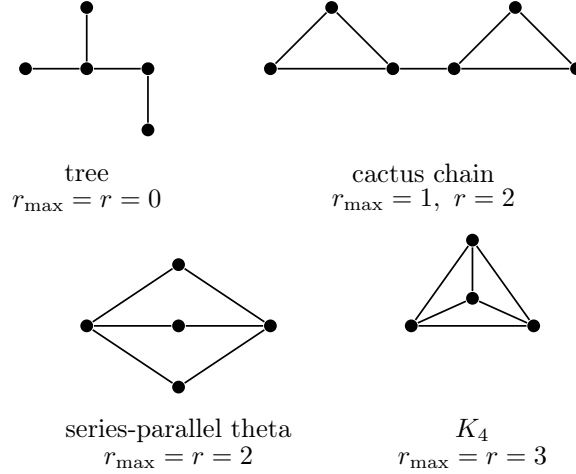


Figure 1: Four graph structures used in the complexity boundaries. Adding cactus blocks raises total rank without raising maximum block rank. Adding parallel paths inside one block raises both. The displayed K_4 is the obstruction to universal individual-arc hull equality.

drop are irrational. Rational approximate states therefore need error bounds; in general they cannot satisfy both physical equations exactly.

2.3 Block rank and block decomposition

A block is a biconnected component, with bridges treated as two-vertex blocks. Write $r_B = |E(B)| - |V(B)| + 1$, $r_{\max} = \max_B r_B$, and $r = |E| - |V| + 1 = \sum_B r_B$. The maximum over no blocks is defined to be zero. (Appendices A, C and D write H and r_H for a block and its rank, and $\mathfrak{r}$ for the bound written r_0 here; appendices G and J write H for the nomination matrix M .) A tree has $r_{\max} = 0$; a cactus, in which distinct simple cycles share at most one vertex (equivalently, each edge lies on at most one simple cycle), has $r_{\max} \leq 1$. A chain of k triangles has $r_{\max} = 1$ and $r = k$. Series-parallel means having no K_4 minor; we use this graph property rather than requiring specified decomposition terminals. A series-parallel block can have arbitrarily large rank: $K_{2,k}$ has rank $k - 1$ and treewidth two.

Deleting a block's edges leaves components attached at its vertices. Summing their nominations gives the block's effective nomination, and the block's state depends only on that aggregate and its own laws. Since the block-vertex incidence graph is a tree, a terminal difference is the sum of drops along its

unique block path, while an individual arc belongs to one block. Lemmas A.4 and A.5 prove these facts without assumptions on flow signs. They are why pressure and arc objectives have different exact-output guarantees.

3 Localizing nomination uncertainty and separating arithmetic

Theorem 3.1 (Block-rank pressure and arc algorithms). *Fix r_0 . Let G be connected with $r_{\max} \leq r_0$, and let nominations range over a nonempty balanced rational box. For symmetric quadratic laws with independent positive rational resistance intervals, every prescribed terminal-drop maximum satisfies (3) in time polynomial in N_ε for fixed r_0 . Every individual arc has exactly computable algebraic minimum and maximum of polynomial encoding length, so unfiltered robust signed capacity validation is polynomial-time, including equality.*

The same conclusions hold for laws affine in arbitrarily many independent rational interval parameters, with densely encoded rational continuous piecewise-polynomial basis functions vanishing at zero and fixed rational break-points, provided every allowed complete law is strictly increasing. Degrees and piece counts may grow. The pressure scenario output is rational; the exact arc conclusion asserts algebraic values, without a general rational near-extremal arc-output guarantee for these laws. No additional operating bounds filter the nomination or coefficient scenario domain in either assertion.

This combines theorems C.2, C.3 and D.5. For the independent affine-box representation, the dense-law admissibility conditions can themselves be checked in polynomial time (proposition D.1). The exponent depends on r_0 (an XP bound); implicit laws and sparse binary exponents are not covered. By proposition C.12, based on Thürauf [49], block rank cannot be replaced by treewidth two, series-parallel structure, or feedback vertex number one unless $P = NP$, even with fixed resistances.

3.1 Proof mechanism: adjoint levels and small faces

The first reduction removes components outside the terminal block path and sums their nomination intervals at the attachment vertices. Every feasible aggregate has a rational disaggregation, obtained by filling the original intervals from their lower bounds. Since the domain is unfiltered, no operating constraint is lost.

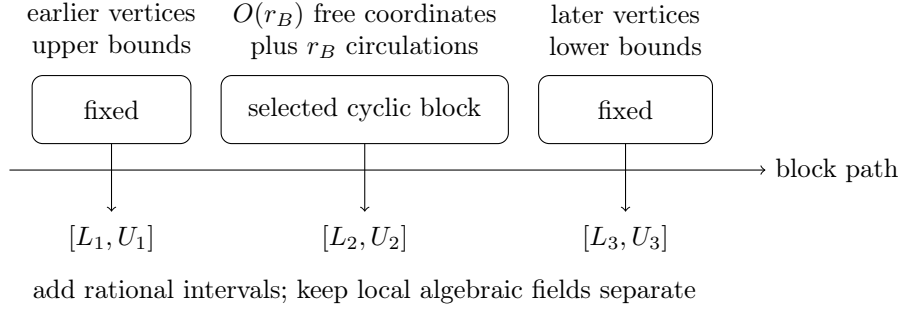


Figure 2: Nomination localization for a terminal difference. Independent coefficients may still be optimized in every block; “fixed” refers to aggregated nominations. The selected block is cyclic; bridge cases are handled directly. The number of blocks is unrestricted.

Temporarily smooth the edge laws so their derivatives are positive. Differentiating (1) and pairing with the terminal objective gives

$$dF_{s,t} = h^\top db, \quad A \operatorname{diag}(1/g'_e(x_e)) A^\top h = e_s - e_t. \quad (4)$$

The adjoint h is the potential of a unit electrical network. At a maximum over a balanced box, a multiplier μ for balance exists such that $h_v > \mu$ forces $b_v = u_v$ and $h_v < \mu$ forces $b_v = \ell_v$, so only nominations at the level $h_v = \mu$ remain free. The maximum principle strictly orders the adjoint ranges of consecutive blocks, so one threshold meets internal vertices of at most one block.

On a cactus, the adjoint is strictly monotone down each of the two branches of that block. At most two nominations remain free, one on each branch, and balance leaves one parameter; there are $O(n^2)$ candidate faces. On a higher-rank block, mark the terminals and the vertices of degree at least three, and suppress the degree-two paths between them; there are at most $2r_B$ marks and $3r_B - 1$ paths. A small positive objective source at every unmarked vertex makes the electrical current strictly increase along each path, so its potential is unimodal and every threshold occurs at most twice. Enumerating the resulting lower/free/upper/free/lower patterns leaves at most $8r_B - 2$ free nominations. Compactness and uniform convergence remove the smoothing and perturbation; the algorithm enumerates closed faces and never computes a perturbation size.

The second reduction handles the many coefficient variables. Inside a selected face, flow is affine in the free nominations and the r_B cycle circulations.

On each flow-sign or law-piece cell, the cycle equations are polynomial in this small core and affine in the coefficient parameters, so the image of a coefficient box in the cycle and objective coordinates is a zonotope of fixed dimension. Its support inequalities can be expanded over the realizable sign patterns, which are polynomially many at fixed dimension, and quantifier elimination removes the remaining fixed-dimensional support variables. The leaf coordinates are recovered from exposed zonotope vertices and a fixed-size convex combination. This avoids enumerating all box corners; the fact that an LP vertex has few non-bound coordinates would not by itself give an efficient algorithm.

The fixed-dimension bounds are classical [8, 46]; we keep the number of real variables bounded *per block*, even when law degrees and numbers of uncertain coefficients grow. Independent block optima are enclosed separately and their rational intervals added. A pressure sensitivity estimate then permits rational rounding inside the original nomination and coefficient constraints. The proofs in appendices C and D handle ties, zero flows, singleton faces, degenerate polytopes, and output bit length.

3.2 An exact arithmetic boundary on a chain of triangles

Theorem 3.2 (Cactus arithmetic and fixed fractional powers). *For symmetric quadratic laws with fixed positive rational resistances, MPD over a balanced rational nomination box on a cactus satisfies (3) in polynomial time. On the restricted box $b = \zeta(e_s - e_t)$, $0 \leq \zeta \leq 1$, exact weak threshold comparison is polynomial-time many-one equivalent to Square-Root Sum. Hardness holds on simple degree-three chains of triangles with positive integer resistances.*

For every fixed finite family $\mathcal{Q} \subset \mathbb{Q} \cap (1, \infty)$, the laws $g_e(t) = \beta_e \operatorname{sgn}(t)|t|^{q_e}$, $q_e \in \mathcal{Q}$, admit the additive value and rational-scenario contract for terminal drops and individual arcs at fixed maximum block rank, over a balanced rational nomination box and independent positive rational resistance intervals, without operating filters. For any fixed reduced exponent greater than one with even denominator, exact arc-capacity comparison is Square-Root-Sum hard already on a single simple cycle with fixed integral nominations and positive integral resistances.

The quadratic statements are theorems B.1 and B.2; the power-law statements are theorems D.6 and D.12. Square-Root Sum asks whether $\sum_i \sqrt{a_i} \leq K$ for positive binary integers a_i, K . Whether it is decidable in polynomial time, or even lies in NP, is open (see also Ajdarów et al. [2, Section 5]); its known upper bounds are discussed in appendix A.5.

The arithmetic gadget uses a triangle whose two terminal branches have total resistances $(a - 1)^2$ and $(a - 1)^2/a$ for $a \geq 2$. Its unit-flow drop is

$$R_a = a + 1 - 2\sqrt{a} > 0.$$

Joining these triangles by unit bridges produces $\Delta = C - 2\sum_i \sqrt{a_i}$, and scaling by a positive common denominator makes every resistance integral without changing the weak comparison. Conversely, each unloaded cycle on a unit source-to-sink cactus contributes a rational number minus one positive radical, so their sum reduces to Square-Root Sum. Multiple-entry cycles have different local formulas, so this converse is not asserted for arbitrary nomination boxes. An existential-real formulation gives the general upper bound $\exists\mathbb{R} \subseteq \text{PSPACE}$ for exact quadratic MPD (proposition B.9).

For fractional powers, the monotone loop equation of one cycle can encode the radical sum directly at a rational circulation threshold. Additive algorithms avoid this exact comparison by using positive rational approximations to the laws. Positive quadrature fractions, with a fixed odd-power factor when necessary, preserve strict monotonicity and have a dense encoding polynomial in accuracy bits, and an explicit monotonicity estimate transfers constitutive error to flow and potential error. This adds network and parameter-recovery guarantees to established fractional-power approximation [12] and rigorous quadrature [31]. The water head-loss exponent $1.852 = 463/250$ [29, Example 2.3] falls under both parts of the theorem.

4 The observable determines the topology boundary

Replacing finite resistance sets by their interval hulls is attractive, since continuous coefficients can be eliminated through convex images, whereas a product of finite sets has exponentially many choices. Whether the replacement preserves an extremum depends on the graph *and* the objective. In this section a universal statement quantifies over every fixed balanced nomination and all independent compact positive resistance sets with attained endpoints, under quadratic laws and without operating filters.

Theorem 4.1 (Universal resistance-hull and region classifications). *For finite connected simple graphs, the universal interval-hull classes for maxima and minima are exactly those in table 1. The same classes characterize separate monotonicity in every resistance coordinate, with the other coordinates fixed. The attainable flow region is convex for every fixed nomination and positive resistance box exactly on cacti. Normalized potential regions and joint flow-potential regions are universally convex exactly on trees.*

Observable whose extrema must be preserved	Universal graph class
Every balanced weighted potential $c^\top \pi$	Trees
Every prescribed difference $\pi_s - \pi_t$	Cacti
Every linear flow objective $d^\top x$	Cacti
Every individual arc flow x_a	Series-parallel

Table 1: Exact equality of resistance-set and interval-hull extrema under fixed nominations, independent compact positive quadratic resistance sets, and no operating filters. Equality of extrema does not assert an exact scalar comparison algorithm.

These claims are theorems F.11 and F.13, with the individual-arc equivalence proved in theorem E.8. Counterexamples use rational data, one varying resistance, and strictly positive resistances on all other edges. The full-region classifications extend qualitatively to every fixed common power $\text{sgn}(x)|x|^q$ with $q > 0$, $q \neq 1$, without a uniform bit bound or algorithm (theorem F.14). With linear laws, by contrast, every graph has the hull property for potential objectives (appendix F).

4.1 Finite comparison and deterministic envelopes

A finite secant argument explains the positive individual-arc result without differentiating a law at zero. The difference of two passive states for the same nomination and two law profiles is an electrical network with positive secant resistances and explicit edge disturbances. For a target edge $a = (u, v)$, let j be the unit electrical current from u to v in that secant network. The target flow difference is a linear combination of the disturbances, with coefficients j_e for $e \neq a$ and $-(1 - j_a)$ on a , divided by the positive target secant resistance. By classical confluence, the signs of these currents do not depend on the positive resistances when the graph is series-parallel.

One may therefore select, pointwise in flow, a maximum or minimum allowed law on each edge according to the sign of its current j_e . The target state of the resulting deterministic envelope attains the uncertainty extremum, and compactness makes the selected law value realizable by an original parameter at that state. The construction is uniform in the nomination, although the realizing profile can depend on the state (theorem E.4, for independent compact continuous law families). For quadratic interval or finite resistance uncertainty the envelope is directional quadratic, and its primitive has a linear-size second-order-cone formulation. Rational convex optimization and recovery then give polynomial additive arc optimization

for fixed nominations on every series-parallel graph, without a rank bound (theorem E.6).

Deterministic feasibility for rated monotone characteristics has related series/parallel propagation methods: Zemanian’s report uses endpoint propagation and operating-point recovery [56, Conditions 2.1, Lemmas 4.1 and 5.1, Procedure 8.1]. Those deterministic procedures do not provide the present uncertainty-envelope and accuracy-bit output guarantees.

4.2 What fails when the objective changes

A triangle already separates weighted potentials from pairwise potentials. With edges oriented $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$, resistances $(1, 1, \theta)$, nominations $(2, -3, 1)$, and $c = (-5, 12, -7)$, the circulation q satisfies $6q + 3 - \theta q^2 = 0$ on $-1/2 < q < 0$. Its weighted potential is

$$W = -\frac{105}{4} - 12(q + 1/4)^2.$$

At $\theta = 24$, $q = -1/4$ and $W = -105/4$. At $\theta = 16/3$ and 144, the circulation is respectively $-3/8$ and $-1/8$ and both values are smaller by $3/16$. Thus a two-point set and its interval hull give different weighted-potential optima on one cycle, although this graph is in the universal class for every individual potential difference. Weighted cancellation changes the extremal geometry.

A rank-two theta block supplies the corresponding linear-flow counterexample; a rank-three K_4 supplies the individual-arc counterexample. Subdividing suitable uncertain paths encodes Subset Sum at the same small block rank, giving the discrete boundaries in table 2, with explicit rational gaps and positive integer resistances where stated. Multiplying all resistances scales a pressure gap but leaves flows unchanged; flow gaps are amplified by scaling nominations instead.

Two further distinctions matter. First, the envelope theorem and hull equality do not settle exact scalar arithmetic on unrestricted series-parallel graphs. A parallel probe on a cactus branch transfers its radical sum to an arc comparison, giving Square-Root-Sum hardness even with fixed unit nominations and fixed integral resistances; a restricted probe family has the converse reduction (theorem E.9). Second, hull preservation for an extremum does not preserve exact realization of a target flow. For a prescribed rational $\bar{x}$, conservation can be checked and the equations $A^\top \pi = (\beta_e \bar{x}_e | \bar{x}_e|)_e$ are linear in π, β , so interval realization is an LP on any graph. With two finite resistance choices per uncertain edge it is NP-complete on one simple cycle, and so is existential unit-capacity design on that cycle (theorem E.11). These are existential questions, whereas robust validation tests every scenario.

Objective and graph	Scenario assumptions	Computational conclusion
Potential difference; one rank-two block	Fixed small nominations; independent two-point resistances	NP-hard, with polynomial-bit gap (theorem E.1).
Individual arc; series-parallel, fixed block rank	Balanced nomination box; independent finite resistances	Exact algebraic extrema and robust capacity decision; includes every rank-at-most-two graph (corollary E.5).
Individual arc; rank-three block	Fixed small nominations; independent two-point resistances	Existential strict or weak threshold is NP-complete; complementary universal test is coNP-complete (theorem E.7).
Weighted potential; one cycle	Fixed nominations and three objective support vertices; independent finite resistances	NP-complete threshold attainment, coNP-complete robust upper bounds (theorem F.7).
Linear flow; rank-two block	Fixed nominations; independent finite resistances	NP-complete threshold attainment; includes unit-cost total absolute flow on a DAG (theorem F.9).
Individual arc; series-parallel with growing block rank	Balanced nomination box; fixed resistances	NP-hard optimization and coNP-hard robust upper bounds; no general membership claim (theorem E.10).

Table 2: Discrete and nomination boundaries for quadratic laws. All rows use unfiltered scenarios. Hardness gaps obstruct polynomial accuracy-bit algorithms unless P equals NP; they are not uniformly strong-hardness claims.

5 Weighted objectives and many interacting block responses

A terminal difference is a weighted potential with two nonzero coefficients. General weights create competing local contributions, so localization to one active block no longer suffices. This is a genuine complexity barrier: with growing objective support, weighted-potential threshold attainment is NP-complete on degree-three trees with fixed resistances and a balanced nomination box, and absolute-error $1/8$ approximation is NP-hard on this unnormalized family. At fixed support p , trees admit exact rational extrema, scenarios, flows, and potentials in $N^{O(p)}$ time (theorems F.2 and F.3).

5.1 Three accuracy-bit extensions

Theorem 5.1 (Weighted accuracy-bit algorithms). *Let $W = c^\top \pi$ with $c \in \mathbb{Q}^V$ and $\mathbf{1}^\top c = 0$. Consider any of: fixed positive rational asymmetric quadratic coefficients; independent positive rational symmetric coefficient intervals; or independent positive rational directional coefficient intervals. Scenarios have no operating filters. In each of the following cases, both extrema satisfy the rational value and original-scenario contract (3) in time polynomial in N_ε for the fixed parameters:*

1. *The graph is a cactus. Nominations form a rational balanced affine family on a nonempty compact rational polytope of fixed dimension d ; objective support and total cycle rank are unrestricted. Alternatively, nominations range over a balanced rational box and $p = |\text{supp } c|$ is fixed.*
2. *Maximum block rank is at most a fixed r_0 , and nominations range on one nonempty compact rational affine line segment. Objective support, number of noncactus blocks, and total cycle rank are unrestricted.*
3. *Put $\kappa(G) = \sum_{B: r_B \geq 2} r_B$. The total noncactus rank $\kappa(G)$ is bounded by a fixed κ_0 , and the balanced affine nomination domain has fixed dimension d . Alternatively, use a balanced rational nomination box and fixed objective support p . Cactus rank is unrestricted.*

In case 2 with fixed laws, the conclusion also holds for admissible densely encoded rational continuous piecewise-polynomial laws vanishing at zero, with fixed rational breakpoints and growing degrees and piece counts. That extension returns rational nominations and does not assert uncertain polynomial coefficients.

Case 1 is theorem G.1 and corollary G.9; case 2 is theorem J.4 and corollary J.5; case 3 is theorem J.6 and corollary J.7. By comparison, fixed *global* cycle rank r and fixed support p permit exact algebraic weighted extrema and algebraic scenarios under independent continuous resistance intervals (theorem F.4), because all nonlinear variables then fit in one common algebraic field. Fixed rank per block allows an unbounded number of independent fields and needs the separate summation arguments.

The support parameter can be refined. On a tree, after contracting edges with zero induced objective cut flow and orienting the rest along positive cut flow, only the k vertices without exactly one incoming and one outgoing edge matter, and exact optimization takes $N^{O(k)}$ time even with growing support

(theorem F.3). At fixed global rank, a supplied marked set with a constant source sign on each suppressed path gives a similar signed-path refinement, including returning paths (theorem F.4). Both conditions can be checked directly and do not require all objective weights to have one sign.

5.2 A face theorem and two ways to sum

Choose a rational tree flow w satisfying $Aw = c$. Then

$$W = c^\top \pi = w^\top A^\top \pi = \sum_e w_e g_e(x_e).$$

This places each contribution in its physical block. Prune subtrees without objective support, and contract a block if every induced objective sum at its attachments vanishes; both operations preserve the objective and admit rational nomination disaggregation. The incidence subtree spanning p objective vertices has $O(p)$ branching locations and ordinary chains, and an adjoint threshold can leave one selected block per ordinary chain plus the special blocks. Path suppression and a second positive-source perturbation then leave $O(r_0 p)$ free nominations on $n^{O(r_0 p)}$ closed faces, for a positive bound $r_0 \geq 1$ with $r_{\max} \leq r_0$ (theorem G.2). This face reduction does not depend on the coefficients. It is structural: it does not give an additive algorithm for arbitrarily many higher-block value functions of several parameters.

For quadratic laws on a cactus, every local circulation equation is piecewise quadratic. With uncertain coefficients, a single equality LP first eliminates the resistance choices: reduced costs are ordered by one weight threshold, and tied weights remain as a recoverable interval sum. The circulation candidates include boundaries, stationary points, flat pieces, and zero-flow cases, so a local maximum is selected from polynomially many rational or quadratic-radical formulas in the nomination parameters. Geometrically spaced rational polynomial panels approximate the square roots, and a common sign decomposition allows all block surrogates to be added in the *same* fixed-dimensional nomination coordinates. The surrogate is optimized exactly in a fixed number of variables, without one value variable per cycle. Global continuity then recovers original rational parameters, even if rounding changes local flow signs.

For higher-rank blocks on one nomination line, the local algebraic degree need not be two. The interface is instead a rational quantifier-free graph of a scalar Lipschitz semialgebraic function. Lemma J.2 isolates exceptional parameters, uses the Lipschitz bound on small neighborhoods, and constructs rational analytic interpolants on geometric panels elsewhere, with explicit

polynomial dependence on dense degree, coefficient bits, and accuracy bits. The resulting rational polynomials keep the original parameter t , so arbitrarily many block responses can be added and optimized without forming their joint algebraic field. This is why the one-line result allows an unrestricted number of higher-rank blocks.

For several nomination parameters, case 3 keeps all noncactus circulations in one exact core of dimension $d + \kappa(G)$ and compiles the cactus contributions. Here $\kappa(G)$ is a *sum*, not a maximum: a graph with a thousand rank-two blocks does not have small $\kappa(G)$, although the one-line theorem applies to it.

5.3 Exact parameters, filters, and limits

At fixed rational nominations on a cactus, the quadratic weighted algorithm gives an exactly optimizing rational coefficient profile, even with optional signed arc capacities and arbitrary objective support. Its local values have algebraic degree at most two, and the global value may be kept as their separate sum (theorem G.12). An exact profile need not be an endpoint profile: on the four-cycle in example G.13, the unique optimum is $(1, 2, 1, 1)$, with the second coefficient strictly inside $[1, 3]$. A polynomial exact scenario output is thus compatible both with an interior optimum and with the absence of an efficient exact comparator for its value.

For fixed-dimensional affine nominations, local capacity filters permit exact feasibility and additive algebraic scenarios of polynomial representation size. With fixed laws, closed bounded-degree semialgebraic filters in the nomination parameters and the flows of one block are also allowed; the joint coefficient theorem permits only the stated affine arc bounds. A supplied positive tightening margin gives a rational scenario feasible for the original capacities and near-optimal relative to the *tightened* optimum (theorem G.10). The reference optimum matters, because a small coefficient perturbation need not preserve an optimizer on an exact capacity boundary.

Indeed, rational inputs can force an irrational nomination. The four-cycle in example G.11 has flows $(q + t - 1, q - 2, q, q - 2)$ and resistances $(1, 1, 1, 2)$. The bounds $|x_3| \leq 1$ and $|x_4| \leq 1$ force $q = 1$; the cycle equation becomes $t^2 = 2$ for $1 \leq t \leq 2$. No rational feasible nomination exists. This is an output obstruction for filtered domains; it does not conflict with the unfiltered rational approximation theorems.

The open case, equation (87), is accuracy-bit optimization of a growing sum of bounded-rank block values in several shared nomination coordinates; a Lipschitz grid is polynomial only in $1/\varepsilon$, and this delimits the present methods rather than proving hardness.

6 Continuous design, correlations, and the direction of optimization

With fixed nominations, every cactus cycle has one scalar circulation. This permits richer coefficient dependence than independent edge boxes and separates correlations within a cycle from those across cycles.

For each edge let $g_e(x; \theta) = g_{e0}(x) + \sum_j \theta_j g_{ej}(x)$, with densely encoded rational polynomial pieces, fixed rational breakpoints, and admissible continuous strictly increasing complete laws vanishing at zero throughout the parameter domain; admissibility is a promise in this correlated model. The degree, number of pieces, and parameter dimension may grow. Bridge flows are fixed by the nomination. On a cycle C with edge signs $\chi_e \in \{\pm 1\}$ relative to a cyclic traversal, $x_e = x_e^0 + \chi_e q$ and the loop equation is

$$H_C(q, \theta) = \sum_{e \in C} \chi_e g_e(x_e^0 + \chi_e q; \theta) = 0.$$

(Reorienting reversed edges with the law $-g_e(-x; \theta)$ makes every $\chi_e = 1$; see appendix H.) This function is continuous, strictly increasing in q , and affine in θ ; rational bounds on total injection give a root bracket.

Theorem 6.1 (Cycle-local design and global capacity recovery). *Let G be a connected simple cactus with fixed rational balanced nominations and the admissible affine polynomial-law model above.*

1. *If each cycle has its own independent nonempty bounded rational parameter polytope, the attainable flow region is $x = x^0 + Zq$ with independent algebraic circulation intervals. Rational cycle-local signed capacities clip these intervals exactly. Feasibility, each interval endpoint, and a rational original parameter profile attaining it are computable in polynomial bit time. Linear flow objectives admit exactly optimizing rational profiles and additive scalar values, with no exact-comparison claim for a growing sum.*
2. *In the same independent-cycle model with those capacities, a densely encoded rational polynomial flow objective promised convex on $[-B - 1, B + 1]^m$, $B = \sum_v |b_v|$, admits an exactly feasible rational parameter scenario within ε of its minimum in polynomial accuracy-bit time. Degrees and number of cycles are unrestricted. For maximization of $g(Rx)$, the analogous additive scenario guarantee holds when $R \in \mathbb{Q}^{k \times m}$ has fixed row count k and the dense rational polynomial g is convex on a box containing the attainable measurements.*

3. *If one nonempty bounded rational parameter polytope instead couples all cycles, feasibility of signed arc capacities is rational linear feasibility and has a rational witness. Robust capacity testing uses rational LPs and returns rational violating profiles when needed. Individual arcs have exact algebraic extrema and rational optimizing profiles, including when other such capacities filter the domain.*

All complexity bounds count dense input and output bits. These statements exclude variable nominations and potential filters; part 2 excludes flow constraints coupling distinct cycle coordinates.

These are theorems H.2, H.3, H.5 and H.8. The few-measurement maximization also holds for independent finite quadratic resistance sets at fixed nominations, without capacity filters. Its running time is $N^{O(k)}$ times a polynomial in accuracy bits, and it promises a near-optimal scenario, not exact comparison of candidate values. A positive-semidefinite quadratic of fixed rank r uses at most $r + 1$ measurements, including its linear term (corollary H.9).

6.1 Why scalar roots admit exact rational parameter recovery

At a rational z , strict increase gives

$$\max_{\theta \in P} q(\theta) \geq z \iff \min_{\theta \in P} H_C(z, \theta) \leq 0.$$

The right side is an LP of any parameter dimension. A largest root is attained at a vertex of P : writing a profile as a convex combination of vertices writes its zero loop value as the same combination, so some vertex has a root no smaller. Vertex coordinates have polynomial bit bounds from determinants, so after substitution all vertex-piece roots satisfy a common degree and coefficient-height bound. Root separation then gives a polynomial number of bisection bits that distinguishes unequal candidate roots without enumerating vertices, and an LP vertex at the final threshold attains the exact extreme root. Theorem H.1 proves this for general monotone dense piecewise-polynomial root families, including repeated roots, roots at breakpoints, and lower-dimensional polytopes.

Capacities are equally direct: $q \geq a$ is equivalent to $H_C(a, \theta) \leq 0$, and $q \leq b$ to $H_C(b, \theta) \geq 0$. Both are linear in the original parameters, also under global correlations; independence across cycles is needed only to combine the scalar attainable intervals into a box. Another rational LP realizes a rational circulation target. With exact endpoint profiles, this permits recovery even when a clipped interval is an irrational singleton.

Dense convex minimization uses rational inner boxes approximating the algebraic circulation box, a controlled convex extension of a polynomial promised convex only on the stated cube, and exact rational oracles; the optimizer is mapped back to original coefficients with exact capacity feasibility. For convex maximization through k measurements, the circulation box projects to a zonotope with polynomially many exposed candidates at fixed k ; their algebraic values are compared approximately and their endpoint profiles recovered separately, as in the classical zonotope approach with explicit handling of irrational endpoints.

6.2 Independence and convexity do not make every objective easy

Independent coefficient intervals can realize a circulation cube on a bounded-data cactus. A coupled convex quadratic on that cube encodes Max-Cut. Convex maximization in all flows is therefore strongly NP-hard, even with unit terminal nominations, maximum degree three, and resistances in $[1, 16]$, and absolute-error $1/4$ approximation remains hard (theorem H.7). The fixed-measurement theorem thus depends on the dimension of the measured image, not only on the degree or convexity of the objective.

Global correlations can destroy even linear-flow tractability. Two cycles sharing one resistance already have a curved attainable image: in example I.1, two selected flows satisfy $x_2 = 2x_1/(1 + x_1)$ on $1/3 \leq x_1 \leq 1/2$. Their difference has its strict maximum at the interior shared resistance $\theta = 2$, although its two endpoint values agree. This example proves no hardness, but shows that independent-cycle endpoint reasoning does not carry over. A separate paired-triangle construction proves strong NP-hardness of total-flow maximization and minimum dissipation over a global resistance polytope, with all resistances in $[1, 3]$, positive flows, and a fixed absolute gap (theorem I.2).

For symmetric quadratic laws write

$$D(\beta) = b^\top \pi(\beta) = \sum_e \beta_e |x_e(\beta)|^3 = 3 \min_{Ax=b} \sum_e \beta_e |x_e|^3 / 3.$$

This is the constitutive dissipation; the primitive energy is $D/3$ at the physical state. As a pointwise infimum of linear functions of β , D is concave, so the direction of optimization is decisive.

Theorem 6.2 (Maximum dissipation over a global polytope). *On any connected graph with fixed rational balanced nominations and symmetric quadratic laws, let resistances range over a nonempty bounded rational polytope with*

supplied positive rational lower and upper bounds. Maximum physical dissipation has an exact formulation with two second-order cones per edge and a rational admissible profile within ε of its maximum computable in polynomial accuracy-bit time. No operating filters are imposed. On a cactus, rational signed arc capacities may be added: infeasibility is decided, or the same guarantee holds with every original capacity satisfied exactly.

This is theorem I.3 and corollary I.4. Cubic energy duality is classical and is developed for the symmetric network model in Groß et al. [29]; concavity of effective resistance in linear networks is also classical [48]. Our contribution includes the global resistance-polytope formulation, the explicit rational output and precision argument, and the original-instance certificate of proposition I.5. The theorem does not extend to arbitrary potential filters: on a triangle, an upper bound on one terminal drop already makes the feasible resistance set nonconvex (appendix I).

6.3 Rational witnesses need margins beyond the scalar setting

Beyond cacti, exact capacities can force irrational coefficients even when every capacity is an ordinary positive-width upper bound; a simple rank-two series-parallel instance with one uncertain resistance is an example (proposition I.6). Rational feasibility on a cactus thus reflects the scalar loop equation, not a general property of rational physical models.

Strict slack restores a quantitative rational witness statement on any graph. For fixed nonzero nominations put $M = \frac{1}{2}\|b\|_1$. If two symmetric or directional coefficient profiles have lower bound $\beta_L > 0$, and q_e is the absolute symmetric coefficient change or the larger of the two directional changes on edge e , then

$$\|x' - x\|_\infty \leq \frac{M}{2\beta_L} \sum_e q_e. \quad (5)$$

Proposition I.8 proves this bound, including zero flows and reversals. Consequently a feasible real profile with a supplied rational capacity slack can be rounded safely (corollary I.9). This is a conditional existence and rounding guarantee, not an algorithm for finding the initial strictly feasible profile.

7 Certificates for physical outputs and original scenarios

A numerical solution raises a question that the preceding algorithms do not: what does its actual output prove? For a fixed directional quadratic network

with at least one edge, rational nominations, and positive rational coefficients, rational witnesses answer this independently of a solver's stopping rule and can be attached to a verified uncertainty-envelope reduction.

Write $E_e(t) = \beta_e^{\text{sgn}(t)} |t|^3/3$ and $E = \sum_e E_e$, where the positive coefficient is selected at zero by convention. Let x^* minimize E on $Ax = b$, and let $\beta_L = \min_{e,\sigma} \beta_e^\sigma > 0$. The statement below uses a connected graph for simplicity; the appendix and the deterministic certificate verifiers also allow several components with feasible nominations and separate gauges. The original-envelope reader and the numerical conservation-repair producer require a connected simple loopless graph.

Theorem 7.1 (Rational energy and conserved-curvature certificates). *Suppose $y \in \mathbb{Q}^m$ satisfies $Ay = b$, $p \in \mathbb{Q}^V$, and rational $\zeta_e \geq 0$ satisfy $\zeta_e^2 \beta_e^{\text{sgn}((A^\top p)_e)} \geq |(A^\top p)_e|^3$. Then*

$$L = b^\top p - \frac{2}{3} \sum_e \zeta_e \leq E(x^*) \leq E(y), \quad \delta = E(y) - L \geq 0.$$

Every rational $\eta \geq 0$ with $\beta_L \eta^3 \geq 6\delta$ certifies $\|x^ - y\|_\infty \leq \eta$.*

More generally, suppose verified rational intervals $[l_e, u_e]$ contain both y_e and x_e^ . Set $h_e = 2\beta_e^+ l_e$ when $l_e > 0$, $h_e = -2\beta_e^- u_e$ when $u_e < 0$, and $h_e = 0$ otherwise. For a rational linear goal w , let v be rational and put $s = w - A^\top v$. If $s_e = 0$ wherever $h_e = 0$, define $C(v) = \sum_{h_e > 0} s_e^2/h_e$. Every rational $r \geq 0$ with $r^2 \geq 2\delta C(v)$ certifies $|w^\top(x^* - y)| \leq r$. The least factor C_* is attained by rational linear algebra whenever the zero-edge condition is feasible. For $\delta > 0$, $\sqrt{2\delta C_*}$ is the exact supremum of $|w^\top d|$ over $Ad = 0$, $\frac{1}{2} \sum_e h_e d_e^2 \leq \delta$. If the zero-edge condition is infeasible, that supremum is infinite. All displayed witness conditions are checkable by rational arithmetic in polynomial time in their encoded lengths.*

The first assertion is proposition K.1; the goal assertions are theorem K.5 and proposition K.6. Both require the trial flow to be exactly conserved, $Ay = b$; a small conservation residual cannot replace this. Sharpness refers to the *conserved quadratic error set* just stated, not to every physical error or to the additional coordinate intervals. At zero gap, $x^* = y$ and all goals are exact without a curvature condition.

7.1 From energy to intervals and linear goals

The scalar conjugate is $E_e^*(d) = \frac{2}{3} \sqrt{|d|^3 / \beta_e^{\text{sgn}(d)}}$. Rational upper root bounds make the dual lower energy check exact, and uniform cubic convexity converts

the gap into a flow radius. This specializes classical Fenchel duality and primal-dual error control [6, 14].

Separate-edge bounds can be sharpened using the reverse Bregman expression

$$D_e(y_e, z) = E_e(y_e) - E_e(z) - E'_e(z)(y_e - z).$$

At the physical state, conservation cancels the linear terms in the sum, so $\sum_e D_e(y_e, x_e^*) = E(y) - E(x^*) \leq \delta$ with every term nonnegative. Rational outward boundaries at which $D_e(y_e, z) \geq \delta$ therefore enclose x_e^* and can be refined by rational bisection (proposition K.2). This is the convex-function divergence of Bregman [16] with the unknown state in the second argument; exchanging the arguments does not preserve the gap identity. Monotone law evaluation and path sums then give deterministic potential-difference and weighted-potential intervals.

Separate edges discard conservation information. A stronger object is the compact conserved energy set $S(y) = \{x : Ax = b, E(x) \leq E(y)\}$, which contains x^* . For a linear goal, its support dual uses rational multipliers $\lambda > 0, v$ and upper roots satisfying $R_e^2 \lambda \beta_e^{\text{sgn}(s_e)} \geq |s_e|^3$, $s = w - A^\top v$. They certify

$$w^\top x \leq \lambda E(y) + v^\top b + \frac{2}{3} \sum_e R_e \quad (x \in S(y)).$$

Applying the construction to $-w$ gives an interval. If $E(y) > E(x^*)$, the infimum of the rational bounds equals the exact support value, by Slater duality and rational approximation (theorem K.4). This is a completeness statement for witnesses, not a polynomial-time generic producer. The curvature certificate of theorem 7.1 is a directly computable quadratic alternative based on the classical electrical cut/cycle projection [37, Chapter 2, Section 4]. Its rational KKT system remains valid with singular gauges and zero-curvature edges; it fails exactly for a goal that detects a circulation supported on the zero-curvature subgraph.

For a worked comparison, take two source-to-sink paths each of length L , coefficient one, and total transfer two. The physical flow is one on every edge. A rational conserved trial with $1 + \varepsilon$ on one path and $1 - \varepsilon$ on the other, $0 < \varepsilon < 1$, has exact gap $2L\varepsilon^2$. The exact support interval for a selected first-path edge has width 2ε ; rational upper and lower dual witnesses attain it when ε is rational. Separate-edge Bregman intervals have width $2\sqrt{2L}\varepsilon + o(\varepsilon)$, whereas the optimized curvature interval has width $2\varepsilon + o(\varepsilon)$ for fixed L as $\varepsilon \downarrow 0$. Example K.7 gives the exact witnesses and asymptotics. The example isolates the information recovered by conservation; its physical state is elementary.

7.2 Certifying the original uncertainty question

A correct certificate for a deterministic network does not by itself prove that this network is the envelope of a stated uncertainty instance. The integrated verifier therefore checks the original graph, nominations, allowed resistance intervals or finite sets, target edge, target sense, absence of a K_4 minor, and exact electrical signs, and reconstructs the target-specific envelope from these data. It then verifies one conserved-flow certificate for that envelope and a second for the chosen original resistance scenario. Rational interval subtraction gives a certified scenario loss; optional curvature witnesses sharpen the target bounds.

Resolved envelope signs give an exactly optimal original endpoint scenario, even when its state is irrational. At an unresolved sign, intervals bound how much choosing the wrong directional coefficient can matter, and the loss is not set to zero. Exact optimality also holds when tied coefficients or an exact zero flow make the choice immaterial (proposition K.3). Original-scenario optimality is thus separated from exact evaluation of its flow. Common rational scaling of nominations and resistances preserves all energy, interval, and goal inequalities exactly (equation (109)).

7.3 What the implementation and saved evidence establish

The supplementary archive contains the source, saved rational witnesses, exact verifiers, and reproduction scripts; the core certificate replay needs only Python’s standard library, with checks active under `python -S -O`. The numerical producer uses CVXPY and Clarabel to propose flows and potentials, then repairs conservation and constructs rational bounds. It accepts no requested accuracy guarantee and does not implement every algorithm of this paper; the supplied-block weighted solver likewise implements neither graph decomposition nor the varying-nomination quantifier-elimination algorithms. A Lean formalization proves the deterministic certificate guarantees, but not the envelope mapping, the software, or the bit-complexity claims; appendix K states its exact scope.

The retained six-edge witness has uniform width below $3.405 \cdot 10^{-4}$, maximum Bregman flow width below $2.894 \cdot 10^{-6}$, and maximum conservation-aware goal width below $9.447 \cdot 10^{-7}$. These are verified enclosure widths, not observed true errors. The original-instance benchmark contains ten certified cases, nine with exact endpoint optimality. It includes an 80-edge graph of rank 39, a full coefficient ratio of $3 \cdot 10^6$, supply 10^{-8} , reversed orientations, zero nominations, and unresolved signs. The ambiguous-sign case keeps a

positive loss bound; goal information improves its verified target width and loss bound without resolving the sign exactly. Appendix K gives the full values, conditions, and implementation roles.

One case uses the topology and pipe geometry of the open dataset accompanying Vrachimis et al. [53], distributed at doi:10.5281/zenodo.1185136. The packaged instance contains synthetic quadratic coefficients, nominations, and intervals. It does not reproduce the source hydraulic laws or leak estimates, and the archive does not redistribute the original INP file. Saved exact checks establish the particular witnesses; the proofs establish the universal statements.

8 Conclusion

The results separate three sources of difficulty: interacting cycles fix the dimension of the local algebra, objective support and independence decide whether local responses can be optimized separately, and exact comparison is an arithmetic question even on simple graphs. Rational scenarios, exact local states, and additive intervals are therefore different outputs, and the certificates keep them apart in computation.

Three questions lie beyond the theorems proved here. The exact threshold complexity of multiple-entry cactus MPD is not settled by the Square-Root-Sum equivalence for a single source and sink. The unrestricted several-parameter sum of bounded-rank weighted block responses still needs an accuracy-bit algorithm or a matching obstruction. The fixed-rank algorithms have parameter-dependent exponents; whether FPT bounds hold is separate from the proved XP guarantees.

References

- [1] Akshay Agrawal and Stephen Boyd. Disciplined quasiconvex programming. *Optimization Letters*, 14:1643–1657, 2020. doi: 10.1007/s11590-020-01561-8. URL <https://web.stanford.edu/~boyd/papers/pdf/dqcp.pdf>.
- [2] Michal Ajdarów, James C. A. Main, Petr Novotný, and Mickael Randour. Taming infinity one chunk at a time: Concisely represented strategies in one-counter MDPs. In *52nd International Colloquium on Automata, Languages, and Programming (ICALP 2025)*, volume 334 of *Leibniz International Proceedings in Informatics*, pages 138:1–138:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025. doi: 10.4230/LIPIcs.ICALP.2025.138. URL <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ICALP.2025.138>.
- [3] Eric Allender, Peter Bürgisser, Johan Kjeldgaard-Pedersen, and Peter Bro Miltersen. On the Complexity of Numerical Analysis. *SIAM Journal on Computing*, 38(5):1987–2006, 2009. doi: 10.1137/070697926. URL <https://doi.org/10.1137/070697926>.
- [4] Denis Aßmann. *Exact Methods for Two-Stage Robust Optimization with Applications in Gas Networks*. PhD thesis, Friedrich-Alexander-Universität Erlangen-Nürnberg, 2019. URL <https://d-nb.info/1196351791/34>. Published by FAU University Press, FAU Studies Mathematics & Physics 16.
- [5] Denis Aßmann, Frauke Liers, Michael Stingl, and Juan C. Vera. Deciding Robust Feasibility and Infeasibility Using a Set Containment Approach: An Application to Stationary Passive Gas Network Operations. *SIAM Journal on Optimization*, 28(3):2489–2517, 2018. doi: 10.1137/17M112470X. URL <https://doi.org/10.1137/17M112470X>. Locators refer to arXiv:1808.10241v1.
- [6] Sören Bartels and Marijo Milicevic. Primal-dual gap estimators for a posteriori error analysis of nonsmooth minimization problems. *ESAIM: Mathematical Modelling and Numerical Analysis*, 54(5):1635–1660, 2020. doi: 10.1051/m2an/2019074. URL <https://numdam.org/articles/10.1051/m2an/2019074/>.
- [7] Saugata Basu. Algorithms in Real Algebraic Geometry: A Survey. arXiv:1409.1534 [math.AG, cs.CC, cs.CG, cs.SC], 2014. URL <https://arxiv.org/abs/1409.1534>.

- [8] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, 2 edition, 2006. doi: 10.1007/3-540-33099-2. URL <https://doi.org/10.1007/3-540-33099-2>.
- [9] Gal Binyamini and Dmitry Novikov. Complex cellular structures. *Annals of Mathematics*, 190(1):145–248, 2019. doi: 10.4007/annals.2019.190.1.3. URL <https://arxiv.org/abs/1802.07577>.
- [10] Garrett Birkhoff and J. B. Diaz. Non-linear network problems. *Quarterly of Applied Mathematics*, 13(4):431–443, 1956. doi: 10.1090/qam/77398. URL <https://doi.org/10.1090/qam/77398>.
- [11] Johannes Blömer. Computing sums of radicals in polynomial time. In *Proceedings of the 32nd Annual Symposium on Foundations of Computer Science*, pages 670–677, 1991. doi: 10.1109/sfcs.1991.185434. URL <https://doi.org/10.1109/SFCS.1991.185434>.
- [12] Andrea Bonito and Joseph E. Pasciak. Numerical Approximation of Fractional Powers of Elliptic Operators. *Mathematics of Computation*, 84(295):2083–2110, 2015. doi: 10.1090/s0025-5718-2015-02937-8. URL <https://doi.org/10.1090/S0025-5718-2015-02937-8>.
- [13] Julius Borcea, Rikard Bøgvad, and Boris Shapiro. On rational approximation of algebraic functions. *Advances in Mathematics*, 204(2):448–480, 2006. doi: 10.1016/j.aim.2005.06.002. URL <https://arxiv.org/abs/math/0409353>. Theorem locators refer to arXiv:math/0409353v2.
- [14] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004. URL https://www.seas.ucla.edu/~vandenbe/cvxbook/bv_cvxbook.pdf.
- [15] René Brandenberg and Paul Stursberg. Extremal solutions for network flow with differential constraints. *Journal of Optimization Theory and Applications*, 207(3):70, 2025. doi: 10.1007/s10957-025-02792-4. URL <https://arxiv.org/abs/2308.07396>. Locators refer to the August 16, 2023 preprint, titled Extremal solutions for Network Flow with Differential Constraints—A Generalization of Spanning Trees.
- [16] Lev M. Bregman. The relaxation method of finding the common point of convex sets and its application to the solution of problems in convex programming. *USSR Computational Mathematics and Mathematical*

- Physics*, 7(3):200–217, 1967. doi: 10.1016/0041-5553(67)90040-7. URL <https://www.lix.polytechnique.fr/~nielsen/Bregman1966.pdf>.
- [17] John F. Canny. Some Algebraic and Geometric Computations in PSPACE. Technical Report UCB/CSD-88-439, EECS Department, University of California, Berkeley, August 1988. URL <https://www2.eecs.berkeley.edu/Pubs/TechRpts/1988/6041.html>.
 - [18] P. Chaffoureaux and M. Hasler. Monotonicity in nonlinear resistive circuits. In *IEEE International Symposium on Circuits and Systems*, pages 395–398, 1990. doi: 10.1109/ISCAS.1990.112055.
 - [19] Enric Cosme Llópez and Damien Pous. K_4 -free graphs as a free algebra. In *42nd International Symposium on Mathematical Foundations of Computer Science*, volume 83 of *Leibniz International Proceedings in Informatics*, pages 76:1–76:14. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2017. doi: 10.4230/LIPIcs.MFCS.2017.76. URL <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.MFCS.2017.76>.
 - [20] Daniel Nicolas Dadush. *Integer Programming, Lattice Algorithms, and Deterministic Volume Estimation*. PhD thesis, Georgia Institute of Technology, 2012. URL <https://homepages.cwi.nl/~dadush/papers/dadush-thesis.pdf>.
 - [21] Alberto Del Pia, Santanu S. Dey, and Marco Molinaro. Mixed-integer quadratic programming is in NP. *Mathematical Programming*, 162(1–2):225–240, 2017. doi: 10.1007/s10107-016-1036-0. URL <https://arxiv.org/abs/1407.4798>. Theorem locators refer to the author manuscript dated October 10, 2018.
 - [22] R. J. Duffin. Nonlinear networks. IIa. *Bulletin of the American Mathematical Society*, 53(10):963–971, 1947. doi: 10.1090/s0002-9904-1947-08917-5. URL <https://doi.org/10.1090/S0002-9904-1947-08917-5>.
 - [23] R. J. Duffin. Topology of series-parallel networks. *Journal of Mathematical Analysis and Applications*, 10:303–318, 1965. doi: 10.1016/0022-247x(65)90125-3. URL [https://doi.org/10.1016/0022-247X\(65\)90125-3](https://doi.org/10.1016/0022-247X(65)90125-3).
 - [24] David Eppstein. Parallel recognition of series-parallel graphs. *Information and Computation*, 98(1):41–55, 1992. doi: 10.1016/0890-5401(92)90041-d. URL [https://doi.org/10.1016/0890-5401\(92\)90041-D](https://doi.org/10.1016/0890-5401(92)90041-D).

- [25] Kousha Etessami and Mihalis Yannakakis. On the Complexity of Nash Equilibria and Other Fixed Points. *SIAM Journal on Computing*, 39(6): 2531–2597, 2010. doi: 10.1137/080720826. URL <https://doi.org/10.1137/080720826>.
- [26] Jean-Albert Ferrez, Komei Fukuda, and Thomas M. Lieblich. Solving the fixed rank convex quadratic maximization in binary variables by a parallel zonotope construction algorithm. *European Journal of Operational Research*, 166(1):35–50, 2005. doi: 10.1016/j.ejor.2003.04.011. URL <https://www.cs.mcgill.ca/~fukuda/download/paper/qpzono040429.pdf>. Author revision dated April 29, 2004.
- [27] Tatiana González Grandón, Holger Heitsch, and René Henrion. A joint model of probabilistic/robust constraints for gas transport management in stationary networks. *Computational Management Science*, 14(3): 443–460, 2017. doi: 10.1007/s10287-017-0284-7. URL https://www.wias-berlin.de/preprint/2401/wias_preprints_2401.pdf. Section locators refer to WIAS Preprint 2401.
- [28] Claudia Gotzes, Holger Heitsch, René Henrion, and Rüdiger Schultz. On the quantification of nomination feasibility in stationary gas networks with random load. *Mathematical Methods of Operations Research*, 84(2):427–457, 2016. doi: 10.1007/s00186-016-0564-y. URL https://www.wias-berlin.de/people/heitsch/GHHS16_Preprint.pdf. Author manuscript: WIAS Preprint 2158, titled Feasibility of nominations in stationary gas networks with random load (2015).
- [29] Martin Groß, Marc E. Pfetsch, Lars Schewe, Martin Schmidt, and Martin Skutella. Algorithmic Results for Potential-Based Flows: Easy and Hard Cases. Technical report, Optimization Online, 2017. URL <https://optimization-online.org/2017/08/6185/>. Published in Networks 73(3):306–324, 2019, doi:10.1002/net.21865.
- [30] M. Hasler and C. Wang. Parameter tolerances in non-linear resistive circuits: worst case analysis based on monotonicity. In *International Symposium on Nonlinear Theory and its Applications*, pages 841–846, Hawaii, USA, 1993.
- [31] Fredrik Johansson and Marc Mezzarobba. Fast and rigorous arbitrary-precision computation of Gauss–Legendre quadrature nodes and weights. *SIAM Journal on Scientific Computing*, 40(6):C726–C747, 2018. doi:

- 10.1137/18M1170133. URL https://marc.mezzarobba.net/ecrits/JohanssonMezzarobba_Legendre_v3_2018.pdf.
- [32] Max Klimm, Marc E. Pfetsch, Martin Skutella, and Lea Strubberg. Approximating the Network Design Problem for Potential-Based Flows. arXiv (cs.DM), 2026. URL <https://arxiv.org/abs/2604.26882>.
 - [33] Martine Labbé, Fränk Plein, and Martin Schmidt. Bookings in the European gas market: Characterisation of feasibility and computational complexity results. *Optimization and Engineering*, 21(1):305–334, 2020. doi: 10.1007/s11081-019-09447-0. URL <https://doi.org/10.1007/s11081-019-09447-0>.
 - [34] Martine Labbé, Fränk Plein, Martin Schmidt, and Johannes Thürauf. Deciding feasibility of a booking in the European gas market on a cycle is in P for the case of passive networks. *Networks*, 78(2):128–152, 2021. doi: 10.1002/net.22003. URL <https://optimization-online.org/wp-content/uploads/2019/11/7472.pdf>.
 - [35] Retsef Levi, Georgia Perakis, and Gonzalo Romero. A continuous knapsack problem with separable convex utilities: Approximation algorithms and applications. *Operations Research Letters*, 42:367–373, 2014. doi: 10.1016/j.orl.2014.06.007. URL <https://www-2.rotman.utoronto.ca/facbios/file/CKP%20ORL.pdf>.
 - [36] Miguel Sousa Lobo, Lieven Vandenbergh, Stephen Boyd, and Hervé Lebret. Applications of second-order cone programming. *Linear Algebra and its Applications*, 284:193–228, 1998. doi: 10.1016/S0024-3795(98)10032-0. URL <https://web.stanford.edu/~boyd/papers/pdf/socp.pdf>.
 - [37] Russell Lyons and Yuval Peres. *Probability on Trees and Networks*. Cambridge University Press, 2016. URL https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf. Cited text: corrected paperback version of 19 August 2026.
 - [38] Nimrod Megiddo. Combinatorial optimization with rational objective functions. *Mathematics of Operations Research*, 4(4):414–424, 1979. URL <https://theory.stanford.edu/~megiddo/pdf/rational.pdf>.
 - [39] Maurice Mignotte. An inequality about factors of polynomials. *Mathematics of Computation*, 28(128):1153–1157, 1974. URL <https://www.jstor.org/stable/2005373>.

- [40] G. J. Minty. Monotone networks. *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences*, 257(1289): 194–212, 1960. doi: 10.1098/rspa.1960.0144. URL <https://doi.org/10.1098/rspa.1960.0144>.
- [41] Sidhant Misra, Marc Vuffray, and Michael Chertkov. Maximum Throughput Problem in Dissipative Flow Networks with Application to Natural Gas Systems. arXiv preprint arXiv:1504.02370, 2015. URL <https://arxiv.org/abs/1504.02370>.
- [42] Shmuel Onn and Uriel G. Rothblum. Convex combinatorial optimization. *Discrete & Computational Geometry*, 32(4):549–566, 2004. doi: 10.1007/s00454-004-1138-y. URL <https://arxiv.org/abs/math/0309083>. Locators refer to arXiv:math/0309083.
- [43] Knut Petras. Self-validating integration and approximation of piecewise analytic functions. *Journal of Computational and Applied Mathematics*, 145(2):345–359, 2002. doi: 10.1016/S0377-0427(01)00586-6. URL [https://doi.org/10.1016/S0377-0427\(01\)00586-6](https://doi.org/10.1016/S0377-0427(01)00586-6).
- [44] Marc E. Pfetsch, Martin Schmidt, Martin Skutella, and Johannes Thürauf. Potential-Based Flows—An Overview. Optimization Online preprint, 2026. URL <https://optimization-online.org/2026/01/potential-based-flows-an-overview/>.
- [45] Rico Konrad Béla Raber. *Optimization, Reduction, and Robustness of Potential-Based Flow Networks*. PhD thesis, Technische Universität Berlin, 2022. URL <https://d-nb.info/1258349914/34>.
- [46] James Renegar. On the computational complexity and geometry of the first-order theory of the reals. Part III: Quantifier elimination. *Journal of Symbolic Computation*, 13(3):329–352, 1992. doi: 10.1016/s0747-7171(10)80005-7. URL [https://doi.org/10.1016/S0747-7171\(10\)80005-7](https://doi.org/10.1016/S0747-7171(10)80005-7).
- [47] Martin Robinius, Lars Schewe, Martin Schmidt, Detlef Stolten, Johannes Thürauf, and Lara Welder. Robust optimal discrete arc sizing for tree-shaped potential networks. *Computational Optimization and Applications*, 73(3):791–819, 2019. doi: 10.1007/s10589-019-00085-x. URL <https://optimization-online.org/wp-content/uploads/2018/02/6447.pdf>. Locators refer to the November 24, 2018 author manuscript.

- [48] Claude E. Shannon and D. W. Hagelbarger. Concavity of resistance functions. *Journal of Applied Physics*, 27(1):42–43, 1956. doi: 10.1063/1.1722193.
- [49] Johannes Thürauf. Deciding the feasibility of a booking in the European gas market is coNP-hard. *Annals of Operations Research*, 318(3):591–618, 2022. doi: 10.1007/s10479-022-04732-1. URL <https://doi.org/10.1007/s10479-022-04732-1>. Locators refer to the author manuscript dated April 6, 2022.
- [50] Johannes Thürauf, Julia Grübel, and Martin Schmidt. Adjustable robust nonlinear network design without controllable elements under load scenario uncertainties. *Mathematical Programming*, 218(1–2):403–437, 2026. doi: 10.1007/s10107-025-02207-2. URL <https://doi.org/10.1007/s10107-025-02207-2>.
- [51] Stephen A. Vavasis. Quadratic programming is in NP. *Information Processing Letters*, 36(2):73–77, 1990. doi: 10.1016/0020-0190(90)90100-C.
- [52] Antoine Vigneron. Geometric optimization and sums of algebraic functions. *ACM Transactions on Algorithms*, 10(1):1–20, 2014. doi: 10.1145/2532647. URL <https://antoinevigneron.github.io/manuscripts/rational.pdf>. Theorem and section locators refer to the author manuscript dated October 21, 2011.
- [53] Stelios G. Vrachimis, Demetrios G. Eliades, and Marios M. Polycarpou. Real-time hydraulic interval state estimation for water transport networks: a case study. *Drinking Water Engineering and Science*, 11:19–24, 2018. doi: 10.5194/dwes-11-19-2018. URL <https://dwes.copernicus.org/articles/11/19/2018/>.
- [54] Changlu Wang and Martin Hasler. Convexity of resistive circuit characteristics. Technical report, Department of Electrical Engineering, EPFL, n.d. URL <https://infoscience.epfl.ch/bitstreams/d2fb4e43-bb7e-42b7-977c-39560d9e7d46/download>. Undated report; Theorem 4, p. 13, and conclusion, p. 21.
- [55] Yosef Yomdin. Smooth parametrizations in dynamics, analysis, diophantine and computational geometry. arXiv:1406.1719v2, 2014. URL <https://arxiv.org/abs/1406.1719>.
- [56] A. H. Zemanian. Does the operating point of a series-parallel network of monotone resistors satisfy all resistor ratings? *IEEE*

Transactions on Circuits and Systems I: Fundamental Theory and Applications, 46(10):1291–1295, 1999. doi: 10.1109/81.795844.
URL <https://dspace.sunyconnect.suny.edu/server/api/core/bitstreams/def0da05-b31a-4254-8d24-457cf05b544d/content>.
Locators refer to the expanded CEAS Technical Report 736, April 4, 1996.

Complete technical development

The appendices contain complete statements and proofs of all results used in the main text, in dependency order. Each appendix states its local conventions.

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A Preliminaries

The algorithms below separate two questions: how the graph restricts an optimizing scenario, and how accurately its physical state can be computed from binary input. This section fixes the passive-flow model, proves the block decomposition used in the structural arguments, and states the arithmetic and output conventions that separate exact comparison from high-precision approximation.

A.1 Networks, laws, and passive states

Let $G = (V, E)$ be a finite connected loopless graph, with $n = |V|$ and $m = |E|$. Parallel edges are allowed unless simplicity is stated. Choose an orientation for each edge. An arc $e = (u, v)$ has tail u and head v ; its column in the incidence matrix $A \in \mathbb{R}^{V \times E}$ has $+1$ at u and -1 at v . A nomination $b \in \mathbb{R}^V$ specifies injections (positive entries) and withdrawals (negative entries), and is balanced: $\mathbf{1}^\top b = 0$. Flows $x \in \mathbb{R}^E$ are signed relative to the orientation. A passive state consists of a flow and potentials $\pi \in \mathbb{R}^V$ satisfying

$$Ax = b, \quad A^\top \pi = g(x) := (g_e(x_e))_{e \in E}. \quad (6)$$

A general edge law $g_e : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and strictly increasing, with $g_e(0) = 0$; neither surjectivity nor a growth condition is assumed. The symmetric quadratic law is $g_e(t) = \beta_e t|t|$, with $\beta_e > 0$. The asymmetric quadratic law has two positive coefficients:

$$g_e(t) = \begin{cases} \beta_e^+ t^2, & t \geq 0, \\ -\beta_e^- t^2, & t \leq 0. \end{cases}$$

The two formulas agree at zero. Reversing an edge replaces x_e by $-x_e$ and its law by $\tilde{g}_e(t) = -g_e(-t)$, so symmetric coefficients are unchanged and asymmetric coefficients are swapped.

Theorem A.1 (Existence and uniqueness). *For every balanced nomination on a connected graph and every family of general edge laws as above, a passive state exists. Its flow is unique, and its potentials are unique up to an additive constant. The flow is exactly the minimizer of the primitive energy*

$$\mathcal{E}(x) = \sum_{e \in E} F_e(x_e), \quad F_e(t) = \int_0^t g_e(u) du, \quad \text{over } \{x : Ax = b\}. \quad (7)$$

Proof. Choose a rooted spanning tree. Route each subtree's total nomination through its parent edge, with the sign prescribed by the orientation, and set other flows to zero. Conservation holds at every nonroot vertex; balance gives it at the root. This constructs a feasible x^0 .

Each F_e is continuously differentiable with strictly increasing derivative g_e , hence strictly convex. Also $F_e(t) \geq 0$. For $t \geq 1$,

$$F_e(t) \geq g_e(1)(t - 1),$$

and for $t \leq -1$,

$$F_e(t) = - \int_t^0 g_e(u) du \geq [-g_e(-1)](|t| - 1).$$

With $a_e = \min\{g_e(1), -g_e(-1)\} > 0$ we get $F_e(t) \geq a_e|t| - a_e$ for all t , including $|t| \leq 1$. If $m > 0$, put $a = \min_e a_e > 0$; then

$$\mathcal{E}(x) \geq a \|x\|_1 - \sum_e a_e.$$

For $x = x^0 + z$ with $z \in \ker A$, this lower bound is at least $a \|z\|_1 - a \|x^0\|_1 - \sum_e a_e$, so sublevel sets on the closed affine feasible space are bounded. The nonempty sublevel set at $\mathcal{E}(x^0)$ is compact, and continuity gives a minimizer. Strict convexity gives uniqueness. If $m = 0$, the connected graph has one vertex and the assertion is immediate.

At the minimizer every directional derivative along $\ker A$ vanishes, so $g(x) \in (\ker A)^\perp = \text{im } A^\top$, giving a multiplier π with $A^\top \pi = g(x)$. No constraint qualification is needed, since $(\ker A)^\perp = \text{im } A^\top$ holds for every matrix, including the rank-deficient incidence matrix. Conversely, if equation (6) holds, convexity gives, for every feasible y ,

$$\mathcal{E}(y) \geq \mathcal{E}(x) + g(x)^\top (y - x) = \mathcal{E}(x) + \pi^\top A(y - x) = \mathcal{E}(x).$$

Finally, $A^\top(\pi - \pi') = 0$ makes $\pi - \pi'$ equal at adjacent vertices, hence constant by connectivity. $\square$

The variational principle is classical. In resistance variables, Birkhoff and Diaz [10, Section 5, Theorems 2' and 3'] characterize the Neumann solution by stationarity of the sum of primitives and obtain existence when those primitives tend to $+\infty$ in both directions; the bound above verifies that condition here. Duffin [22, Theorems 2 and 3] proves existence and uniqueness in conductivity variables (current as a function of potential drop).

His hypotheses apply when each g_e is onto $\mathbb{R}$, for then the conductivity g_e^{-1} is defined on all of $\mathbb{R}$, continuous, strictly increasing, and unbounded in both directions; we do not assume this. Minty [40, Theorem 6.4 with its Corollary I, and the corollary to Theorem 8.1] works more generally with maximal monotone resistor relations. By Corollary I, any two solutions agree in current and voltage on each “real resistor” branch, whose relation contains no horizontal or vertical segment. His existence criterion is the absence of unbalanced cycles and cocycles in the associated admissible voltage and current intervals. The proof above is self-contained for the present prescribed-nomination model.

Remark A.2 (Bounded laws and normalization). Bounded laws do not obstruct existence even with a nontrivial cycle space. Take two parallel edges from s to t with laws $g_1(u) = \arctan u$ and $g_2(u) = 2 \arctan u$, whose suprema are different. For nomination $(B, -B)$, conservation gives flows q and $B - q$. Their drops agree exactly when

$$\arctan q - 2 \arctan(B - q) = 0.$$

For every real B the left side is continuous and strictly increasing in q . As $q \rightarrow +\infty$, both $\arctan q$ and $-\arctan(B - q)$ have positive limits; as $q \rightarrow -\infty$, both have negative limits. The left side therefore has exactly one root. This is the general-law cycle equation of lemma A.6 after reversing the second edge. Superlinear primitive growth, such as the cubic primitive of a quadratic law, also gives coercivity directly. Potentials are gauge invariant: adding $\alpha \mathbf{1}$ changes no equation. We fix a reference vertex v_0 and impose $\pi_{v_0} = 0$ whenever a potential vector is output.

At every passive state the dissipation identity is

$$b^\top \pi = (Ax)^\top \pi = x^\top A^\top \pi = \sum_e x_e g_e(x_e). \quad (8)$$

For symmetric quadratic laws,

$$b^\top \pi = \sum_e \beta_e |x_e|^3, \quad \mathcal{E}(x) = \frac{1}{3} \sum_e \beta_e |x_e|^3.$$

For asymmetric quadratic laws use β_e^+ on positive flows and β_e^- on negative flows in both sums. In either quadratic model, dissipation is three times the primitive energy.

Lemma A.3 (Homogeneity). *Fix symmetric or asymmetric quadratic laws. If (x, π) is the state for b , then for $q \geq 0$ the nomination qb has flow qx and potential differences q^2 times those for b .*

Proof. For $q > 0$ scaling preserves the sign of each nonzero flow, and both law pieces vanish at $q = 0$. Thus in either quadratic model $A(qx) = qb$ and $g_e(qx_e) = q^2 g_e(x_e)$, so $(qx, q^2\pi)$ is a state. Uniqueness in theorem A.1 gives the claim, also for $q = 0$. $\square$

A.2 Structure lemmas

A block is a maximal biconnected subgraph, with every bridge included as a two-vertex block. Blocks have edges; the one-vertex graph has no blocks. The block–vertex incidence graph has a node for each block and each original vertex, with an edge whenever a vertex belongs to a block. For $n > 1$, deleting the nodes of noncut vertices (these are leaves by lemma A.4) gives the block–cut tree.

Lemma A.4 (Block incidence tree). *Every edge belongs to exactly one block, every simple cycle lies in one block, and distinct blocks share at most one vertex. The block–vertex incidence graph of a connected graph is a tree, so it defines a unique block path between any two vertices.*

Proof. A cycle is biconnected and extends to a maximal biconnected subgraph because the graph is finite. Every edge on no cycle is a bridge: otherwise an alternative path between its endpoints would close a cycle. Thus every edge belongs to a block. Two distinct blocks cannot share two vertices: otherwise deleting any one vertex from their union would leave each block connected (a bridge block leaves a connected singleton) and would leave a shared vertex, so the union would be biconnected, contrary to maximality. This also excludes shared edges, so each edge has a unique block.

Connectivity of G gives connectivity of its incidence graph: replace each edge along a graph path by the two incidences with its block. Suppose the incidence graph has a simple cycle with distinct block nodes $H_1, \dots, H_j$ and distinct vertex nodes $w_1, \dots, w_j$, where H_i contains w_{i-1}, w_i cyclically. The preceding intersection property gives $j \geq 3$. The union of these blocks remains connected after deletion of any vertex w : each $H_i - w$ is nonempty and connected, and at most one of the distinct joining vertices w_i has been deleted. The remaining joins connect all these subgraphs in a path or a cycle. Their union is therefore biconnected, contradicting the maximality of the distinct H_i . This proves acyclicity; if $n = 1$, the incidence graph is a single vertex. Finally, let $n > 1$ and suppose v belongs to just one block H . Every edge incident to v lies in H , since an edge’s block contains both endpoints and H is the only block containing v . Thus any path between other vertices that passes through v enters and leaves through neighbors in $H - v$. Rerouting

that segment inside the connected graph $H - v$ gives a walk avoiding v , and hence a path avoiding v . Therefore v is not a cut vertex. A vertex in several blocks is a cut vertex, since a path between the incident branches after its deletion would contradict the incidence tree. Thus the noncut vertices are precisely the original-vertex leaves. $\square$

Lemma A.5 (Block decomposition). *For the general laws of theorem A.1 the following hold.*

1. *If $e = (u, v)$ is a bridge and S is the component of $G - e$ containing its tail u , then $x_e = \sum_{w \in S} b_w$.*
2. *Fix a block H . Delete its edges, retaining all vertices, and let $S_{H,v}$ be the component of $G - E(H)$ containing $v \in V(H)$. These components contain exactly one vertex of H each and partition V . Define the aggregated nomination by*

$$b_v^H = \sum_{w \in S_{H,v}} b_w = b_v + \sum_{w \in S_{H,v} \setminus \{v\}} b_w. \quad (9)$$

Then $x|_{E(H)}$ is the unique passive flow on H for b^H , and $\pi|_{V(H)}$ is a potential vector for that block.

3. *Let the block path from s to t traverse $H_1, \dots, H_j$ with successive boundary vertices $v_0 = s, v_1, \dots, v_j = t$. If π^{H_i} is any potential vector for H_i at b^{H_i} , then*

$$\pi_s - \pi_t = \sum_{i=1}^j (\pi_{v_{i-1}}^{H_i} - \pi_{v_i}^{H_i}). \quad (10)$$

For $s = t$ the sum is empty.

Proof. Summing conservation over the tail component of a bridge cancels every internal edge and leaves x_e , proving (a). If a component of $G - E(H)$ contained two vertices of H , choose a path in it and take the segment between two consecutive visits to H . This segment has no internal vertex in H and uses no edge of H . Closing it by a path inside H gives a simple cycle with an edge outside H and an edge inside H . By lemma A.4, every cycle lies in one block and every edge has a unique block, so the outside edge would belong to H , a contradiction. For a bridge H , the same cycle would contain the bridge, which is impossible. Connectivity ensures that every component meets H . Thus the sets in (b) partition V , and b^H is balanced. The aggregate b_v^H

includes all branches hanging at v away from H and counts b_v exactly once, even when v belongs to several blocks.

Solve each block with its balanced b^H , using theorem A.1, and assemble their edge flows. We check global conservation without assuming the restriction property. At a noncut vertex v , its only block has $b_v^H = b_v$. At a cut vertex v , let $H_1, \dots, H_\varkappa$ be its incident blocks. The components U_i of $G - v$ are indexed by these blocks; write $B_i = \sum_{w \in U_i} b_w$. Balance gives $b_v + \sum_i B_i = 0$, while

$$b_v^{H_i} = b_v + \sum_{h \neq i} B_h = -B_i.$$

Hence the assembled divergence at v is $\sum_i b_v^{H_i} = -\sum_i B_i = b_v$. Root the block-cut tree at any block. Shift the potential constant on each successive block to match the already assigned potential at its parent cut vertex; since the incidence graph is a tree, no conflict arises. All edge laws now hold globally. Global uniqueness identifies this constructed state with the given state and proves (b), including the potential restriction claim. Finally, the shifted block differences telescope along the unique block path, proving (c). The one-vertex case is immediate. $\square$

Lemma A.6 (Cactus cycle equation). *Let C be a cycle block of a cactus with general laws as in theorem A.1, and use its aggregated nomination b^C . Orient the cycle consistently and designate one cycle edge e_0 as the chord relative to the spanning path $C - e_0$. There are linear functions λ_e of b^C , with $\lambda_{e_0} = 0$, such that every conservation-feasible flow is $x = \lambda + q\mathbf{1}$. A potential exists for this flow if and only if*

$$\Phi(q) := \sum_{e \in E(C)} g_e(q + \lambda_e) = 0.$$

The continuous function Φ is strictly increasing and has exactly one real root. For symmetric quadratic laws the specialization is

$$\Phi(q) = \sum_{e \in E(C)} \beta_e(q + \lambda_e)|q + \lambda_e|.$$

Proof. Route b^C on the spanning path with zero chord flow. Subtree sums show that its flows λ_e are linear in b^C . The kernel of the consistently oriented cycle incidence matrix is the span of the all-ones edge vector: zero divergence equates the two incident circulation values at each vertex. Thus every feasible

flow is $\lambda + q\mathbf{1}$. A potential exists exactly when the sum of the oriented drops around the cycle is zero: necessity follows by telescoping, and sufficiency by integrating drops along the spanning path and checking the chord. This condition is $\Phi(q) = 0$. Every summand is continuous and strictly increasing. In the extended real sense,

$$\lim_{q \rightarrow +\infty} \Phi(q) = \sum_e \sup_{u \in \mathbb{R}} g_e(u) > 0, \quad \lim_{q \rightarrow -\infty} \Phi(q) = \sum_e \inf_{u \in \mathbb{R}} g_e(u) < 0.$$

Indeed $\sup g_e \geq g_e(1) > 0$ and $\inf g_e \leq g_e(-1) < 0$ for each edge. All terms in each limit have the same strict sign, so the finite sums are well defined even when infinite. Hence a finite sign-changing bracket exists, also when all laws are bounded, and the intermediate value theorem and strict increase prove the assertion. $\square$

Rational data can yield an irrational root. Take two parallel edges between s and t , of resistances 1 and $a > 0$, with unit through-flow. Orient the first from s to t and the second from t to s , and choose the first as chord. The flows are q and $q - 1$; $\Phi(0) = -a < 0 < 1 = \Phi(1)$, and on $(0, 1)$ the equation is $q^2 - a(1 - q)^2 = 0$. Its root is

$$q = \frac{\sqrt{a}}{1 + \sqrt{a}}, \quad 1 - q = \frac{1}{1 + \sqrt{a}}, \quad q^2 = a(1 - q)^2.$$

For rational a that is not a rational square, q is irrational since $\sqrt{a} = q/(1 - q)$. Subdividing the second edge into two edges of resistance $a/2$ gives the same example on a simple triangle.

Lemma A.7 (Degree count). *For a connected block H write $n_H = |V(H)|$, $m_H = |E(H)|$, and $r_H = m_H - n_H + 1$. Then*

$$\sum_{v \in V(H)} (\deg_H v - 2) = 2r_H - 2.$$

If H is not a bridge, at most $2r_H - 2$ vertices have degree at least three.

Proof. The degree sum is $2m_H$, so subtracting $2n_H$ gives $2r_H - 2$. Every vertex of a nonbridge block has degree at least two, and every vertex of degree at least three contributes at least one to that sum. For a bridge, the two degree-one vertices contribute -2 in total, and the second assertion fails at rank zero. $\square$

A.3 Graph parameters

A cactus is a graph in which distinct simple cycles share at most one vertex. For multigraphs a pair of parallel edges is a cycle of length two.

Lemma A.8 (Cactus blocks). *A connected graph is a cactus if and only if every block is a bridge or a simple cycle.*

Proof. A nonbridge block has minimum degree at least two, so following edges until a vertex repeats produces a simple cycle C . If the block is not exactly C , it has an ear relative to C : a path with distinct endpoints on C , with no internal vertex on C , and with edges outside C . Indeed, an extra edge between vertices of C is already an ear. Otherwise a component outside $V(C)$ must have at least two distinct neighbors on C , since a unique neighbor would be a cut vertex of the block. A path through that component between two such neighbors is an ear. The ear together with either of the two paths on C between its endpoints gives two distinct simple cycles sharing both endpoints. Hence a cactus has no such block.

Conversely, two distinct simple cycles sharing at least two vertices have a biconnected union: after any one vertex is deleted, each cycle remains connected and at least one common vertex survives. The union therefore lies in one block. That block is neither a bridge nor a simple cycle, since a simple cycle contains no distinct simple cycle. Thus the stated block condition implies the cactus property. $\square$

Series-parallel means K_4 -minor-free; a prescribed terminal pair need not admit a two-terminal series-parallel decomposition. Universal characterizations by graph class state when the graph must be simple. The cycle rank of a block H is $r_H = m_H - n_H + 1$. Define

$$r_{\max} = \max_H r_H, \quad r = m - n + 1,$$

where the maximum is zero if there are no blocks. For a graph with k connected components the total cycle rank is $m - n + k$.

Lemma A.9 (Total and block cycle rank). *For a connected graph, $r = \sum_H r_H$. Consequently bounded total cycle rank implies bounded block rank, but the converse fails. Trees have block rank zero. Cacti have block rank at most one, and exactly one when they contain a cycle. Series-parallel graphs can have unbounded block rank.*

Proof. By lemma A.4, if there are h blocks and $n > 1$, the block–vertex incidence tree has $n + h$ nodes and $\sum_H n_H$ edges, so $\sum_H (n_H - 1) = n - 1$. Every edge is in exactly one block, so

$$\sum_H (m_H - n_H + 1) = m - (n - 1) = r.$$

The one-vertex case has both sides zero. Nonnegativity of block ranks gives $r_{\max} \leq r$. A chain of k triangles joined at successive cut vertices has block rank one and total rank k . The tree statement follows from its bridge blocks, and the cactus statement follows from lemma A.8. Finally, $K_{2,k}$, for $k \geq 2$, has no K_4 minor. Such a minor would require four disjoint, connected, pairwise adjacent branch sets. At most two could contain a hub vertex. A connected hub-free branch set must be a singleton, because the non-hub vertices are independent. The two hub-free branch sets would therefore be nonadjacent, a contradiction. Deleting any one vertex of $K_{2,k}$ leaves it connected, so it is one block; it has $k + 2$ vertices, $2k$ edges, and rank $k - 1$. Subdivision preserves this example’s rank. $\square$

A.4 Objectives, scenario domains, and problems

Algorithmic inputs have rational data. A balanced nomination box is the nonempty compact set

$$\mathcal{B} = \{b \in \mathbb{R}^V : \ell \leq b \leq u, \mathbf{1}^\top b = 0\}, \quad \ell, u \in \mathbb{Q}^V, \quad \ell \leq u.$$

Its nonemptiness is equivalent to $\sum_v \ell_v \leq 0 \leq \sum_v u_v$: necessity follows by summation; sufficiency follows by interpolating between ℓ and u . Alternatively a nomination family is $\{b^0 + Mz : z \in P\}$, where b^0, M are rational and $P \subseteq \mathbb{R}^k$ is a nonempty compact polytope given by an explicit list of rational linear equalities and inequalities. We require $\mathbf{1}^\top (b^0 + Mz) = 0$ for every $z \in P$. Fixed dimension means that k is fixed independently of the input length, not that $|V|$ is fixed. Either nonemptiness is promised, or the algorithm first reports an empty domain.

Unless specified otherwise, the computational problems use symmetric quadratic laws. Independent resistance intervals form

$$\mathcal{R} = \prod_{e \in E} [L_e, U_e], \quad L_e, U_e \in \mathbb{Q}, \quad 0 < L_e \leq U_e < \infty.$$

Finite resistance uncertainty instead uses $\mathcal{R} = \prod_e R_e$ with each R_e an explicitly listed nonempty finite subset of $\mathbb{Q}_{>0}$. Fixed resistances are singleton

domains. A scenario is $\theta = (b, \beta) \in \Theta := \mathcal{B} \times \mathcal{R}$ (or the stated affine-family replacement). All nomination and resistance choices are independent apart from nomination balance and the stated parameter polytope. Finite sets are never implicitly replaced by interval hulls. Asymmetric models specify domains for both positive coefficients. Other law representations and their encodings are stated where used; an arbitrary continuous law satisfying the existence theorem is not thereby a finite algorithmic input.

Write $(x(\theta), \pi(\theta))$ for the unique normalized passive state. The objectives considered here are

$$\begin{aligned} F_{s,t}(\theta) &= \pi_s(\theta) - \pi_t(\theta), & F_c(\theta) &= c^\top \pi(\theta), \\ X_e(\theta) &= x_e(\theta), & F_d(\theta) &= d^\top x(\theta). \end{aligned}$$

The ordered pair (s, t) is prescribed. The rational potential weights satisfy $\mathbf{1}^\top c = 0$, ensuring gauge invariance, and their support size is $p = |\text{supp } c|$. Flow weights $d \in \mathbb{Q}^E$ have no balance requirement.

Lemma A.10 (Continuity and attainment). *On the scenario domains $\Theta = \mathcal{B} \times \mathcal{R}$ defined above, including the affine nomination families and positive asymmetric quadratic coefficients, the map $\theta \mapsto (x(\theta), \pi(\theta))$ with normalized potentials is continuous. The same holds on the nomination domains with fixed general laws, omitting the resistance coordinate. Each of $F_{s,t}, F_c, X_e, F_d$ attains its maximum and minimum on its domain.*

Proof. For quadratic laws let $\mathcal{E}_\beta$ denote the primitive energy with coefficient vector β : it is $\sum_e \beta_e |x_e|^3/3$ in the symmetric case and uses β_e^+ or β_e^- according to the sign of x_e in the asymmetric case. For a convergent sequence $(b_j, \beta_j) \rightarrow (b, \beta)$ with all limiting resistances positive, spanning-tree routings are uniformly bounded and provide uniformly bounded comparison energies. A common positive lower bound on the coefficients makes the minimizing flows uniformly bounded. If a subsequence converges to x_* , then $Ax_* = b$. For any y with $Ay = b$, add the tree routing of $b_j - b$ to obtain $y_j \rightarrow y$ with $Ay_j = b_j$. Passing to the limit in $\mathcal{E}_{\beta_j}(x_j) \leq \mathcal{E}_{\beta_j}(y_j)$ proves that x_* is the unique minimizer at (b, β) . Every convergent subsequence therefore has the same limit, so the flow is continuous. Summing continuous edge drops along a fixed spanning tree gives continuity of normalized potentials. The same argument works for fixed general laws, using the coercivity bound in theorem A.1, and for positive asymmetric coefficients. The four objectives are continuous functions of the state, and compactness of each nonempty scenario domain gives attainment of both maximum and minimum, including for finite resistance sets. $\square$

Definition A.11 (Optimization, validation, and approximation). For any one of the objectives F above, set $\text{OPT}_F(\Theta) = \max_{\theta \in \Theta} F(\theta)$, which exists by lemma A.10.

1. MPD is the computation of $\text{OPT}_{F_{s,t}}$ over $b \in \mathcal{B}$ with fixed resistances, or jointly over $(b, \beta) \in \mathcal{B} \times \mathcal{R}$ when stated. Weighted potential optimization computes OPT_{F_c} ; linear flow optimization computes OPT_{F_d} . No maximization over the terminals is implicit.
2. Arc extremum computation returns $\min_{\theta \in \Theta} x_e(\theta)$ and $\max_{\theta \in \Theta} x_e(\theta)$ for a prescribed arc e . Minima may equivalently be computed by maximizing $-X_e$.
3. Given tested arcs $T \subseteq E$ and rational signed capacities $\underline{x}_e \leq \bar{x}_e$, robust arc-capacity validation asks whether

$$\forall \theta \in \Theta \ \forall e \in T : \quad \underline{x}_e \leq x_e(\theta) \leq \bar{x}_e.$$

Each scenario's passive state is evaluated before testing the bounds; capacities do not filter Θ . Its complement is $\exists \theta \in \Theta \ \exists e \in T$ with $x_e(\theta) < \underline{x}_e$ or $x_e(\theta) > \bar{x}_e$.

4. The exact threshold version of a maximization problem, for rational γ , asks

$$\text{OPT}_F(\Theta) \geq \gamma, \quad \text{equivalently} \quad \exists \theta \in \Theta : F(\theta) \geq \gamma.$$

The weak inequality includes equality; it is distinct from a strict capacity violation. For a minimum, the threshold convention is applied to $-F$ if it is expressed as a maximization problem.

5. Given rational $\varepsilon > 0$, the additive task returns rational $\underline{v}, \bar{v}$ and a rational scenario $\hat{\theta} \in \Theta$ such that

$$\underline{v} \leq \text{OPT}_F(\Theta) \leq \bar{v}, \quad \bar{v} - \underline{v} \leq \varepsilon, \quad F(\hat{\theta}) \geq \text{OPT}_F(\Theta) - \varepsilon.$$

These are certified guarantees, not floating-point estimates. For an affine family a rational parameter $\hat{z} \in P$ may encode the nomination. Rational points are dense in each of the rational polytopes in the domain: express a point as a convex combination of rational vertices and approximate its weights by rational simplex weights. With continuity from lemma A.10, applied separately for each finite resistance choice, this proves that rational ε -optimal scenarios exist; it gives no polynomial bound on finding them.

An additive interval certifies a threshold answer when $\underline{v} \geq \gamma$ or $\overline{v} < \gamma$. An interval straddling γ need not decide the answer; in particular, arbitrarily narrow intervals alone do not resolve equality.

A.5 Computational conventions

We use the Turing bit model with total input length N : rational data are binary encoded, and running time includes bit costs and output length. Polynomial additive time means polynomial in N and $\max\{0, \log_2(1/\varepsilon)\}$, with the rational encoding of ε counted in N . Tolerances larger than one may be replaced by one. A bound of the form $N^{\varphi(k)}$ for parameter k is XP-type (polynomial for fixed k); an FPT bound is $\varphi(k)N^C$ with a constant exponent C independent of k . The same distinction applies with the accuracy bits included.

Square-Root Sum. SRS takes positive integers $a_1, \dots, a_k, K$ in binary and asks whether

$$\sum_{i=1}^k \sqrt{a_i} \leq K.$$

If zero radicands are allowed, remove them; an empty sum and a nonpositive threshold are handled directly. It is open whether SRS is in P or even in NP; Etessami and Yannakakis [25, Section 1] state these open questions and the PSPACE upper bound. Allender et al. [3, Corollary 1.5] give the stronger counting-hierarchy bound. They use $\geq$; the weak $\leq$ version follows by complementing and accounting for equality. For positive integer radicands, equality to an integer occurs only if every radicand is a perfect square: after grouping square factors, distinct square-free radicals, including 1, are linearly independent over $\mathbb{Q}$ [11, Section 2, Theorem 4]. For each square-free integer $f > 1$ occurring after this grouping, the coefficient of $\sqrt{f}$ is positive. Thus equality is testable by integer square roots. The iterated-PP definition recalled by Allender et al. [3, Section 4, citing Wagner and Torán] also gives closure of the counting hierarchy under complement and polynomial-time Boolean combinations: a deterministic polynomial-time computation with queries to a fixed level belongs to the next PP-oracle level. Complementing the weak $\geq$ test and adjoining the equality case therefore preserves membership in the hierarchy. Blömer [11, Sections 1 and 7] give a polynomial-time Monte Carlo test for whether a sum of real radicals lies in a specified real algebraic number field, including zero testing, and explicitly distinguish this from determining its sign; it is not an efficient general ordering algorithm.

Real algebraic computation. The class $\exists\mathbb{R}$ consists of decision problems polynomial-time reducible to the existence of a real solution to a finite Boolean combination of polynomial equalities and inequalities with rational coefficients. Clearing positive denominators gives integer coefficients. Allowing only conjunctions gives the same class: Boolean choices and strict inequalities can be encoded with additional existential variables. For decision complexity, Renegar [46, Part III, Proposition 2.3 (restating Part I, Proposition 4.1)] constructs all consistent sign vectors of σ integer polynomials of degree at most D in k variables with $(\sigma D)^{O(k)}$ arithmetic operations and $\tau \log \tau \log \log \tau (\sigma D)^{O(k)}$ bit operations for coefficient bit bound τ (replace small τ by a fixed constant). Evaluating the input Boolean formula on these vectors decides existential feasibility, with its formula-evaluation cost included.

We use the following fixed-dimension consequence of Basu [7, Theorem 2.27, Section 2.5.2], which restates the quantifier-elimination theorem from Basu et al. [8, Chapter 14]. For σ polynomials of degree at most $D \geq 2$ in $k + \nu$ variables, partition the quantified variables into ω nonempty blocks of sizes $k_1, \dots, k_\omega$. A prenex formula with these blocks and ν free variables has an equivalent quantifier-free formula computable with

$$\sigma^{(k_\omega+1)\cdots(k_1+1)(\nu+1)} D^{O(k_\omega)\cdots O(k_1)O(\nu)} \quad (11)$$

arithmetic operations in the coefficient ring. Output polynomial degrees are at most $D^{O(k_\omega)\cdots O(k_1)}$; for integer input coefficients of bit size at most τ , all intermediate and output integer bit sizes are at most $\tau D^{O(k_\omega)\cdots O(k_1)O(\nu)}$. Zero free dimension is handled by adding one unused free variable, so we take $\nu \geq 1$ in these bounds. The source gives output as a disjunction of conjunctions of disjunctions of sign tests, with the outer count bounded by equation (11), the conjunction count by $\sigma^{(k_\omega+1)\cdots(k_1+1)} D^{O(k_\omega)\cdots O(k_1)}$, and the inner count by $D^{O(k_\omega)\cdots O(k_1)}$. Thus for fixed total dimension the arithmetic work and coefficient bit sizes are polynomial in σ, D, τ , and ordinary integer arithmetic turns these into polynomial bit costs; reading and evaluating the Boolean input formula adds its encoding length. These bounds use numerical degree and dense polynomial encoding; they do not give polynomial time for sparse binary exponents or for an unbounded number of real variables.

Fixed-dimension sample points. Given an explicit quantifier-free formula in a fixed number k of real variables, using integer polynomials of degree at most D and coefficient bit size at most τ , one can compute in polynomial bit time a finite set of sample points meeting every semialgebraically connected component of every realizable sign condition of the input

polynomials. Testing the formula at these points supplies a point in every nonempty semialgebraic set it defines. Each sample is given by a *real univariate representation*: a common defining univariate polynomial, an isolating interval selecting its real root θ_{alg} , and all coordinates as rational functions of that root, with denominators nonzero there. Degrees and encoding lengths are polynomial for fixed k . See Basu et al. [8, Chapter 13, Sections 13.1 (finding realizable sign conditions) and 13.3 (sample points), and Chapter 12, Section 12.4 (univariate representations)]. All coordinates of a sample lie in the single real algebraic field $\mathbb{Q}(\theta_{\text{alg}})$, of polynomial degree; this is what a *common algebraic representation* means here. The subscript separates the algebraic generator from a scenario θ .

Exact and additive outputs. A real algebraic number α is represented by a nonzero square-free integer polynomial P_α and rational endpoints $\eta_- < \eta_+$ with exactly one real root of P_α in (η_-, η_+) , namely α , and no endpoint root. A Thom encoding (signs of successive derivatives at the root) is an alternative [8, Chapter 10]. The size counts the polynomial's degree, all coefficient bits, and the endpoint or encoding bits. For a degree- D dense polynomial with coefficient bit bound τ and endpoints of total bit length κ , it is $O((D+1)(\tau+1) + \kappa)$. The polynomial need not be minimal. Repeated roots can be removed using $\gcd(P_\alpha, P'_\alpha)$. Refining an isolating interval by rational bisection and exact univariate root counts permits rational rounding to prescribed accuracy, with bit costs measured in the representation size and accuracy bits. Root counting via Sturm sequences and sign determination are described in Basu et al. [8, Chapters 2 and 10]; see also Basu [7, Section 2.5.1] for sign determination. Algebraic output claims apply to semialgebraic input models; arbitrary continuous laws need not give algebraic states or values.

The output contract distinguishes three objects.

- An exact value is the optimum itself in the stated exact encoding; an additive value is a certified rational interval containing it. A sum of separately encoded block values is an expression encoding, not automatically a single polynomial-size isolating representation or a polynomial-time exact comparison procedure.
- An exact optimizing scenario is an admissible parameter choice attaining the optimum, represented by rational or algebraic coordinates as specified. An additive optimizing scenario is an exactly admissible rational choice with the objective guarantee in definition A.11; it need not be close in coordinates to an exact optimizer.

- An exact state satisfies both passive equations exactly, with normalized potentials. An additive state output consists of rational coordinate intervals of width at most ε containing that scenario's unique exact state, or rational coordinates with certified sup-norm error at most ε . Rounded coordinates need not themselves satisfy the equations; a small residual alone is not this error guarantee.

Rational scenarios may have irrational states and values, as the parallel example shows. No output kind is inferred from another without an explicit argument and a bound on its encoding size.

A.6 Notation

Symbol	Meaning
$G = (V, E), n, m$	Oriented network and its vertex and edge counts
$A, \mathbf{1}$	Tail-positive incidence matrix; all-ones vector
b, x, π	Balanced nominations, signed flows, potentials
$g_e, \beta_e, \beta_e^\pm$	Edge law and positive quadratic coefficients
$\mathcal{E}, b^\top \pi$	Primitive energy; constitutive dissipation
H, b^H, r_H	Block, aggregated nomination, cycle rank
$\varkappa, \lambda, q$	Incident-block count, spanning-path flow, circulation
$r_{\max}, r$	Maximum block rank; total cycle rank
$\mathcal{B}, \mathcal{R}, \Theta$	Nomination, resistance, and scenario domains
$(s, t), c, p, d$	Ordered terminals, potential weights/support, flow weights
b^0, M, z, P	Affine offset, matrix, parameter, parameter polytope
ℓ, u, L_e, U_e	Nomination bounds; resistance bounds
$\text{OPT}_F, \gamma, \varepsilon$	Maximum objective value, threshold, additive tolerance
$\underline{v}, \bar{v}$	Certified additive value interval endpoints
σ, D, τ, ν	Polynomial count, degree/bit bounds, free-variable count
f, κ	Square-free integer; isolating-endpoint bit length
$N, \text{SRS}, \exists \mathbb{R}$	Input bit length, Square-Root Sum, existential real class

The distinction between block rank and total cycle rank is central below. For a prescribed terminal pair, the block path decomposes the objective into local drops, which must still be combined in the Turing model. The next two sections use nomination structure and separate approximation of local values to avoid a potentially large algebraic field. Later sections examine which parts of this approach survive changes in edge laws, resistance domains, and objectives.

B Cactus MPD: additive optimization and exact comparison

A cactus separates its cycles topologically, but this does not make all forms of optimization equally easy. A maximum terminal drop can be approximated to any prescribed number of bits in polynomial time, whereas exact comparison already contains Square-Root Sum for one source and one sink. The positive result rests on the existence of an optimum with at most two unsaturated nomination coordinates, both in one cycle; all other block contributions are then constants that can be approximated separately. We prove this reduction, then give the algorithm and the exact-arithmetic classification of the single-source/sink subclass.

B.1 Setting and results

Throughout this section, G is a connected cactus, $s \neq t$ are prescribed vertices, and the laws are symmetric quadratic laws with positive rational resistances β_e . The domain is the nonempty balanced box $\mathcal{B} = \{b : \ell \leq b \leq u, \mathbf{1}^\top b = 0\}$ with finite rational bounds. The intervals may be shifted and may be singletons. Write $F_{s,t}(b) = \pi_s(b) - \pi_t(b)$ and $\text{OPT} = \text{OPT}_{F_{s,t}}(\mathcal{B})$. There are no additional flow or potential bounds. Existence, uniqueness, and attainment follow from theorem A.1 and lemma A.10.

Theorem B.1 (Additive optimization on cacti). *For rational $0 < \varepsilon \leq 1$, one can compute a certified rational interval $[\underline{v}, \bar{v}]$ containing OPT , of width at most ε , and a rational $\hat{b} \in \mathcal{B}$ satisfying $F_{s,t}(\hat{b}) \geq \text{OPT} - \varepsilon$, in time polynomial in the input bit length N and $\log(1/\varepsilon)$. As in appendix A.5, N includes the encoding of ε .*

Theorem B.2 (Exact single-source/sink arithmetic). *Let $\mathcal{B}_{st}$ be the balanced box with $b_s \in [0, 1]$, $b_t \in [-1, 0]$, and $b_v = 0$ for $v \notin \{s, t\}$. The exact threshold problem $\text{OPT}_{F_{s,t}}(\mathcal{B}_{st}) \geq \gamma$, for rational γ , is polynomial-time many-one equivalent to SRS. Hardness already holds on simple cacti of maximum degree three, all of whose cycles are triangles, with positive integer resistances.*

Neither theorem asserts NP-hardness, a polynomial exact algorithm for multiple-entry cactus MPD, or rational physical flows or potentials. The additive theorem addresses the approximation version of the cactus question recorded by Pfetsch et al. [44, pp. 9–10]; the single-source/sink theorem classifies an exact subclass.

B.2 Aggregation and electrical sensitivity

The *core* is the union of the blocks on the s – t path in the block–vertex incidence tree. Order these blocks from s to t , calling their two path vertices the entrance and exit.

Lemma B.3 (Exact aggregation to the core). *Deleting the core edges partitions V into groups S_v , one for each core vertex v , with S_v meeting the core precisely at v . Replacing its nominations by*

$$\bar{b}_v = \sum_{a \in S_v} b_a, \quad \bar{\ell}_v = \sum_{a \in S_v} \ell_a, \quad \bar{u}_v = \sum_{a \in S_v} u_a$$

preserves both the attainable balanced core nominations and their terminal potential differences exactly. Every rational feasible core nomination has a rational disaggregation computable in polynomial bit time. The same statements hold for the smoothed laws used below.

Proof. By lemma A.4, any block outside the path belongs to a subtree attached at a unique path-block vertex. A component after removing core edges cannot meet two core vertices: the argument in lemma A.5(b) would close a cycle. Connectivity gives an attachment for every component. Thus the groups partition V . Summing conservation over S_v cancels all its internal flows and leaves exactly the divergence of the core edges at v . Restriction of the potential equations therefore gives the unique core state for $\bar{b}$, also by lemma A.5.

Conversely, sums of independent intervals fill their sum interval: start every coordinate of S_v at its lower bound and allocate the remainder $\bar{b}_v - \bar{\ell}_v$ successively, adding at most $u_a - \ell_a$ to coordinate a . The remainder starts between zero and the sum of these widths and finishes at zero. Different groups share no variables, so global balance is preserved. Existence of the full state and the restriction argument show that its objective is the prescribed core objective. Rational additions and minima give the encoding and running-time claim. The argument does not use the quadratic form, so it also applies to the smoothed laws. $\square$

Potential or flow bounds would in general restrict which disaggregations are admissible; they are excluded here. Henceforth $G, \ell, u, b, n, m, \mathcal{B}$ denote the aggregated core and its data. By lemma B.3, OPT is unchanged. Aggregation has polynomial encoding length and does not increase the number of vertices or edges.

Lemma B.4 (Smoothed adjoint and its order). *For $\rho > 0$, set $\phi_{e,\rho}(x) = \beta_e(x|x| + \rho x)$ and normalize $\pi_t = 0$. The state is continuously differentiable on the space of balanced nominations. For $F_\rho(b) = \pi_s(b)$ and balanced db ,*

$$\begin{aligned} dF_\rho &= h^\top db, & L_\rho h &= e_s - e_t, & h_t &= 0, \\ L_\rho &= A \operatorname{diag}(1/r_e) A^\top, & r_e &= \beta_e(2|x_e| + \rho) > 0. \end{aligned} \quad (12)$$

The unit electrical current associated with h traverses the core blocks in series. The values of h strictly decrease across each bridge and along both entrance-to-exit branches of each cycle. The closed block ranges are ordered; consecutive ones meet just at their common articulation value and nonconsecutive ones are disjoint. In the unaggregated network, h is constant on every off-core block.

Proof. The law $\phi_{e,\rho}$ is onto, continuously differentiable, and has strictly positive derivative, so its inverse $\psi_{e,\rho}$ is continuously differentiable, with derivative $1/r_e$ at the physical drop. Delete row t of A to obtain $\tilde{A}$. In the coordinates $\tilde{\pi}$ and $\tilde{b}$ obtained by deleting coordinate t , conservation is

$$\tilde{A} \psi_\rho(\tilde{A}^\top \tilde{\pi}) - \tilde{b} = 0.$$

Its potential Jacobian is $\tilde{L}_\rho = \tilde{A} \operatorname{diag}(1/r_e) \tilde{A}^\top$. For a grounded nonzero vector y , $y^\top L_\rho y = \sum_e (y_u - y_v)^2 / r_e > 0$: equality would make y constant by connectivity, hence zero by grounding. The implicit function theorem therefore gives a local continuously differentiable state, and uniqueness makes these local maps agree on overlaps, hence on the entire balanced space. Differentiation gives $\tilde{L}_\rho d\tilde{\pi} = d\tilde{b}$. Symmetry implies $dF_\rho = \tilde{h}^\top d\tilde{b}$, where $\tilde{L}_\rho \tilde{h} = \tilde{e}_s$. Restoring $h_t = 0$ gives equation (12); the deleted row follows because both sides sum to zero.

Put $j_e = (h_u - h_v)/r_e$. This is the passive linear state with nomination $e_s - e_t$. An off-core subtree has zero load away from its attachment. Summing conservation gives zero divergence there too, so its restriction has zero nomination and, by uniqueness, zero current and constant potential. Across each cut separating earlier from later core blocks, conservation gives total electrical current one. In a bridge $j = 1$ in the entrance-to-exit direction. In a cycle let $R_1, R_2 > 0$ be the sums of the electrical resistances along its two branches. Conservation at interior vertices gives constant branch currents j_1, j_2 ; equality of the branch drops and their sum give $R_1 j_1 = R_2 j_2$ and $j_1 + j_2 = 1$, so both are positive and every edge on either branch has strictly positive drop. Each interior value lies strictly between the block endpoints, and adding successive positive block drops proves the range claims. $\square$

The energy/Hessian identity underlying this adjoint is classical: Misra et al. [41, Section IV-A, Lemma 1(a)–(b), equations (19) and (21)] identify potentials as the conjugate-energy gradient and its Hessian as the inverse grounded weighted Laplacian. Positive smoothing ensures that all the derivative weights in our application are finite and strictly positive, including at zero physical flow.

Lemma B.5 (Saturation faces with at most two pivots). *There is a family $\mathcal{F}$ of $O(n^2)$ closed faces of $\mathcal{B}$, enumerable with rational arithmetic in polynomial time and independent of resistances, such that one original MPD maximizer lies in their union. Each face fixes all coordinates at bounds except at most two pivots. Two pivots lie in one cycle, one strictly inside each of its entrance-to-exit branches. A sole pivot may also be an articulation or terminal. Intersecting a face with balance leaves dimension at most one.*

Proof. Fix $\rho > 0$ and a maximizer b^* of F_ρ , which exists by the continuity just proved and compactness. For every $b' \in \mathcal{B}$ the one-sided derivative along $b^* + \vartheta(b' - b^*)$ is nonpositive. Hence b^* maximizes $h^\top b$ over the same polytope, and linear-programming duality supplies a balance multiplier μ_0 with

$$h_v > \mu_0 \implies b_v^* = u_v, \quad h_v < \mu_0 \implies b_v^* = \ell_v. \quad (13)$$

The multiplier also follows directly from exchanges: if $b_i^* < u_i$ and $b_j^* > \ell_j$, the feasible direction $e_i - e_j$ gives $h_i \leq h_j$. Choose μ_0 between the maximum h_i over the former indices and the minimum h_j over the latter. If either set is empty, choose μ_0 beyond the other extreme; if both are empty any value works. These inequalities imply equation (13). Fixed coordinates belong to neither set, so this argument holds even when the feasible polytope has smaller affine dimension or is a singleton.

By lemma B.4, a value strictly inside a block range can equal h_v at at most one interior vertex per cycle branch, and at no vertex in any other block. Equality with an articulation or terminal value singles out only that core vertex. A threshold above h_s forces all lower bounds; one below h_t forces all upper bounds. These vectors need not be zero and are retained if balanced.

The following enumeration is independent of resistances. Include the all-lower and all-upper patterns. At each articulation and at s, t , leave that vertex free, put every preceding vertex at its upper bound, and every following vertex at its lower bound. For each bridge also include a gap threshold, with its entrance and all preceding vertices upper and its exit and all following vertices lower. For each cycle, put the entrance upper and the exit lower, and fix vertices strictly outside it upper before and lower after. On each

ordered list of branch interior vertices include all gap positions (the prefix upper, the suffix lower) and all vertex positions (the preceding prefix upper, the vertex free, and the following suffix lower). Take every pair of branch choices. A branch with no interior vertices has one gap choice. This covers a threshold in the open block range, including equality with one or two interior vertex values. For a cycle of k_C vertices there are $O(k_C^2)$ choices. Since $\sum_H (|V(H)| - 1) = n - 1$, their total is $O(n^2)$. Replace free coordinates with $\ell_v = u_v$ by their fixed value, discard empty patterns by balance and interval comparisons, and optionally discard duplicates. Each retained set is a face defined by bound equalities inside $\mathcal{B}$; extraneous feasible patterns are harmless. The enumeration covers every smoothed maximizer.

It remains to pass to $\rho = 0$ without assuming differentiability of the unsmoothed state. A fixed spanning-tree routing of b is bounded uniformly on the compact box. The smoothed energy is

$$\mathcal{E}_\rho(x) = \sum_e \beta_e \left(\frac{|x_e|^3}{3} + \frac{\rho x_e^2}{2} \right).$$

For $0 < \rho \leq 1$, comparison with this routing uniformly bounds the energy minimizers, since the positive cubic terms bound each flow. If $\rho_i \rightarrow 0$ and $b_i \rightarrow b$, a cluster point x of the minimizing flows satisfies $Ax = b$. For any competitor y with $Ay = b$, add the tree routing of $b_i - b$ to obtain competitors $y_i \rightarrow y$ for b_i . Passing to the limit in $\mathcal{E}_{\rho_i}(x_i) \leq \mathcal{E}_{\rho_i}(y_i)$ gives $\mathcal{E}_0(x) \leq \mathcal{E}_0(y)$. Strict convexity identifies x with the original flow, and tree sums of edge drops give convergence of normalized potentials. This is the continuity argument of lemma A.10, with a uniform energy comparison for the extra smoothing term.

This joint limit also gives uniform convergence $F_\rho \rightarrow F_0$ on the compact box: otherwise a sequence of nominations with errors bounded away from zero would have a convergent subsequence, contradicting this joint limit and continuity of F_0 . Take a convergent subsequence of smoothed maximizers and, since $\mathcal{F}$ is finite, a further subsequence in a single face. Uniform convergence makes the limit an original maximizer, and closedness keeps it in that face. Neither strict inequalities for a limiting adjoint nor positive unsmoothed derivatives are needed. $\square$

B.3 Additive value computation

On a face with two pivots v_1, v_2 , let $\bar{s}$ be minus the sum of the fixed coordinates. Balance gives $b_{v_1} = z$, $b_{v_2} = \bar{s} - z$ and

$$z \in I = [\max\{\ell_{v_1}, \bar{s} - u_{v_2}\}, \min\{u_{v_1}, \bar{s} - \ell_{v_2}\}]. \quad (14)$$

An empty interval is discarded and a singleton is a fixed nomination. With one pivot balance fixes it to $\bar{s}$; with none it tests the fixed sum. All these operations are exact rational operations.

For a nonconstant face only the cycle containing v_1, v_2 is active: the total nomination of the pivots is fixed, and they belong to the same component after deleting the edges of any other block, so every other block has fixed aggregated nominations by lemma A.5. The objective splits as $F_0(z) = D_0 + G_C(z)$, where D_0 is the sum of the fixed blocks' entrance-to-exit drops and G_C is the active cycle drop.

Fixed blocks. A bridge flow and its drop are rational. In a fixed cycle, routing on a spanning path and adding a signed circulation gives $x_e = \xi_e + \chi_e w$, with rational ξ_e and $\chi_e \in \{-1, 1\}$. The loop function $\sum_e \chi_e \beta_e x_e |x_e|$ is strictly increasing and has a unique root by lemma A.6. Its rational breakpoints are $-\xi_e/\chi_e$; on each intervening interval it is a polynomial of degree at most two. Exact signs at the breakpoints locate the root, or show it is a breakpoint. A nonempty open piece cannot be identically zero by strict increase. Linear pieces, double roots, and zero-flow breakpoint roots are handled by polynomial isolation rather than by unconditionally dividing by a quadratic leading coefficient. The root and every block drop are algebraic of degree at most two: on its sign piece a drop is a rational polynomial of degree at most two in that root. Rational root isolation and refinement [8, Chapter 10] approximate each drop in polynomial bit time; once the root is bracketed, interval evaluation of the drop converges at a rate bounded by its polynomial derivative on the bracket. All these bounds have polynomial bit length. The sum D_0 is never isolated as one algebraic number.

Uniform bounds. Set

$$B = \sum_v \max\{|\ell_v|, |u_v|\}, \quad V_{\max} = B^2 \sum_e \beta_e. \quad (15)$$

Every physical flow, also for $\rho > 0$, satisfies $|x_e| \leq B$. To see this, orient each nonzero edge along its actual flow. The potential strictly decreases along it, so there are no directed cycles. Add a super source supplying each positive nomination and a super sink collecting each negative nomination. Conservation permits repeatedly extracting a positive source-to-sink path, subtracting the minimum remaining flow on it, and deleting an exhausted edge. There is no residual circulation because the orientation is acyclic. The path weights sum to total injection, at most B , and each edge flow is a

sub-sum of these weights. Aggregated block nominations are sums over a partition, so their ℓ_1 norm is at most B , and so is the magnitude of their spanning-path routing. Subtracting it from the physical cycle flow proves $|w| \leq 2B$. Finally any simple path drop, including an active block drop or the full objective, has absolute value at most $V_{\max}$. If $B = 0$, the only core nomination and all core flows are zero, and the algorithm returns $[0, 0]$ and a disaggregation immediately.

Active cycle. For a k_C -edge active cycle, rational tree routing now gives

$$x_e(z, w) = \xi_e + \omega_e z + \chi_e w. \quad (16)$$

The k_C flow-zero lines form an arrangement with $O(k_C^2)$ cells. One can enumerate it by inserting lines successively: a new line has at most $k_C - 1$ distinct intersection points and hence at most k_C segments splitting existing cells. Intersections, ordering along lines, and side tests are rational computations; coincident lines can be merged. Use closures of the two-dimensional cells, retaining each original line's sign. These closures cover the whole plane, including all lower-dimensional intersections. Intersect with $z \in I$ and $-2B \leq w \leq 2B$; this also covers singleton I and zero flows.

On a cell let $\epsilon_e \in \{-1, 1\}$ specify the weak sign of x_e . The physical loop condition and the active drop along one fixed entrance-to-exit branch are respectively

$$\begin{aligned} \Phi_C(z, w) &= \sum_e \chi_e \beta_e \epsilon_e x_e(z, w)^2 = 0, \\ G_C(z, w) &= \sum_{e \text{ on branch}} \delta_e \beta_e \epsilon_e x_e(z, w)^2. \end{aligned}$$

where δ_e is the branch traversal sign. They have degree at most two. The cell inequalities $\epsilon_e x_e \geq 0$ and the compact box are linear. Conservation is already built into equation (16); the loop condition is sufficient for a potential by lemma A.6. At a zero flow both polynomial pieces agree, so closed cells add no false states.

For rational γ_* , feasibility with the further inequality $G_C(z, w) \geq \gamma_*$ is an existential semialgebraic problem in *two* real variables, with $O(k_C)$ constraints of degree at most two per cell. Clear positive denominators first. Aggregation, routing, squaring affine forms, and summing at most m terms use sums and fixed powers of input rationals; a common denominator gives coefficient bit length $O(N + \iota)$ after ι threshold bisections. The displayed estimates use the safe upper bound $O(N^4 + \iota)$. By the Renegar bound in

appendix A.5, a cell decision costs at most

$$O(k_C^{a_{\text{dec}}}(N^4 + \iota + 1)^2)$$

bit operations, for an absolute constant exponent a_{dec} (the exponent for two variables and degree two). Formula evaluation is included. Alternatively equation (11) with two quantified variables and one unused free variable gives polynomial-count exponent six and an absolute constant degree factor; its coefficient-bit bound is polynomial as well. Neither exponent depends on cactus size.

Every $z \in I$ has a feasible point in the union of the cells. Binary search on $[-V_{\max}, V_{\max}]$ using these exact tests gives an interval for $\max_I G_C$ of width $\varepsilon/8$. Keep a feasible lower threshold and a cell witnessing it. The initial lower endpoint $-V_{\max}$ is feasible; the upper endpoint need only bound the optimum, so equality at $V_{\max}$ causes no problem. At most $O(1 + \log^+(V_{\max}/\varepsilon))$ iterations suffice, where $\log^+ a = \max\{0, \log_2 a\}$. Approximate each fixed block drop by a rational interval of width at most $\varepsilon/(8n)$. There are at most $n - 1$ blocks, so the resulting interval for D_0 has width at most $\varepsilon/8$. Adding gives a face interval $[\underline{v}_J, \bar{v}_J]$ of width at most $\varepsilon/4$; fixed nominations use only the fixed-block procedure. Then

$$[\underline{v}, \bar{v}] = [\max_{J \in \mathcal{F}} \underline{v}_J, \max_{J \in \mathcal{F}} \bar{v}_J] \quad (17)$$

contains the global optimum by lemma B.5. Its width is at most $\varepsilon/4$: choose a face supplying the largest upper endpoint, and note that the largest lower endpoint is at least its lower endpoint.

Algorithm B.6 (Cactus additive MPD). Input the data of theorem B.1.

1. Find the block path, aggregate intervals, and retain the groups for disaggregation. Enumerate the saturation faces of lemma B.5, discard empty ones, and parameterize them by equation (14). Handle $B = 0$ directly.
2. For each face, isolate and refine fixed-block drops to total interval width $\varepsilon/8$. If the face varies, enumerate its active cycle's sign cells and bisect the active optimum to width $\varepsilon/8$, retaining a feasible lower threshold and cell.
3. Form the face intervals and return the value interval equation (17). Select a face with greatest lower endpoint.

4. On that face recover a feasible algebraic parameter at the retained lower threshold, round it within $\varepsilon/(8C)$ as in the proof of theorem B.1, recompute the other pivot by balance, and greedily disaggregate. A fixed face already has a rational nomination. Return this rational nomination together with the value interval.

B.4 Rational nomination recovery

Lemma B.7 (Lipschitz estimate). *For any simple s - t core path P_{st} and balanced $b, b' \in \mathcal{B}$,*

$$|F_0(b) - F_0(b')| \leq C \|b - b'\|_1, \quad C = 2B \sum_{e \in P_{st}} \beta_e.$$

This estimate also holds on any connected graph with the same data assumptions, using a simple terminal path and its nomination bound B .

Proof. The flow bound in equation (15) gives derivative resistances $r_e \leq \beta_e(2B + \rho)$. We prove the two electrical facts needed to bound the adjoint. At a vertex other than s, t , $L_\rho h = 0$ expresses h_v as a weighted average of its neighbors with positive weights. A global maximum at such a vertex propagates to its neighbors until it reaches a terminal. The sink cannot be a maximum since $(L_\rho h)_t = -1$, and the source cannot be a minimum since $(L_\rho h)_s = 1$. Applying the same argument to the minimum shows $0 = h_t \leq h_v \leq h_s$ everywhere.

For $j = \text{diag}(1/r_e)A^\top h$, we have $Aj = e_s - e_t$ and $\sum_e r_e j_e^2 = h^\top Aj = h^\top (e_s - e_t) = h_s$. For any other unit terminal flow y , write $y = j + \Delta x$, with $A\Delta x = 0$. Then

$$\sum_e r_e y_e^2 = \sum_e r_e j_e^2 + \sum_e r_e (\Delta x_e)^2,$$

because the cross term is $2h^\top A\Delta x = 0$. Testing with the signed unit flow on P_{st} proves the effective-resistance/path comparison

$$0 \leq h_v \leq h_s \leq \sum_{e \in P_{st}} r_e \leq (2B + \rho) \sum_{e \in P_{st}} \beta_e.$$

Integrate $h^\top(b' - b)$ along the balanced feasible segment from b to b' , using lemma B.4. The absolute value is bounded by the last upper bound times $\|b' - b\|_1$. Uniform convergence of the smoothed objectives, proved in lemma B.5 by an argument valid on any connected graph, permits $\rho \downarrow 0$. $\square$

Proof of theorem B.1. It remains to justify witness extraction, error budgets, and bit costs in algorithm B.6. For a selected nonconstant face, take the cell and feasible lower threshold γ_- from its bisection. Eliminate w from the cell constraints, $\Phi_C = 0$, and $G_C \geq \gamma_-$, leaving a quantifier-free formula in z . The bound equation (11) has one quantified and one free variable here, so its polynomial-count exponent is four, with constant output degree and polynomial coefficient bits. Isolate the roots of these univariate output polynomials and test their signs at the roots and in the intervening intervals, using Basu et al. [8, Chapter 10]. This is the one-variable case of the fixed-dimension sample-point primitive in appendix A.5. This yields an algebraic feasible z^* of polynomial encoding length; a feasible interval may instead supply a rational point. Identically zero polynomials and singleton feasible sets are covered by sign tests. A corresponding w exists by the projected formula and is the unique physical circulation, so neither the algebraic constant sum nor w needs to be compared or output. The resulting objective is at least the selected face's certified lower endpoint: its active drop is at least γ_- , and its constant contribution is at least the lower endpoint of the interval for D_0 .

If $C > 0$, refine z^* and clip a rational approximation to its rational interval I , obtaining $|\hat{z} - z^*| \leq \varepsilon/(8C)$. Clipping cannot increase distance to a point in I . Set the other pivot to $\bar{s} - \hat{z}$. The nomination changes in ℓ_1 norm by at most $\varepsilon/(4C)$, so lemma B.7 limits the loss to $\varepsilon/4$. All interval endpoints, fixed coordinates, and singleton cases remain exactly feasible. Since $s \neq t$ and resistances are positive, $C = 0$ means $B = 0$, already handled. Greedy disaggregation from lemma B.3 gives rational original loads satisfying all original bounds and balance exactly, with the same objective.

Selecting the greatest lower endpoint loses at most $\varepsilon/4$ relative to the global optimum: if face J contains a global maximizer, then $\text{OPT} \leq \bar{v}_J \leq \underline{v}_J + \varepsilon/4 \leq \underline{v}_{\text{selected}} + \varepsilon/4$. The extracted parameter attains at least $\underline{v}_{\text{selected}}$, and rounding loses at most $\varepsilon/4$. Thus the nomination loss is at most $\varepsilon/2 \leq \varepsilon$, while equation (17) is a certified value interval of width at most $\varepsilon/4 \leq \varepsilon$. This is precisely the value and exactly admissible rational-scenario contract of definition A.11; no rational state output is asserted.

Let $N_\varepsilon = N + \lceil \log_2(1/\varepsilon) \rceil + 1$. The rational bounds $B, V_{\max}, C$, interval endpoints, and isolated quadratic roots have polynomial bit length; value bisection uses $O(N_\varepsilon)$ iterations and rounding uses polynomially many bits. There are $O(n^2)$ faces and $O(n^2)$ cells per face. With the absolute constant a_{dec} above, their decision cost is bounded by

$$O(N^{a_{\text{dec}}+4}(N^4 + N_\varepsilon)^2 N_\varepsilon).$$

All other rational work, fixed quadratic isolation/refinement, and the one-

variable projection and recovery have cost $O((N+N_\varepsilon)^{a_{\text{rec}}})$ for another absolute constant a_{rec} , by the stated fixed-dimension bounds and univariate isolation. These exponents are independent of the number of blocks, vertices, and accuracy bits. The sum is a polynomial in N and $\log(1/\varepsilon)$, as required. $\square$

B.5 Exact arithmetic and the existential-real upper bound

Lemma B.8 (Parallel branches and the radical gadget). *Two internally unloaded branches of total quadratic resistances $\varrho_1, \varrho_2 > 0$ have effective quadratic resistance*

$$\begin{aligned} R(\varrho_1, \varrho_2) &= \frac{\varrho_1 \varrho_2}{(\sqrt{\varrho_1} + \sqrt{\varrho_2})^2} \\ &= \begin{cases} \frac{\varrho_1 \varrho_2 (\varrho_1 + \varrho_2)}{(\varrho_1 - \varrho_2)^2} - \frac{2\varrho_1 \varrho_2}{(\varrho_1 - \varrho_2)^2} \sqrt{\varrho_1 \varrho_2}, & \varrho_1 \neq \varrho_2, \\ \varrho_1/4, & \varrho_1 = \varrho_2. \end{cases} \end{aligned} \quad (18)$$

For integer $a \geq 2$, the choice $\varrho_1 = (a-1)^2$, $\varrho_2 = (a-1)^2/a$ has unit-flow drop $a+1-2\sqrt{a} > 0$.

Proof. Internal conservation makes each branch flow constant. For total $\zeta > 0$ both flows have the sign of their common potential drop, so both are positive. Equal drops and conservation give

$$x_1 = \frac{\zeta \sqrt{\varrho_2}}{\sqrt{\varrho_1} + \sqrt{\varrho_2}}, \quad x_2 = \frac{\zeta \sqrt{\varrho_1}}{\sqrt{\varrho_1} + \sqrt{\varrho_2}}.$$

Their common drop is $\zeta^2 R(\varrho_1, \varrho_2)$. At $\zeta = 0$ uniqueness gives zero; symmetry treats negative through-flow if needed. For $\varrho_1 \neq \varrho_2$, multiply $1/(\varrho_1 + \varrho_2 + 2\sqrt{\varrho_1 \varrho_2})$ by $(\varrho_1 + \varrho_2 - 2\sqrt{\varrho_1 \varrho_2})/(\varrho_1 + \varrho_2 - 2\sqrt{\varrho_1 \varrho_2})$; the denominator becomes $(\varrho_1 - \varrho_2)^2$. If $\varrho_1 = \varrho_2$, the equal split gives $\varrho_1/4$ directly. For the gadget $x_1 = 1/(1+\sqrt{a})$ at unit flow, whence $(a-1)^2 x_1^2 = (\sqrt{a}-1)^2 = a+1-2\sqrt{a} > 0$. $\square$

The parallel formula is classical; it is the quadratic specialization of the homogeneous reduction in Groß et al. [29, Lemma 4.2]. Its use for the following arithmetic reductions is what matters here.

Proof of theorem B.2. Write Δ_{st} for the unit-flow terminal drop on the network under consideration. For $\mathcal{B}_{st}$, balance writes every nomination as

$\zeta(e_s - e_t)$ with $0 \leq \zeta \leq 1$. Homogeneity (lemma A.3) gives a drop equal to ζ^2 times the unit-flow drop. The latter is positive since

$$\pi_s - \pi_t = \sum_e \beta_e |x_e|^3 > 0$$

at unit nomination by equation (8). Hence the maximum is attained at $\zeta = 1$, and $\text{OPT}_{F_{s,t}}(\mathcal{B}_{st}) = \Delta_{st}$.

Reduction from SRS. Start with $\sum_i \sqrt{a_i} \leq K$ in the weak convention of appendix A.5. Remove any zero radicands if allowed; remove each radicand one and subtract one from K . If no terms remain, output a fixed yes/no MPD instance according to $0 \leq K$: a resistance-one edge with threshold one/two. Otherwise all remaining $a_i \geq 2$; denote their number by $\mu \geq 1$. Make μ disjoint triangles. Triangle i has direct-edge resistance $(a_i - 1)^2$ and two alternate-path edges each of resistance $(a_i - 1)^2 / (2a_i)$. Join successive exits and entrances by unit-resistance bridges, with s the first entrance and t the last exit. This is a simple cactus with 3μ vertices and $4\mu - 1$ edges. Every triangle terminal has two triangle edges and at most one bridge, so maximum degree is three. Every block carries unit through-flow at $\zeta = 1$, giving

$$\Delta_{st} = C_0 - 2 \sum_{i=1}^{\mu} \sqrt{a_i}, \quad C_0 = \sum_{i=1}^{\mu} (a_i + 1) + (\mu - 1).$$

Set $\gamma = C_0 - 2K$. Then $\Delta_{st} \geq \gamma$ if and only if the preprocessed SRS inequality holds, including equality. If the preprocessed $K \leq 0$, it is a no instance, since a positive radical remains, and one may output the fixed no instance. If $\gamma \leq 0$, the constructed instance already answers yes, and one may instead output the fixed yes instance: the constructed unit-flow drop is strictly positive, so $C_0 > 2 \sum_i \sqrt{a_i}$ and $K \geq C_0/2$ imply yes. This handles nonpositive thresholds without changing the weak convention.

In the remaining construction multiply every resistance and γ by $\Lambda = 2 \prod_i a_i$. Flows are unchanged, since multiplying the potential vector by Λ gives precisely the scaled laws; uniqueness then applies. All resistances become positive integers and the threshold becomes an integer. The bit length of Λ is $O(1 + \sum_i \log a_i)$, and each of the linearly many output numbers has polynomial bit length. Only rational arithmetic, integer multiplication, and exact division are required.

Reduction to SRS. On an arbitrary single-source/sink cactus, every off-path block has zero aggregated nomination and therefore zero flow by lemma A.5 and theorem A.1. The path blocks form a series chain. A bridge contributes its resistance at unit flow; a cycle contributes equation (18),

where ϱ_1, ϱ_2 are sums of its branch resistances. Equal branches contribute the rational $\varrho_1/4$. For each unequal pair absorb the positive radical coefficient into its radicand, obtaining

$$\Delta_{st} = Q - \sum_{j=1}^{\mu'} \sqrt{r_j}, \quad r_j = \frac{4\varrho_{1,j}^3 \varrho_{2,j}^3}{(\varrho_{1,j} - \varrho_{2,j})^4} > 0, \quad (19)$$

where $Q > 0$ is rational. If $\mu' = 0$, decide the rational comparison and output the fixed SRS yes instance $(a_1, K) = (1, 1)$ or no instance $(4, 1)$. Otherwise put $T = Q - \gamma$. If $T \leq 0$, output that fixed no instance, because every radical is positive. This includes $T = 0$ and respects the positive-integer threshold convention for SRS. For $T > 0$, write $r_j = \mathfrak{a}_j/\mathfrak{b}_j$ and $T = \mathfrak{p}/\mathfrak{q}$ with positive integer denominators and define

$$\mathcal{D} = \mathfrak{q} \prod_{j=1}^{\mu'} \mathfrak{b}_j, \quad a'_j = \mathcal{D}^2 \mathfrak{a}_j / \mathfrak{b}_j \in \mathbb{Z}_{>0}, \quad K' = \mathcal{D}T \in \mathbb{Z}_{>0}.$$

Multiplication by positive $\mathcal{D}$ gives the exact equivalence $\Delta_{st} \geq \gamma \iff \sum_j \sqrt{a'_j} \leq K'$. Branch sums, their differences, and the fixed powers in equation (19) have polynomial rational encoding length. Denominator products have bit length at most the sum of factor bit lengths, so $Q, \mathcal{D}, a'_j, K'$ and the complete output also have polynomial encoding length. A small nonzero $\varrho_1 - \varrho_2$ changes magnitudes but not this bound. Testing $\varrho_1 = \varrho_2$ is exact rational comparison. No factorization or square-root evaluation occurs in either encoder. $\square$

Proposition B.9 (General exact MPD is existential-real). *For any connected graph with positive rational symmetric quadratic resistances, rational γ , and a balanced rational nomination box, the exact threshold problem $\exists b \in \mathcal{B} : \pi_s(b) - \pi_t(b) \geq \gamma$ is in $\exists\mathbb{R}$, and therefore in PSPACE.*

Proof. Introduce real variables b_v, π_v and p_e, n_e and impose

$$\begin{aligned} \ell_v &\leq b_v \leq u_v \quad (v \in V), & \mathbf{1}^\top b &= 0, & A(p - n) &= b, & \pi_t &= 0, \\ p_e &\geq 0, \quad n_e \geq 0, \quad p_e n_e &= 0, & \pi_u - \pi_v &= \beta_e(p_e^2 - n_e^2) \quad (e = (u, v)), \\ \pi_s - \pi_t &\geq \gamma. \end{aligned}$$

For nonnegative complementary p_e, n_e , at least one is zero. If $p_e > 0$ then $x_e = p_e - n_e = p_e$ and $x_e|x_e| = p_e^2$; if $n_e > 0$ then $x_e = -n_e$ and $x_e|x_e| = -n_e^2$; when both vanish both sides are zero. Conversely every signed x_e

has the unique such split $p_e = \max\{x_e, 0\}$, $n_e = \max\{-x_e, 0\}$. Thus the constraints encode exactly the passive equations and the threshold, and existence and uniqueness from theorem A.1 identify their potential difference with $F_{s,t}(b)$. There are $2n + 2m$ variables and $O(n + m)$ constraints of degree at most two and polynomial coefficient bit length; clearing positive rational denominators preserves polynomial size. Existentially quantifying all variables gives the reduction to $\exists\mathbb{R}$. The real-algebraic procedures in appendix A.5 give decidability; for the stronger space bound with unbounded dimension we use the polynomial-space existential-real decision theorem of Canny [17, report version, Theorem 3.3]. $\square$

Consequently exact cactus MPD threshold comparison is SRS-hard, already for one source and one sink, and is in $\exists\mathbb{R} \subseteq \text{PSPACE}$. Whether the problem with multiple entries is polynomial-time equivalent to SRS remains open here.

B.6 Precision and earlier algorithms

Additive precision does not settle an equality-sensitive threshold: when the certified interval straddles γ , narrowing it need not terminate with an answer, particularly if $\gamma = \text{OPT}$. Theorem B.1 gives no bound on the number of accuracy bits needed for a general exact comparison.

Labbé et al. [33, Section 6], by Labbé, Plein, and Schmidt, leaves booking feasibility on a single cycle with nonlinear potentials open. The later four-author paper of Labbé, Plein, Schmidt, and Thürauf [34, Section 6, Theorems 6.4–6.5 and Corollary 6.6 (numbering of the November 2020 manuscript)] gives polynomial Turing-time booking validation on a passive cycle under rational-data Assumption 6.1. Its structural configurations reduce the decision systems to at most nine variables and 42 constraints. Thus both structural nomination restrictions and fixed-dimensional real algebraic computation have direct precedents. With arbitrarily many cactus cycles, even fixed single-source/sink nominations can accumulate an unbounded number of independent radicals. Exact local solutions and a compact sum expression do not themselves supply polynomial exact threshold comparison.

The series-parallel reductions of Groß et al. [29, Lemma 4.2 and Theorem 4.3] (later published in *Networks*, 2019) construct an equivalent arc with linearly many reduction applications. Their Section 2.1 explicitly uses the Turing model and discusses root approximation with a fixed error, so their result is not merely a real-arithmetic statement. The stronger zero-gap threshold guarantee needs a separate argument, which theorem B.2 identifies with

SRS on the stated subclass. Thürauf [49, Lemma 4.20 and Section 5] proves general-network MPD hardness, also with predetermined flow directions, and proposes both cacti and approximate MPD as further directions; the overview retains the nonlinear cactus question [44, pp. 9–10]. The results here give polynomial additive optimization for balanced shifted boxes and an exact arithmetic classification for a single source and sink, while leaving the multiple-entry exact boundary open.

C Bounded block rank

The cactus argument extends to networks with interacting cycles when each biconnected block has bounded cycle rank; the number of blocks and the total cycle rank may grow with the input. The extension has two steps: localize free nominations to one block, then show that a small objective perturbation leaves only a bounded number of free coordinates inside it. Independent resistance intervals are then eliminated by a parametric linear-programming argument in fixed dimension.

We write $\mathfrak{r}$ for the fixed block-rank bound, retaining r for total cycle rank. A selected block H has entrance a and exit c_H ; the subscript distinguishes the exit vertex from the weighted-objective vector c . Other local notation is defined where used.

C.1 Results and scope

Throughout, the graph is connected, the laws are symmetric quadratic, and $\mathcal{B} = \{b : \ell \leq b \leq u, \mathbf{1}^\top b = 0\}$ is a nonempty balanced rational box. There are no additional operating constraints on scenarios. All intervals are finite; singleton intervals are allowed.

Theorem C.1 (Bounded block rank with nomination uncertainty). *Fix an integer $\mathfrak{r} \geq 0$. If $r_{\max} \leq \mathfrak{r}$, resistances are positive rationals, and $s \neq t$, then MPD admits a certified rational value interval of width at most ε and an exactly feasible rational ε -optimal nomination, in time polynomial in N and $\log(1/\varepsilon)$ for fixed $\mathfrak{r}$. The exponent may depend on $\mathfrak{r}$; this is an XP-type guarantee.*

Theorem C.2 (Joint nomination and resistance intervals). *The same conclusion holds jointly over $\mathcal{B} \times \mathcal{R}$, where $\mathcal{R} = \prod_{e \in E} [L_e, U_e]$ has rational $0 < L_e \leq U_e$. The output scenario consists of rational nomination and resistance vectors satisfying their bounds and nomination balance exactly.*

Theorem C.3 (Exact arc extrema and capacity validation). *Under the joint hypotheses of theorem C.2, the maximum and minimum signed flow on any prescribed arc are real algebraic numbers of polynomial encoding length, computable exactly in polynomial bit time for fixed $\mathfrak{r}$. Thus robust validation of all rational signed arc capacities is decidable exactly in polynomial time for fixed $\mathfrak{r}$, including equality at a capacity.*

The joint uncertainty model is not new: Aßmann [4, Section 3.3, equations (3.2)–(3.4), p. 56] uses the product of positive coefficient intervals and a balanced nomination box. The resistance-only model and the circulation sign-region construction appear in Aßmann et al. [5, Sections 4.1.4 and 4.3.2, Proposition 4.7 (arXiv v1 numbering)]. The hypotheses here bound only the maximum rank of a block, allowing unbounded total cycle rank and arbitrarily many uncertain coordinates. The pressure and arc conclusions differ for arithmetic reasons: a terminal drop can add values from many blocks, while an arc belongs to one block and admits one algebraic representation of polynomial size. Pressure is therefore handled by summing certified intervals, consistent with the cactus SRS barrier. The exponent depends on the rank bound; the theorems give tractability at each fixed rank, not a practically small exponent.

C.2 One active block and suppressed paths

Lemma C.4 (Block ordering and localization). *Let the resistance vector be positive and real. The aggregation statement of lemma B.3 holds on any connected graph. On the resulting terminal core, the ordinary smoothed adjoint has strictly ordered block ranges: every interior vertex of a nonbridge block has value strictly between its entrance and exit values. There is an $O(n)$ family of closed rational faces containing an MPD maximizer such that each face either fixes the nomination or leaves free coordinates in just one selected block. Vertices strictly before that block are upper-saturated and those strictly after it are lower-saturated. This family is independent of resistances.*

Proof. The aggregation proof, including its rational greedy disaggregation, uses only the incidence tree and state uniqueness, so it applies verbatim. The implicit-function and adjoint calculations in lemma B.4 use only connectivity. The linear passive state with nomination $e_s - e_t$ restricts in each path block to the unit entrance–exit state, by lemma A.5. Its entrance value exceeds its exit value: their difference equals $\sum_{e \in E(H)} r_e \mathcal{J}_e^2 > 0$, since a unit flow is nonzero. The weighted-average maximum principle bounds every other value by the two boundary values. For strictness, suppose an interior vertex has

the entrance value. The block with the entrance deleted is connected, so a path from this vertex to the exit avoids the entrance. The weighted-average equality at successive interior vertices propagates the maximum along this path to the exit, a contradiction. Deleting the exit proves the strict minimum statement. A two-vertex block has no interior vertices; a bridge has positive drop directly. Hence open block ranges are disjoint and successive closed ranges meet only at their shared articulation.

At a smoothed maximizer, feasible exchanges give the multiplier and saturation rule equation (13); its proof allows fixed coordinates and singleton polytopes. If the multiplier is inside a block range, only vertices of that block can be free. At an articulation or terminal value only that vertex can be free. A gap inside a bridge leaves none. Above or below all ranges gives respectively the all-lower or all-upper vector. Enumerate one face for each selected block, leaving all its coordinates unrestricted and fixing vertices strictly outside as stated, together with these vertex, gap, and extreme patterns. Extra points in these faces are admissible and harmless; empty faces are discarded by rational balance and interval tests. The compactness and uniform smoothing argument in lemma B.5 works on every connected graph. A subsequence of smoothed maximizers stays in one of the finitely many closed faces; its limit is an unsmoothed maximizer in that face. $\square$

Fix a selected block. All other core nominations are fixed, and their sums shift the two terminal intervals of the block to give the balanced local box of aggregated nominations b^H . The sum of the block's original core nominations is fixed by balance. Hence every other block's aggregated nomination is fixed, even as the individual nominations of this block vary, and for fixed resistances its contribution is a constant by lemma A.5. The local objective is $\pi_a - \pi_{c_H}$.

Lemma C.5 (Suppression topology). *For a nonbridge block put $K_H = \{v : \deg_H v \geq 3\} \cup \{a, c_H\}$. Its edges partition into p_H maximal paths whose endpoints are in K_H and whose internal vertices have degree two. The interiors partition $V(H) \setminus K_H$, and*

$$|K_H| \leq 2r_H, \quad p_H = r_H + |K_H| - 1 \leq 3r_H - 1.$$

Proof. The degree count lemma A.7 bounds the number of high-degree vertices by $2r_H - 2$; the two terminals give the first bound. Every component after removing K_H has maximum degree two. It cannot be a cycle, since it would have no edge leaving it in the connected block. Thus it is a path, possibly a singleton, with its two missing incidences attached to K_H , and following a degree-two vertex ends at marked vertices in both directions.

They are distinct: a path returning to one marked vertex would make that vertex a cut vertex, since another marked terminal exists. Include also direct marked-to-marked edges as paths without internal vertices. This partitions all edges. Suppressing each interior removes equal numbers of edges and vertices, preserves connectedness and cycle rank, and leaves a multigraph with $|K_H|$ vertices and p_H edges. This proves the equality and bound. Parallel paths and rank-one blocks are included; bridges are excluded. $\square$

Lemma C.6 (Positive forcing and unimodality). *Use smoothed laws and perturb the local objective to*

$$F_{\delta,\rho}(b^H) = \pi_a - \pi_{c_H} + \delta \sum_{v \in V(H) \setminus K_H} (\pi_v - \pi_{c_H}), \quad \delta, \rho > 0.$$

On each suppressed path, its adjoint values first strictly increase and then strictly decrease, allowing either part to be empty and a maximum plateau of at most two adjacent vertices. Every level occurs at most twice, and each strict upper-level set is an interval in path order.

Proof. Linearity of the differentiated objective and the grounded calculation of lemma B.4 give

$$L_\rho h = e_a - e_{c_H} + \delta \sum_{v \in V(H) \setminus K_H} (e_v - e_{c_H}), \quad h_{c_H} = 0.$$

Orient a path as $v_0, \dots, v_j$. If $R_i > 0$ is the derivative resistance on $v_i v_{i+1}$, set $j_i = (h_{v_i} - h_{v_{i+1}})/R_i$. Conservation at an internal vertex v_i says exactly

$$j_i - j_{i-1} = \delta > 0,$$

so the currents strictly increase. A negative current means an increase in h , a positive current a decrease, and at most one current is zero. A zero current separates these phases and gives a two-vertex maximum plateau. Strictness on either side proves both level claims, also for the subsequence of internal vertices and for coincident endpoint values. $\square$

Lemma C.7 (Bounded nomination faces). *There is a resistance-independent family $\mathcal{F}$ of $n^{O(\tau)}$ closed rational nomination faces containing an MPD maximizer for every positive real resistance vector. In a selected nonbridge block, each face has at most $f_H \leq |K_H| + 2p_H \leq 8r_H - 2$ free coordinates. Balance leaves at most $8r_H - 3$ independent nominations if $f_H > 0$; if $f_H = 0$ it only tests feasibility. All other block nominations are fixed after aggregation. Bridges and faces with a sole free coordinate can be reduced to fixed rational nominations.*

Proof. Refine each local box from lemma C.4 separately. At a maximizer of $F_{\delta,\rho}$ the same feasible-exchange argument gives $h_v > \mu_0 \Rightarrow b_v^H = u_v^H$ and $h_v < \mu_0 \Rightarrow b_v^H = \ell_v^H$. Consequently the internal nominations of every path have the pattern

lower / optional free pivot / upper / optional free pivot / lower.

Every segment may be empty. A two-vertex plateau at the threshold is two adjacent free pivots with no intervening upper vertex. For a path with j internal vertices use positions $0, 1, \dots, 2j$, with vertices at odd positions and gaps at even positions. Choose positions $i_1 \leq i_2$. Odd chosen positions are free, vertices strictly between them upper, and the rest lower. These $O((j+1)^2)$ patterns cover all thresholds, including all-upper, all-lower, and monotone cases. Each marked vertex independently has status lower, upper, or free. Enumerating all combinations gives at most $3^{2r_H}(2n+1)^{2p_H}$ local patterns and the claimed free count. Intersect with exact balance and remove empty patterns. Singleton bounds are treated as fixed, regardless of the enumerated status.

The perturbed smoothed objectives converge uniformly. Orient each nonzero physical flow by decreasing potential. It has no directed cycle; successively subtracting the minimum flow on a source–sink path produces a finite nonnegative path decomposition. Its total weight is the total positive nomination. Thus $|x_e| \leq B$, where on the aggregated core we use the rational bound

$$B = \sum_v \max\{|\ell_v|, |u_v|\}.$$

Block aggregation does not increase this bound. For $0 < \rho \leq 1$, normalized potentials are bounded in absolute value by $(B^2 + B) \sum_e \beta_e$, by summing edge drops on a simple path. Therefore the perturbation is bounded by $\delta n(B^2 + B) \sum_e \beta_e$ uniformly on the local box. The energy compactness argument of lemma B.5 gives uniform convergence of the unperturbed smoothed objective. Letting both parameters tend to zero and taking a subsequence in one closed pattern face gives an original local maximizer in that face. Refining a coarse face that already contains a global maximizer preserves its optimal value because the other drops are constant. This is an existence proof; the algorithm needs no perturbation size.

A bridge’s local nomination is $(\zeta, -\zeta)$ with a rational feasible interval for ζ . Its entrance–exit drop $\beta\zeta|\zeta|$ is strictly increasing, so choose the largest endpoint. A sole free coordinate is fixed by balance. Include these rational choices and the fixed patterns from lemma C.4. Altogether the face count

is at most $C_{\mathfrak{r}}(n+1)^{6\mathfrak{r}+1}$ for a constant $C_{\mathfrak{r}}$ when $\mathfrak{r} \geq 1$; at rank zero it is $O(n)$. $\square$

C.3 Fixed resistances: algebraic local optimization

On a face eliminate one free nomination by balance, if any. Let z consist of the remaining nominations and the circulations in a fundamental cycle basis of the selected block. Tree routing is zero on every chord, and the basis has the identity matrix on chord rows. Consequently

$$x_e = X_e(z), \quad d_H = \max\{0, f_H - 1\} + r_H \leq 9r_H - 3 \quad (r_H \geq 1), \quad (20)$$

where X_e is rational affine, and each circulation is a chord flow in $[-B, B]$. The nomination inequalities and this circulation box form a compact rational polytope. An inactive block has only r_H circulation variables. The dimension bounds for nonbridge blocks are

Quantity	Bound or definition
$ K_H $	$\leq 2r_H$
p_H	$\leq 3r_H - 1$
f_H	$\leq 8r_H - 2$
d_H	$\leq 9r_H - 3$
d_*	$= \max\{1, 9\mathfrak{r} - 3\}$

The circulation-coordinate cell count at fixed total rank is given by Aßmann et al. [5, Proposition 4.7 (arXiv v1 numbering)]. Here the coordinates also include free nominations, and the rank bound is imposed separately on each block.

The affine hyperplanes $X_e = 0$ have at most $\sum_{i=0}^{d_H} \binom{m_H}{i}$ full-dimensional cells, by the standard recurrence. They can be constructed by inserting hyperplanes successively and testing the two strict sides of each current cell by rational linear feasibility; identically zero forms are omitted. Thereafter use closed cell inequalities. Their union covers all of space, even if the feasible polytope is lower dimensional or lies on a zero-flow hyperplane. On a cell choose signs $\epsilon_e \in \{-1, 1\}$ and put $Q_e = \epsilon_e X_e^2$ (zero forms contribute zero). These formulas agree on shared boundaries.

Lemma C.8 (Affine encoding on a sign cell). *For fixed positive β , a core point z satisfies the cell and cycle constraints if and only if $X(z)$ is its nomination's physical flow with those cell signs.*

Proof. Write $Z_{H,ie}$ for the signed cycle incidence. Let $\omega_{H,e} \in \{-1, 0, 1\}$ describe a fixed path from a to c_H . The complete physical and objective equations on a cell are

$$\sum_e Z_{H,ie} \beta_e Q_e(z) = 0 \quad (1 \leq i \leq r_H), \quad G_H(z, \beta) = \sum_e \omega_{H,e} \beta_e Q_e(z). \quad (21)$$

Indeed conservation is built into X , and orthogonality of the drop vector to a basis of $\ker A_H$ is equivalent to its membership in $\text{im } A_H^\top$. Integrating along the tree constructs the potentials, and uniqueness identifies these equations with the physical state. They have degree two at fixed resistances. $\square$

Proof of theorem C.1. Aggregate the terminal core. Enumerate the faces in lemma C.7. At a rational threshold, equation (21) and the cell and nomination constraints are a fixed-dimensional existential-real system. Renegar’s consistent-sign algorithm in appendix A.5 decides it in $\tau^2(\sigma D)^{O(d_H+1)}$ bit operations, where here $D = 2$, $\sigma = O(N)$, and clearing denominators gives the safe bound $\tau = O(N^4 + \iota)$ at bisection step ι . Binary search with a retained feasible lower threshold gives an active-block value interval of any requested width. Use the rational absolute drop bound $V_{\max} = B^2 \sum_e \beta_e$ for the initial bracket.

For each inactive block the same equations have only r_H variables and a unique physical solution. Projection onto a circulation or drop coordinate, using equation (11), and univariate isolation therefore give exact algebraic coordinates and drops, with degrees bounded in terms of $\mathfrak{r}$ and coefficient bits polynomial in N . This also works for a solution on several sign boundaries: project each closed cell, retain its unique real physical solution, and discard empty cells. No nonsingular Jacobian or complex isolated-root hypothesis is needed. A fixed bridge contributes a rational constant. Isolate and refine block drops separately; never form the algebraic field generated by all blocks.

Give each block an interval of width $\varepsilon/(8n)$; summing gives each face a certified interval of width at most $\varepsilon/8$, and taking separately the largest lower and upper endpoints gives a global interval of that width. Choose a face with largest lower endpoint. Its retained active threshold, together with the inactive block lower endpoints, is at least that face lower endpoint. Recover an algebraic feasible active core at this threshold by the fixed-dimension sample-point primitive in appendix A.5, applied to the cell, cycle, nomination, and threshold constraints. It supplies all d_H coordinates in one common local algebraic representation, with polynomial degree and encoding length for fixed $\mathfrak{r}$.

Put $C = 2B \sum_{e \in P_{st}} \beta_e$ for any simple terminal path. If $B = 0$, all core flows and the objective vanish; greedily disaggregate zero and finish. Otherwise refine each free nomination to a rational isolating interval of width at most $\varepsilon/(8nC)$. Intersect these intervals with the face box and exact balance, keeping its fixed coordinates. This rational polytope is nonempty because it contains the algebraic nomination. Rational linear feasibility, or greedy allocation between its coordinate bounds, produces a rational point. Its ℓ_1 error is at most $\varepsilon/(8C)$, so lemma B.7 bounds the objective loss by $\varepsilon/8$. Choosing the face lost at most $\varepsilon/8$; disaggregation loses nothing. This proves both output guarantees, including fixed and singleton faces.

For explicit parameter-dependent bounds, set $d_* = \max\{1, 9r - 3\}$ and $N_\varepsilon = N + \lceil \log_2(1/\varepsilon) \rceil + 1$, with $0 < \varepsilon \leq 1$. There are at most $C_r(N+1)^{6r+1}$ faces, at most $(N+1)^{d_*}$ cells per local problem, and at most N blocks per face. Bisection takes $O(N_\varepsilon)$ steps. For an absolute constant A_0 , the total decision work is bounded by

$$C_r(N+1)^{6r+2+d_*+A_0(d_*+1)}(N^4 + N_\varepsilon)^2 N_\varepsilon.$$

Projection of d_H variables with one free value coordinate has polynomial-count exponent $2(d_H + 1)$ in equation (11). A fixed number of such projections, sample-point calls, and root refinements bounds all remaining work by $C_r(N + N_\varepsilon + 1)^{2^{A_1(r+1)^2}}$ for an absolute constant A_1 . These loose bounds include rational arithmetic, output length, and rank-zero cases. $\square$

C.4 A parametric box lemma

Lemma C.9 (Parametric box LP). *Fix the core dimension d_0 , the number k of equations, and an input degree bound D_{in} . Let z range over a compact rational semialgebraic set $\mathcal{C} \subset \mathbb{R}^{d_0}$ described by an explicit quantifier-free formula whose polynomials have degree at most D_{in} . Let $\alpha_e(z), c_0(z) \in \mathbb{Q}[z]^k$ and $g_e(z), g_0(z) \in \mathbb{Q}[z]$ have degree at most D_{in} , and let $\beta_e \in [L_e, U_e]$ ($1 \leq e \leq m_0$) be independent finite rational intervals. The attainable set*

$$\mathcal{V} = \{(z, v) : z \in \mathcal{C}, \exists \beta : \sum_e \beta_e \alpha_e(z) = c_0(z), \quad g_0(z) + \sum_e \beta_e g_e(z) = v\}$$

has a quantifier-free description computable in polynomial bit time. Feasibility and exact optimization of v over this set, including an algebraic optimizing core and leaf vector of polynomial encoding length, are computable in polynomial bit time in the total rational input size.

Proof. Put $W_e(z) = (\alpha_e(z); g_e(z))$ and $w(z, v) = (c_0(z); v - g_0(z))$. For fixed z the image

$$\mathcal{Z}(z) = \sum_e W_e(z)[L_e, U_e]$$

is compact, as the continuous image of the compact product box under the sum map; convex, because coordinatewise convex combinations stay in the box and the map is linear; and nonempty, also with zero columns and singleton intervals. Its support function is

$$\max_{y \in \mathcal{Z}(z)} \eta^\top y = \sum_e \max\{L_e \eta^\top W_e(z), U_e \eta^\top W_e(z)\}.$$

The equality follows by independently choosing a maximizing endpoint of each interval; ties allow either endpoint. A point w belongs to a nonempty compact convex set S if and only if it satisfies every support inequality. Necessity is immediate; for sufficiency, if $w \notin S$, choose a closest point $y_0 \in S$. For each $y \in S$, differentiation of the squared distance along $y_0 + \vartheta(y - y_0)$ at $\vartheta = 0+$ gives $(w - y_0)^\top (y - y_0) \leq 0$. Taking $\eta = w - y_0$ gives $\eta^\top w = \eta^\top y_0 + \|w - y_0\|_2^2 > \max_{y \in S} \eta^\top y$, a contradiction. Hence membership is exactly

$$z \in \mathcal{C} \quad \wedge \quad \forall \eta \in \mathbb{R}^{k+1} : \quad \eta^\top w(z, v) \leq \sum_e \max\{L_e \eta^\top W_e(z), U_e \eta^\top W_e(z)\}. \quad (22)$$

We expand the maxima without introducing one variable per maximum or listing all formal sign assignments. Enumerate only the realizable sign vectors of the polynomials $t_e(z, \eta) = \eta^\top W_e(z)$ in $d_0 + k + 1$ variables, using Renegar's bound in appendix A.5. For each realized vector choose U_e for positive sign and L_e for negative or zero sign. On its sign condition replace the right side by the resulting sum, and take the disjunction of these guarded inequalities. This is exact at zero as well. With m_0 leaves there are $(m_0(D_{\text{in}} + 1))^{O(d_0 + k + 1)}$ such vectors, polynomially many terms, and degree at most $\max\{2, D_{\text{in}} + 1\}$. Summation and rational denominator clearing have polynomial coefficient bit length. This supplies a polynomial-size first-order formula with $d_0 + 1 + (k + 1)$ variables. Equation (11), with one quantified block of size $k + 1$ and $d_0 + 1$ free variables, has polynomial-count exponent $(k + 2)(d_0 + 2)$ and fixed degree and bit factors. It gives the asserted quantifier-free description.

The full feasible set is closed in the product of the compact core and the boxes, hence compact, and so is its image $\mathcal{V}$. If $\mathcal{V}$ is nonempty, the maximum is attained. Append to the quantifier-free formula $\Psi(z, v)$ the assertion

$$\neg \exists z', v' : \Psi(z', v') \wedge v' > v.$$

This again has a fixed number of variables. Eliminate the quantifiers using equation (11), then apply the sample-point primitive in appendix A.5 to obtain an exact optimal pair (z^*, v^*) in the common real algebraic field $\mathbb{Q}(\theta_{\text{alg}})$, with polynomial degree and encoding length. Empty sets are detected first.

The leaves are recovered constructively, so that no exponential enumeration of box corners is hidden in an LP step. All $W_e(z^*)$ and $w(z^*, v^*)$ lie in that same ordered algebraic field. Enumerate the full-dimensional cells of the central hyperplanes $\eta^\top W_e(z^*) = 0$ for nonzero columns with $L_e < U_e$. There are $m_0^{O(k+1)}$ cells. Fixed-dimensional exact linear sign tests can be done in $\mathbb{Q}(\theta_{\text{alg}})$, or by adjoining its single defining root and isolating inequalities to the rational tests. For each cell record the maximizing endpoint vector $\beta^{(j)}$ and its image $y_j = \sum_e W_e(z^*)\beta_e^{(j)}$. Zero columns and singleton boxes have a fixed arbitrary endpoint choice. These images include every vertex of $\mathcal{Z}(z^*)$. A singleton zonotope is immediate: every support choice has its sole image. Otherwise, the image of a box is the convex hull of the images of its finitely many corners, hence a polytope. A vertex can be strictly separated from the convex hull of the other distinct corner images. Its exposing directions contain an open set; choosing one outside the finitely many nonzero hyperplanes yields a cell in the enumeration. Conversely its cell support choice gives that vertex. Thus their convex hull is the whole zonotope, also when its affine dimension is smaller than $k + 1$.

Every point in a convex hull in $\mathbb{R}^{k+1}$ is a convex combination of at most $k + 2$ affinely independent vertices. Indeed, in a representation with a dependent support, an affine dependence lets one change the weights in both directions while preserving their sum and the point; move until one weight vanishes and repeat. Enumerate subsets of at most $k + 2$ of the y_j , solve their affine systems, and test nonnegative weights. At least one passes for $w(z^*, v^*)$. For its weights ϖ_j return $\beta = \sum_j \varpi_j \beta^{(j)}$, which lies in the box and has exactly the required image, so attains v^* . Only fixed-size linear systems and polynomially many field operations are used. Exact sign tests and this fixed-size linear algebra take polynomial bit time in $\mathbb{Q}(\theta_{\text{alg}})$, using the root's isolating representation and rational polynomial arithmetic. The solutions are determinant ratios of fixed size, and the final sums have polynomial encoding length in the same field. Ties, redundant equations, zero columns, and singleton zonotopes require no genericity.

An optimal vertex of the original box LP with k equations has at most k non-bound coordinates: otherwise a dependence of the corresponding columns permits a two-sided feasible motion. This fact alone is not a polynomial algorithm, because choosing those coordinates still leaves exponentially many

endpoint assignments. The recovery above avoids that enumeration: it uses only the polynomially many exposed zonotope vertices and need not output an LP vertex. $\square$

C.5 Joint optimization and rational recovery

Lemma C.10 (Joint Lipschitz estimate). *On any connected graph with the stated boxes, put $B = \sum_v \max\{|\ell_v|, |u_v|\}$ and fix a simple terminal path P_{st} . For all admissible scenarios in these unfiltered product boxes,*

$$|F_{s,t}(b, \beta) - F_{s,t}(b', \beta')| \leq C_b \|b - b'\|_1 + B^2 \|\beta - \beta'\|_1, \quad C_b = 2B \sum_{e \in P_{st}} U_e.$$

Proof. The nomination estimate lemma B.7 is uniform after replacing each resistance by its upper bound. For resistance sensitivity hold b fixed and differentiate the smoothed equations. With $R_\rho = \text{diag}(\beta_e(2|x_e| + \rho))$ and $T_e = x_e|x_e| + \rho x_e$, they give

$$A dx = 0, \quad A^\top d\pi = R_\rho dx + \text{diag}(T_e) d\beta.$$

Multiply by AR_ρ^{-1} , then pair with the ordinary unit adjoint $L_\rho h = e_s - e_t$. Symmetry yields

$$dF_\rho = \sum_e j_e T_e d\beta_e, \quad \frac{\partial F_\rho}{\partial \beta_e} = j_e(x_e|x_e| + \rho x_e).$$

There is no sign reversal, since the parameter term is on the right of the differentiated potential equation. The nonzero electrical currents follow strictly decreasing h , so they are acyclic, and conservation has just one unit source and one unit sink. Following positive edges from the source to the sink and subtracting the minimum edge current repeatedly decomposes them into paths of total weight one. Therefore $|j_e| \leq 1$. The physical path decomposition in lemma C.7 gives $|x_e| \leq B$, whence the derivative is bounded by $B^2 + \rho B$. Integrate on the resistance-box segment. Uniform convergence as $\rho \downarrow 0$ holds jointly over both compact boxes: in the energy argument use the common positive lower bound $\min_e L_e$ and the upper bounds U_e for the tree comparison energies. Uniqueness identifies every subsequential limit as before, and passing to the limit gives the resistance estimate. Changing nominations first and resistances second, the triangle inequality completes the proof. $\square$

Proof of theorem C.2. A joint maximizer exists by lemma A.10. With its resistances fixed, lemma C.7 supplies an equally good nomination in the

resistance-independent family, which is jointly optimal, since an improvement would contradict the original joint maximum. On any selected face, inactive aggregated nominations are fixed. Lemma A.5 and independence of resistance intervals make the face maximum the sum of independently optimized block drops.

For an active nonbridge cell use equation (20). Apply lemma C.9 with $k = r_H$, $\alpha_{e,i} = Z_{H,ie}Q_e$, $c_0 = 0$, $g_e = \omega_{H,e}Q_e$, and $g_0 = 0$. The core is the compact rational nomination/circulation polytope intersected with the closed sign cell. All coefficient degrees are at most two. The cycle equations are exactly the linking equations, so both directions of the mapping follow from equation (21). An inactive block uses only its r_H circulations and the same lemma. For a bridge, write its flow in entrance–exit orientation as ζ . At fixed ζ the best resistance is U_e if $\zeta \geq 0$ and L_e if $\zeta < 0$. The best value remains strictly increasing in ζ , so an active bridge takes the largest feasible ζ ; zero flow permits either endpoint. This covers fixed flows and singleton intervals.

Optimize each cell exactly, compare its algebraic values within that block, and retain a local optimizer. Approximate each block optimum by an interval of width $\varepsilon/(8n)$, sum by face, and select the largest face lower endpoint. The resulting global interval has width at most $\varepsilon/8$, and the selected face has an exact optimizing scenario with loss at most $\varepsilon/8$. Each block’s algebraic data are stored separately; sums are never compared and fields never combined.

If $B = 0$ on the aggregated core, its objective is zero: disaggregate zero nominations and choose rational resistance endpoints everywhere (off-core flows need not vanish). Otherwise $C_b > 0$. Refine every free core nomination coordinate to an interval of width $\varepsilon/(8nC_b)$ and recover a rational balanced point in its face by the rational box/balance feasibility procedure above. Refine each relevant resistance coordinate to width $\varepsilon/(8mB^2)$, intersect with $[L_e, U_e]$, and choose any rational point there. These operations give nomination loss at most $\varepsilon/8$ and resistance loss at most $\varepsilon/8$ by lemma C.10. The total loss, including selection, is at most $3\varepsilon/8$. Greedy disaggregation and arbitrary rational off-core resistance choices complete the output. Circulations need not survive rounding: the rounded scenario has its own physical state, and the Lipschitz estimate compares the objectives.

For bit costs, the face, block, and cell counts above still apply. In the box lemma, $d_0 \leq d_*$ and $k \leq \mathfrak{r}$, so the sign construction has $N^{O(d_* + \mathfrak{r} + 1)}$ terms and its first QE has count exponent $(\mathfrak{r} + 2)(d_* + 2)$. Its output degrees depend only on $\mathfrak{r}$. The optimal-pair formula adds one fixed-size quantifier block; the sample-point primitive in appendix A.5 then supplies the optimal core and value in fixed dimension. These sampling bounds and equation (11) bound

the full algorithm, including algebraic recovery and refinement, by

$$C_{\mathbf{r}}(N + N_{\varepsilon} + 1)^{2^{A_2(\mathbf{r}+1)^2}}$$

for an absolute constant A_2 , which is polynomial for every fixed rank bound. All constants used in rounding have polynomial rational bit length; the factors n, m cost only logarithmic extra precision. $\square$

C.6 Exact local arc values

Proof of theorem C.3. Let $e = (u, v)$ belong to block H . The block path between its endpoints consists of H alone. Aggregate all outside components into their attachments. The resulting balanced interval box has exactly the attainable local nominations, and outside resistances have no effect on x_e , by lemmas A.5 and B.3. At fixed resistances,

$$x_e = \text{sgn}(\pi_u - \pi_v) \sqrt{|\pi_u - \pi_v| / \beta_e}$$

is strictly increasing in its endpoint drop. Fixing the resistances of a joint flow maximizer therefore transfers a maximizer to the faces of lemma C.7 for terminals (u, v) . On each cell the target flow $X_e(z)$ is affine. In lemma C.9 use the cycle coefficients as before, but now $g_0 = X_e$ and all $g_f = 0$. This gives an exact algebraic cell optimum of polynomial encoding length. Compare the polynomially many cell and face values exactly by real-algebraic sign determination. To minimize, reverse the ordered terminals and the target orientation and maximize $-X_e$ with the corresponding face family.

For a bridge let S_e be its tail component and $\bar{S}_e = V \setminus S_e$. Writing $\ell(S_e) = \sum_{v \in S_e} \ell_v$, and similarly for the other sums, the exact attainable flow interval is

$$[\ell(S_e), u(S_e)] \cap [-u(\bar{S}_e), -\ell(\bar{S}_e)].$$

Interval addition and greedy allocation prove attainability of every point, and nonemptiness follows from the balanced-box hypothesis. The rational endpoints are independent of resistances.

Repeat for every tested edge and check whether its maximum is at most its rational upper capacity and its minimum at least its lower capacity; exact sign tests include equality. There are at most $2m$ optimizations with the same fixed-rank polynomial bound. Capacities test all original scenarios and never filter the optimization domain. Unlike the pressure span in theorem B.2, an arc objective involves no sum of independently encoded block values: all its algebraic computation occurs in one bounded-dimensional block core. $\square$

Remark C.11 (Optional rational arc witnesses). Round an algebraic *arc-flow* optimizer, with target flow x , and let y be the target flow of the rounded scenario. If their endpoint drops are Δ, Δ' and target resistances β_e, β'_e , then

$$|x - y|^2 \leq \frac{2}{L_e} (|\Delta - \Delta'| + B^2 |\beta_e - \beta'_e|).$$

Indeed $|x|x - |y|y$ has magnitude at least $|x - y|^2/2$: for equal signs use $|x+y| \geq |x-y|$, and for opposite signs use $x^2 + y^2 \geq (|x| + |y|)^2/2$. Subtracting $\Delta = \beta_e x|x|$ and $\Delta' = \beta'_e y|y|$ gives $\beta_e |x|x - \beta_e y|y| = \Delta - \Delta' + (\beta'_e - \beta_e)y|y|$; take absolute values and use $|y| \leq B$ and $\beta_e \geq L_e$. The joint Lipschitz estimate for (u, v) then permits rational scenario rounding to a desired flow error, using its square in the error budget and only polynomially many precision bits. A pressure-maximizing resistance scenario cannot replace the arc optimizer: on two parallel edges with unit total flow, other resistance one, and target resistance in $[1, 4]$, the target flow is $1/(1 + \sqrt{\beta_e})$, decreasing in β_e , whereas its endpoint drop $\beta_e/(1 + \sqrt{\beta_e})^2$ is increasing.

C.7 Limits of the structural and arithmetic claims

Proposition C.12 (Broader graph parameters do not suffice). *Unless $P = NP$, a polynomial additive algorithm with polynomial dependence on accuracy bits cannot hold on all networks of treewidth two, on all series-parallel networks, or on all networks of feedback vertex number one, even with fixed positive rational resistances.*

Proof. Use Thürauf [49, Section 4, Figure 2, equation (3), and Lemmas 4.3 and 4.17 (numbering of the April 6, 2022 manuscript)]. For a Partition instance with $j \geq 3$ positive integers S_i and $K_0 = \sum_i S_i$, his graph has edges $sz_i^+, z_i^+t, z_i^+z_i^-$. The bookings are $K_0/2$ at s, t and S_i at z_i^+, z_i^- ; translate entries to intervals $[0, b_v]$ and exits to $[-b_v, 0]$. The main-edge resistances are $1/S_i^2$ and pendant resistances one. These are admissible rational data here, with a nonempty balanced box, and his objective (2) imposes no potential bounds. The cited lemmas give maximum drop at least one for a yes instance and strictly below $\Theta_0(K_0) < 1$ for a no instance. For an explicit encoding gap, equation (3) defines, in renamed notation,

$$\begin{aligned} \vartheta_0 &= 1 - \left(1 - \frac{1}{8K_0^2}\right)^2, & \Pi_0 &= \max\{1 - \vartheta_0 + \vartheta_0^2, 1 - \vartheta_0^2/K_0^2\}, \\ \tilde{\vartheta}_0 &= (1 - \Pi_0)/5, & \Theta_0(K_0) &= \max\{1 - \tilde{\vartheta}_0^2/(K_0^2 j^2), \Pi_0 + 4\tilde{\vartheta}_0\}. \end{aligned}$$

Here $0 < \vartheta_0 < 1$, $0 < \Pi_0 < 1$, and both terms defining $\Theta_0(K_0)$ are strictly below one. Fixed powers, rational arithmetic, and maxima give polynomial encoding length, so $\mathfrak{g} = 1 - \Theta_0(K_0) > 0$ is rational with polynomially bounded $\log(1/\mathfrak{g})$. A certified interval of width $\mathfrak{g}/4$ lies entirely above $(1 + \Theta_0(K_0))/2$ in the yes case and below it in the no case, deciding Partition.

By lemma A.9, the main block $K_{2,j}$ has rank $j - 1$ and is series-parallel; adding pendant edges preserves that property. Bags $\{s, t, z_i^+\}$ in a path, with each bag $\{z_i^+, z_i^-\}$ attached to its corresponding main bag, give a width-two tree decomposition. There is a cycle, so the treewidth is exactly two. Deleting s leaves the tree with branches $t - z_i^+ - z_i^-$, and the cycle shows that the feedback vertex number is exactly one. These class bounds and the gap prove the proposition. This does not contradict the bounded-block-rank results, since $j - 1$ grows. $\square$

Remark C.13 (Convex fibers and exact comparison). Convexity alone cannot replace the box hypothesis in the exact-output lemma without encountering the SRS barrier. With no nonlinear core, let leaf i be the singleton $\{y_i : 0 \leq y_i \leq a_i + 1, y_i^2 = a_i\}$ for a positive integer a_i . Each fiber is compact and convex and has a degree-two rational description, but its unique point is $\sqrt{a_i}$. Exact optimization and rational-threshold comparison of $\sum_i y_i$ would decide $\sum_i \sqrt{a_i} \leq K$, and these independent fibers supply no fixed common algebraic field of bounded degree. Thus the box lemma's proof does not extend merely by assuming convex quadratic fibers. This is an exact-arithmetic obstruction, not an NP-hardness claim or a proof that every larger leaf class is intractable.

D Representation of edge laws

The structural reduction of appendix C uses monotonicity, whereas its exact algorithm uses a polynomial representation. We separate these roles for continuous piecewise-polynomial laws with independent uncertain coefficients, show that a fixed fractional power changes exact arc comparison even on one cycle, and use positive rational approximations to recover an algorithm polynomial in accuracy bits for a fixed finite family of fractional powers. All optimization domains in this section are unfiltered product domains: operating capacities are tested after optimization, not imposed as scenario constraints.

D.1 Dense polynomial laws and their admissibility

An edge has the law

$$g_e(x, \theta_e) = f_{e0}(x) + \sum_{j=1}^{h_e} \theta_{ej} f_{ej}(x), \quad \theta_{ej} \in [\underline{\theta}_{ej}, \bar{\theta}_{ej}]. \quad (23)$$

The intervals are finite and rational, and all their coordinates are independent, including across edges. Every basis function is continuous on $\mathbb{R}$, vanishes at zero, and is specified by a finite list of rational breakpoints and rational polynomial pieces, including its two unbounded outer intervals. Polynomials are densely encoded: coefficients through the stated degree are listed, including zeros. Degrees, piece counts, and the h_e may grow with input size. The complete law must be strictly increasing for every allowed coefficient vector. Individual bases need not be increasing, and coefficient intervals need not be positive. There is no derivative-matching condition at breakpoints.

This affine notation is also used later for explicitly stated correlated models; the results proved here use independent boxes. In particular, a parameter shared by different edges is not a coordinate θ_{ej} in this model. Fixed laws correspond to $h_e = 0$. Positive uncertain mixtures of x , $x|x|$, x^3 , and $\max\{x - a, 0\}$ with rational $a \geq 0$ give examples whenever the complete law is strictly increasing. Signed even integer powers use two polynomial pieces.

Proposition D.1 (Validation of the law family). *All the representation and admissibility conditions above can be checked in polynomial bit time in the dense rational input size.*

Proof. Check breakpoint order and discard empty pieces. Rational evaluation checks continuity at every breakpoint and the value at zero of every basis. Refine each edge's pieces by the union of its basis breakpoints. The total number of refined descriptions is polynomial in the explicit input size; this operation does not enumerate combinations of pieces. On an open refined interval J , the minimum derivative over the box is

$$f'_{e0}(x) + \sum_j \min\{\underline{\theta}_{ej} f'_{ej}(x), \bar{\theta}_{ej} f'_{ej}(x)\}. \quad (24)$$

Discard identically zero f'_{ej} when forming a root partition. Isolate and order the real roots in J of the product of all remaining f'_{ej} . Its degree and coefficient bits are polynomial in dense input length. Between consecutive roots each endpoint choice in equation (24) is fixed, and univariate polynomial sign

tests check nonnegativity of the resulting rational polynomial on each such interval, including unbounded intervals. At partition roots use continuity of this minimum of finitely many continuous polynomials. One-sided limits suffice at original breakpoints. These tests are polynomial-time instances of the real-algebraic primitives in appendix A.5. They decide whether every complete law is nondecreasing.

For each original refined interval J , also test the rational linear system that sets every coefficient of $f'_{e0} + \sum_j \theta_{ej} f'_{ej}$ to zero, together with the coefficient box; rational LP decides its feasibility in polynomial time. A feasible system supplies an allowed law constant on J , which fails strict increase. Conversely, a continuous nondecreasing function failing strict increase is constant on a nontrivial interval. Some open subinterval lies in one refined piece, and a polynomial derivative vanishing there vanishes identically on that piece, so one of these LPs is feasible. Reject exactly in this case. There are polynomially many LPs and sign tests, and no field containing all isolated roots is needed. $\square$

The strictness test is essential. The family $x + \theta|x|$ with $\theta \in [-1, 1]$ is nondecreasing but has a flat half-line at either endpoint; the smaller box $[-1/2, 1/2]$ is admissible. In contrast, $(1 - \theta)x^3 + \theta x$, $\theta \in [0, 1]$, is admissible despite a zero derivative at $x = 0, \theta = 0$: isolated derivative zeros are permitted.

D.2 Structural transfer, dense degrees, and coefficient recovery

Lemma D.2 (Law-independent nomination faces). *For any fixed continuous strictly increasing laws vanishing at zero, the graph-defined face family of lemma C.7 contains a maximizer of a prescribed terminal drop. Its size and free-coordinate bounds are unchanged. The same family contains a joint maximizer when the laws are jointly continuous in flow and parameters in a compact parameter domain and remain admissible throughout that domain.*

Proof. Existence follows from theorem A.1. Acyclic physical flows give $|x_e| \leq B := \sum_v \max\{|\ell_v|, |u_v|\}$, as in the proof of lemma C.7. On this flow interval, compactness and continuity of the laws bound every normalized potential by a sum of edge-drop bounds along a path. If inputs converge, every subsequential state limit satisfies the limiting equations, and uniqueness identifies it. This proves continuity of the state and attainment on the compact input domain.

To handle corners, fix a nonnegative smooth kernel κ supported on $[-1, 1]$ with integral one, and put

$$\begin{aligned} S_\rho f(x) &= \int \kappa(t) f(x - \rho t) dt - \int \kappa(t) f(-\rho t) dt, \\ g_{e,\rho}(x) &= S_\rho g_e(x) + \rho x, \quad 0 < \rho \leq 1. \end{aligned}$$

Convolution preserves increase, centering enforces value zero at zero, and $g'_{e,\rho} \geq \rho$. The smoothed laws converge uniformly on the flow interval to the original laws; for a compact continuous parameter family this convergence is uniform in the parameters too. The preceding bounded-state subsequence argument gives uniform convergence of physical states; centering preserves the sign of flow and its acyclic bound in this argument.

For these smooth laws the differentiated network has positive diagonal resistances $g'_{e,\rho}(x_e)$. The proofs of lemmas C.4 and C.6 therefore apply: the unit adjoint has ordered block ranges, and positive forcing δ at the internal vertices of a suppressed path gives $j_i - j_{i-1} = \delta$. The same threshold patterns leave at most $8r_H - 2$ free nominations in the selected nonbridge block. The perturbing sum of normalized potentials is uniformly bounded, so its contribution tends uniformly to zero with δ . Let ρ, δ tend to zero and take a subsequence in one of the finitely many closed faces; its limit maximizes the original objective. This argument only proves existence of an optimizer on a face; neither smoothing nor its rate is part of the algorithm. Finally, at a joint maximizer fix its parameters and replace its nomination by such a face maximizer, which remains jointly optimal. The face family depends on the graph and nomination bounds, not on the chosen laws. $\square$

Lemma D.3 (Growing dense degree in the box lemma). *In lemma C.9, fix only the core dimension and the number of linking equations. Allow the degree bound to grow, with all core polynomials densely encoded. Its exact optimization and recovery conclusions still have polynomial bit complexity. The returned common local algebraic field has polynomial degree and encoding length, not necessarily constant degree.*

Proof. The support scores there are $\eta^\top W_j(z)$, of degree at most $D + 1$ in a fixed number of variables. The number of realizable sign vectors is $(h(D + 1))^{O(d_0 + k + 1)}$, so listing only these vectors and forming their guarded support sums gives a polynomial-size formula, and every sum has polynomial coefficient bits. The fixed-dimensional quantifier-elimination bound equation (11) is polynomial in polynomial count, numerical degree, and coefficient bits; it does not require degree to be fixed. Apply it also to the optimal-pair formula, then

use appendix A.5. Degrees and encoding lengths of the resulting algebraic sample are polynomial for fixed dimensions.

For recovery, all columns evaluated at this sample lie in its one ordered algebraic field. The construction in lemma C.9 enumerates polynomially many support cells in fixed dimension, then subsets of at most $k + 2$ image vertices and fixed-size affine systems. Exact field arithmetic and sign determination are polynomial in the field degree and coefficient bits. Fixed-size determinants and the final polynomially many sums keep these sizes polynomial. Thus growing local degree does not create a compositum of independently represented roots. Dense encoding bounds D by the input size; sparse binary-encoded exponents would not give this bound. $\square$

For later rational recovery define basis bounds $N_{ej}, M_{ej} \in \mathbb{Q}_{\geq 0}$ on $[-B - 1, B + 1]$ by taking, over all pieces $\sum_i a_i x^i$, the maxima of

$$\sum_i |a_i| T^i, \quad \sum_{i \geq 1} i |a_i| T^{i-1}, \quad T = \max\{1, B + 1\},$$

respectively. Zero bounds are allowed. Put $U_{ej} = \max\{|\underline{\theta}_{ej}|, |\bar{\theta}_{ej}|\}$ and $M_e = M_{e0} + \sum_j U_{ej} M_{ej}$. These rationals have polynomial bit length even for growing dense degrees.

Lemma D.4 (Pressure sensitivity for affine law boxes). *For a simple terminal path P_{st} put $C_b = \sum_{e \in P_{st}} M_e$. Writing $F_{s,t}(b, \theta) = \pi_s - \pi_t$, we have*

$$|F_{s,t}(b, \theta) - F_{s,t}(c, \eta)| \leq C_b \|b - c\|_1 + 2 \sum_{e,j} N_{ej} |\theta_{ej} - \eta_{ej}|. \quad (25)$$

Proof. Use the centered smoothing from lemma D.2. A continuous piecewise polynomial is locally Lipschitz; its convolution derivative is the convolution of its almost-everywhere derivative. Hence the complete smoothed derivative lies in $[\rho, M_e + \rho]$ on physical flows. Testing the minimum energy of the unit s - t electrical adjoint against a unit path flow bounds its effective resistance by $\sum_{e \in P_{st}} (M_e + \rho)$. Grounding at t , the maximum principle puts its potentials between zero and that resistance, so the nomination sensitivity $dF = h^\top db$ is bounded by this path constant times $\|db\|_1$.

Linearity of centered convolution preserves the affine coefficients. At fixed b , differentiating the state equations and pairing with the unit adjoint gives

$$\frac{\partial F_{s,t,\rho}}{\partial \theta_{ej}} = j_e S_\rho f_{ej}(x_e).$$

As in lemma C.10, the electrical current is acyclic and decomposes into paths of total weight one, so $|j_e| \leq 1$. Also $|S_\rho f_{ej}(x_e)| \leq 2N_{ej}$. Integrate first on the balanced nomination segment and then on the coefficient-box segment; every intermediate law is admissible. Uniform state convergence as $\rho \downarrow 0$ proves the claimed estimate. $\square$

Theorem D.5 (Independent polynomial-law uncertainty). *Fix $\mathfrak{r}$. On a connected graph of block rank at most $\mathfrak{r}$, take the law model equation (23) and a nonempty balanced rational nomination box. In polynomial time in the dense input size and requested accuracy bits one can compute a rational interval of width at most ε containing the maximum of any prescribed terminal drop, and rational admissible nomination and coefficient inputs whose objective is within ε of that maximum. Maximum and minimum signed flow on any arc are exactly computable real algebraic numbers of polynomial encoding length. Consequently unfiltered robust validation of rational arc capacities is polynomial-time decidable, including equality.*

Proof. The case $s = t$, an edgeless graph, or $B = 0$ is immediate for the relevant zero objective; rational feasible inputs are obtained by box/balance allocation. Otherwise use lemma D.2. On each face all but one block have fixed aggregated nominations. Since coefficients are independent across blocks, the maximum face value is the sum of the independently optimized block drops.

In a block, use the affine flow coordinates $X_e(z)$ of equation (20). Pull back every basis breakpoint a to the affine hyperplane $X_e(z) = a$. Their arrangement has polynomial size in the fixed core dimension. Use its closed cells, intersected with the compact nomination and circulation polytope. Constant affine forms are assigned their appropriate piece, and at a breakpoint both adjacent formulas agree, so boundary states are neither lost nor added. On a cell every $f_{ej}(X_e(z))$ is a rational polynomial. Substitution of a degree- D polynomial into an affine form in d variables has at most $\binom{D+d}{d}$ monomials and polynomial coefficient bits when d is fixed.

For a cycle-basis row $Z_{H,i}$ and a path vector ω_H , the linking equations and objective are

$$\sum_e Z_{H,ie} \left(f_{e0}(X_e) + \sum_j \theta_{ej} f_{ej}(X_e) \right) = 0,$$

$$G_H(z, \theta) = \sum_e \omega_{H,e} \left(f_{e0}(X_e) + \sum_j \theta_{ej} f_{ej}(X_e) \right).$$

Their equivalence to the physical equations follows from orthogonality to the cycle space, exactly as in equation (21). There are r_H linking equations; each scalar θ_{ej} is one interval leaf. Move the fixed terms to the right side and apply lemma D.3. A bridge uses its conserved affine flow and no cycle equation, so it is covered as well. Compare local cell optima exactly.

Keep algebraic data for distinct blocks separate. Approximate each block maximum by an interval of width at most $\varepsilon/(8n)$, sum by face, and take the maximum lower and upper endpoints. This brackets the global maximum with width at most $\varepsilon/8$. Selecting the face with largest lower endpoint loses at most $\varepsilon/8$; store its exact local optimizers. To recover a rational input, enclose their free nomination coordinates in rational intervals and intersect with their rational face and exact balance. This polytope contains the algebraic nomination, so rational LP (or box/balance allocation) returns a feasible rational nomination. Round each coefficient in its own isolating interval intersected with its input interval. Choose widths so that the two terms in equation (25) are each at most $\varepsilon/8$, ignoring zero sensitivity constants rather than dividing by them; only polynomially many precision bits are required. Greedy disaggregation and arbitrary rational coefficients off the terminal core complete the scenario. Its physical state need not have the sampled circulations; the sensitivity estimate controls its pressure directly.

For an arc $e = (u, v)$, aggregate into its one block. At fixed coefficients, strict increase of g_e makes maximizing x_e equivalent to maximizing $\pi_u - \pi_v$ over nominations. Thus a joint arc maximizer lies on one of the same faces after its coefficients are fixed. Use the affine objective $X_e(z)$ in the box lemma. A minimum uses $-X_e$ and reversed ordered terminals; oddness of the law is unnecessary. Exact comparison of the polynomially many local algebraic values gives each extremum, and rational-capacity comparison includes equality; no sum of independent block values arises. This conclusion neither requires nor claims a rational near-extremal arc scenario for arbitrary polynomial laws. $\square$

The general monotone-flow model and uncertainty in physical coefficients are established; for example, Aßmann et al. [5, Sections 3 and 4] study polynomial robust feasibility, with explicit tree and cycle reductions. Thürauf et al. [50, Sections 1 and 3] reduce robust-feasibility checking for a fixed design to polynomially many nonlinear worst-case problems and use them in an iterative adversarial design algorithm; that reduction is not a polynomial-time solver for each problem. The contribution of theorem D.5 is the fixed-block-rank bit bound with growing dense degrees and arbitrarily many independent coefficients, using the face reduction and fixed-dimensional support elimina-

tion. It does not extend by analogy to uncertain breakpoints, sparse binary exponents, or shared coefficient constraints.

D.3 A fractional-power exact arc obstruction

Write $\phi_q(x) = \text{sgn}(x)|x|^q$ for $q > 0$, with $\phi_q(0) = 0$. The law is algebraic for rational q , but this alone does not provide a polynomial-size dense description of a sum of many such laws in a bounded-dimensional core.

Theorem D.6 (Fractional-power arc comparison). *For the fixed common law $g_e(x) = \beta_e \phi_{3/2}(x)$, exact comparison of one signed arc flow with a rational capacity is SRS-hard already on a simple cycle with fixed integral nominations and positive integral resistances. The same holds for every fixed reduced rational exponent $p/q > 1$ whose denominator q is even. It also holds for robust validation when singleton scenario boxes are allowed.*

Proof. Take positive integers $a_1, \dots, a_k, K$ and the question $\sum_i \sqrt{a_i} \leq K$. If necessary append unit radicands and increase K accordingly to ensure $k \geq 2$. On a consistently oriented simple cycle with $k+1$ edges prescribe a base flow and resistances by

$$c_i = a_i, \quad \beta_i = 1/a_i \quad (1 \leq i \leq k), \quad c_{k+1} = -1, \quad \beta_{k+1} = K.$$

Set the nomination at each vertex to its outgoing base flow minus its incoming base flow. It is integral and balanced, and every conserved flow is $x_i = c_i + z$ for one circulation z . The physical equation is

$$h(z) := \sum_{i=1}^{k+1} \beta_i \phi_{3/2}(c_i + z) = 0.$$

It is continuous, strictly increasing, and has opposite infinite limits, so its root z^* is unique. Since $h(0) = \sum_i \sqrt{a_i} - K$, including the equality case,

$$\sum_i \sqrt{a_i} \leq K \iff h(0) \leq 0 \iff z^* \geq 0 \iff x_{k+1} \geq -1.$$

This is one lower capacity. Since the law is odd, reversing that arc gives the upper capacity $-x_{k+1} \leq 1$. Multiplying all resistances by $\prod_i a_i$ makes them positive integers without changing the root. The product and every nomination have polynomial binary encoding length.

For a fixed reduced $p/q > 1$ with even q , coprimality makes p odd. Replace the positive-edge data by

$$c_i = a_i^{q/2}, \quad \beta_i = a_i^{-(p-1)/2}.$$

Then $\beta_i \phi_{p/q}(c_i) = a_i^{1/2}$. Retain the final edge and the same capacity. The preceding equivalences hold unchanged. All exponents in these rational data are fixed integers, so their encoding lengths are polynomial. A common positive integer clears the resistance denominators. Singleton boxes give the robust statement. If a convention requires bounds on every arc, bound all other flows by the rational acyclic-flow bound B ; those bounds impose no further restriction. $\square$

This is a lower bound for exact comparison, not NP-hardness or an additive-approximation obstruction. No converse reduction is asserted, and the proof says nothing about fixed exponents of odd denominator. General homogeneous laws and their arithmetic evaluation issues already appear in Groß et al. [29, Section 2.1, May 2018 manuscript]; the specific contribution here is the capacity predicate on a single cycle with fixed data. Our limited source comparison does not establish priority over all older nonlinear-circuit or hydraulic literature.

D.4 Positive rational approximation with an explicit encoding

Positive rational approximation of fractional powers is established; for example, the uniform half-line estimate in Bonito and Pasciak [12, Section 3.3, equation (37), Lemma 3.4, and Remark 3.1, arXiv version] for $\lambda^{-\alpha}$ becomes one for t^α on $(0, 1]$ under $\lambda = 1/t$. Its positive fractions extend continuously by zero at $t = 0$. Their Section 3.2 also uses Gaussian quadrature on dyadic panels. The following construction supplies the rational encoding and monotonicity needed by the network algorithm, with explicit error and bit bounds.

Lemma D.7 (Positive rational scalar approximation). *Fix a rational $0 < \alpha < 1$. Given a positive rational $\tau \leq 1$, one can construct positive rational a_ν, s_ν such that*

$$R_\alpha(t) = \sum_{\nu=1}^J a_\nu \frac{t}{t + s_\nu}, \quad R_\alpha(1) = 1, \quad \sup_{0 \leq t \leq 1} |R_\alpha(t) - t^\alpha| \leq \tau.$$

The term count is $O_\alpha((1 + \log(1/\tau))^2)$. Construction time, coefficient bits, and the expanded dense numerator and denominator sizes are polynomial in the tolerance encoding and accuracy bits.

Proof. For $t \geq 0$ set

$$I_\alpha(t) = \int_0^\infty \frac{ts^{\alpha-1}}{t+s} ds.$$

It converges for $t > 0$, and its integrand is zero for $t = 0$. Substitution $s = tu$ for $t > 0$ proves $I_\alpha(t) = t^\alpha I_\alpha(1)$, also valid at zero. For $0 \leq t \leq 1$, truncating to $[2^{-L}, 2^L]$ loses at most

$$\alpha^{-1}2^{-\alpha L} + (1 - \alpha)^{-1}2^{-(1-\alpha)L}.$$

On panel $[2^k, 2^{k+1}]$, substitute $s = 2^k u$ and use the n -point Gauss–Legendre rule on $[1, 2]$, with nodes $u_j \in (1, 2)$ and positive weights w_j summing to one; it is exact through degree $2n - 1$. The resulting approximation is

$$Q(t) = \sum_{k=-L}^{L-1} \sum_{j=1}^n w_j 2^{k\alpha} u_j^{\alpha-1} \frac{t}{t + 2^k u_j}.$$

The error is uniform, even at $t = 0$. The panel integrand

$$F_{k,t}(u) = 2^{k\alpha} u^{\alpha-1} \frac{t}{t + 2^k u}$$

is analytic on a neighborhood of $|u - 3/2| \leq 1$. On this disk $\operatorname{Re} u \geq 1/2$, so the principal power is analytic, $|u^{\alpha-1}| \leq 2$, and $|t/(t + 2^k u)| \leq 1$. Cauchy’s coefficient bound makes the degree- $(2n - 1)$ Taylor remainder on $[1, 2]$ at most $4 \cdot 2^{k\alpha} 2^{-2n}$. Exactness and positivity of the quadrature bound its integration error by twice that remainder. Thus

$$\sup_{[0,1]} |I_\alpha - Q| \leq \alpha^{-1}2^{-\alpha L} + (1 - \alpha)^{-1}2^{-(1-\alpha)L} + 16L 2^{L-2n}. \quad (26)$$

For target unnormalized error ζ , take $L = O_\alpha(1 + \log(1/\zeta))$ and $n = O_\alpha(1 + \log(1/\zeta))$ so the right side is at most $\zeta/2$. This gives $J = 2Ln$ terms. Integer choices can be made from rational upper bounds on the displayed powers, or from these fixed constant inequalities with additional slack.

It remains to round effectively. The Legendre recurrence constructs a rational polynomial of degree n and polynomial coefficient bits. Root isolation and the positive weight formula $2/((1 - v_j^2)P'_n(v_j)^2)$ on $[-1, 1]$, followed by the affine change of interval, give rational enclosures for each node and weight. Their fixed rational powers are obtained by isolating the corresponding positive algebraic roots. Each individual number has polynomial algebraic degree and encoding length; one never forms their joint algebraic field. Polynomial-time arbitrary-precision enclosures also follow directly from Johansson and Mezzarobba [31, Sections 1 and 7.2].

Before rounding, $s = 2^k u_j \geq 2^{-L}$ and $0 < a = w_j 2^{k\alpha} u_j^{\alpha-1} \leq 2^L$. Enclose and round each a, s upward to positive dyadic rationals with error at most

$h \leq 1$; this works even for an arbitrarily small weight, because the upper endpoint of a narrow enough enclosure is positive. Along these rounding segments, $s \geq 2^{-L}$ and $a \leq 2^{L+1}$. For every $t \geq 0$,

$$\left| \frac{\partial}{\partial a} \frac{at}{t+s} \right| \leq 1, \quad \left| \frac{\partial}{\partial s} \frac{at}{t+s} \right| = \frac{at}{(t+s)^2} \leq \frac{a}{s} \leq 2^{2L+1}.$$

Choose a dyadic h with $Jh(1 + 2^{2L+1}) \leq \zeta/2$. Its required precision is $O(\log(1/\zeta) + L + \log J)$ bits, and the rounded sum $Q_{\mathbb{Q}}$ has uniform error at most ζ and positive coefficients.

On $[1, 2]$, $s^{\alpha-1}/(1+s) \geq 1/6$, so $I_{\alpha}(1) \geq 1/6$. If $\zeta \leq 1/12$, then $Q_{\mathbb{Q}}(1) \geq 1/12$. Set $R_{\alpha}(t) = Q_{\mathbb{Q}}(t)/Q_{\mathbb{Q}}(1)$. Since $t^{\alpha} \leq 1$,

$$|R_{\alpha}(t) - t^{\alpha}| \leq \frac{|Q_{\mathbb{Q}}(t) - I_{\alpha}(t)| + t^{\alpha}|I_{\alpha}(1) - Q_{\mathbb{Q}}(1)|}{Q_{\mathbb{Q}}(1)} \leq 24\zeta.$$

Take $\zeta \leq \min\{1/12, \tau/24\}$. Normalization is exact rational arithmetic. Products of the J positive linear denominators have degree J and polynomial coefficient bits: denominator bit lengths add, and coefficient growth adds at most polynomially many further bits. Summing the corresponding numerators obeys the same bounds, which proves the dense encoding and construction claims. $\square$

Lemma D.8 (Rational constitutive approximation). *Fix a nonempty finite family $\mathcal{Q} \subset \mathbb{Q} \cap (1, \infty)$. Given rational $B > 0$ and $\delta > 0$, one can construct for every $q \in \mathcal{Q}$ a rational function ψ_q which is smooth, strictly increasing on $\mathbb{R}$, vanishes at zero, and has a denominator positive on all of $\mathbb{R}$, such that*

$$\sup_{|x| \leq B} |\psi_q(x) - \phi_q(x)| \leq \delta.$$

Its expanded dense encoding and construction time are polynomial in the input and accuracy bits, with constants depending on $\mathcal{Q}$.

Proof. For odd integer q use $\psi_q(x) = x^q$ exactly. Otherwise write uniquely

$$q = 2j + 1 + 2\alpha, \quad j \geq 0 \text{ an integer}, \quad 0 < \alpha < 1.$$

Choose $T = 2^h \geq \max\{1, B\}$ with h a nonnegative multiple of the denominator of 2α . The rational $T^{2\alpha}$ has polynomial bit length because the exponent family is fixed. Put

$$\psi_q(x) = T^{2\alpha} x^{2j+1} R_{\alpha}(x^2/T^2).$$

The corresponding expression with $R_\alpha(t)$ replaced by t^α is exactly $\phi_q(x)$. Apply lemma D.7 with $\tau \leq \min\{1, \delta/(T^{2\alpha}B^{2j+1})\}$. Every term of ψ_q is a positive rational multiple of $x^{2j+3}/(x^2 + c)$ for a rational $c > 0$, and

$$\frac{d}{dx} \frac{x^{2j+3}}{x^2 + c} = \frac{x^{2j+2}((2j+1)x^2 + (2j+3)c)}{(x^2 + c)^2}.$$

This is positive for $x \neq 0$ and nonnegative at zero. Therefore the sum is strictly increasing, odd, vanishes at zero, and is unbounded in both directions. Its denominator is a product of positive quadratics. The number of terms, their bit lengths, and expansion sizes are polynomial as in the preceding lemma; j and all rational exponent denominators are fixed. The range $1 < q < 3$ is the case $j = 0$; the extra odd power covers fixed exponents above three. $\square$

Lemma D.9 (Rational-law local optimization). *Suppose $g_e(x) = \beta_e \psi_e(x)$, where the densely encoded rational $\psi_e = P_e/Q_e$ are strictly increasing, vanish at zero, and $Q_e(x) > 0$ for all real x . Resistances have independent positive rational intervals. At fixed block rank, the face construction and exact local algebraic optimization of theorem D.5 remain polynomial in the rational encoding, provided polynomial-bit rational bounds on the laws on $[-B, B]$ are supplied. Thus terminal-drop optima have arbitrarily accurate rational intervals; each nomination face has separate exact local algebraic optimizers, and global face selection is additive. Arc-flow optima have exact local algebraic values and scenarios.*

Proof. The face reduction is lemma D.2. In a local core z , substitute $X_e(z)$ into P_e, Q_e , and form the positive product $Q_*(z) = \prod_e Q_e(X_e(z))$. Multiplying every cycle equation by Q_* gives polynomial coefficients of the resistance leaves and introduces no spurious roots. The total degree is at most the sum of the substituted degrees, and in a fixed number of variables the expansion has polynomially many monomials and polynomial coefficient bits.

For a path-drop objective introduce one extra core coordinate v and the equation

$$\sum_e \omega_{H,e} \beta_e P_e(X_e) \prod_{f \neq e} Q_f(X_f) - v Q_*(z) = 0.$$

Bound v by the supplied rational path-drop bound. Treat this as one extra linking equation and maximize the core-only objective v using lemma D.3; the core dimension and equation count each increase by one. For an arc use the affine objective $X_e(z)$ without that extra equation. Keep independent block values separate and sum rational intervals for the pressure objective as before. $\square$

D.5 Physical error and rational scenario recovery

Fix a nonempty finite $\mathcal{Q} \subset \mathbb{Q} \cap (1, \infty)$. Throughout this subsection let

$$\begin{aligned} H &= \lceil \max \mathcal{Q} \rceil, & c_* &= 2^{1-H}, & T_0 &= \max\{1, B\}, \\ M &= HT_0^{H-1}, & N_0 &= BT_0^{H-1}. \end{aligned}$$

Thus H is a fixed integer, $c_* > 0$ is rational, and on $[-B, B]$ every ϕ_q and its derivative are bounded in absolute value by N_0 and M , respectively; these bounds have polynomial rational bit length.

Lemma D.10 (Constitutive error controls the physical state). *Let $m > 0$, $\beta_L = \min_e L_e$, and $\beta_U = \max_e U_e$. Suppose $|\psi_e - \phi_{q_e}| \leq \delta$ on $[-B, B]$, where the ψ_e are continuous strictly increasing laws vanishing at zero. For the same admissible (b, β) , let x, π be the original state and $y, \hat{\pi}$ the state with laws $\beta_e \psi_e$. For $0 < \eta \leq 1$, the condition*

$$\delta \leq \frac{c_* \beta_L \eta^H}{m \beta_U} \quad (27)$$

implies

$$\|x - y\|_\infty \leq \eta, \quad |(\pi_s - \pi_t) - (\hat{\pi}_s - \hat{\pi}_t)| \leq m \beta_U (M \eta + \delta). \quad (28)$$

Both estimates are uniform on the full scenario domain.

Proof. For $q > 1$ the signed power satisfies

$$(\phi_q(a) - \phi_q(b))(a - b) \geq 2^{1-q} |a - b|^{q+1} \geq c_* |a - b|^{q+1} \quad (q \in \mathcal{Q}). \quad (29)$$

For equal signs assume $a = b + d \geq b \geq 0$ after symmetry. The function $(b + d)^q - b^q$ is increasing in b , so it is at least d^q . For opposite signs convexity gives $|a|^q + |b|^q \geq 2^{1-q}(|a| + |b|)^q$. Multiplying these increment bounds by $|a - b|$ proves the inequality.

Both flows lie in $[-B, B]^E$ by acyclicity, and $A(x - y) = 0$. Pairing this circulation with the difference of the two potential gradients yields

$$\sum_e \beta_e [\phi_{q_e}(x_e) - \phi_{q_e}(y_e)](x_e - y_e) = \sum_e \beta_e [\psi_e(y_e) - \phi_{q_e}(y_e)](x_e - y_e).$$

Let $D = \|x - y\|_\infty$. If $D = 0$ the flow bound is immediate. Otherwise select an edge e attaining D . Nonnegativity of all terms on the left, equation (29), and the uniform error on the right give $c_* \beta_L D^{q_e+1} \leq m \beta_U \delta D$, hence $D^{q_e} \leq \eta^H$ under equation (27). If $\eta < D \leq 1$, then $D^{q_e} \geq D^H > \eta^H$; if $D > 1$, then $D^{q_e} > 1 \geq \eta^H$. Both contradict the bound, so $D \leq \eta$. Each edge-drop error is at most $\beta_U (M \eta + \delta)$. Summing on a simple terminal path proves the second estimate. $\square$

Lemma D.11 (Original-law input rounding). *For two admissible scenarios (b, β) and (b', β') , original-law terminal drops satisfy*

$$|F_{s,t}(b, \beta) - F_{s,t}(b', \beta')| \leq M \sum_{e \in P_{st}} U_e \|b - b'\|_1 + N_0 \|\beta - \beta'\|_1. \quad (30)$$

For a target arc $e = (u, v)$ with original flows x_e, x'_e , write the corresponding endpoint drops as Δ, Δ' . Then

$$|x_e - x'_e|^{q_e} \leq \frac{|\Delta - \Delta'| + N_0 |\beta_e - \beta'_e|}{c_* L_e}. \quad (31)$$

In particular, a right side at most ξ^H , $0 < \xi \leq 1$, guarantees flow error at most ξ . Polynomially many input precision bits suffice for any prescribed pressure or flow accuracy.

Proof. The original laws are continuously differentiable. Add ρx to each complete law and use the ordinary electrical adjoint. Its nomination path bound is $\sum_{e \in P_{st}} (U_e M + \rho)$. At fixed nomination, resistance differentiation gives $\partial F_\rho / \partial \beta_e = j_e \phi_{q_e}(x_e)$, because the added linear term is independent of β_e . This is bounded by N_0 . Integrating over the two box segments and passing to the uniform zero-smoothing limit as before proves equation (30).

Subtracting $\Delta = \beta_e \phi_{q_e}(x_e)$ and $\Delta' = \beta'_e \phi_{q_e}(x'_e)$ gives

$$\beta_e [\phi_{q_e}(x_e) - \phi_{q_e}(x'_e)] = \Delta - \Delta' + (\beta'_e - \beta_e) \phi_{q_e}(x'_e).$$

Use the increment version of equation (29), the bound N_0 , and $\beta_e \geq L_e$. The final implication follows from $q_e \leq H$ exactly as in lemma D.10. For the endpoint drop, take the one-edge path in equation (30); its nomination constant is $U_e M$. Thus a desired flow error ξ requires only the fixed power ξ^H on the right side. All constants have polynomial rational bit length. $\square$

Theorem D.12 (Fixed fractional powers: polynomial accuracy-bit algorithms). *Fix a block-rank bound $\mathfrak{r}$ and a nonempty finite family $\mathcal{Q} \subset \mathbb{Q} \cap (1, \infty)$. On each edge choose $q_e \in \mathcal{Q}$ and the law $\beta_e \phi_{q_e}$, with independent positive rational resistance intervals. For a nonempty balanced rational nomination box, one can compute a rational interval of width at most ε containing the maximum prescribed terminal drop, and rational admissible nomination and resistance inputs within ε of that maximum. The same guarantees hold for the maximum and minimum signed flow on any prescribed arc. Running time is polynomial in input size and accuracy bits, with its exponent and constants allowed to depend on $\mathfrak{r}, \mathcal{Q}$. No exact threshold comparison or rational physical-state output is claimed.*

Proof. Replace ε by $\min\{\varepsilon, 1\}$. The edgeless and $B = 0$ cases are immediate. Put $a = \varepsilon/8$ and choose rational

$$\eta = \min\{1, a, a/(2m\beta_U M)\}, \quad \delta = \min\{1, a/(2m\beta_U), c_*\beta_L\eta^H/(m\beta_U)\}.$$

Construct the rational ψ_{q_e} by lemma D.8. By lemma D.10, their flow objectives and terminal-drop objectives each differ from the original ones by at most a , uniformly over the scenario domain. A rational absolute drop bound is $m\beta_U(N_0 + \delta)$, so lemma D.9 applies. All construction degrees and bit lengths are polynomial in the original input and accuracy bits; H and exponent denominators are fixed.

Compute each surrogate block optimum exactly, retaining its local algebraic scenario. Enclose each block value to width $\varepsilon/(8n)$, sum by face, and select the face with greatest lower endpoint. This gives an interval $[L, U]$ of width at most $\varepsilon/8$ containing the surrogate maximum, and a separately encoded algebraic scenario whose surrogate value loses at most $\varepsilon/8$. For an arc only one block is relevant and exact local comparison is possible, but the same interval budget may be used. A minimum is obtained by maximizing the negative objective. The interval $[L - a, U + a]$ contains the original maximum and has width at most $3\varepsilon/8$.

Apply lemma D.11 to recover rational inputs for the *original* laws. For pressure, choose sufficiently narrow nomination and resistance isolating boxes that equation (30) is at most $\varepsilon/8$. For an arc, use equation (31) and require its right side to be at most $(\varepsilon/8)^H$. Nomination-box intersection with its rational face and balance contains the exact algebraic point; rational LP returns an exactly feasible rational point. Resistance coordinates are rounded in their original intervals independently. All needed enclosures have polynomial precision. On a terminal core, round its aggregated nomination first and then greedily disaggregate; its objective is determined by that aggregate, so the sensitivity bound is applied on that core. Complete irrelevant coefficients with rational endpoints. The recovery needs no exact original-law flow computation.

The output's original objective loses at most a in comparing the two optima, $\varepsilon/8$ in face selection, a in comparing the two laws at the selected algebraic scenario, and $\varepsilon/8$ in rounding, in total at most $\varepsilon/2$. For arc optimization, rounding starts from an arc optimizer, not from a pressure optimizer, whose target resistance may differ. $\square$

The theorem covers any fixed finite family in $(1, 3)$, and the odd-power factor in lemma D.8 extends it to all fixed rational exponents greater than one. The approximation principle itself is not claimed as new; the contribution is its

use with the block-rank reduction, rational encoding, uniform physical-error transfer, and rational scenario recovery. If exponents are arbitrary binary input, α or $1 - \alpha$ can be exponentially small, the dyadic construction can require exponentially many panels, and the odd power and the exponent H are no longer fixed. No algorithm polynomial in such an exponent encoding is asserted. Exact integer power laws have the polynomial representations treated above, including the linear law at the endpoint $q = 1$. Theorem D.6 is compatible with these additive guarantees: an accuracy interval need not decide an exact equality or a threshold closer than the requested error.

For example, the water head-loss exponent 1.852 in Groß et al. [29, Example 2.3, May 2018 manuscript] is the fixed reduced rational 463/250. Its even denominator puts that mathematical law within both the single-cycle exact barrier and the fixed-family additive theorem.

E Discrete and series-parallel boundaries

Independent finite resistance choices behave differently from intervals. For pressure, two cycles in one block already permit a Subset-Sum encoding. For an individual arc, classical electrical current signs preserve interval-hull extrema on series-parallel graphs, and the first hard block rank is three. At unbounded block rank, fixed nominations still admit high-precision arc optimization, but exact arithmetic and nomination uncertainty impose separate limits. Finally, a worst-case hull theorem does not decide whether some finite resistance choice realizes a prescribed flow.

Throughout this section, series-parallel means that the undirected graph has no K_4 minor; equivalently, its biconnected components admit series-parallel decompositions [19]. No terminals are designated globally. Except in the explicit continuous-law statements, all laws are $g_e(x) = \beta_e x |x|$, all input data are rational, and every allowed resistance is positive. Finite sets are listed explicitly. Scenario domains contain all passive states for the allowed nominations and resistances; capacities are tested afterwards and do not filter that domain.

E.1 A pressure maximum encoding Subset Sum

Theorem E.1 (Discrete pressure hardness at block rank two). *Maximizing a prescribed potential difference over independent two-point resistance sets is NP-hard on simple graphs of maximum degree three with one rank-two block and one bridge. Nominations are fixed small integers. The reduction has a rational separation gap of polynomial encoding length. It excludes*

an algorithm polynomial in input length and accuracy bits unless $P = NP$. With unnormalized positive integer resistances, even absolute-error-one value approximation is NP-hard. These are not strong-hardness claims.

Proof. Take positive Subset-Sum integers $a_1, \dots, a_n, K$, put $S = \sum_i a_i$, and preprocess $K > S$ directly. Use vertices $0, 1, 2, 3$ and arcs

$$(0, 2), (2, 1), (0, 3), (3, 1) \quad \text{of resistances} \quad \frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{2}.$$

Replace $(2, 3)$ by a directed path of n edges, whose internal nominations vanish, with choices

$$\beta_i \in \left\{ \frac{1}{2n}, \frac{1}{2n} + \frac{a_i}{2K} \right\}. \quad \tau = \sum_i \beta_i = \frac{1}{2} + \frac{\sum_i a_i \sigma_i}{2K}, \quad \sigma_i \in \{0, 1\}.$$

The resulting theta block has rank two. First set the nominations at $0, 1, 2, 3$ to $(4, -4, 3, -3)$. The unique positive root q of

$$(12\tau + 1)q^2 + 10q - 23 = 0 \tag{32}$$

lies in $(0, 2)$, since the polynomial is negative at zero, positive at two, and strictly increasing on $[0, \infty)$. Assign cross-path flow q and, in the displayed order, outer flows

$$\frac{q+1}{2}, \quad \frac{7-q}{2}, \quad \frac{7-q}{2}, \quad \frac{q+1}{2}.$$

Their divergences are the stated nominations. The two outer path drops agree by symmetry, and the potential difference between their intermediate vertices is $[23 - 10q - q^2]/12 = \tau q^2$. Thus every cycle equation holds, and theorem A.1 identifies this as the physical state. The common outer drop is

$$\pi_0 - \pi_1 = \frac{(q+1)^2}{8} + \frac{(7-q)^2}{24} = 2 + \frac{(q-1)^2}{6}.$$

Add a leaf 4 and an arc $(4, 1)$ of resistance three. Give 4 nomination one and change b_1 to -5 . The new arc carries flow one, and block aggregation preserves the theta state. Hence

$$F := \pi_4 - \pi_0 = 1 - \frac{(q-1)^2}{6} \leq 1, \tag{33}$$

with equality exactly when $q = 1$, or equivalently $\tau = 1$ and $\sum_i a_i \sigma_i = K$. The fixed nominations are $(4, -5, 3, -3, 1)$. The graph is simple, has maximum

degree three, and has just the asserted blocks. Fixed yes/no instances of Subset Sum, for example $(a_1, K) = (1, 1)$ and $(2, 1)$, handle preprocessing within the same graph class.

If a choice does not sum to K , then $|\tau - 1| \geq 1/(2K)$. Subtracting the polynomial at 1 from its value at q gives

$$|q - 1| = \frac{12|\tau - 1|}{(12\tau + 1)(q + 1) + 10} \geq \frac{6}{31K + 18S}.$$

Here $q < 2$ and $\tau \leq (K + S)/(2K)$ bound the denominator by $31 + 18S/K$. Consequently every non-target choice satisfies

$$F \leq 1 - \Delta, \quad \Delta = \frac{6}{(31K + 18S)^2}. \quad (34)$$

An absolute-error estimate below $\Delta/3$, a requested accuracy of polynomial binary length, distinguishes the two optimum values. In a yes instance, a discrete choice with error below Δ must itself encode a target subset, which integer arithmetic can check.

Multiply every resistance by $6nK$. The four fixed coefficients become $3nK, nK, nK, 3nK$, the path choices become $\{3K, 3K + 3na_i\}$, and the bridge coefficient becomes $18nK$. Flows are unchanged and potentials scale by $6nK$. Multiplying again by $T = (31K + 18S)^2$ gives an integer peak $6nKT$ and a gap $6nKT\Delta = 36nK \geq 36$. An absolute-error-one estimate compared with the peak minus 18 decides Subset Sum. These coefficients have polynomial binary length, but need not have polynomial numerical magnitude; this scaling proves neither strong nor relative-error hardness. $\square$

The binary path sum is the usual Subset-Sum mechanism; the network feature is the strict interior pressure maximum in equation (33). The individual outer and cross flows in this gadget are monotone in τ , so pressure hardness alone does not establish arc hardness. The interval version of this same rank-two problem remains covered by theorem C.2. Existing discrete sizing hardness has a different scope: Robinius et al. [47, Theorem 4.7] treats minimum-cost arc sizing subject to potential feasibility, already on trees, and credits the earlier water-network hardness of Yates, Templeman, and Boffey. Here resistance choices are adversarial parameters for one unfiltered pressure objective, without a cost budget or potential bounds. On a tree with fixed nominations this objective is linear in the resistances, so that design result neither proves nor conflicts with the rank-two boundary above.

E.2 Finite comparison, electrical signs, and resistance hulls

Lemma E.2 (Finite secant comparison). *On any connected graph, let x, π and y, ρ be the passive states for the same nomination and two families of continuous strictly increasing laws g_e, G_e vanishing at zero. Set*

$$r_e = \frac{G_e(y_e) - G_e(x_e)}{y_e - x_e} \quad (y_e \neq x_e), \quad d_e = G_e(x_e) - g_e(x_e),$$

and choose any $r_e > 0$ when $y_e = x_e$. For a target arc $a = (u, v)$, let j be the unit electrical flow from u to v with resistances r . Then

$$(\rho_u - \rho_v) - (\pi_u - \pi_v) = j^\top d, \tag{35}$$

$$y_a - x_a = \frac{\sum_{e \neq a} j_e d_e - (1 - j_a) d_a}{r_a}. \tag{36}$$

Moreover, $0 \leq j_a \leq 1$.

Proof. Strict increase makes every secant positive; the arbitrary choice multiplies a zero flow difference. Subtracting the physical systems gives

$$A(y - x) = 0, \quad A^\top(\rho - \pi) = \text{diag}(r)(y - x) + d.$$

Write $j = \text{diag}(1/r)A^\top h$ with $Aj = e_u - e_v$. Pairing the second identity with j makes its first right-hand term $h^\top A(y - x) = 0$, proving equation (35). Its a coordinate gives equation (36). The electrical maximum principle puts $h_u \geq h_v$, so $j_a \geq 0$; decomposition of the acyclic unit electrical flow into source-to-sink paths gives $j_a \leq 1$, as in the proof of lemma B.7. $\square$

Lemma E.3 (Classical adjacent-terminal signs). *In a series-parallel graph, the sign of every unit electrical current between the endpoints of an existing target edge is independent of all positive electrical resistances. Edges off the target block have zero current. The signs are computable by one rational electrical solve with unit resistances.*

Proof. Block decomposition makes every off-block current zero. The target block is a bridge or is two-terminal series-parallel with the target endpoints as terminals, by Eppstein [24, Lemma 9]. In a two-terminal decomposition, series components transmit the same positive current and parallel components split it into positive currents, so induction fixes each edge's direction. The bridge case is immediate. A reduced rational Laplacian solve at unit resistances has polynomial bit complexity and recovers these signs, including zeros. $\square$

This is classical confluence theory: Duffin [23, Theorems 0 and 1] relates resistance-independent current directions to graph structure, and his Section 5 treats monotone nonlinear resistor characteristics. The restriction to adjacent terminals is essential: arbitrary terminal pairs in a theta block need not have fixed signs.

Theorem E.4 (Continuous-law envelopes). *Fix a balanced nomination on a series-parallel graph. For each edge let Θ_e be a nonempty compact parameter space, with a jointly continuous law $g_e(t, \theta)$ such that each member is strictly increasing in t and vanishes at zero. Edge choices are independent. For each target arc, one deterministic envelope network attains its maximum flow over all scenarios; a second attains its minimum. The two envelope profiles are independent of the nomination.*

Proof. Define $g_e^{\max}(t) = \max_{\theta \in \Theta_e} g_e(t, \theta)$ and $g_e^{\min}(t) = \min_{\theta \in \Theta_e} g_e(t, \theta)$. Compactness and joint continuity give continuous envelopes and attained extrema. For $s < t$, evaluate the maximum at a member attaining the maximum at s , and the minimum at a member attaining the minimum at t ; strict increase of the members then gives strict increase of both envelopes. They vanish at zero, so theorem A.1 applies even if they are bounded.

Let σ_e be the signs in lemma E.3. For maximization choose $G_a = g_a^{\min}$ on the target, choose $G_e = g_e^{\max}$ when $e \neq a$ and $\sigma_e > 0$, and choose $G_e = g_e^{\min}$ when $e \neq a$ and $\sigma_e < 0$. On zero-current edges choose any member. Compare its physical flow y with any original flow x using equation (36). Every numerator term is nonnegative, so $y_a \geq x_a$. At y_e choose a member attaining the selected pointwise envelope value, independently on every edge. The complete state y, ρ satisfies those original laws, hence is their unique state, and the upper bound is attained. Interchange minima and maxima for the lower bound. Neither envelope construction uses b . $\square$

The theorem allows coupling among the parameters of one edge; only independence between edges is required. For general continuous families it is structural: without an input representation it implies no algorithm. The fixed envelope flow is continuous in b : bounded nominations give bounded acyclic flows, and summing the continuous edge drops along a spanning tree bounds normalized potentials. Every subsequential limit satisfies the limiting state equations and hence is unique. Therefore, for any nonempty compact balanced nomination set $\mathcal{U}$,

$$\max_{b \in \mathcal{U}, \theta \in \prod_e \Theta_e} x_a(b, \theta) = \max_{b \in \mathcal{U}} x_a^{\text{env}}(b).$$

A realizing member profile may depend on b , and different target arcs usually require different envelope networks. Neither a simultaneous vector optimum nor preservation of existential feasibility follows. The pressure identity equation (35) also gives a cactus version: for any terminal pair, orient the unit electrical flow along its block path. Bridge currents and the positive split between the two paths in every traversed cycle have fixed signs. Choosing the upper envelope on positive-current edges and the lower on negative-current edges, with the same attainment argument, maximizes that potential difference. Reverse the choices to minimize it.

Corollary E.5 (Separate monotonicity and exact finite-set validation). *On a series-parallel graph, a target arc flow is separately monotone in each scalar parameter that changes only one law affinely, provided all intermediate complete laws are admissible. Thus products of compact scalar sets with attained endpoints and admissible interval hulls have the same target-flow extrema as their interval hulls. This equality continues to hold jointly over any compact nomination set.*

For quadratic laws, finite rational resistance sets, a balanced rational nomination box, and fixed maximum block rank, exact arc extrema and robust capacity validation are polynomial-time on series-parallel graphs. In particular this holds whenever $r_{\max} \leq 2$.

Proof. Write the changing law as $g_e(t, \theta) = g_e^0(t) + \theta f_e(t)$. If $f_e(x_e(\theta_0)) = 0$ at some parameter, the entire physical state at θ_0 satisfies the equations for every parameter value, and uniqueness makes it constant. Otherwise $f_e(x_e(\theta))$ has a fixed strict sign by continuity. State continuity follows directly from bounded flows, locally uniform convergence of the laws, and uniqueness: along a convergent parameter sequence, any flow subsequence has a convergent subsequence, normalized potentials are bounded by summing edge drops along a spanning tree, and its limit satisfies the limiting equations. The flow bound is the fixed total positive nomination, by acyclicity.

For $\theta' < \theta$ compare the two states by the secant lemma. Only $d_e = (\theta - \theta')f_e(x_e(\theta'))$ is nonzero. Its coefficient is j_e/r_a for $e \neq a$ and $-(1 - j_a)/r_a$ for $e = a$, and its weak sign is fixed by lemmas E.2 and E.3. This proves monotonicity without smoothness or a nonzero-derivative assumption. Compactness and the continuity just proved give an optimizer. Move it one coordinate at a time to an appropriate endpoint; its objective stays optimal and the final parameters belong to the original compact sets. The same argument holds with an optimizing nomination fixed.

For the computational claim, replace each resistance set by its endpoint interval and use theorem C.3. An exact finite resistance optimizer can also be

recovered. Compute the optimum V , fix one resistance to its lower endpoint, and reoptimize the remaining interval problem. Keep that endpoint if its exact optimum equals V , and otherwise fix the upper endpoint; separate monotonicity ensures that one endpoint retains V . Polynomially many fixed-rank optimizations and algebraic comparisons suffice. At the end the fixed-resistance algorithm returns an algebraic nomination optimizer; no rational exact physical state is promised.

Finally, a K_4 minor must occur within a block. To see this, take a subdivision: as K_4 is cubic, prune each connected minor branch set to a tree joining its at most three attachment points; the three paths meet at one vertex, giving a subdivision. This biconnected subdivision lies in one block. Cycle rank cannot increase under edge deletion or contraction, and K_4 has rank three, so $r_{\max} \leq 2$ excludes such a minor. $\square$

E.3 An additive algorithm without a block-rank bound

Theorem E.6 (Fixed-nomination series-parallel arc optimization). *Fix rational balanced nominations on a series-parallel graph, and allow independent resistance intervals or explicitly listed finite positive resistance sets. For any target arc and rational $\varepsilon > 0$, its maximum and minimum have rational enclosing intervals of width at most ε , and allowed rational endpoint scenarios attaining each extremum within ε can be computed in time polynomial in input length and $\log^+(1/\varepsilon)$. Block rank is unrestricted.*

Proof. Write L_e, U_e for the resistance endpoints. The pointwise maximum of $L_e t|t|$ and $U_e t|t|$ uses U_e for $t \geq 0$ and L_e for $t \leq 0$; the minimum reverses them. Thus theorem E.4 constructs one asymmetric quadratic network for each extremum, with positive rational coefficients $c_e^+, c_e^- \in \{L_e, U_e\}$. Its state x^* minimizes

$$E(x) = \frac{1}{3} \sum_e [c_e^+(x_e^+)^3 + c_e^-(x_e^-)^3], \quad Ax = b, \quad (37)$$

where $x_e^+ = \max\{x_e, 0\}$ and $x_e^- = \max\{-x_e, 0\}$.

This is a rational second-order cone program of linear size. Introduce nonnegative p_e, n_e with $x_e = p_e - n_e$; at a minimum both cannot be positive, since subtracting their minimum preserves flow and decreases energy. For $z \geq 0$, the cubic epigraph $z^3 \leq u$ lifts to

$$z^2 \leq w, \quad w^2 \leq uz, \quad u, w, z \geq 0.$$

The first two constraints are rotated second-order cones. They imply $z^4 \leq uz$, and conversely $w = z^2$ is feasible whenever $u \geq z^3$; the zero case is immediate.

Apply this to every p_e, n_e . A conic representation alone does not guarantee the required output accuracy, so we give explicit bit bounds.

If $b = 0$, all flows vanish. Otherwise put $B = \sum_v |b_v| > 0$, $\beta_L = \min_e L_e$, and $\beta_U = \max_e U_e$. The passive state has $|x_e^*| \leq B/2$. Choose a rational tree-supported flow x^0 with $|x_e^0| \leq B/2$, and a fundamental-cycle matrix $C \in \{0, 1, -1\}^{E \times k}$ whose chord submatrix is the identity. Every feasible flow is $x^0 + Cz$ and z^* is the vector of chord flows, so $|z_i^*| \leq B/2$. When $k = 0$, return the tree flow. Otherwise minimize $f(z) = E(x^0 + Cz)$ on $[-B, B]^k$. On this cube,

$$\begin{aligned} |x_e^0 + (Cz)_e| &\leq R := (m+1)B, \\ |\partial_i f| &\leq G_0 := m\beta_U R^2, \quad \|\nabla f\|_2 \leq G := 1 + mG_0. \end{aligned}$$

Rational evaluation and gradients use the sign of each affine flow to select its cubic piece, and have polynomial bit cost. A coordinate outside the cube yields a separating coordinate hyperplane. At a point with $f(z) > t$, its rational gradient separates that point from $\{f \leq t\}$; a zero gradient there certifies that this sublevel set is empty.

For any $\delta > 0$, a sublevel with $t \geq f(z^*) + \delta$ contains the ball about z^* of radius

$$r_\delta = \min\{B/4, \delta/(2G)\}.$$

The cube lies in the ball of radius mB . Thus the rational ellipsoid weak-feasibility algorithm, using the above exact separation oracle, finds a feasible point whenever the sublevel has this ball, in a number of bit operations polynomial in the input and $\log((1+mB)/r_\delta)$. Threshold bisection between zero and $f(0)$, with inner accuracy $\delta/4$, returns a rational $\hat{z}$ with $f(\hat{z}) \leq f(z^*) + \delta$: a failed ball search at threshold t implies $t < f(z^*) + \delta/4$, while a successful search supplies a point with value at most t , so bisection terminates with objective error δ . All ellipsoid arithmetic uses polynomial-length rounding, as in the standard rational weak-optimization theorem [20, Theorem 2.5.9]. Its bounded-body hypotheses are met by the cube; if a globally Lipschitz objective is used in that formulation, extend each scalar primitive beyond $[-R, R]$ by its tangent affine functions; the resulting convex rational oracle agrees on the entire cube and has the same gradient bound. The returned flow $\hat{x} = x^0 + C\hat{z}$ satisfies conservation exactly.

It remains to convert energy error to flow and scenario error. For an envelope law G_e , the function $G_e(t) - \beta_L t|t|$ is nondecreasing. The scalar inequality

$$(u|u| - v|v|)(u - v) \geq \frac{1}{2}|u - v|^3$$

follows by factoring on each same-sign half-line and by $u^2 + v^2 \geq (|u| + |v|)^2/2$ for opposite signs. Integrating this monotonicity along a segment, and using $G(x^*)^\top(\hat{x} - x^*) = 0$, gives

$$E(\hat{x}) - E(x^*) \geq \frac{\beta_L}{6} \sum_e |\hat{x}_e - x_e^*|^3. \quad (38)$$

Hence energy error $\delta = \beta_L \eta^3/6$ gives $\|\hat{x} - x^*\|_\infty \leq \eta$, and $[\hat{x}_a - \eta, \hat{x}_a + \eta]$ encloses the exact extremum.

Choose each original resistance endpoint using the sign of $\hat{x}_e$ and the envelope rule; either endpoint is allowed at zero. A wrong sign choice can occur only when $|x_e^*| \leq \eta$, so the selected law g'_e has residual

$$|g'_e(x_e^*) - G_e(x_e^*)| \leq \beta_U \eta^2.$$

Let x' be its true passive flow. Pair the two potential systems with the circulation $x' - x^*$ and apply the same scalar inequality to g' . With $D = \|x' - x^*\|_\infty$ this gives

$$\frac{\beta_L}{2} D^3 \leq m \beta_U \eta^2 D, \quad D \leq \sqrt{\frac{2m\beta_U}{\beta_L}} \eta.$$

Set $C_0 = 1 + 2m\beta_U/\beta_L$ and $\eta = \min\{\varepsilon/2, \varepsilon/C_0\}$. The interval then has width at most ε and the selected allowed scenario has target flow within ε of optimal. All internal tolerances have polynomial binary length. Neither an exact sign decision for an irrational flow nor a numerical solver oracle is assumed. $\square$

The convex energy principle and weak convex optimization are classical; the quantitative conclusion combines them with an envelope construction and control of endpoint selection near zero flow. We do not claim priority for the envelope identity itself: nonlinear circuit tolerance analysis predates this formulation, and we could not read in full the directly relevant 1993 paper of Hasler and Wang, *Parameter tolerances in non-linear resistive circuits: worst case analysis based on monotonicity* [30], and we draw no priority conclusion from it. The claims here are the explicitly proved model and output guarantees.

More specifically, Chauffoureaux and Hasler [18, Theorem 3] gives a structural source-to-resistor monotonicity criterion through a linear circuit test, motivates endpoint evaluation for worst-case analysis, and sketches extensions to resistor parameters and the observed branch itself. Our finite

secant argument accommodates zero derivatives and merely continuous laws, while the quadratic algorithm specifies binary input, accuracy bits, and an allowed endpoint scenario. Zemanian [56, Sections 4–5 and 8–9 of the report version] composes finite rated characteristic segments on series-parallel networks, tests existence of a within-ratings operating point, and recovers a designated branch by choosing the reduction order. That is a deterministic rated-characteristic problem, whereas the uncertainty here varies independently selected complete constitutive laws; the report’s formulas, tables, and interpolation do not provide the binary-precision guarantee proved above.

E.4 Arc hardness at block rank three

A high-resistance probe converts a pressure gap into a rational arc threshold. The following monotonicity controls its feedback on the original network. If a probe (s, t) carries flow z , its removal changes the old nomination to $b - ze_s + ze_t$. Let $F(z)$ be the old-network pressure $\pi_s - \pi_t$ at this nomination. For two distinct values z, w , strict increase of every law gives

$$\begin{aligned} & (b(z) - b(w))^{\top} (\pi(z) - \pi(w)) \\ &= \sum_e (g_e(x_e(z)) - g_e(x_e(w))) (x_e(z) - x_e(w)) > 0. \end{aligned}$$

The left side is $-(z - w)(F(z) - F(w))$, so F is strictly decreasing. A probe of resistance M obeys $F(z) = Mz|z|$; hence its flow has the sign of $F(0)$, and it reduces the absolute terminal pressure. All these comparisons concern the unique full physical state, which exists by theorem A.1.

Theorem E.7 (Discrete arc classification). *At fixed maximum block rank, deciding whether a finite resistance choice and a nomination in a bounded rational balanced box can give $x_a > c$ is in NP; deciding whether every scenario satisfies $x_a \leq c$ is in coNP. Both are complete for these classes already with fixed small integer nominations, two-point resistance choices, and a simple biconnected graph of maximum degree three and rank three. The same completeness holds for existence of $x_a \geq c$ and its complementary universal strict inequality. All thresholds are rational. The reduction has a polynomially encoded positive gap, excluding polynomial-time optimization in accuracy bits unless $P = NP$.*

Proof. Start with theorem E.1, before integer scaling, and set

$$D = \left\lceil \frac{10000}{\Delta} \right\rceil, \quad H = 1 - \Delta/2, \quad M = HD^2, \quad c = 1/D.$$

Add the target edge $(4, 0)$ of resistance M . The former leaf and this edge form an ear between 1 and 0 in the theta block. The entire graph is biconnected, has rank three, and still has maximum degree three. For each discrete choice the old pressure is $F(0) = 1 - (q - 1)^2/6 \in (5/6, 1]$. The new edge flow z is positive, and monotonicity gives

$$0 < Mz^2 = F(z) \leq 1, \quad z \leq 2/D < 1,$$

since $H \geq 1/2$. The old nomination changes by $ze_0 - ze_4$. Both nominations lie in a box with total absolute coordinate bound at most 18. The fixed path 4, 1, 2, 0 has resistance sum $11/3$. Thus lemma B.7 gives, uniformly in all uncertain cross-path resistances,

$$0 \leq F(0) - F(z) \leq 264z \leq 528/D < \Delta/8. \quad (39)$$

For a target subset, $F(0) = 1$, whence $Mz^2 > 1 - \Delta/8 > H = Mc^2$ and $z > c$. For every non-target choice, $Mz^2 \leq 1 - \Delta < H$ and $z < c$. These strict gaps prove the claimed reductions for both weak and strict existential comparisons, without changing equality conventions, and their complements give the two universal versions. If all edges require capacities, bounds $[-16, 16]$ on every other edge are harmless by the total-nomination flow bound, and the target can have lower bound -16 .

For a target choice the pressure excess over H is greater than $3\Delta/8$; for a non-target choice the deficit is at least $\Delta/2$. Since $z + c \leq 3/D < 4/D$ and $M \leq D^2$,

$$|z - c| = \frac{|F(z) - H|}{M(z + c)} \geq \frac{3\Delta}{32D}. \quad (40)$$

An additive estimate of error below $\Delta/(64D)$ separates yes and no instances, and near-optimal discrete scenarios identify target subsets at this accuracy. Every number has polynomial binary length. Writing $A_0 = 31K + 18S$, multiplication of all resistances by $6nKA_0^2$ makes them positive integers, including the probe coefficient $6nK(A_0^2 - 3)D^2$. This changes no flow or capacity; in particular, resistance scaling does not amplify equation (40).

For membership, guess one input resistance index per edge. With these coefficients fixed, theorem C.3 computes the exact target extrema over the continuous balanced nomination box in polynomial bit time at fixed block rank. Compare the algebraic maximum with c using the required weak or strict inequality; the verifier needs no rational nomination or physical-state witness. For several capacities, also guess an offending edge and direction. Complementation gives universal membership, and with the rank-three reduction this proves completeness. The argument gives no pressure-threshold membership theorem for sums over arbitrarily many blocks. $\square$

Corollary E.5 and theorem E.7 yield a sharp block-rank boundary: finite-resistance robust arc validation is polynomial at rank at most two and coNP-complete when rank three is permitted. Not every rank-three graph is hard; series-parallel graphs can have arbitrarily large block rank and still satisfy theorem E.6 for fixed nominations. The reduction is from Subset Sum with a small flow gap, so it establishes neither strong hardness nor fixed-error hardness under the stated fixed-nomination normalization.

E.5 The universal graph class for individual-arc hulls

Theorem E.8 (Universal arc-hull characterization). *For a finite connected simple graph G , the following are equivalent:*

1. *G is series-parallel.*
2. *For every fixed balanced nomination and positive quadratic resistance interval family, each target arc flow is separately monotone in every resistance coordinate.*
3. *For every fixed balanced nomination and independent compact positive resistance sets with attained endpoints, each target-flow maximum and minimum equals its interval-hull counterpart.*

If G has a K_4 minor, failure can be witnessed by rational data, small integral nominations, and one two-point uncertain resistance. Given a K_4 subdivision, the witness has polynomial encoding length and an explicit rational gap.

Proof. The positive implications are corollary E.5. For the converse, take the theta in equation (32). Keep nominations $(4, -4, 3, -3)$ and a single cross edge of resistance θ . Its reverse pressure is

$$F_\theta(0) = \pi_1 - \pi_0 = -2 - (q - 1)^2/6.$$

At $\theta = 1$ it equals -2 . At $\theta_L = 23/108$ and $\theta_U = 71/12$, substitution in equation (32) gives $q = 3/2$ and $q = 1/2$, respectively, and both pressures equal $-49/24$.

Add the target edge $(1, 0)$ of resistance $M = 10^8$. This is exactly K_4 . Its flow $z < 0$ obeys $F_\theta(z) = Mz|z|$ with old nominations $b + ze_0 - ze_1$. The preceding probe monotonicity gives

$$F_\theta(0) < F_\theta(z) < 0, \quad Mz^2 < -F_\theta(0) \leq 49/24 < 4.$$

Thus $|z| < 2/\sqrt{M} < 1$. Use the nomination bound $B = 18$ and path 1, 2, 0 of resistance sum $2/3$ in lemma B.7:

$$|F_\theta(z) - F_\theta(0)| \leq 48|z| < 96/\sqrt{M} < 1/96.$$

The interior probe pressure is greater than -2 , whereas both endpoint pressures are less than $-49/24 + 1/96 = -65/32$, so the interior target flow is strictly larger than both endpoint flows. Quantitatively, the pressure advantage exceeds $1/32$, and all three probe flows have magnitude below $2/\sqrt{M}$. Since they are negative,

$$F_I - F_E = M(z_I - z_E)(|z_I| + |z_E|), \quad z_I - z_E > g := \frac{1}{128\sqrt{M}} = \frac{1}{1280000}.$$

Here I denotes the interior scenario and E either endpoint.

Every K_4 minor contains a subdivision, by the branch-tree argument in corollary E.5. Embed the above six-edge model on that subdivision H . Distribute each fixed branch resistance equally over its path, and set every internal nomination to zero. On a subdivided uncertain path use a fixed total $d = \theta_L/2$ on all but one edge, distributed equally, and resistance $\theta - d$ on the remaining edge. If the path has length one, use $d = 0$. All endpoints are positive. Series aggregation preserves the three states and their target gap; choose any actual edge of the probe path as target.

It remains to restore all edges of G outside H . Put zero nominations at extra vertices and give every extra edge the same positive integer resistance R . Let $m = |E(G)|$. For each of the three scenarios, the theta flow with $q = 1$, zero probe flow, and zero extra-edge flows is conservation-feasible. Its energy is $(10 + \theta)/3 < 6$, independently of subdivisions. Hence the complete physical energy minimum has every extra-edge flow bounded by $u = (18/R)^{1/3}$. The total positive nomination is seven, so every physical edge flow has magnitude at most seven.

Restriction of that state to H is the passive state at induced balanced nominations $b' = A_H x_H$. They obey $|b'_v| \leq 7 \deg_H(v)$ and $\|b' - b\|_1 \leq 2mu$. The original nominations obey the same coordinate bounds, so a containing box has total absolute bound at most $14m$. For the actual target edge of resistance β_a , apply lemma B.7 to the single-edge terminal path, and use $|s|s| - t|t|| \geq |s - t|^2/2$. This gives

$$|x'_a - x_a|^2 \leq \frac{2}{\beta_a} |\Delta(\pi_u - \pi_v)| \leq 56m \|b' - b\|_1 \leq 112m^2 u.$$

Take

$$R = 18 \left(\frac{2000m^2}{g^2} \right)^3.$$

It is an integer, has polynomial binary length, and gives $|x'_a - x_a| < g/4$ in each scenario because $112/2000 < 1/16$. The restored interior flow still exceeds both restored endpoint flows by more than $g/2$. Every other resistance remains fixed, so this violates both separate monotonicity and the two-point hull equality. All path splits also have polynomial rational encoding length. The characterization itself needs only some sufficiently large R ; the displayed bound proves the stronger encoding claim once a subdivision has been selected. $\square$

E.6 Exact arithmetic and nomination uncertainty

The additive guarantee in theorem E.6 does not extend to exact threshold comparison, even without uncertainty.

Theorem E.9 (A probe-cactus arithmetic equivalence). *Exact weak lower comparison of one quadratic arc flow with a rational threshold is SRS-hard on simple series-parallel graphs of maximum degree three, with fixed unit source-sink nominations and fixed positive integer resistances. The threshold $1/2$ suffices for hardness. On the restricted family consisting of a two-terminal cactus branch with two terminal bridges in parallel with one probe edge, weak lower comparison of the designated probe flow oriented from the unit source to the unit sink is polynomial-time many-one equivalent to SRS.*

Proof. The integer cactus encoder in theorem B.2 produces a chain of triangles that is simple and has maximum degree three, with unit-demand effective resistance $D > 0$ and positive integer threshold H , such that

$$\sum_i \sqrt{a_i} \leq K \iff D \geq H.$$

Its terminals have degree two. Add new terminals S, T , each joined to an old terminal by a unit-resistance bridge, and add the probe (S, T) of resistance $H + 2$. The cactus branch has effective resistance $D + 2$. The graph is simple, maximum degree three, and two-terminal series-parallel. Unit demand splits into positive branch flows $1 - x$ and x ; homogeneity gives

$$(D + 2)(1 - x)^2 = (H + 2)x^2, \quad x = \frac{\sqrt{D + 2}}{\sqrt{D + 2} + \sqrt{H + 2}}.$$

Thus $x \geq 1/2$ if and only if $D \geq H$, including equality; the construction evaluates no square root. For preprocessed trivial cases use a triangle whose direct resistance is two and whose two-edge path has coefficients one and

one. Its effective resistance is $1/2$; after two unit terminal bridges the branch has resistance $5/2$. A probe of resistance one yields a fixed yes instance, and resistance nine a fixed no instance, in the same literal family.

Conversely, let $R > 0$ be the effective resistance of any such cactus branch and $\beta > 0$ the probe resistance. The probe flow is strictly between zero and one. Thresholds $c \leq 0$ are trivial yes instances and $c \geq 1$ are trivial no instances. For $0 < c < 1$,

$$x \geq c \iff R \geq \beta \left(\frac{c}{1-c} \right)^2.$$

This is precisely the rational weak lower cactus-pressure comparison reduced to SRS in theorem B.2. The equivalence keeps weak inequalities distinct from strict capacity violations. $\square$

Adding the probe merges the growing number of cactus cycles into one block, so the construction does not contradict the exact fixed-block-rank algorithm. Since SRS is not known to be in P or to be NP-hard (appendix A.5), theorem E.9 is an arithmetic classification of the specified family, not an NP-hardness result or an upper bound for all series-parallel graphs.

Allowing a nomination box creates a different obstruction. The pressure hardness used next is due to Thürauf; we add only its quantitative transfer to an arc objective.

Theorem E.10 (Nomination hardness on series-parallel graphs). *With fixed positive rational quadratic resistances, maximizing one signed arc flow over a balanced rational nomination box is NP-hard on series-parallel graphs. Robust upper-capacity validation is coNP-hard. The reduction has a positive rational gap of polynomial encoding length, so a polynomial algorithm in input length and accuracy bits would imply $P = NP$. Block rank grows with the instance; no NP or coNP membership claim is made for this class.*

Proof. We use the pressure instance of Thürauf [49, Figure 2, equation (3), and Lemmas 4.3 and 4.17]. For positive Partition numbers $S_1, \dots, S_n$ with $n \geq 3$ and $K = \sum_i S_i$, it has arcs $(s, z_i^+), (z_i^+, t), (z_i^+, z_i^-)$, with respective resistances $1/S_i^2, 1/S_i^2, 1$. Its signed nomination box, intersected with balance, is

$$b_s \in [0, K/2], \quad b_t \in [-K/2, 0], \quad b_{z_i^+} \in [0, S_i], \quad b_{z_i^-} \in [-S_i, 0].$$

Its total absolute coordinate bound is $3K$. The main block is $K_{2,n}$, and all

(z_i^+, z_i^-) edges are pendant bridges. Define

$$\begin{aligned}\varepsilon_0 &= 1 - \left(1 - \frac{1}{8K^2}\right)^2, \\ M_0 &= \max\{1 - \varepsilon_0 + \varepsilon_0^2, 1 - \varepsilon_0^2/K^2\}, \quad \varepsilon_1 = (1 - M_0)/5, \\ T &= \max\{1 - \varepsilon_1^2/(K^2n^2), M_0 + 4\varepsilon_1\}, \quad \gamma = 1 - T > 0.\end{aligned}$$

These are rational numbers of polynomial encoding length, with $0 < T < 1$. Thürauf proves that a yes Partition instance admits a nomination with $\pi_s - \pi_t \geq 1$, while in a no instance every nomination has $\pi_s - \pi_t < T$. The cited statements concern his pressure subproblem (2), which imposes conservation and passive laws but no potential bounds, so no feasibility filtering is imported. Small Partition instances can be decided directly and replaced by a fixed yes or no instance with at least three positive items.

Add a direct probe (s, t) with

$$D = \left\lceil \frac{1000K(3K+2)}{\gamma} \right\rceil, \quad H = 1 - \gamma/2, \quad M = HD^2, \quad c = 1/D.$$

It is one more parallel branch, so the graph remains series-parallel; the main block rank increases from $n - 1$ to n . For an original nomination b , let $F_b(z)$ be the old pressure at $b - ze_s + ze_t$. The probe relation is $F_b(z) = Mz|z|$ and F_b is strictly decreasing. If $F_b(0) \leq 0$, then $z \leq 0 < c$. Otherwise $F_b(z) \leq F_b(0)$. A two-edge s - t path has resistance sum at most two, and the uniform flow bound $3K$ gives $|F_b(0)| \leq 18K^2$. For a yes witness,

$$0 < z \leq 6K/D < 1.$$

The original and induced nominations lie in an enlarged balanced box with absolute coordinate bound at most $3K + 2$. By lemma B.7 on that two-edge path,

$$0 \leq F_b(0) - F_b(z) \leq 8(3K + 2)z \leq \frac{48K(3K + 2)}{D} < \gamma/4.$$

The yes witness therefore has $F_b(z) > 1 - \gamma/4 > H$ and $z > c$. In a no instance, each positive probe flow satisfies $F_b(z) \leq F_b(0) < T < H$, so $z < c$, and nonpositive flows satisfy $z < c$ trivially. The perturbed nominations need not retain the entry/exit sign restrictions: the sensitivity estimate is valid on their enlarged signed box, and the source theorem is applied only to original nominations.

The yes-case pressure margin exceeds $\gamma/4$. Rationalizing the probe equation and using $z + c \leq (6K + 1)/D$ gives the flow gap

$$g_{\text{arc}} = \frac{\gamma}{4(6K + 1)D}.$$

For a positive no-case flow, $z < 2/D$ and $H - F_b(z) > \gamma/2$, so $c - z > \gamma/(6D) \geq g_{\text{arc}}$. A nonpositive flow has deficit at least $1/D$. Hence the optimum is above $c + g_{\text{arc}}$ in a yes instance and below $c - g_{\text{arc}}$ in a no instance. All encoding lengths, including that of the required approximation error, are polynomial. Other capacities may be set to $[-3K, 3K]$ and do not restrict the original scenarios. $\square$

This is a consequence of Thürauf's Partition pressure construction, not a new source of pressure hardness. It is compatible with theorem E.4: eliminating resistance uncertainty leaves a nomination optimization that can still be hard. The gap does not establish strong hardness or rule out running time polynomial in reciprocal error.

E.7 Prescribed-flow realization and existential capacity design

Theorem E.11 (An inverse-design quantifier boundary). *For independent quadratic resistance intervals on any graph, realizing a prescribed rational flow is rational linear feasibility and is polynomial-time. For finite resistance sets, prescribed rational flow realization is NP-complete, even on one simple cycle with fixed nominations $(2, -2)$, unit prescribed flow in an acyclic orientation, positive integer resistances, and two choices per uncertain edge. Existence of a finite resistance choice satisfying unit arc capacities is also NP-complete on the fixed-nomination single-cycle class.*

Proof. For a target $\bar{x}$, first test $A\bar{x} = b$. Put $t_e = \bar{x}_e |\bar{x}_e| \in \mathbb{Q}$. The remaining interval problem is

$$A^\top \pi = \text{diag}(t) \beta, \quad L \leq \beta \leq U.$$

After fixing a potential reference, these are rational linear constraints; zero target flows simply impose zero endpoint potential drop. Linear programming recovers a feasible rational solution of polynomial length. With finite sets, guessing one resistance index per edge leaves rational edge drops, whose membership in $\text{im } A^\top$ is checked by rational linear algebra. This proves NP membership for prescribed-flow realization on any graph, without a physical-state algebraicity assumption.

For hardness take positive Subset-Sum integers $a_1, \dots, a_n, K$, $n \geq 2$, $S = \sum_i a_i$, and $0 < K \leq S$. Join source to sink by a long directed path of n edges and one direct edge. Give the long edges choices $\{1, 1 + a_i\}$ and the direct edge fixed resistance $D = n + K$. Prescribe flow one on every arc and nominations $2, -2$ at the terminals, zero elsewhere. This is a simple cycle, maximum degree two, with an acyclic arc orientation. The long-path total resistance is

$$\theta = n + \sum_i a_i \sigma_i.$$

The two path drops at the target flow are θ and D , so the conserved target is physical exactly when the chosen subset sums to K . Trivial source cases use a triangle: both long-edge option sets $\{1, 2\}$ and direct resistance three give a yes instance; both sets $\{1, 3\}$ and direct resistance three give a no instance.

For any resistance choice, every long edge has the same physical flow p , and the direct edge has flow z , with $p + z = 2$. Both are strictly positive because their path drops agree and their total flow is positive. Unit upper capacities imply $p, z \leq 1$ and hence force $p = z = 1$; conversely the target satisfies every capacity. Absolute capacities give the same equivalence. For general fixed-nomination single-cycle inputs, guess the resistance choices and verify all capacities by the exact scalar piecewise-quadratic cycle algorithm, or by theorem C.3. Thus this existential capacity problem is in NP on the stated class; no membership claim for capacity design at general rank is made. $\square$

The two-path quadratic split gives an explicit approximation gap:

$$p = \frac{2\sqrt{D}}{\sqrt{\theta} + \sqrt{D}}, \quad z = 2 - p, \quad |p - 1| = \frac{|D - \theta|}{(\sqrt{D} + \sqrt{\theta})^2}.$$

If no subset sums to K , then $|D - \theta| \geq 1$. With $M_1 = n + S + K$, both coefficients are at most M_1 , so $|p - 1| \geq 1/(4M_1)$. The minimum maximum arc load is therefore one in a yes instance and at least $1 + 1/(4M_1)$ in a no instance. The minimum maximum deviation from the all-one target is zero or at least $1/(4M_1)$, respectively. Both objectives are convex functions of the flow, but minimizing them over this finite scenario set is hard in accuracy bits. The gap depends on precision; it gives neither fixed-error nor strong hardness under the fixed-small-nomination normalization.

Replacing $\{1, 1 + a_i\}$ by $[1, 1 + a_i]$ lets θ range throughout $[n, n + S]$. Every nontrivial instance then realizes the target because $D = n + K$ lies in this interval. Moreover, the all-low and all-high finite scenarios have $p \geq 1$ and

$p \leq 1$, respectively; since the complete flow vector is affine in p , their segment contains the all-one target. The target thus belongs both to the interval flow region and to the convex hull of finite-scenario flows, but to no finite scenario. Containment of every scenario in a convex capacity set is preserved by convexification; intersection with that set is not. This explains why robust validation on a cycle is polynomial by corollary E.5, while existence of a capacity-feasible finite design is NP-complete.

F Weighted objectives and attainable regions

A linear combination of potential differences need not inherit the properties of a single difference, and a linear combination of arc flows need not inherit those of an individual arc. This section makes both distinctions precise. Unless stated otherwise, the law is $g_e(x) = \beta_e x |x|$, all input data are rational, and the connected graph is simple. Potential objectives are $F = c^\top \pi$ with $\mathbf{1}^\top c = 0$; flow objectives are $d^\top x$. Nomination uncertainty means the balanced box $\mathcal{B} = \{b : l \leq b \leq u, \mathbf{1}^\top b = 0\}$ with finite bounds. Resistance choices are independent positive compact intervals, or, where specified, explicitly listed nonempty finite sets. No operating bounds filter these scenarios. Empty nomination boxes are detected first.

F.1 Tree hardness and exact rational algorithms

On a tree, let w be the unique vector satisfying $Aw = c$. Conservation likewise fixes x from b , independently of resistance. Consequently

$$F = \sum_e w_e \beta_e x_e |x_e|, \quad \beta_e^{\text{opt}} = \begin{cases} U_e, & w_e x_e |x_e| \geq 0, \\ L_e, & w_e x_e |x_e| < 0, \end{cases} \quad (41)$$

where L_e, U_e are the attained resistance extrema. Both choices belong to a finite set and to its interval hull. Each w_e is an objective cut sum, and each x_e is a nomination cut sum.

We use the following elementary form of the classical certificate theorem for rational quadratic programming, which goes back to Vavasis [51]; see also Del Pia et al. [21, Section 2.2, Theorem 3].

Lemma F.1 (Rational quadratic maxima). *A rational quadratic polynomial on a nonempty bounded rational polytope has a rational maximizer of polynomial encoding length. In fixed dimension d , its maximum and a rational maximizer can be found in $N^{O(d)}$ bit time, where N is the total input length. Singular stationary systems require no nonsingularity assumption.*

Proof. Parameterize the rational affine hull first. Write the quadratic as $Q(z) = \frac{1}{2}z^\top Hz + a^\top z + a_0$ with H symmetric. Let a maximum z^* lie in the relative interior of its minimal face, whose affine hull is $Ez = f$ and whose tangent space has rational basis columns T . Stationarity is the linear system

$$Ez = f, \quad T^\top(Hz + a) = 0.$$

If z, z' satisfy it, then $v = z' - z$ is tangent, $v^\top(Hz + a) = 0$, and subtraction of the stationary equations gives $v^\top Hv = 0$. Thus $Q(z') = Q(z)$. The nonempty intersection of this affine stationary set with the polytope is bounded; each of its vertices is rational, has polynomial encoding length by determinant bounds, and has value $Q(z^*)$. This proves the certificate assertion in any dimension. For the algorithm, enumerate all subsets of at most d independent tight inequalities, including the empty subset; they give equations for the affine hull of every face. Solve each stationary system, test its intersection with the polytope by rational linear programming, and retain one rational point when it is feasible. Every retained point is feasible, and the list includes a global maximum; compare their rational values. This handles positive-dimensional stationary sets as well as isolated stationary points and vertices, using $N^{O(d)}$ operations of polynomial bit length. $\square$

Theorem F.2 (Arbitrary support on trees). *For quadratic trees with a balanced nomination box, arbitrary rational zero-sum c , and independent finite or interval resistance sets, deciding whether $\max F \geq T$ is NP-complete. Deciding whether $F \leq J$ for every scenario is coNP-complete. Hardness already holds with fixed resistances, maximum degree three, and nonzero objective coefficients in $\{-1, 1\}$. Approximating the maximum to absolute error at most $1/8$, or returning a feasible rational nomination within $1/8$ of it, is NP-hard on that family. These are absolute-error statements on an unnormalized class, not strong or relative-error hardness.*

Proof. Take positive Subset Sum integers $w_1, \dots, w_n, K$ with $0 < K < \sum_i w_i$; the other, trivial targets are decided in advance and mapped to fixed yes or no instances. Attach a leaf ℓ_i to vertex v_i of the path $v_1, \dots, v_n$. Give backbone edges resistance one and edge $v_i \ell_i$ resistance $1/w_i$. Inject K at v_1 , put zero nomination at other backbone vertices, and withdraw $y_i \in [0, w_i]$ at ℓ_i . Balance is exactly $\sum_i y_i = K$. All leaf flows are y_i , and

$$F = \sum_i (\pi_{v_i} - \pi_{\ell_i}) = \sum_i y_i^2 / w_i \leq K,$$

with equality exactly when all y_i are endpoints. This is the classical convex knapsack endpoint mechanism of Levi et al. [35, Proposition 1], realized here within the stated passive-network restrictions.

A convex maximum over this capped simplex occurs at a vertex. At most one coordinate of a vertex is strictly between its bounds; it is an integer because K, w_i are integers. In a no instance it is some $r \in \{1, \dots, w_i - 1\}$, and the deficit is $r - r^2/w_i \geq (w_i - 1)/w_i \geq 1/2$. Thus yes maxima equal K and no maxima are at most $K - 1/2$. The robust limit $J = K - 1/4$ gives the complementary hardness. Since a returned rational nomination can be evaluated exactly, the same gap rules out both additive outputs. More constructively, with $m_i = \min\{y_i, w_i - y_i\}$ we have $K - F \geq \frac{1}{2} \sum_i m_i$. If $K - F < 1/4$, nearest-endpoint rounding changes the sum by less than $1/2$, so the resulting integer sum equals K . Multiplying all resistances by $\prod_i w_i$ gives positive integer resistances of polynomial binary length if desired; the unscaled gap already proves the approximation assertion.

For membership on the full tree class, eliminate resistance by equation (41). On every closed flow-sign cell of $\mathcal{B}$, the resulting objective is a rational quadratic with polynomial-size data. A global maximizer lies in one such cell, so by lemma F.1 there is a polynomial-bit rational maximizing nomination. Together with the selected allowed endpoints it is a certificate: cut sums and path integration recover rational x, π and verify either $F \geq T$ or the strict violation $F > J$. The certificate exists without enumerating the exponentially many cells. $\square$

The comb family also has an exact pseudopolynomial algorithm. For each possible nonendpoint coordinate i , run Subset Sum reachability on the other capacities up to K , and for each reachable s with $0 \leq K - s \leq w_i$ evaluate $s + (K - s)^2/w_i$. The maximum over these candidates is exact because every capped-simplex vertex has at most one nonendpoint coordinate; endpoint vertices are included as well. Backtracking recovers the rational state. This takes $O(n^2 K)$ arithmetic operations on polynomial-bit numbers, which is not polynomial in the binary input length.

Theorem F.3 (Objective support and cut directions on trees). *On the tree model of theorem F.2, both extrema and a rational optimizing nomination, resistance, flow, and normalized potential can be computed in $N^{O(p)}$ bit time when $p = |\text{supp } c|$ is fixed. A stronger parameter is obtained by contracting all edges with $w_e = 0$ and orienting every remaining edge so that $w_e > 0$. Let k be the number of vertices that do not have exactly one incoming and one outgoing edge. The running time is then $N^{O(k)}$, regardless of objective*

support. If no edge remains, $F = 0$. These bounds are polynomial for each fixed parameter; no bound of the form $f(p)N^C$ or $f(k)N^C$ is asserted.

Proof. Contracting zero-weight edges is exact: sum the nomination intervals and objective coefficients in each cluster. Every nomination in a summed interval can be disaggregated within the original intervals by greedy filling from the lower bounds, with polynomial bit complexity. Retained cut flows and all retained terms of equation (41) are unchanged; contracted terms are zero.

The directed contracted tree has k marked vertices and $k - 1$ maximal paths whose internal vertices each have one incoming and one outgoing edge. For fixed resistance, smooth the laws to $\beta_e(x|x| + \rho x)$, $\rho > 0$. The nomination gradient h of the weighted potential objective, defined modulo a constant on the balance space, satisfies

$$h_{\text{tail}(e)} - h_{\text{head}(e)} = w_e \beta_e(2|x_e| + \rho) > 0,$$

by direct differentiation of the tree cut-sum formula or by the adjoint identity in lemma B.4 with source c . Thus h strictly decreases along each directed path, even if the magnitudes w_e vary along it. The polyhedral normal cone for a maximum over the balanced box supplies a common scalar λ with

$$h_v > \lambda \Rightarrow b_v = u_v, \quad h_v < \lambda \Rightarrow b_v = l_v, \quad l_v < b_v < u_v \Rightarrow h_v = \lambda. \quad (42)$$

These conditions remain valid on a lower-dimensional nomination domain; coordinates with equal bounds are already fixed. Along each path the internal nominations are upper bounds followed by lower bounds, with at most one free coordinate between them. Enumerate every such pivot, and every pivot-free cut, including both extremes. Leave all marked nominations free. This gives $N^{O(k)}$ closed faces with at most $2k - 1$ free coordinates before balance.

The face family depends neither on ρ nor on resistance. On the compact nomination domain, the smoothed objective converges uniformly to F : the added term is $\rho \sum_e w_e \beta_e x_e$, and tree flows are uniformly bounded. A sequence of smoothed maximizers has a convergent subsequence on one fixed closed face, and its limit maximizes F . For joint optimization, first fix the resistance of a joint maximizer and then apply this argument. Thus some joint maximum lies on the same finite family, also for finite resistance sets.

On a face, parameterize balance and the face's feasible rational affine hull by at most $2k - 1$ variables; each flow is rational affine in them. Enumerate the polynomially many cells cut out by the hyperplanes $x_e = 0$, omitting identically zero forms and retaining closed cells. Boundaries cause no problem

because a zero-flow term vanishes for either endpoint choice. The resistance rule equation (41) leaves a rational quadratic on each cell. Apply lemma F.1, compare all rational values, and recover the endpoint resistances and original nominations. Cut sums and path integration give the whole rational physical state. Minimization follows by replacing c by $-c$.

For the support bound, note that contraction cannot increase the number of nonzero objective clusters. Every leaf of the contracted tree has nonzero objective coefficient, so there are at most p leaves and at most $p - 2$ branching vertices. A marked degree-two vertex has both currents entering or both leaving and hence a nonzero coefficient. Therefore $k \leq 2p - 2$. $\square$

On a path, w_e is the cumulative sum of c on one side of the cut. After zero-weight contraction, t sign changes give $k = t + 2$. In particular, a single common sign gives a polynomial algorithm even with unbounded support. The parameter measures cumulative cut directions, not signs of the individual coefficients. The pairwise tree algorithms of Labbé et al. [33, Section 5 of the author manuscript] precede this weighted formulation. Fixed-nomination resistance linearity on trees is also explicit in Afkman et al. [5, Section 4.2]; the additional content here is the weighted nomination-face parameter and its exact rational output guarantee.

F.2 A single algebraic core at fixed global cycle rank

Theorem F.4 (Fixed global rank and supplied signed paths). *Fix global cycle rank r and objective support p . Under independent positive resistance intervals, joint optimization of $c^\top \pi$ over the balanced nomination box admits exact algebraic extrema and an algebraic optimizing scenario in polynomial bit time for fixed r, p . More generally, after repeatedly removing zero-objective leaves and aggregating their nomination intervals at their neighbors, suppose a nonempty marked set K contains every vertex of degree different from two. Suppose also that on each maximal path with internal vertices outside K , the internal coefficients c_v are all nonnegative or all nonpositive. Returning paths are allowed. With $k = |K|$ and r fixed, the same conclusion holds even when $|\text{supp } c|$ grows. A supplied K and its sign condition can be checked directly. A pure cycle must have at least one marked vertex. Finite resistance sets are excluded.*

Proof. Removing a leaf with $c_v = 0$ preserves objective values: its nomination is aggregated at the neighbor, its resistance can be any allowed value, and its potential has coefficient zero. Repeat, retaining the aggregation data for recovery. The case $c = 0$ is handled directly. Suppressing the remaining

unmarked degree-two vertices produces a connected multigraph with k vertices and $P = k - 1 + r' \leq k - 1 + r$ paths, where r' is its actual rank. For the support specialization, mark all nonzero coefficients and all vertices of degree at least three. Every remaining leaf has a nonzero coefficient, and the degree identity gives at most $2r' + p - 2$ branching vertices. Hence

$$k \leq 2r' + 2p - 2, \quad P \leq 3r' + 2p - 3,$$

counting parallel and returning paths in the suppressed graph.

First hold resistance fixed. Choose $v_0 \in K$, give each path a sign $\sigma \in \{1, -1\}$ agreeing with its internal coefficient signs (either sign for an all-zero path), and smooth the laws as above. Perturb the *potential objective* to

$$c^\top \pi^\rho + \delta \sum_{v \notin K} \sigma_{\text{path}(v)} (\pi_v^\rho - \pi_{v_0}^\rho), \quad \delta > 0.$$

The adjoint has positive differential resistances $R_e = \beta_e(2|x_e| + \rho)$ and internal source $c_v + \delta\sigma$, strictly of sign σ . On an ordered path, $j_i = (h_{v_i} - h_{v_{i+1}})/R_i$ satisfies $j_i - j_{i-1} = c_{v_i} + \delta\sigma$. Thus the currents strictly increase on positive-source paths and strictly decrease on negative-source paths. The potential h has a single peak or valley, respectively; at most one current vanishes, allowing a plateau of two adjacent vertices at the extremum. Every horizontal level meets at most two internal vertices, also on a returning path. By equation (42), positive-source paths have lower/upper/lower endpoint order and negative-source paths have upper/lower/upper order, apart from at most two free internal coordinates. Enumerating these patterns, including missing phases and zero or one pivot, gives $N^{O(k+P)}$ closed faces with at most

$$k + 2P = 3k + 2r' - 2$$

free nominations. Marked coordinates are left free.

This family is independent of both perturbations and of resistance. The uniform limiting argument of lemma C.7 applies with unchanged analytic assumptions: the total possible nomination bounds all flows; for $0 < \rho \leq 1$ a path estimate bounds normalized potentials by sums of $\beta_e(B^2 + B)$, where $B = \sum_v \max\{|l_v|, |u_v|\}$. Compactness and uniqueness of the energy minimizer give uniform state convergence as $\rho \downarrow 0$, and the δ term tends uniformly to zero. A convergent subsequence of perturbed maximizers lies on one fixed closed face and yields an original maximum. Fixing the resistance of a joint maximizer then shows that the same face family is complete for joint optimization. The algorithm never chooses a perturbation size.

On one face, eliminate balance and use its rational affine hull with $d \leq k + 2P$ variables z . A spanning-tree flow gives $x = x^0(z) + Cq$, with all r' circulation coordinates q retained in the same core. Chord coordinates equal chord flows and are bounded by B ; free nominations inherit their bounds. Enumerate the polynomially many flow-sign cells in this fixed-dimensional compact core. On each closed cell write $x_e|x_e| = p_e(z, q)$, a rational quadratic. The r' cycle equations and the objective have the form

$$\sum_e C_{ej} \beta_e p_e(z, q) = 0 \quad (1 \leq j \leq r'), \quad F = \sum_e w_e \beta_e p_e(z, q),$$

where w is any rational solution of $Aw = c$, for example one supported on the spanning tree. This is the setting of lemma C.9, with fixed core dimension, fixed aggregate count, degree two, and one scalar interval leaf β_e per edge. That lemma supplies the exact optimum and all optimizing leaves in a common polynomial-size algebraic representation. The bound $|F| \leq B^2 \|c\|_1 \sum_e U_e$ gives an explicit objective bound if needed; $B = 0$ is immediate. Compare the polynomially many algebraic candidates. Disaggregate removed nominations greedily in the winning field, recover removed flows by conservation, choose any allowed removed resistance, and integrate drops to recover normalized potentials. All arithmetic stays in that single field. Replacing c by $-c$, with path signs reversed, proves minimization. $\square$

Corollary F.5 (Fixed nominations). *At fixed global rank and fixed rational nominations, independent positive resistance intervals permit exact algebraic optimization of any rational zero-sum potential objective or any rational linear flow objective, without a support bound. With fixed rational resistances the corresponding value can be compared exactly with a rational threshold in polynomial time. On trees and with fixed nominations, finite resistance choices also permit exact rational weighted-potential optimization.*

Proof. With fixed nominations, the core of the preceding algebraic construction contains only the r circulation variables. A potential objective is the same aggregate with arbitrary rational w ; a flow objective is affine in the core. Singleton resistance intervals give exact evaluation. The tree conclusion follows directly from equation (41). $\square$

These are global-rank algorithms: they never sum values from independent algebraic fields, so they do not by themselves give a weighted algorithm when only block rank is bounded. Nor does the theorem give rational optimal scenarios when cycles remain. The next constructions show that excluding finite resistance sets when cycles remain is substantive.

F.3 Discrete weighted objectives at ranks one and two

Lemma F.6 (A three-terminal cycle). *Orient a triangle $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$, with resistances $(1, 1, \theta)$, $\theta > 0$, nominations $(2, -3, 1)$, and objective coefficients $c = (-5, 12, -7)$. Its flow is $(q + 2, q - 1, q)$, where*

$$6q + 3 - \theta q^2 = 0, \quad q = \frac{-6}{6 + \sqrt{36 + 12\theta}} \in (-1/2, 0).$$

Its objective is

$$F = -5(q + 2)^2 - 7(q - 1)^2 = -\frac{105}{4} - 12(q + 1/4)^2. \quad (43)$$

The maximum is $-105/4$ at $\theta = 24$, $q = -1/4$. At $\theta = 16/3$ and 144 , respectively, $q = -3/8$ and $-1/8$; both endpoint values are $-423/16$, below the interior value by $3/16$.

Proof. The displayed flows satisfy conservation. On $(-1/2, 0)$ their signs are $(+, -, -)$, so the cycle equation is $(q + 2)^2 - (q - 1)^2 - \theta q^2 = 0$. Its values at $-1/2$ and 0 have opposite signs, and the sum of signed edge laws is strictly monotone, so this is the unique physical root. Substitution and completing the square prove every stated value. $\square$

Theorem F.7 (One-cycle weighted-potential hardness). *With fixed nominations, independent explicitly listed resistance choices, and one simple cycle, existential weighted-potential threshold attainment is NP-complete and robust upper bounds are coNP-complete. Hardness holds with two-point choices, only three nonzero nominations $(2, -3, 1)$, and only three nonzero objective coefficients $(-5, 12, -7)$. Positive integer resistances suffice. There is a gap of polynomial binary length, and absolute-error-one value approximation is NP-hard after polynomially encoded resistance scaling. No strong-hardness or relative-error claim follows.*

Proof. Take positive Subset Sum integers a_i, K , put $S = \sum_i a_i$, and decide the case $K > S$ in preprocessing. Replace the triangle's $2 \rightarrow 0$ edge by n series edges, with zero internal nominations and objective coefficients, and independent choices

$$\beta_i \in \{12/n, 12/n + 12a_i/K\}.$$

Their effective resistance is $\theta = 12 + (12/K) \sum_i a_i \sigma_i$, so equation (43) reaches its peak exactly for a target subset. The graph remains a simple cycle, also

for one-item instances. Preprocessed instances use the fixed outputs $\theta = 24$ or 12, with the same peak threshold.

In a no instance $|\theta - 24| \geq 12/K$. Subtracting the equation at $q_0 = -1/4$ gives

$$|q + 1/4| = \frac{|\theta - 24|}{16(6 + \theta(1/4 - q))} \geq \frac{1}{20K + 12S}.$$

Here $-1/2 < q < 0$ and $\theta \leq 12 + 12S/K$ bound the denominator by $240 + 144S/K$. Hence the no maximum is at most

$$-105/4 - \Delta, \quad \Delta = \frac{12}{(20K + 12S)^2} = \frac{3}{4(5K + 3S)^2}.$$

A strict violation of $J = -105/4 - \Delta/2$ exists exactly in the yes case, and value error below $\Delta/4$ separates the cases. In the yes case, a discrete scenario within $\Delta/4$ of the maximum also reveals a target subset, checkable with integer arithmetic.

Scaling every resistance by $4nK$ produces first-edge resistances $4nK$ and path choices $\{48K, 48K + 48na_i\}$, all positive integers. Flows are unchanged, and the objective and gap scale by $4nK$. A further factor $(5K + 3S)^2$ makes the gap $3nK \geq 3$, so an absolute-error-one estimate separates maxima by comparison with the scaled peak minus $3/2$. Both scalings have polynomial encoding length.

For membership, guess one allowed resistance per edge and write $x_e = q + d_e$ in a consistent cycle orientation. Sort the rational breakpoints $-d_e$ of the strictly increasing cycle equation $\sum_e \beta_e(q + d_e)|q + d_e| = 0$; on each interval it is a rational quadratic. Rational breakpoint comparisons locate its unique root; a linear piece or a breakpoint root is handled directly. The root has degree at most two, and all potentials and the weighted objective lie in the same quadratic field. Exact comparison verifies either a weak threshold or a strict violation in polynomial bit time, which proves both membership statements. The interval counterpart is polynomial by corollary F.5. $\square$

Lemma F.8 (A rank-two weighted-flow peak). *Use edges $0 \rightarrow 2, 2 \rightarrow 1, 0 \rightarrow 3, 3 \rightarrow 1, 2 \rightarrow 3$, with resistances $(1, 1, 1, 2, \theta)$ and nominations $(4, -4, 3, -3)$ at $(0, 1, 2, 3)$. Writing $a = x_{02}$ and $q = x_{23}$, the unique physical state has*

$$\begin{aligned} a(q) &= q + 9 - 2\sqrt{2q + 18}, \\ x_{21} &= a + 3 - q, \quad x_{03} = 4 - a, \quad x_{31} = 1 - a + q, \\ \theta q^2 &= 16 - 8a(q), \quad 0 < q < 3. \end{aligned}$$

All five flows are positive. With $u = \sqrt{2q + 18}$,

$$F = -9a + 5q = -\frac{9}{2} - 2(u - 9/2)^2. \quad (44)$$

The peak $-9/2$ occurs at $a = q = 9/8$, $\theta_* = 448/81$. At $\theta_L = 1792/5329$ the pair $(q, a) = (73/32, 57/32)$; at $\theta_U = 12032$ it is $(1/32, 17/32)$. Both endpoint values equal $-37/8$, giving interior advantage $1/8$.

Proof. Balance gives the four outer flows. Equal outer-path drops require $a^2 + (a + 3 - q)^2 = (4 - a)^2 + 2(1 - a + q)^2$, which the stated branch solves. On $[0, 3]$, a increases from $9 - 6\sqrt{2} > 0$ to $12 - 4\sqrt{6} < 4$; also $a + 3 - q = 12 - 2\sqrt{2q + 18} > 0$ and $1 - a + q = 2\sqrt{2q + 18} - 8 > 0$. The cross drop is $(4 - a)^2 - a^2 = 16 - 8a$. The function $\theta q^2 - 16 + 8a(q)$ is strictly increasing on $[0, 3]$, negative at zero and positive at three. Its unique root gives positive flows satisfying both cycle equations, hence the passive state. Completing the square proves the objective identity, and substitution proves the three rational states. $\square$

Theorem F.9 (Rank-two weighted-flow and total-flow hardness). *At fixed nominations and global rank two, existential linear-flow threshold attainment with independent finite resistance sets is NP-complete and robust upper bounds are coNP-complete. Hardness holds on maximum-degree-three graphs with two-point choices, the four fixed nominations in lemma F.8, and two fixed objective coefficients $(-9, 5)$, or $(9, 5)$ with a shifted threshold. It also holds for unit-cost total absolute flow on a directed acyclic graph with those same small nominations. Positive integer resistances suffice. These reductions have polynomial-bit gaps, which rule out polynomial dependence on accuracy bits unless $P = NP$. Absolute-error-one hardness follows separately by scaling nominations; common resistance scaling alone does not enlarge a flow gap.*

Proof. Take positive Subset Sum data a_i, K . In the theta of lemma F.8, replace the cross edge by an n -edge path with choices

$$\left\{ \frac{224}{81n}, \frac{224}{81n} + \frac{224a_i}{81K} \right\}.$$

Its total resistance is $\theta = (224/81)(1 + \sum_i a_i \sigma_i / K)$, which equals $448/81$ exactly for a target subset. Use any cross-path edge for the $5q$ term. The global rank is two, the degree is at most three, and all subdivision nominations are zero. Preprocessed instances with $K > S = \sum_i a_i$ use the fixed yes/no theta instances $\theta = 448/81$ and $224/81$.

For the gap put $H = 23K + 15S$. In a no instance, $|\theta - \theta_*| \geq 224/(81K) > 2/K$ and $\theta \leq 3(1 + S/K)$. Subtract the cross equations at q and $q_* = 9/8$:

$$(\theta - \theta_*)q_*^2 = -(q - q_*) \left(\theta(q + q_*) + 8 \frac{a(q) - a(q_*)}{q - q_*} \right).$$

The quotient is in $(0, 1)$ since $0 < a'(q) < 1$; also $q + q_* < 5$ and $q_*^2 > 1$. Hence $|q - q_*| > 2/H$. Since $u + 9/2 < 10$, $|u - 9/2| > 2/(5H)$. The no maximum is therefore at most $-9/2 - \Delta$, where $\Delta = 1/(4H^2)$. Use the strict robust limit $-9/2 - \Delta/2$ or value error below $\Delta/4$. As before, a near-optimal discrete scenario reveals a target subset in the yes case.

Scaling all resistances by $81nK$ multiplies the four outer coefficients $(1, 1, 1, 2)$ by $81nK$ and gives cross choices $\{224K, 224K + 224na_i\}$; it changes no flows, values, or gaps. For absolute-error-one hardness, multiply all nominations by $M = 16H^2$ after this integer conversion. Homogeneity (lemma A.3) scales flows and F by M , so the gap is at least four and comparison with the scaled peak minus two suffices. In this last assertion the nominations are no longer fixed small numbers.

For positive coefficients, conservation at 0 gives $9x_{03} + 5x_{23} = F + 36$, with target $63/2$ and the same gap. For unit coefficients on *every* edge, subdivide $0 \rightarrow 3$ into $9n + 1$ edges of total resistance one. Keep the other three outer edges. Use $5n$ cross-path edges: $4n$ fixed edges of resistance $28/(81n)$ and n uncertain edges with choices

$$\left\{ \frac{112}{81n}, \frac{112}{81n} + \frac{224a_i}{81K} \right\}.$$

The effective cross resistance is unchanged. All flows are positive, and the orientation is acyclic (order 0, 2, 3, 1, with path vertices inserted). Thus total absolute flow equals total oriented flow, namely

$$(9n + 1)(4 - a) + a + (a + 3 - q) + (1 - a + q) + 5nq = nF + 36n + 8.$$

Its peak is $(63n + 16)/2$ and its gap at least $n\Delta$. There are $14n + 4$ edges and $14n + 3$ vertices, preserving rank two and maximum degree three. Common resistance scaling by $81nK(9n + 1)$ makes all data integral: the long outer-path edges have resistance $81nK$, fixed cross edges $28K(9n + 1)$, and uncertain options $112K(9n + 1)$ and $112K(9n + 1) + 224n(9n + 1)a_i$. No constant-gap claim with the small nominations is inferred.

For membership, guess the finite choices and use corollary F.5 with singleton intervals: its fixed-rank algebraic evaluation verifies weak attainment or strict violation exactly. For total absolute flow, flow-sign cells again make the objective affine in the core. This proves completeness, not just hardness. $\square$

F.4 Cactus flow boxes and the difference between scenarios and values

Theorem F.10 (Flow region and exact optimizing resistance scenario). *Fix rational balanced nominations on a cactus. With independent resistance intervals, its attainable flow region is an affine image of independent scalar circulation intervals. With finite resistance sets, the convex hull of the attainable flow region is that same image for the interval hulls. For any rational linear flow objective, a rational endpoint resistance scenario attaining the exact optimum can be found in polynomial bit time, even with arbitrarily many cycles. The value admits an additive estimate in time polynomial in input size and accuracy bits. Exact comparison of that scalar value with a rational threshold is nevertheless SRS-hard, already with fixed positive integer resistances and unit source/sink nominations. No NP-hardness claim is made for this arithmetic barrier.*

Proof. By lemmas A.5 and A.6, bridge flows are fixed and a consistently oriented cycle has flows $q + d_e$, with rational fixed offsets. Its physical circulation solves $H_\beta(q) = \sum_e \beta_e(q + d_e)|q + d_e| = 0$. Define

$$H_-(q) = \sum_e \min\{L_e(q + d_e)|q + d_e|, U_e(q + d_e)|q + d_e|\},$$

$$H_+(q) = \sum_e \max\{L_e(q + d_e)|q + d_e|, U_e(q + d_e)|q + d_e|\}.$$

Both are continuous and strictly increasing, with opposite infinite limits. Let q^+ be the root of H_- and q^- the root of H_+ . Since $H_- \leq H_\beta \leq H_+$, all attainable circulations lie in $[q^-, q^+]$. At either root, choosing for each summand the endpoint that attains it gives an actual endpoint scenario realizing that root, also for finite sets. Sorting the rational breakpoints $-d_e$ makes both equations piecewise quadratic. Rational sign tests locate the appropriate piece; its root, including linear or breakpoint cases, has a polynomial-size algebraic encoding of degree at most two. Comparing it with the breakpoints recovers the realizing resistance endpoints exactly in polynomial time. This scalar root principle is consistent with the cycle-interval linearization in Aßmann et al. [5, Lemma 4.10].

For intervals, the continuous image of the connected compact resistance box is the whole interval $[q^-, q^+]$. Different blocks have fixed effective nominations and disjoint resistance variables. Thus, for a rational particular flow x^0 and signed cycle matrix Z ,

$$\mathcal{X} = \{x^0 + Zq : q_C \in [q_C^-, q_C^+] \text{ for all cycles } C\}. \quad (45)$$

Cycle supports are disjoint, so this is a possibly degenerate parallelotope. For finite sets the attainable circulation set is a Cartesian product, and the convex hull of a finite Cartesian product is the product of the convex hulls. Each scalar hull is the same interval, which proves the second assertion.

A linear flow objective is $D + \sum_C A_C q_C$, with rational D, A_C . Choose the endpoint scenario for q_C^+ when $A_C > 0$, for q_C^- when $A_C < 0$, and any allowed setting when $A_C = 0$. This requires no comparison of sums of independent radicals. Approximating each selected root within $\varepsilon/[2(1 + \sum_C |A_C|)]$ and summing gives value error below ε ; quadratic root enclosure takes polynomial time in the input and accuracy bits. Separate quadratic encodings also represent the physical flow; no common field containing all cycle roots is required.

For the arithmetic obstruction take a SRS instance $\sum_i \sqrt{a_i} \leq K$ with positive integer data. Remove the terms with $a_i = 1$ and subtract their count from K . If no terms remain or the adjusted $K \leq 0$, decide directly and use the fixed outputs below. Otherwise build a chain of separate triangles joined by resistance-one bridges, with unit injection at the first and unit withdrawal at the last terminal. In triangle i , a direct edge of resistance $a_i > 1$ is parallel to a two-edge path of resistances $1/2, 1/2$, so the direct flow is $x_i = 1/(1 + \sqrt{a_i})$. Give it objective coefficient $a_i - 1$ and give all other edges coefficient zero. With m triangles,

$$F = \sum_i (a_i - 1)x_i = \sum_i \sqrt{a_i} - m.$$

Thus $F \leq K - m$ is exactly the source comparison, including equality. Doubling every resistance makes all data positive integers and leaves flows unchanged. Distinct triangles joined by bridges give a simple maximum-degree-three cactus. For the fixed yes/no outputs use the triangle with direct resistance 8, alternate resistances 1, 1, and direct-edge objective coefficient 3: its value is 1, so thresholds 1 and 0 serve. The SRS convention and its unresolved complexity are as in appendix A.5. The scenario here is unique, so scenario selection is trivial while exact scalar comparison still contains SRS. $\square$

For rank at most one, the objective in the proof has at most one quadratic root, so even exact scalar comparison is polynomial. Thus rank two is the lowest rank at which the discrete weighted-flow hardness of theorem F.9 can occur, while arbitrary cactus rank creates a separate arithmetic issue. Rational membership in equation (45) is decided by conservation and separate quadratic interval comparisons. For finite sets this tests convex-hull membership, not

finite attainability, which is already hard on one cycle by theorem E.11. Every continuous convex function on the flow box has a maximizing vertex, since its value at a convex combination is bounded by the largest vertex value, and each vertex has an endpoint resistance realization. This structural fact is not an efficient algorithm for arbitrary convex maximization.

F.5 The objective determines the universal resistance-hull class

Call a graph a *universal hull graph* for an objective class if, for every fixed balanced nomination and every objective in that class, independent nonempty compact positive resistance sets have the same maximum and minimum as their interval hulls. By compactness the sets attain their endpoints; the definition needs no representation of their interiors. *Separate monotonicity* means monotonicity of every one-coordinate section after all other coordinates are fixed; the direction may depend on those fixed coordinates and on the nomination.

Theorem F.11 (Universal hull hierarchy). *Under quadratic laws, finite connected simple graphs have the following universal hull classes:*

<i>objective class</i>	<i>graph class</i>
<i>all zero-sum linear potential objectives</i>	<i>trees</i>
<i>all individual potential differences</i>	<i>cacti</i>
<i>all linear flow objectives</i>	<i>cacti</i>
<i>all individual arc flows</i>	<i>series-parallel graphs.</i>

In each row the same graph class is characterized by universal separate monotonicity over positive resistance boxes. Each negative direction has a witness with one varying resistance, rational nominations and objective coefficients, and strictly positive rational resistances, all of polynomial encoding length, given the relevant cycle or theta subdivision (or the K_4 subdivision in the last row).

Proof. The final row is theorem E.8. We prove the other three rows, including the restoration bounds needed on graphs larger than the basic witnesses.

Positive directions. On trees, fixed nominations fix every flow, and equation (41) is affine in resistance. For pairwise potentials on cacti, the cactus part of theorem E.4 gives the required one-coordinate monotonicity: cactus terminal adjoint currents have fixed signs along their unique block route. The sign of the law change also stays fixed. For an affine one-edge

parameter $g_e(x, \alpha) = g_e^0(x) + \alpha f_e(x)$, a zero of $f_e(x_e(\alpha))$ makes the entire state at that parameter satisfy the laws at every parameter value, so by uniqueness the state is constant. Otherwise continuity fixes the sign of $f_e(x_e)$, and the finite-secant identity fixes the direction of the terminal objective. This argument also proves the positive direction for the admissible continuous affine edge-law families of theorem E.4, including several independent coordinates on one edge.

For linear flows on a cactus, block independence makes the objective $D + \sum_C A_C q_C$. Vary only one resistance β_e in one cycle. At $q = -d_e$ the strictly increasing cycle function $H_\beta(q)$ is independent of β_e , so the physical sign of $q + d_e$ is fixed as β_e varies. If this sign is zero for one value, the root is constant. Otherwise increasing β_e changes the old root's equation by a term of fixed nonzero sign, so root comparison makes q monotone, and so is its affine objective contribution.

In every positive direction, start at an optimizer over the compact interval box and move each coordinate in turn to a maximizing endpoint, without reducing the objective. The value remains the global optimum after every move, even if later monotonicity directions change, and the final corner belongs to the original product of compact sets. Applying the same argument to the negative objective gives minima. Thus separate monotonicity implies the stated hull equality.

Weighted-potential converse. On any non-tree choose a simple cycle and three distinct vertices on it, and represent its three connecting paths as subdivisions of lemma F.6. Split each resistance-one side into equal positive shares. On the uncertain side, if it is subdivided, reserve a fixed total $8/3$ on all but one edge and put resistance $\theta - 8/3$ on the remaining edge; on a single edge use θ . The three values $\theta \in \{16/3, 24, 144\}$ keep these resistances positive and reproduce the triangle because internal nominations vanish. Put the three stated nominations and objective coefficients on the branch vertices and zero elsewhere.

Let $m = |E|$ and restore every other edge with resistance

$$R = 12(10000m^2)^3.$$

The conserved comparison flow with $q = -1/4$ on the selected cycle and zero elsewhere has energy $[(7/4)^3 + (5/4)^3 + \theta(1/4)^3]/3 = (468 + \theta)/192 < 4$ at all three settings. Hence every extra-edge flow of the full physical minimizer has magnitude at most $u = (12/R)^{1/3}$. Total positive nomination is three, so every full flow has magnitude at most three. Restrict the full state to the cycle; its induced nomination b' is balanced, has $|b'_v| \leq 6$, and satisfies

$\|b' - b\|_1 \leq 2mu$. The original and induced nominations lie in a common balanced box with total absolute bound $B \leq 6m$. Apply lemma B.7 to the two paths of total resistance one in $F = -5(\pi_0 - \pi_1) + 7(\pi_1 - \pi_2)$. The objective error at each setting is at most

$$(5 + 7) 2(6m) (2mu) = 288m^2u = 288/10000 < 3/64.$$

The restored interior value thus exceeds both endpoints by more than $3/16 - 2(3/64) = 3/32$, violating both properties.

Theta subdivisions. Every noncactus simple connected graph contains three internally disjoint paths between two vertices. Indeed, take a block other than an edge or cycle and a cycle in it. A chord gives such a theta. Otherwise some component of the block minus the cycle has at least two distinct neighbors on the cycle, since a single neighbor would be an articulation vertex; a connecting path through that component and the two cycle arcs form the theta. By simplicity, at most one theta path is a single edge. Designate branch vertices 2, 3, choose internal vertices 0, 1 on the other two paths, and split the four outer segments at them. This realizes either four-vertex theta gadget by positive resistance subdivisions with zero internal nominations. Keep only one variable edge on the cross path, reserving on all other cross edges a fixed total less than its lower resistance endpoint.

Pairwise-potential converse. Use the pressure theta already computed in equation (32), with outer resistance totals $(1/2, 1/6, 1/6, 1/2)$ and nominations $(4, -4, 3, -3)$; the leaf from that hardness reduction is not needed. For cross totals $\tau \in \{23/108, 1, 71/12\}$ the cross flows are respectively $3/2, 1, 1/2$, and

$$\pi_1 - \pi_0 = -2 - (q - 1)^2/6.$$

The interior value exceeds both endpoint values by $1/24$. For a subdivided cross path reserve a fixed total $23/216$ and vary the remaining edge, which keeps it positive. Restore all extra edges with

$$R = 18(10000m^2)^3.$$

The comparison flow at $q = 1$ has outer flows $(1, 3, 3, 1)$ and energy $(10 + \tau)/3 < 6$, so extra flows are at most $u = (18/R)^{1/3}$. The full-flow bound 14 gives $|b'_v| \leq 14 \deg_{\text{theta}}(v)$ and $B \leq 28m_{\text{theta}}$ on the selected subgraph, with $\|b' - b\|_1 \leq 2m_{\text{extra}}u$. The terminal path through 2 has resistance $2/3$, so lemma B.7 bounds the pressure error by

$$\frac{224}{3} m_{\text{theta}} m_{\text{extra}} u \leq 75m^2u = 75/10000 < 1/96.$$

Thus the strict interior advantage survives.

Linear-flow converse. Use lemma F.8 with its three cross resistances 1792/5329, 448/81, 12032. Split the outer totals (1, 1, 1, 2) on the theta segments and, if necessary, reserve a fixed cross total 896/5329. Choose one edge of segment $0 \rightarrow 2$ for coefficient -9 and one cross-path edge for coefficient 5; on the isolated theta their flows are a, q . Restore all other edges with

$$R = 18000(10^9 m^2)^3.$$

The conserved comparison flow $a = q = 9/8$ has other path flows (3, 23/8, 1) and energy

$$[(9/8)^3 + 3^3 + (23/8)^3 + 2 + \theta(9/8)^3]/3 < 6000$$

at all three settings (at the upper setting it is 91657/16). Hence extra flows are at most $u = (18000/R)^{1/3}$, and all full flows are bounded by the total positive nomination seven. The induced theta nominations obey $|b'_v| \leq 21$, $B \leq 21m$, and $\|b' - b\|_1 \leq 2mu$. For either selected edge, scalar quadratic monotonicity gives

$$\begin{aligned} |x_e(b') - x_e(b)|^2 &\leq \frac{2}{\beta_e} |(\pi_{\text{tail}} - \pi_{\text{head}})(b') - (\pi_{\text{tail}} - \pi_{\text{head}})(b)| \\ &\leq 4B\|b' - b\|_1 \leq 168m^2u, \end{aligned}$$

using lemma B.7 on the single-edge path. This also follows directly from the fact that $|s|s - |t|t$ has magnitude at least $|s - t|^2/2$. The resistance cancels even on a subdivided uncertain edge. The objective error is at most $14\sqrt{168m^2u} < 1/32$, since its square is at most $196 \cdot 168/10^9 < 1/1024$. The restored interior advantage therefore exceeds $1/8 - 2(1/32) = 1/16$. Alternatively, select the $0 \rightarrow 3$ segment with coefficient 9 and the cross edge with coefficient 5. On the isolated theta the objective is then $F + 36$, and after restoration the same error estimate applies directly to those selected edges. Thus positive objective coefficients suffice.

All induced nominations above are balanced because the added vertices have zero original nomination and aggregate extra-edge exchanges cancel. The restricted physical state satisfies the selected subgraph's laws, so uniqueness justifies each comparison with its perturbed-nomination state. All resistance splits and displayed restoring integers have polynomial encoding length. Each witness has a strict interior value above both endpoints, so its endpoint-only set has a smaller maximum than its interval hull. $\square$

The positive hull statements also hold jointly over any compact nomination set on which the state is continuous: fix the nomination of a joint optimizer, then move resistance coordinates to endpoints. This does not assert separate monotonicity of the nomination-optimized value. For pairwise potentials, the continuous affine one-edge parameter version proved above extends beyond quadratic resistances; a parameter shared by several edges is not covered.

When the scalar sets have rational attained endpoints, endpoint recovery is constructive in the additive sense. Suppose the applicable interval-hull algorithm returns a rational feasible nomination and parameter scenario within $\varepsilon/2$ of its maximum. Keep the nomination fixed and, for each of $M > 0$ parameters in turn, evaluate the two endpoint settings within $\delta = \varepsilon/(4M)$ and retain the one with the larger estimate. By separate monotonicity the better endpoint is no worse than the current value, and selecting by these estimates loses at most 2δ . After $2M$ fixed-scenario evaluations the total loss is at most $\varepsilon/2$, and every parameter belongs to its original compact set. Minima use the smaller estimate. For $M = 0$ no rounding is needed. The polynomial and fixed fractional law algorithms of appendix D supply the needed accuracy-bit evaluations within their scope; the argument presupposes such an evaluation algorithm and does not provide one for an arbitrary continuous law. In particular, endpoint recovery hides no exact comparison of a sum of independent pressure radicals.

F.6 Convexity of entire attainable state regions

Normalize one potential to zero throughout this subsection. An attainable region varies resistance in a positive interval box while fixing balanced nominations; the output may be the whole flow vector, the whole normalized potential vector, or their joint vector.

Lemma F.12 (One varying cyclic resistance). *For any strictly increasing continuous passive edge laws of the form $\beta_e f(x_e)$ with $f(0) = 0$, vary just one resistance on a cyclic edge $e = (u, v)$. Either the entire state is constant, or its own-edge flow q is a continuous strictly monotone parameter for the attainable state. In the latter case the terminal drop $\pi_u - \pi_v$ is also an injective linear coordinate on the potential region. If either the flow region or the potential region is convex, that region equals the segment between its endpoint states, and every linear objective has endpoint extrema.*

Proof. Deleting e leaves a connected graph. Prescribing the flow q on e gives remaining nominations $b - q(e_u - e_v)$. Let $P(q)$ be the remaining network's terminal drop. For $q_2 > q_1$ the two conserved remaining flows differ, and

strict monotonicity implies

$$0 < \sum_{f \neq e} [g_f(x_f(q_2)) - g_f(x_f(q_1))] [x_f(q_2) - x_f(q_1)] = -(q_2 - q_1) [P(q_2) - P(q_1)].$$

Thus P is continuous and strictly decreasing. The full state solves $P(q) = \beta_e f(q)$, an equation with a unique root, whose sign is determined by $P(0)$ independently of resistance. If $P(0) = 0$, then $q = 0$ for all resistances and the remaining state is fixed. Otherwise changing β_e changes the old root's equation by a nonzero term of fixed sign, so root comparison makes $q(\beta_e)$ strictly monotone, and state continuity gives a continuous bijection onto its interval of flows. The remaining state is uniquely determined by q , and P is strictly monotone. Therefore each flow vector is uniquely determined by its coordinate q , and each potential vector by its linear coordinate $\pi_u - \pi_v = P(q)$. If the region is convex, it contains the chord between its endpoint states, which already takes every intermediate coordinate value. Since each coordinate value determines a unique point of the region, the entire region is that chord. $\square$

Theorem F.13 (Universal convex-region classes). *For the quadratic law on a finite connected simple graph, the attainable flow region is convex for every fixed balanced nomination and every positive resistance box if and only if the graph is a cactus. The analogous class for normalized potential regions, and for joint flow-potential regions, is exactly the trees. Negative witnesses can have one varying resistance, rational nominations, and strictly positive rational resistances, all of polynomial encoding length. The same universal classes hold if every interval is required to have strictly positive width, but the positive-width extension below is existential and supplies no polynomial encoding bound on the widths.*

Proof. The cactus flow region is the affine box in theorem F.10. On a tree, flows are fixed and normalized potentials are linear in resistance, so the potential and joint regions are affine images of the box. On every noncactus, the one-coordinate linear-flow witness of theorem F.11 contradicts the endpoint conclusion of lemma F.12, so its flow region is nonconvex. On every non-tree, the weighted-potential witness contradicts the corresponding potential conclusion, and the joint region cannot be convex because its linear projection onto potentials is nonconvex. In both applications the strict interior advantage excludes the constant-state exception in the deletion lemma.

For strictly positive widths, take one of these compact nonconvex regions S and a point $z \notin S$ on a chord between two of its states. Compactness gives $\text{dist}(z, S) > 0$. Thicken all fixed resistance coordinates to small positive-width

rational intervals containing their original positive values; the two original states remain attainable. Uniform continuity on a compact positive resistance neighborhood implies that all states of a sufficiently small thickening lie within less than half that distance of S . They still exclude z , so the new region remains nonconvex. This applies to flows, normalized potentials, and joint states. Continuity gives the existence of rational widths, not their bit complexity. $\square$

Theorem F.14 (Qualitative extension to every nonlinear power). *For each fixed real exponent $\gamma > 0$, $\gamma \neq 1$, the classifications in theorem F.13 remain valid for the common law $g_e(x) = \beta_e \operatorname{sgn}(x)|x|^\gamma$. Witnesses with rational nominations and strictly positive rational resistances exist. This is a geometric statement; no uniform encoding bound, exact algebraic algorithm, or accuracy-bit algorithm is asserted for this extension. The linear exponent $\gamma = 1$ is excluded.*

Proof. Strict convexity and coercivity of $\sum_e \beta_e |x_e|^{\gamma+1}/(\gamma+1)$ give a unique continuous state. On cacti, the image of each independent compact connected resistance box in its scalar circulation coordinate is an interval, which proves the positive direction for flows. On trees, the fixed flow makes potentials linear in resistance, which proves the other two positive directions.

For a flow obstruction use the theta edges of lemma F.8, now with outer resistances $(1, 3, 2, 2)$ and nominations $(2, -2, 0, 0)$. Conservation gives outer flows $(a, a - q, 2 - a, 2 - a + q)$. Near $(a, q) = (1, 0)$ all are positive, and their outer-cycle equation is

$$\Phi(a, q) = a^\gamma + 3(a - q)^\gamma - 2(2 - a)^\gamma - 2(2 - a + q)^\gamma = 0.$$

At that point $\Phi_a = 8\gamma > 0$. Implicit differentiation gives $a'(0) = 5/8$. Moreover the second-derivative numerator divided by $\gamma(\gamma - 1)$ is

$$(1 - 2)(5/8)^2 + (3 - 2)(5/8 - 1)^2 = -1/4,$$

so $a''(0) = (\gamma - 1)/32 \neq 0$. The cross drop $P(q) = 2(2 - a(q))^\gamma - a(q)^\gamma$ equals one at zero and stays positive for small positive q , so the resistance $\beta_{23}(q) = P(q)/q^\gamma$ realizes these states. The deletion lemma makes this continuous parametrization injective. Choose a compact nontrivial positive q interval on which a'' has strict sign; the graph of a over it is not a segment. If $L(q)$ is the endpoint secant of a , then either $a - L(q)$ or its negative is a linear flow objective plus a constant, with strictly positive interior value and zero endpoint values. The constant is immaterial.

For potentials take the triangle formed by the path $0 \rightarrow 1 \rightarrow 2$, with both resistances one, and a variable direct edge $0 \rightarrow 2$, with nominations $(2, 1, -3)$. The path flows are $a, a + 1$, and the direct flow is $2 - a$. For $0 < a < 2$ choose

$$\beta_{02}(a) = \frac{a^\gamma + (a + 1)^\gamma}{(2 - a)^\gamma}.$$

It is strictly positive and strictly increasing. Normalizing $\pi_2 = 0$ gives $(\pi_0, \pi_1) = (a^\gamma + (a + 1)^\gamma, (a + 1)^\gamma)$ and

$$\frac{d\pi_0}{d\pi_1} = 1 + \left(\frac{a}{a + 1} \right)^{\gamma-1}.$$

This expression has nonzero derivative on an interior interval when $\gamma \neq 1$. The potential graph therefore has strict curvature, so some linear potential objective (made zero-sum by its coefficient at 2) has a strict interior advantage over its endpoints. Its potential region is nonconvex by lemma F.12, and so is its joint region.

Subdivide these witnesses on a theta in any noncactus, or on a cycle in any non-tree, exactly as in theorem F.11. Series resistance coefficients add for every common exponent, and all new internal nominations are zero. Reserve a small fixed positive cross-path total to keep only one resistance variable. Restore all other edges with a common resistance R . A conserved comparison flow supported on the selected subgraph has uniformly bounded energy over the compact interval of varying resistance, so energy minimality makes every extra flow $O(R^{-1/(\gamma+1)})$ uniformly. The full physical flow bound is independent of R . Restricting to the selected subgraph therefore induces nomination perturbations tending uniformly to zero. Uniqueness and continuous dependence of that subgraph's state give uniform convergence of its selected flows and normalized potentials, so the strict interior objective advantage survives for some finite R . The variable edge is cyclic in the full graph, so the deletion lemma again turns that advantage into nonconvexity of the corresponding entire region.

Finally, strict inequalities and continuity allow rational approximations of the two resistance endpoints, the chosen interior setting, and a sufficiently large rational R . Resistance subdivisions can use positive rational shares, and the displayed nominations are already integers. If the separating objective initially has real coefficients, approximate these as well, preserving its strict gap and, when required, its zero sum. This proves existence with rational data, without bounding their sizes. The positive-width argument of theorem F.13 applies unchanged. At $\gamma = 1$ both curvature computations vanish, so this proof gives neither converse. $\square$

These classifications concern convex sets of full vectors under *resistance variation*. They differ from convexity of scalar transfer characteristics under source variation, for which Wang and Hasler [54, Theorem 4, p. 13, and conclusion, p. 21] give topological conditions and propose extension to other circuit parameters as further work (on Hasler and Wang [30] see appendix E). The curvature, energy, and convex-graph arguments used here are elementary; the statements established here are the specific universal output classes, the restricted converse witnesses, and the quantified quadratic restoration.

A related cactus characterization appears in Brandenburg and Stursberg [15, Definitions 1–2 and Theorem 18 of the preprint]: each linear differential-flow polytope fixes its positive elasticity vector, while their universal nondegeneracy characterization quantifies over all positive elasticity vectors and all nomination/capacity bounds. Those quantifiers and that property differ from resistance-attainable nonlinear state regions. In contrast, with linear Ohmic laws *every* graph has the potential-objective hull property. Indeed, ground one vertex, vary one conductance from t_0 to t , and write $L(t) = L_0 + (t - t_0)aa^\top$. For any grounded objective vector c , put $U = c^\top L_0^{-1}a$, $V = a^\top L_0^{-1}b$, and $r = a^\top L_0^{-1}a$. The rank-one inverse identity gives

$$c^\top \pi(t) = c^\top \pi(t_0) - \frac{(t - t_0)UV}{1 + (t - t_0)r}.$$

Positive definiteness makes the denominator positive for every positive conductance in the interval, so the derivative has fixed sign $-UV$. Passing to the reciprocal resistance reverses the parameter direction and preserves monotonicity. Coordinatewise endpoint selection then proves the linear hull assertion on all connected graphs. This familiar sensitivity calculation says nothing about convexity of the entire state regions.

G Weighted cactus optimization in accuracy bits

A cactus may have arbitrarily many cycle blocks. Even when its nominations depend on one scalar, a weighted potential objective can therefore contain arbitrarily many independent radicals. The algorithms in this section approximate the local formulas before adding them, so that the final semialgebraic optimization has only the nomination parameters as variables. A structural theorem reduces an unrestricted balanced nomination box to polynomially many such parameter families when the objective support and maximum block rank are fixed.

Throughout, $c \in \mathbb{Q}^V$ satisfies $\mathbf{1}^\top c = 0$ and the objective is $W = c^\top \pi$. Accuracy-bit complexity means polynomial dependence on the binary input

length and $\max\{0, \log(1/\varepsilon)\}$ for positive rational ε ; we may replace ε by $\min\{\varepsilon, 1\}$. The exponents of our polynomial bounds may depend on the fixed parameters, so no fixed-parameter tractability is claimed.

Theorem G.1 (Affine nominations on a cactus). *Fix d . Let G be a connected cactus and let $b(z) = b^0 + Hz$ be a rational affine balanced nomination family on a nonempty compact rational polytope $P \subseteq \mathbb{R}^d$. The support of c , cycle lengths, and number of cycles are unrestricted. Consider either*

1. *fixed positive rational asymmetric quadratic edge laws; or*
2. *uncertain quadratic laws in either of two models: one shared coefficient $\beta_e \in [L_e, U_e]$ per symmetric edge, or two independent directional coefficients $\beta_e^\pm \in [L_e^\pm, U_e^\pm]$ per asymmetric edge. All intervals have positive rational endpoints and are independent across edges.*

For each maximum and minimum of W , a rational interval of width at most ε containing the optimum and a rational ε -optimal original parameter scenario can be computed in accuracy-bit polynomial time. In case 2 the extrema are joint over z and all resistances. No operating constraints filter the scenario set in this statement.

The local quadratic and radical formulas used below have direct predecessors. Gotzes, Heitsch, Henrion, and Schultz derive piecewise quadratic-radical and rational circulation formulas along affine nomination rays in Theorem 6 of the author manuscript of Gotzes et al. [28]; that manuscript also extends the method to node-disjoint cycles with attached trees (printed pp. 18 and 24). Vigneron optimizes sums of algebraic functions by common arrangements and approximate summation [52, Theorem 6 and Section 3.2 of the October 21, 2011 author manuscript]. His Section 2.3 includes a bit model, in which the approximation schemes are polynomial in $1/\varepsilon$ rather than in accuracy bits. Thus neither the local formulas nor the use of approximation to avoid exact radical-sum comparisons is a new principle here, and we make no claim of exhaustive literature priority. The guarantee established below combines arbitrary weighted objectives, fixed nomination dimension, unbounded cactus cycle count, the stated law models, explicit accuracy-bit bounds, and parameter recovery.

G.1 A resistance-independent nomination-face theorem

A *nomination face* fixes some coordinates at their box endpoints, lets the others keep their intervals, and imposes exact total balance. Faces in the

next theorem are formed after the stated objective-preserving reductions, so they need not be faces of the original box.

Theorem G.2 (Small nomination faces at bounded block rank). *Fix integers $r \geq 1$ and $p \geq 2$. Suppose every block of a connected graph has cycle rank at most r , the nominations range over a nonempty rational balanced box, and $|\text{supp } c| \leq p$. For fixed positive quadratic laws, including asymmetric laws, one can construct $n^{O(rp)}$ rational faces, each with $O(rp)$ unfixed nomination coordinates, such that a maximum of W lies on at least one face after objective-preserving reductions. Every feasible reduced nomination has a rational lift when it is rational, with polynomial encoding length.*

The reductions and face family depend on the graph, nomination bounds, and c , but not on the positive law coefficients. Consequently the same family contains an optimum for joint optimization over any independent compact positive coefficient sets. This last assertion is structural; it does not compute the variable-coefficient block objectives.

Proof. If $c = 0$, start at all nomination lower bounds and raise coordinates in a fixed order until their sum is zero; feasibility ensures that this is possible, and all but at most one coordinate end at endpoints. Fixing those endpoint coordinates and retaining the possible remaining interval with exact balance gives one rational face with at most one unfixed coordinate, using rational arithmetic of polynomial encoding length.

Otherwise use the block-vertex incidence tree, with a node for every block and every original vertex. A subtree with no objective support except possibly at its single attachment can be pruned by adding all its nomination intervals to the attachment interval: its exterior state depends only on that aggregate, and its removed vertices have zero objective coefficient. A prescribed aggregate in a sum of intervals can be split by starting at all lower endpoints and filling the remaining amount in a fixed order, with rational arithmetic of polynomial encoding length.

A second reduction is needed. For a block B , remove its edges and write G_v for the component rooted at $v \in V(B)$. Put $\gamma_v = \sum_{u \in V(G_v)} c_u$. If every $\gamma_v = 0$, contract all of B to one vertex, summing its vertices' nomination intervals and objective coefficients. To prove that the objective is preserved, write

$$W = \sum_{v \in V(B)} \sum_{u \in V(G_v)} c_u (\pi_u - \pi_v) + \sum_{v \in V(B)} \gamma_v \pi_v. \quad (46)$$

The last sum vanishes. The flows and relative potentials in each G_v are determined by the nominations away from its root, and the root supplies

their balancing nomination. Translating these components to a common root potential and summing their root nominations produces the contracted state and leaves (46) unchanged. Conversely, disaggregate the contracted nomination within the original root intervals. Subtracting the already determined exterior flows leaves balanced effective injections into B . Solve the passive block and translate each G_v to its recovered root potential. Existence and uniqueness follow from theorem A.1, and the objective is again unchanged. Any admissible coefficients on the contracted block suffice for this lift.

Repeat pruning and zero- γ contraction until neither applies. No surviving block loses two distinct vertices to one contraction, since that would create a cycle in the incidence tree. Thus surviving block ranks are unchanged, and objective support cannot increase. The induced coefficient sums on surviving blocks are unaffected, because contraction only regroups terms in the same components. All tests use rational sums of c and do not inspect resistances. The reductions use only nomination boxes and balance; flow or potential filters would invalidate them.

Let T be the minimal incidence subtree spanning the remaining support. Retain each physical block represented in T in its entirety, including its nonsupport vertices and their nominations. Every leaf of T is a support vertex. A tree with at most p leaves has at most $p - 2$ branching nodes and total branching degree $O(p)$. Mark the support vertex nodes and branching vertex nodes. Call a block *special* if its block node branches in T or is adjacent to a marked vertex node. There are $O(p)$ special blocks, with $O(p)$ total incidences in T , and all other blocks form $O(p)$ *ordinary chains*. Each ordinary block has two retained ports and no internal objective source or retained side attachment. Its induced sources for an electrical adjoint are $I, -I$, with the same I along the chain in a consistent orientation. This I is a known sum of objective coefficients, and it is nonzero because otherwise the block would have been contracted.

First smooth the laws to $g_e^\rho(t) = g_e(t) + \rho t$, $\rho > 0$, whose differential resistances are positive, also at zero flow. Implicit differentiation as in lemma B.4 gives the nomination gradient h , modulo a constant, from $A \text{diag}(1/(g_e^\rho)'(x_e))A^\top h = c$. By the two-terminal maximum principle, the internal h values of an ordinary nonbridge block lie strictly between its port values. Indeed an internal extremum propagates through all neighbors unless it reaches a terminal, and two-vertex-connectivity gives a path to the other terminal avoiding the first. The two terminal values differ because the injected current I is nonzero. Bridge blocks have no internal vertices. Along each chain the port values are strictly ordered, in the direction fixed by I , and consecutive open internal ranges are disjoint.

At a maximum on the balanced box the polyhedral normal-cone condition supplies a scalar λ such that

$$h_v > \lambda \implies b_v = u_v, \quad h_v < \lambda \implies b_v = \ell_v. \quad (47)$$

This condition remains valid for singleton intervals and a lower-dimensional balanced box. On each ordinary chain, at most one block's open range contains λ . If λ equals an articulation level, choose either adjacent block and leave that block's endpoints free. If λ is outside the chain range, all internal coordinates saturate on one side. Enumerate a selected block on each chain, or one of the two all-saturated alternatives. Outside the selected blocks the endpoint choices are fixed by the known sign of I ; all vertices of special blocks remain free. A shared endpoint is left free whenever a selected or special block contains it, so no inconsistent assignment arises. This gives $n^{O(p)}$ closed *coarse faces*, each involving only $O(p)$ free blocks.

We next justify taking limits before the second perturbation. All physical flows are bounded by a common total-positive-nomination bound B_0 ; for $0 < \rho \leq 1$, edge drops are bounded by $\beta_U B_0^2 + B_0$, and path sums bound normalized potentials. States are jointly continuous in nominations and $\rho \geq 0$: a fixed spanning-tree right inverse of A corrects a feasible flow when nominations vary, and energy minimality and strict convexity identify every subsequential limit as the unique limiting physical flow. Edge laws and path integration then give convergence of potentials, and compactness gives uniform convergence on the nomination box as $\rho \downarrow 0$. A subsequence of smoothed maximizers belongs to one fixed closed coarse face, and its limit is an original maximizer. Fix this coarse face.

Within its special and selected nonbridge blocks, mark every retained port, every objective-support vertex, and every vertex of internal block degree at least three. A block of rank r_B has at most $2r_B - 2$ vertices of the last kind, since $\sum_v (\deg_B v - 2) = 2r_B - 2$. The total port count is $O(p)$ and there are $O(p)$ such blocks, so $O(rp)$ vertices are marked. Suppress unmarked degree-two interiors into paths. By Euler's formula, a block with k_B marked vertices has $r_B + k_B - 1$ such paths; this includes a cycle with two marked ports. Every retained block has at least two ports. Mark bridge endpoints directly. In total there are $O(rp)$ paths and marked coordinates.

Smooth again and, only on the fixed coarse face, perturb the objective by

$$\delta \sum_{v \in U} (\pi_v - \pi_a), \quad \delta > 0,$$

where U consists of these path-interior vertices and a is marked. Vertices of U have no original objective sources or off-block attachments, so each has

adjoint source exactly δ . Orient a suppressed path from left to right and let j_i be its successive adjoint currents. Conservation gives $j_i - j_{i-1} = \delta$. Because voltage decreases in the direction of positive current, the adjoint values first increase and then decrease, with at most a plateau at two consecutive vertices, so a horizontal level contains at most two interior vertices. Applying (47) on this coarse face shows that each path has lower endpoints, at most one free pivot, upper endpoints, at most one free pivot, and lower endpoints; empty runs are allowed. Enumerate the two pivot positions and endpoint runs, and leave marked coordinates free. This gives $n^{O(rp)}$ faces, each with $O(rp)$ free coordinates.

The perturbation is uniformly at most $\delta|U|$ times the uniform potential-difference bound. Letting $\delta, \rho \downarrow 0$ and passing to a fixed-face subsequence again preserves a maximum on the coarse face, hence a global maximum. The coarse face must be fixed before the positive interior sources are added: perturbing all vertices at the first stage would destroy the zero-source chain argument.

Every construction and endpoint pattern is independent of the coefficients. After balance elimination, each face is a compact rational polytope with $O(rp)$ coordinates and a rational affine nomination map. Interval sums and their greedy disaggregation have polynomial bit length. For joint coefficient optimization, choose an attained joint maximum, freeze its coefficients, and apply the same family. The reductions preserve attainable objective values at those surviving coefficients, and arbitrary admissible removed coefficients permit a lift. This proves the final assertion. $\square$

G.2 Fixed-law cycle formulas and rational square-root panels

Choose a rational spanning-tree flow w with $Aw = c$, putting zero on chords. Then

$$W = w^\top A^\top \pi = \sum_e w_e g_e(x_e). \quad (48)$$

A spanning-tree particular flow for $b(z)$ is affine in z . Coherently orient each cycle, reversing its offsets and weights and swapping asymmetric coefficients and their directional intervals when necessary. On a cycle its flows have form

$$x_e = q + \ell_e(z), \quad F(z, q) = \sum_{e \in C} g_e(q + \ell_e(z)) = 0, \quad (49)$$

with rational affine ℓ_e . Bridges have affine flows. The cycle variables are independent because the cycles are edge-disjoint, and cycle-compatible drops

integrate to a global potential. Each $F(z, \cdot)$ is continuous and strictly increasing, with opposite limits at infinity.

Rational linear programming supplies a box containing P whose endpoint bit lengths are polynomial, even if P has lower dimension. Coefficient bounds on this box supply a polynomial-bit $B_0 \geq 1$ bounding total positive nominations, hence all physical flows, and $R \geq 1$ with $|q| \leq R$ in every physical scenario; if a chord offset is zero, $R = B_0$ suffices. Write β_U for the largest coefficient or allowed coefficient endpoint. The uniform bound

$$|W| \leq \|c\|_1(n-1)\beta_U B_0^2 \quad (50)$$

follows by normalizing one potential and integrating drops along paths.

Lemma G.3 (Local fixed-law representation). *For fixed d , a cycle's contribution to (48) has polynomially many exact semialgebraic charts in z , each represented by a rational function of bounded degree or by*

$$P_2(z) + L_1(z)\sqrt{\Delta(z)},$$

where P_2, Δ are rational polynomials of degree at most two and L_1 is rational affine. Chart domains have polynomial-size bounded-degree rational descriptions. All bit lengths are polynomial. The all-zero-flow stratum is included explicitly.

Proof. Enumerate the realizable sign cells of the affine forms $q + \ell_e(z)$ in $d+1$ variables, and take their closures. For fixed d there are polynomially many, so all $2^{|C|}$ sign vectors need not be enumerated. On each closed cell, (49) is

$$A_2 q^2 + B_1(z)q + C_2(z) = 0,$$

where A_2 is rational constant and the subscripts bound degrees. At a physical solution its derivative satisfies

$$2A_2 q + B_1(z) = 2 \sum_{e \in C} \beta_e^{\text{sgn } x_e} |x_e| \geq 0.$$

For $A_2 \neq 0$ the physical root is therefore

$$q = \frac{-B_1(z) + \sqrt{B_1(z)^2 - 4A_2 C_2(z)}}{2A_2}. \quad (51)$$

The plus sign is correct even when $A_2 < 0$. Substituting into a quadratic weighted drop gives the asserted polynomial plus affine radical term; all

divisions are by the fixed nonzero rational $2A_2$ or its square. For $A_2 = 0$ and $B_1(z) > 0$, substitute $q = -C_2/B_1$ exactly and keep the nonzero-denominator condition; no uniform lower bound on B_1 is needed. If $A_2 = B_1(z) = 0$ at a physical point, the derivative identity forces every flow to vanish. Represent this chart by $q = -\ell_1(z)$ and $\ell_e(z) = \ell_1(z)$ for every e ; its contribution is zero. These charts cover all physical solutions.

For each chart, eliminate its single q from the root equation, the closed sign inequalities, $|q| \leq R$, and, when needed, the root selector $2A_2q + B_1 \geq 0$. In the linear case keep $B_1 > 0$; in the zero case use the displayed affine equalities. The fixed-dimensional elimination bounds in (11) give polynomially many rational polynomials of bounded degree and polynomial coefficient bit length. Disconnected domains are allowed. By uniqueness, several valid charts at a boundary give the same physical contribution. $\square$

Lemma G.4 (Constructive rational square-root panels). *Given a rational $S \geq 1$ of polynomial bit length and $0 < \eta \leq 1$, one can construct a piecewise rational polynomial approximation to $\sqrt{t}$ on $[0, S^2]$, uniformly within η , whose panel count, degrees, dense coefficient sizes, and construction time are polynomial in the input length and $\log(1/\eta)$.*

Proof. First take $S = 1$. Choose J with $2^{-J} \leq \eta$ and use zero on $[0, 4^{-J}]$. For $j = 0, \dots, J-1$, use the panel $[4^{-j-1}, 4^{-j}]$, center $a_j = (9/16)4^{-j}$, and polynomial

$$T_j(t) = \frac{3}{4} 2^{-j} \sum_{k=0}^K \binom{1/2}{k} (t/a_j - 1)^k.$$

All numbers are rational. On the panel $|t/a_j - 1| \leq 7/9$ and the binomial coefficients have magnitude at most one. The convergent binomial series therefore gives

$$|T_j(t) - \sqrt{t}| \leq \frac{9}{2} (7/9)^{K+1}.$$

Choose $K = O(1 + \log(1/\eta))$ to make this at most η . On the zero panel the error is at most $2^{-J} \leq \eta$, the value of $\sqrt{t}$ at its right endpoint. Either adjacent formula is valid at a shared boundary; the approximant need not be continuous. Expanded coefficient bit lengths are polynomial in J, K , including the products jK . For general S , approximate $\sqrt{t/S^2}$ within η/S and multiply by S . This proves all bounds. $\square$

For each discriminant in lemma G.3, absolute coefficient sums on the box for P give a rational power of two S of polynomial bit length bounding

its square root on its domain. Absolute coefficient bounds similarly give $M_i \geq |L_{1,i}(z)|$. Set

$$\eta = \frac{\varepsilon}{16(1 + \sum_i M_i)}, \quad a = \varepsilon/16, \quad (52)$$

where the sum may range over all charts. Replacing every radical by its panels changes any selected sum of local contributions by at most a , and all rational charts are kept exact. Substituting a quadratic discriminant into a panel polynomial doubles its degree; a dense polynomial of polynomial degree in fixed d variables still has polynomially many monomials.

G.3 One common partition and original-parameter recovery

Collect the polynomials appearing in every chart-domain formula, every panel-boundary formula $\Delta - S^2 t_j$, every rational denominator, the bridge zero-flow hyperplanes, and the inequalities for P . Fixed-dimensional sign decomposition, or sampling all realizable sign conditions followed by their Boolean descriptions, gives a polynomial-size common partition. On every part choose an eligible chart and panel for each cycle, and the correct bridge-law branch; eligibility depends only on these signs. Thus every point is covered, including all lower-dimensional boundaries and singular zero-flow strata, and combinations of charts across cycles are never enumerated.

The sum of the chosen surrogates is one rational function $\widehat{W}(z)$ on each part. Multiplying polynomially many bounded-degree denominators and adding numerators gives polynomial degree and polynomial coefficient bit length, and dense expansion takes polynomial time since d is fixed. For threshold decisions one may multiply by the squared common denominator, retaining all nonzero conditions. The real-algebraic bounds from the preliminaries therefore apply, even though the degrees now grow with accuracy bits.

The parts may be open. We therefore optimize a supremum with the existential test $\exists z$ in the part : $\widehat{W}(z) > t$, without assuming attainment on every part. The uniform error a and (50) give a polynomial-bit bound on every surrogate supremum, including near a vanishing chart denominator. Bisection on the union of these tests produces an interval $[L, U]$ of width at most $\varepsilon/8$ containing the largest surrogate supremum. Then

$$[L - a, U + a] \quad (53)$$

contains the physical optimum and has width at most $\varepsilon/4$. Sample on a part at the strictly lower threshold $L - \varepsilon/16$; the existential sampling primitive

appendix A.5 returns an algebraic z_* of polynomial representation size. Its true objective is within $5\varepsilon/16$ of the optimum, since the optimum is at most $U + a$ and the sample has value greater than $L - \varepsilon/16 - a$. The strictly lower threshold handles unattained suprema and exact equality alike.

Lemma G.5 (Nomination recovery bound). *For fixed asymmetric quadratic laws and two balanced nominations whose physical flows have magnitude at most B_0 ,*

$$\|x(b) - x(b')\|_\infty \leq \frac{1}{2}\|b - b'\|_1, \quad (54)$$

$$|W(b) - W(b')| \leq K_b\|b - b'\|_1, \quad K_b = \|c\|_1(n-1)\beta_U B_0. \quad (55)$$

These inequalities hold on every connected graph.

Proof. Put $y = x(b) - x(b')$ and $h = \pi(b) - \pi(b')$. Strict monotonicity gives $\text{sgn } y_e = \text{sgn}(A^\top h)_e$ whenever $y_e \neq 0$. Orient these difference flows positively. The potential h strictly decreases along each such arc, so their directed support is acyclic. A path decomposition from positive to negative entries of $b - b'$ has total weight $\|b - b'\|_1/2$, which proves the first inequality. On $[-B_0, B_0]$ each law is Lipschitz with constant $2\beta_U B_0$, also across zero, so every drop changes by at most $\beta_U B_0\|b - b'\|_1$. Normalize one potential, use paths of length at most $n - 1$, and sum against $|c_v|$. $\square$

Let $H_1 = \sum_{v,j} |H_{vj}|$. Isolate each coordinate of z_* in a rational interval of width h with $K_b H_1 h \leq \varepsilon/4$; this condition is omitted if its coefficient is zero. Rational linear programming on the intersection of that box with P returns a rational $z' \in P$ of polynomial bit length. The intersection contains z_* , even for lower-dimensional P , and $\|b(z') - b(z_*)\|_1 \leq H_1 h$, so the physical objective loss is at most $\varepsilon/4$. Rounding need not preserve the selected chart, panel, or any nonzero denominator; only the global physical estimate matters. At rational z' the cycle roots can be enclosed individually by monotone bisection and their drops summed with rational error bounds, without a common number field. This proves case 1 of theorem G.1; negating c proves the minimum statement.

G.4 Eliminating independent resistance intervals

We now prove case 2 of theorem G.1. Independent positive resistance intervals are an established gas-network uncertainty model. Afkman, Liers, Stingl, and Vera give exact tree reductions by linear programming and a single-cycle reduction of circulation intervals to polyhedral coefficient conditions [5,

Sections 4.2 and 4.3.3, Lemma 4.10]. González Grandón, Heitsch, and Henrion combine random loads and robust roughness uncertainty on trees, with explicit rectangular and ellipsoidal inner optimizations [27, Sections 2–3]. The additional task here is deterministic weighted optimization over independent cycle blocks with the accuracy and recovery guarantees of theorem G.1.

For fixed z , (48) separates into independent cycle maxima and bridge maxima. Each bridge chooses the endpoint maximizing its weighted drop, using its active directional interval in the asymmetric model. On a closed flow-sign cell, write $[L_e, U_e]$ for the active interval $[L_e^{s_e}, U_e^{s_e}]$ in that model; in the symmetric model retain the shared interval. Only the active coefficient enters the local problem: inactive coefficients can take any allowed rational value, and at zero flow neither coefficient contributes. Thus the two models have the same local reduction, although shared symmetric uncertainty is a diagonal constraint and not literally a product of two directional intervals. On a closed flow-sign cell of a cycle, put $f_e(z, q) = s_e(q + \ell_e(z))^2$, where $s_e \in \{-1, 1\}$ is the selected sign; at zero either sign gives the correct value. For fixed (z, q) the resistance problem is the one-equality box linear program

$$\max_{L \leq \beta \leq U} \left\{ \sum_e w_e f_e \beta_e : \sum_e f_e \beta_e = 0 \right\}. \quad (56)$$

Lemma G.6 (Threshold elimination with ties). *For each distinct edge weight $\lambda \in \{w_e : e \in C\}$ let $T = \{e : w_e = \lambda\}$. For $e \notin T$, choose $\hat{\beta}_e = U_e$ when $(w_e - \lambda)s_e > 0$, and L_e otherwise. Define*

$$S = \sum_{e \notin T} \hat{\beta}_e f_e, \\ l_T = \sum_{e \in T} \min\{L_e f_e, U_e f_e\}, \quad u_T = \sum_{e \in T} \max\{L_e f_e, U_e f_e\}.$$

This threshold has a feasible optimizing profile exactly on the domain

$$S + l_T \leq 0 \leq S + u_T. \quad (57)$$

Its value there is the rational quadratic polynomial

$$Q_\lambda(z, q) = \sum_{e \notin T} (w_e - \lambda) \hat{\beta}_e f_e(z, q). \quad (58)$$

If (56) is feasible, at least one threshold applies.

Proof. The tied coordinates fill every aggregate value in $[l_T, u_T]$, so they can supply $-S$ exactly when (57) holds. Every untied coordinate maximizes its reduced-cost expression $(w_e - \lambda)f_e\beta_e$, also in the zero-flow case, where either endpoint is optimal, and the tied reduced costs vanish. The feasible profile therefore attains the Lagrangian upper bound, whose value is (58) because $\sum f_e\beta_e = 0$.

Conversely, the dual function is the convex piecewise-affine function

$$D(\lambda) = \sum_e \max_{L_e \leq \beta_e \leq U_e} (w_e - \lambda)f_e\beta_e.$$

Partition the real line at all distinct listed edge weights, whether or not they are actual breakpoints of D . On every resulting interval or tail, D is affine. Feasibility bounds D below by the primal optimum, so its tails cannot decrease without bound. A minimum therefore occurs at a listed weight: an affine function on a bounded interval has a minimizing endpoint, and a flat tail has a minimizing finite endpoint. This also covers a globally constant dual, even when some flows are nonzero. Box-LP strong duality gives the primal optimum there. Moreover, equality in the sum of reduced-cost upper bounds forces each nonzero untied term to use the chosen endpoint, so the remaining tied aggregate satisfies (57). If every $f_e = 0$, any threshold is feasible with value zero. The argument also covers singleton intervals and arbitrary ties. $\square$

All endpoints in this lemma are fixed on a sign cell. Both inequalities in (57) are quadratic in (z, q) with a constant coefficient of q^2 , and their number does not grow with the number of tied edges. There are polynomially many realizable sign cells in fixed $d + 1$ dimensions, and at most $|C|$ thresholds per cell.

Lemma G.7 (Complete circulation candidates). *The maximum cycle contribution conditional on z is the maximum of polynomially many feasible candidates, each rational of bounded degree or of the form $P_2(z) + L_1(z)\sqrt{\Delta(z)}$. Candidate eligibility has polynomial-size bounded-degree rational semialgebraic descriptions. These assertions remain true with rational signed arc-flow bounds.*

Proof. Fix a closed sign cell and a threshold. Impose $|q| \leq R$, its sign inequalities, (57), and any specified bounds $\underline{x}_e \leq q + \ell_e(z) \leq \bar{x}_e$. For fixed z the feasible q set is compact. Each defining polynomial has form

$$aq^2 + b(z)q + c(z),$$

with rational constant a , affine b , and quadratic c ; linear bounds are special cases. An optimum of the quadratic $Q_\lambda(z, q)$ occurs at a boundary, a stationary point, or a feasible point of a flat objective. A boundary point has at least one active polynomial that is nonconstant in q after specialization at z ; otherwise all specialized constraints would be constant or locally strict there, and it would not be a boundary point. The artificial bounds $q = \pm R$ are included, so this applies also to flat objectives.

For a boundary polynomial with $a \neq 0$, enumerate both roots

$$q = \frac{-b(z) \pm \sqrt{b(z)^2 - 4ac(z)}}{2a}. \quad (59)$$

Substitution in Q_λ gives $P_2 + L_1\sqrt{\Delta}$, absorbing the root sign in L_1 . The denominator $2a$ is constant. For $a = 0$ and $b(z) \neq 0$, substitute $q = -c(z)/b(z)$ and retain the nonzero condition. On $a = b(z) = 0$ the polynomial is constant in q and supplies no boundary candidate: as the remaining inequalities check, it either excludes the parameter or imposes no additional q boundary. This handles vanishing coefficients without dividing by zero.

Write $Q_\lambda = a_Q q^2 + b_Q(z)q + c_Q(z)$. If $a_Q \neq 0$, include the stationary candidate $q = -b_Q(z)/(2a_Q)$, whether or not it is a local maximum; feasibility and later value selection remove harmless extras. If $a_Q = 0$ and $b_Q(z) \neq 0$, there is no interior optimum. On $a_Q = b_Q(z) = 0$, include the flat candidate $c_Q(z)$ with the existential condition that the original q domain is nonempty.

For each nonflat candidate, eliminate q from its defining equation and all original domain inequalities. The plus root in (59) uses the selector $2aq + b \geq 0$ and the minus root uses $2aq + b \leq 0$; these selectors remain correct when $a < 0$ and at a double root. The rational case uses $b \neq 0$. Projection is in $d + 1$ variables with bounded degree, so its output size and coefficient lengths are polynomial. Flat eligibility is the same fixed-dimensional existential projection without a root equation. Thus no feasible candidate value is spurious, and the boundary/stationary/flat list includes a local optimum. $\square$

Apply lemma G.4 to every radical candidate, with error budgets as in (52). This time also add to the common partition the comparison numerators between every pair of surrogate pieces of the *same* cycle, together with their denominators. There are polynomially many pieces and pairs, and their degrees and coefficient lengths remain polynomial. Eligibility and every such comparison are constant on a sign part. Choose a largest eligible surrogate for each cycle and then sum, adding bridge endpoint values. No exact radicals are compared, and no cross-cycle candidate combinations are enumerated.

For precision accounting, if every candidate of cycle C has error at most a_C , then

$$\left| \max_j V_{Cj} - \max_j \widehat{V}_{Cj} \right| \leq a_C.$$

The candidate selected by the surrogate maximum has true value at least $\max_j V_{Cj} - 2a_C$. Our choice of tolerances ensures $\sum_C a_C \leq a = \varepsilon/16$. Hence the summed surrogate approximates the conditional joint optimum within a , and the actual selected scenario approximates that surrogate within a . Apply the same supremum search and sampling as before. The interval (53) again encloses the true joint optimum, and the selected physical scenario at the algebraic sample loses at most $5\varepsilon/16$, including the loss from selecting candidates by approximate comparisons.

To recover that scenario, recover each selected q from its equation and root selector over the algebraic sample z_* . On a flat branch, sample one point in its nonempty compact one-dimensional feasible set. For the tied coordinates, start at the lower end of every interval

$$[\min\{L_e f_e, U_e f_e\}, \max\{L_e f_e, U_e f_e\}]$$

and fill the required residual until the sum is $-S$. Divide the resulting contribution by f_e only if $f_e \neq 0$; at zero choose L_e . At most one tied coordinate needs to be interior. This recovers the original resistance coordinates, not just an eliminated aggregate. In the directional model give each inactive coefficient any rational value in its interval. In the symmetric model the recovered active coefficient is the single shared coefficient; no two independent copies are introduced. At zero flow choose any allowed original coefficient or pair. These choices satisfy the original laws.

Process each cycle over the common algebraic z_* and its own local extension. A nonflat root has degree at most two over the field of z_* , and flat one-dimensional sampling also has a polynomial-size representation by the fixed-dimensional bounds. Rational operations on these representations keep polynomial degree and coefficient size. The output is a list of separate coordinate representations; no primitive element containing the roots of all cycles is constructed. The resulting local states share the same nomination and satisfy every cycle equation, so they form a global physical scenario.

Lemma G.8 (Continuity in the original coefficients). *Let θ list the original uncertain coefficients: m entries in the symmetric model and $2m$ in the directional model. For either model on any connected graph,*

$$|W(b, \theta) - W(b', \theta')| \leq K_b \|b - b'\|_1 + \|c\|_1 B_0^2 \|\theta - \theta'\|_1. \quad (60)$$

For each terminal potential difference at fixed b , its coefficient change alone is at most $B_0^2 \|\theta - \theta'\|_1$.

Proof. For symmetric laws the resistance statement is already lemma C.10; the following argument also covers independent directional coefficients. Smooth by $g_e^\rho(x) = g_e(x) + \rho x$, with ρ independent of the coefficients. For a terminal objective $F = \pi_s - \pi_t$, the electrical adjoint solves $AR^{-1}A^\top h = e_s - e_t$, where $R = \text{diag}((g_e^\rho)'(x_e))$. Its currents $j = R^{-1}A^\top h$ have $|j_e| \leq 1$: nonzero currents follow strictly decreasing h , hence decompose into terminal paths of total weight one. Differentiating the physical equations at fixed b and pairing with this adjoint gives

$$dF_\rho = \sum_e j_e \begin{cases} x_e |x_e| d\beta_e, & \text{symmetric model,} \\ (x_e)_+^2 d\beta_e^+ - (x_e)_-^2 d\beta_e^-, & \text{directional model,} \end{cases}$$

where $u_+ = \max\{u, 0\}$ and $u_- = \max\{-u, 0\}$. The laws are continuously differentiable jointly in flow and coefficients, including at zero, and R is positive, so the differentiation is valid there as well. Every coefficient derivative has magnitude at most B_0^2 . Integrate along the segment inside the original coefficient box and let $\rho \downarrow 0$. The uniform flow and potential bounds and the strict-convexity continuity argument in theorem G.2 hold uniformly on this compact coefficient box, so the terminal estimate survives the limit. Normalize one potential and sum against $|c_v|$. Finally combine this with lemma G.5, whose K_b is uniform over all allowed directional coefficients. Sign crossings require no extra case or nonzero-flow assumption. $\square$

Let k be the length of θ , so $k \leq 2m$. Choose polynomial-bit widths $h, \delta > 0$ with $K_b H_1 h + \|c\|_1 B_0^2 k \delta \leq \varepsilon/4$. Round z_* by rational LP inside P and its isolating box, and round each original coefficient within δ inside its own rational interval. In the symmetric model round the one shared coefficient, which preserves its required equality across the two flow directions. Certified coordinate isolation takes polynomial time, separately for each local extension. Singleton intervals are kept exactly. These original parameters are feasible without maintaining candidate-domain membership, and the continuity estimate supplies the remaining objective guarantee. This completes case 2 of theorem G.1; again negate c for minimization.

G.5 Balanced boxes and exact local capacity filters

Corollary G.9 (Unrestricted nomination boxes at fixed support). *Fix p . On a cactus, for a rational balanced nomination box and $|\text{supp } c| \leq p$, both*

fixed positive asymmetric quadratic laws and joint independent coefficient intervals in either model of theorem G.1 admit the value and rational-scenario guarantees of theorem G.1. Total cycle rank is unrestricted. No operating constraints filter this box.

Proof. Apply theorem G.2 with $r = 1$. Each of the polynomially many faces has $O(p)$ free nominations and hence a fixed-dimensional rational affine parameterization. For joint uncertainty, hold the resistances of a joint maximizer fixed to identify a face containing an equally good nomination; the face family is resistance independent. Apply theorem G.1 on every face with tolerance, for example, $\varepsilon/8$. The maximum of the certified lower endpoints and the maximum of the upper endpoints enclose the global optimum. Select the scenario from a face with the largest lower endpoint. Its optimum is within $\varepsilon/8$ of the best face optimum and its witness has the stated local error, which gives the desired global tolerance. Rational disaggregation reverses all reductions, with arbitrary rational allowed resistances in removed blocks, and objective preservation gives the original-network result. The constant-objective case is solved by rational LP. Minimization follows by changing the sign of c . $\square$

Theorem G.10 (Local filters and the rational-output boundary). *For the fixed-dimensional affine nomination models of theorem G.1, allow rational signed arc bounds $\underline{x}_e \leq x_e \leq \bar{x}_e$. Feasibility is decidable exactly in polynomial bit time. If feasible, optimum intervals of width at most ε and exactly feasible algebraic ε -optimal original parameters of polynomial representation size are computable in accuracy-bit polynomial time.*

In the fixed-law model the same algebraic-output conclusion permits a polynomial-size list of bounded-degree rational polynomial equalities and inequalities defining a closed set, each involving z and the flows of one cycle or one bridge. For the joint resistance model the assertion here is limited to the stated arc bounds.

For either model with arc bounds, let a supplied rational $\sigma > 0$ give a nonempty tightened problem $\underline{x}_e + \sigma \leq x_e \leq \bar{x}_e - \sigma$ on all bounded ends. A rational scenario feasible for the original bounds and ε -optimal relative to the tightened optimum can be computed in accuracy-bit polynomial time, including polynomial dependence on the encoding length of σ .

Proof. In the fixed-law model add the local bounds, or local polynomial constraints, to the chart eligibility formulas before projecting q . The unique physical q and the objective formulas are unchanged. Substituting the affine flows into a bounded-degree polynomial uses only (z, q) , so the local dimension

remains fixed even for long cycles. Bridge constraints involve z directly. In the joint model the arc bounds are affine in (z, q) and already covered by lemma G.7; arbitrary polynomial filters could add higher-degree circulation boundaries and are not asserted there.

A global parameter is feasible exactly when every cycle has an eligible chart or candidate and every bridge bound holds. Take the common partition with these eligibility conditions and discard parts violating them; exact real-algebraic feasibility decides whether any part remains. The parameter and resistance boxes are compact and physical states are continuous, so the closed filters give a compact feasible scenario set with attained extrema. The surrogate supremum algorithm and exact sampling on surviving parts give the stated values and algebraic parameters. In the joint case use the local tied-coordinate recovery; all added flow bounds depend only on (z, q) , so that recovery preserves them. No rational LP rounding is used at this stage.

For the tightening claim, the fixed-law nomination flow bound is (54). For resistance rounding at fixed b , let x, y be the two flows, with $\|\theta - \theta'\|_\infty \leq \delta$, $k = \dim \theta \leq 2m$, and $\beta_L > 0$ the smallest allowed directional coefficient. Applying lemma G.8 to the endpoints of each edge gives $|\Delta_e - \Delta'_e| \leq B_0^2 k \delta$. An inverse-law bound gives

$$\|x - y\|_\infty^2 \leq \frac{2B_0^2(k+1)}{\beta_L} \delta. \quad (61)$$

Indeed, for either fixed quadratic law, $|g_e(u) - g_e(v)| \geq \beta_L |u - v|^2/2$: at equal signs use $|u^2 - v^2| \geq |u - v|^2$, and at opposite signs use $u^2 + v^2 \geq (|u| + |v|)^2/2$. Meanwhile changing the original coefficients within δ changes $g_e(y_e)$ by at most $B_0^2 \delta$, since only one directional coefficient is active at y_e . Subtracting the two endpoint drops and applying the triangle inequality proves the displayed bound, even if x_e, y_e have opposite signs; at zero both branches agree. Combining the nomination and resistance changes bounds the total flow error by

$$\frac{1}{2} H_1 h + B_0 \sqrt{2(k+1)\delta/\beta_L}.$$

Choose h, δ to make this at most $\sigma/2$ and the objective loss in (60) at most $\varepsilon/4$; in the fixed-law case omit the resistance term. For example, requiring $H_1 h \leq \sigma/2$ and $\delta \leq \beta_L \sigma^2 / [32B_0^2(k+1)]$ suffices for the flow condition. These widths have polynomial bit length. Optimize the tightened problem with the remaining error budget, then round inside P and the original resistance intervals. The resulting rational scenario satisfies the original capacities and loses at most ε relative to the tightened optimum. This implies no comparison between the tightened and original optima, and the rounding need not preserve arbitrary local polynomial filters from the fixed-law extension. $\square$

Example G.11 (Exact capacities can force irrational nominations). On a coherently oriented four-cycle take $t \in [1, 2]$ and

$$x = (q + t - 1, q - 2, q, q - 2), \quad \beta = (1, 1, 1, 2).$$

Successive flow differences give the balanced rational affine nomination family $b = (t + 1, -t - 1, 2, -2)$. Impose $|x_3| \leq 1$, $|x_4| \leq 1$, and $|x_1|, |x_2| \leq 10$. The first two bounds imply both $q \in [-1, 1]$ and $q \in [1, 3]$, hence $q = 1$. The cycle equation is then $t^2 - 1 + 1 - 2 = t^2 - 2 = 0$. Its only feasible parameter is $t = \sqrt{2}$, and the other bounds hold. All input data and capacities are rational, and the symmetric capacities are strictly positive. Thus no rational parameter is exactly feasible here, so an exactly feasible rational output cannot be guaranteed, even for fixed resistances.

The box-face corollary does not extend to these filters: pruning, zero-source contraction, and the box normal-cone argument use the full unfiltered nomination domain. Likewise, a bound on a potential difference across many cycles is not a local flow constraint, and exact projection of such a bound can require comparing a sum of independent radicals.

G.6 Exact rational resistance profiles at fixed nominations

Theorem G.12 (Exact rational profile, separately represented value). *On a cactus with fixed rational balanced nominations, independent positive rational coefficient intervals in either model of theorem G.1, and any rational balanced objective vector c , feasibility under optional rational signed arc bounds is decidable in polynomial bit time. If feasible, an exactly maximizing or minimizing rational resistance profile of polynomial encoding length can be computed in polynomial bit time. Each local optimal value has algebraic degree at most two; the global value may be retained as a sum of these separate values. This theorem supplies no exact scalar comparator for that growing sum.*

Proof. Set $d = 0$ in lemma G.7. All flow-sign, capacity, and artificial circulation bounds are rational, so their boundary circulations are rational, as are stationary points of a nonflat rational quadratic objective. At rational q , every f_e is rational, and the tied-aggregate filling rule recovers rational resistances of polynomial bit length.

An irrational boundary circulation can only come from one of the two aggregate quadratics in (57). At $S + l_T = 0$ the tied aggregate must attain its minimum, so every nonzero contribution uses its minimizing resistance endpoint. At $S + u_T = 0$ every nonzero tied contribution analogously uses its

maximizing endpoint. Zero-flow coordinates may use either rational endpoint, and untied coordinates already use rational endpoints. Thus the recovered original resistance profile is rational even when its induced q is irrational. In the directional model choose every inactive coefficient rationally in its own interval; in the symmetric model retain the shared recovered coefficient. If both aggregate bounds coincide, either construction works.

For a flat objective choose a boundary point of its nonempty compact feasible q set, rather than an arbitrary algebraic sample; the artificial bounds make the set bounded, so such a point exists. At least one specialized defining polynomial nonconstant in q is active there (identically zero constraints do not count), which returns to the rational or aggregate-quadratic boundary cases above. There are polynomially many candidates on every cycle, each with a rational or quadratic-algebraic objective value.

Exact comparison of two local values takes polynomial time. One explicit method writes their difference as $U + V$, with $U = a + b\sqrt{d}$ and $V = c\sqrt{e}$ for rational coefficients and nonnegative rational radicands. Decide each sign by rational arithmetic and squared magnitude comparisons. If the signs agree, the answer is immediate. If they are opposite, the sign is $\text{sgn}(U) \text{sgn}(U^2 - V^2)$, and

$$U^2 - V^2 = (a^2 + b^2d - c^2e) + 2ab\sqrt{d}$$

has only one radical. Zero cases are checked first. This recognizes exact equality without factoring radicands or imposing a separation tolerance; alternatively, constant-degree root isolation suffices. Select an exact local winner and retain its rational profile. Bridges check their fixed flow bounds and choose the objective-maximizing endpoint. The independent local winners together form a global physical state and an exactly optimizing rational profile. None of these selections compares the sum of all block values. Individual radical isolation and rational summation can enclose that sum to any requested accuracy in polynomial additional bit time. Negating c yields minimization. $\square$

Example G.13 (A unique interior-resistance optimum). On a coherently oriented four-cycle let $b = (0, 0, 3, -3)$, so $x = (q, q, q + 3, q)$, and let

$$\beta \in [1, 4] \times [1, 3] \times [1, 4] \times [1, 5], \quad c = (2, -3, -2, 3).$$

The edge weights $w = (2, -1, -3, 0)$ satisfy $Aw = c$. Strict monotonicity of the cycle equation and its values at $q = -3, 0$ put every physical circulation in $(-3, 0)$. Subtract $\lambda = -1$ times the cycle equality from the objective. The

three untied reduced costs are strictly negative, so setting their resistances to their lower bounds gives, for every feasible profile,

$$W \leq -4q^2 - 2(q+3)^2 = -6(q+1)^2 - 12 \leq -12.$$

At $q = -1$ the profile $(1, 2, 1, 1)$ gives signed drops $(-1, -2, 4, -1)$. Their sum is zero and their weighted sum is -12 , so the bound is attained. Equality forces $q = -1$; then the strictly negative reduced costs force $\beta_1 = \beta_3 = \beta_4 = 1$, and the cycle equation forces $\beta_2 = 2$. The optimizer is therefore unique and has an interior second resistance, so no box corner is optimal. Stationary circulation candidates and tied resistance recovery cannot be replaced by endpoint-profile enumeration.

The accompanying implementation `exact_weighted_cactus.py`, in the directory `code/potential_flow_mpd/`, implements the symmetric-coefficient case of theorem G.12 for *supplied blocks*: coherent cycle offsets and weights, resistance intervals, optional local flow bounds, and fixed-flow bridges. For both extrema it returns exact rational optimal resistances, separate quadratic encodings of local circulations and values, and a certified rational interval for the total value. It does not compute or verify an original graph's block decomposition, derive its nomination offsets or objective weights, implement independent directional-coefficient intervals, or implement the varying-nomination semialgebraic optimizer. Its exact candidate comparisons handle distinct stored radicands; arithmetic on local expressions stays within one stored radical, and the sum across blocks remains separately represented. Its physical-witness, independent LP-vertex, radical-cancellation, capacity, orientation, and input-schema checks are regression evidence for this supplied-block scope; they supplement, but do not replace, the candidate-completeness proof, and they are not a general untrusted optimality-certificate format.

The structural theorem applies beyond cacti, but the approximation argument here uses the explicit quadratic, rational, and zero cycle formulas. It does not give a weighted accuracy-bit algorithm for general bounded-rank blocks with unbounded total rank. Discrete resistance sets do not satisfy the tied-interval filling argument, and cross-cycle resistance correlations do not admit the independent summation step.

H Continuous design, correlations, and convex performance

Fixed nominations give a different tractability boundary from uncertain nominations. On a cactus, a cycle can have arbitrarily many correlated

law parameters while its physical flow still has only one free coordinate. This section exploits that scalar equation to prove exact local optimization and capacity feasibility, additive convex minimization with rational original parameters, and a complementary boundary for convex maximization. The number and size of the cycles are unrestricted throughout.

The optimization ingredients have direct predecessors. Scalar threshold bisection is classical quasiconvex optimization [1, Sections 2.1 and 3]; the single-piece degree-one case is linear-fractional optimization, and exact recovery from parametric linear optimization goes back to Megiddo [38, Section 2]. For quadratic laws, the cycle capacity inequalities below already appear in Aßmann et al. [5, Proposition 4.9, Lemma 4.10, and Proposition 4.11]. We supply the arithmetic details needed to combine these mechanisms with dense polynomial laws and to return rational admissible parameters, also when an optimal physical flow is irrational.

H.1 Exact recovery for a monotone polynomial root

Theorem H.1 (Monotone root optimization over an H-polytope). *Let $P = \{\theta \in \mathbb{R}^t : A\theta \leq b\}$ be a nonempty bounded rational polytope, with no bound on t . On a supplied rational bracket $a < b$ let*

$$H(q, \theta) = h_0(q) + \sum_{j=1}^t \theta_j h_j(q)$$

be continuous and piecewise polynomial in q . The partition consists of fixed rational breakpoints independent of θ , and every piece has densely encoded rational coefficients and degree at most D . Suppose, for every $\theta \in P$, that $H(\cdot, \theta)$ is strictly increasing and $H(a, \theta) \leq 0 \leq H(b, \theta)$. Then the smallest and largest roots over P are computable exactly in polynomial bit time in the total dense input. Each output consists of a rational optimizing vertex of P and a real-algebraic root of degree at most D and polynomial coefficient bit length. The theorem permits lower-dimensional P , repeated polynomial roots, and roots at breakpoints. Strict increase and the bracket signs are promises.

Proof. Write $q(\theta)$ for the unique root. At every rational $z \in [a, b]$, strict increase gives the exact rational LP tests

$$\max_{\theta \in P} q(\theta) \geq z \iff \min_{\theta \in P} H(z, \theta) \leq 0, \quad (62)$$

$$\min_{\theta \in P} q(\theta) \leq z \iff \max_{\theta \in P} H(z, \theta) \geq 0. \quad (63)$$

These extrema are attained, as the following argument shows. Express any θ as a convex combination of vertices. The same combination of their $H(q(\theta), \cdot)$ values is zero, so at least one vertex has nonpositive value and one has nonnegative value, and their roots bracket $q(\theta)$. The extreme vertex roots are therefore global extrema. In particular the root map is quasilinear: each sublevel and superlevel set is the intersection of P with an affine halfspace. No derivative lower bound is needed.

We now give one separation bound for all vertex roots without enumerating the vertices. The case $t = 0$ is ordinary single-polynomial root isolation, with piece selection as described below; assume $t > 0$. Clear denominators in each row of $A\theta \leq b$, and let $C \geq 1$ bound the absolute integer coefficients and right-hand sides. Every vertex has t independent active rows, also for a lower-dimensional polytope: otherwise a nonzero direction orthogonal to the active normals would admit small feasible displacements in both directions. Cramer's rule gives a common nonzero integer denominator d and numerators n_j with

$$|d|, |n_j| \leq \Delta := t!C^t.$$

Clear every polynomial coefficient denominator, on every piece, by one positive integer R , and let $P_0 \geq 1$ bound the resulting integer coefficients. Substituting a vertex and multiplying by Rd gives an integer polynomial of degree at most D and height at most

$$H_0 = (t + 1)\Delta P_0.$$

Delete duplicate breakpoints and zero-width pieces first. Strict increase implies that every specialized positive-width piece is nonconstant, hence nonzero, and by continuity a breakpoint root belongs to an adjacent closed piece. Thus this height bound covers every vertex root. The logarithms of all these integers are polynomial in the input length; taking products of input denominators only adds their bit lengths.

Take any two of these nonzero integer polynomials, allowing equality, and form their product Q , of degree at most $2D$ and height at most $(D + 1)H_0^2$. Its primitive squarefree part S is an integer factor. The classical factor bound of Mignotte [39] gives $\|S\|_1 \leq 2^{\deg S} \|Q\|_2$, so its height is at most

$$H_s = 2^{2D}(2D + 1)(D + 1)H_0^2.$$

If S has degree $n \geq 2$, Cauchy's root bound puts all its complex roots in $|z| \leq 1 + H_s$, and its discriminant is a nonzero integer. For any two distinct roots at distance δ , the discriminant product formula therefore implies

$$1 \leq H_s^{2n-2} \delta^2 [2(1 + H_s)]^{n(n-1)-2}.$$

Since $n \leq 2D$, the rational number

$$\sigma = [2(1 + H_s)]^{-4D^2} \quad (64)$$

is a lower bound on every nonzero distance between roots of any two vertex-piece polynomials. If $n = 1$ there is no pair to separate. Squarefree extraction handles common factors and repeated roots, and $\log(1/\sigma)$ is polynomial in the dense input length. The assumptions exclude $D = 0$.

Bisect $[a, b]$ using (62), retaining an interval $[l, u]$ containing the largest root, and stop when $u - l < \sigma$. True tests, including equality, raise l ; false tests lower u . Now minimize $H(l, \theta)$ over P and select a vertex optimizer θ_v . Its value is nonpositive, so

$$l \leq q(\theta_v) \leq q_{\max} \leq u.$$

Both middle quantities are vertex roots, so separation forces equality. A vertex optimizer is obtained by imposing the rational optimal objective equality and then minimizing coordinates successively on the remaining face; after t further LPs all coordinates are fixed. Each intermediate set is a face of the original P and contains an original vertex. Hence each newly fixed coordinate has the original Cramer bound, and the initial objective optimum is the objective at an original vertex. All intermediate LP encodings therefore have a common polynomial bound, rather than an iterated unspecified one. For the minimum, bisect with (63) and take a maximizing LP vertex at the final upper bracket endpoint; its root lies between $q_{\min}$ and that endpoint, and the same separation argument applies. Endpoints of the initial bracket cause no exception.

Finally substitute the recovered vertex, evaluate H at the ordered rational breakpoints and at a, b , and return a rational zero if one occurs there. Otherwise their signs locate a positive-width piece with one root, whose polynomial, squarefree part, and rational isolating interval give the exact encoding. Standard univariate isolation and refinement have polynomial bit complexity in the degree, coefficient bits, and requested accuracy bits [8]. All bisection queries have polynomial-bit rational coefficients, and continuity makes a query at a breakpoint unambiguous on P . $\square$

The theorem supplies an explicit exact recovery step for a classical LP-bisection reduction. It is not an algorithm for arbitrary roots of a polynomial family or for a sparsely encoded binary exponent, because the precision in (64) depends polynomially on the numerical degree. Moving breakpoints and selection among nonmonotone branches are also outside its scope.

H.2 Cycle laws and capacities under correlated parameters

Fix a balanced rational nomination b on a connected simple cactus. An admissible law has the form

$$g_e(x; \theta) = g_{e0}(x) + \sum_j \theta_j g_{ej}(x), \quad (65)$$

with fixed rational piece boundaries and dense rational polynomial coefficients. For every admissible parameter vector, the law is promised continuous and strictly increasing on $\mathbb{R}$, with $g_e(0; \theta) = 0$. The total piece count and degrees may grow. These promises do not extend the independent-box validation algorithm of proposition D.1 to general correlations.

We first derive the scalar reduction from these assumptions. For a fixed parameter vector the primitive $\int_0^x g_e(s; \theta) ds$ is nonnegative and strictly convex, and for $|x| \geq 1$ it is at least $c_e(|x| - 1)$, with the positive constant

$$c_e = \min\{g_e(1; \theta), -g_e(-1; \theta)\}.$$

The sum is therefore coercive and has a unique minimum on the nonempty conservation affine space. Stationarity supplies potentials and the edge laws. Each nonzero physical flow points to a lower potential, so the resulting directed graph is acyclic, and source-to-sink flow decomposition bounds every magnitude by $B := \sum_v |b_v|$.

Conservation fixes every bridge flow and every cycle's effective nominations independently of all law parameters. Orient a cycle consistently and let its circulation q_C equal one reference edge flow. Rational conservation elimination gives oriented flows $q_C + d_e$ with one $d_e = 0$. A reversed input edge uses $-g_e(-x; \theta)$, which preserves all law promises, also for nonodd laws. The cycle's consistency equation is

$$H_C(q, \theta) := \sum_{e \in C} g_e(q + d_e; \theta) = 0. \quad (66)$$

It is strictly increasing in q and affine in θ , and its root lies in $[-B, B]$. Its partition is the union of the translated edge breakpoints. Translation expands $(q + d_e)^j$ using only binomial coefficients and powers of rational offsets, with polynomial bit cost and unchanged degree. Thus the union partition and every dense coefficient list have polynomial size; no product of edge polynomials is formed. Conservation and these cycle equations suffice for global potential consistency because the cactus cycles form a basis of its cycle space.

Theorem H.2 (Global correlations: exact capacities and individual arcs). *Let one parameter vector in (65) range over a nonempty bounded rational polytope P , allowing arbitrary correlations between cycles. Feasibility of rational signed arc bounds is one rational linear feasibility problem. A feasible instance has a polynomial-time computable rational admissible parameter witness satisfying all bounds exactly. Robust satisfaction over P is decidable by rational LPs, which also return rational violating scenarios when a bound fails.*

Every individual signed arc has polynomial-time computable exact algebraic extrema and rational optimizing parameter scenarios, also when other such capacities filter the parameter domain. The same statements hold for rational linear flow inequalities whose nonconstant part involves at most one cycle circulation.

Proof. Check bridge bounds directly. In the original orientation a cycle edge has flow $s_e(q_C + d_e)$, $s_e \in \{-1, 1\}$. Collect all its bounds into a rational interval $a_C \leq q_C \leq b_C$, intersecting with $[-B, B]$. A permitted linear constraint is a constant test if its circulation coefficient is zero, and otherwise another scalar bound, with its inequality reversed if that coefficient is negative. Reject any failed constant test or $a_C > b_C$. For every fixed θ , strict increase gives

$$a_C \leq q_C(\theta) \leq b_C \iff H_C(a_C, \theta) \leq 0 \leq H_C(b_C, \theta). \quad (67)$$

Both expressions are rational affine functions of θ . Conjoining all these inequalities with P gives a bounded rational polytope P_{cap} , and rational linear feasibility gives precisely the claimed decision and witness. In particular, irrational physical roots do not require an algebraic parameter certificate.

For robust validation, maximize $H_C(a_C, \theta)$ and minimize $H_C(b_C, \theta)$ over P for each cycle. All lower-bound maxima must be nonpositive and all upper-bound minima nonnegative. These tests include equality, and a strict failure supplies a rational LP optimizer certifying a physical violation by (67). A failed constant bound is violated by any rational point of P .

For the extrema, apply theorem H.1 to the target cycle over P , or over the nonempty P_{cap} when capacities filter the scenarios. Rational shifts and sign changes give the desired signed arc extrema without increasing their algebraic degree. Bridge extrema are fixed rational values. If $B = 0$, all physical flows vanish: check the bounds and return any rational parameter point when feasible. This also handles an edgeless graph before a nontrivial root bracket is needed. $\square$

For quadratic laws, (67) is the interval-to-halfspace reduction of Afmann and coauthors cited above; conjoining it across cactus blocks preserves poly-

hedral uncertainty, and affine polynomial coefficients give the same reduction by rational evaluation. The argument needs no convexity of the globally correlated flow image, and the theorem does not claim that this image is a box or that coupled convex performance can be minimized over it.

Theorem H.3 (Independent cycles: attainable intervals and exact profiles). *Suppose instead that cycle C has its own nonempty bounded rational parameter polytope P_C , independent between cycles. Bridge laws are fixed admissible laws, or have independent nonempty rational parameter polytopes of the same type. Then the attainable flow region is*

$$x = x^0 + Zq, \quad \ell_C \leq q_C \leq u_C, \quad (68)$$

where x^0 is rational and the columns of Z have disjoint cycle supports and entries in $\{0, -1, 1\}$. Each endpoint has an exact algebraic encoding of degree at most the largest law degree on its cycle and a polynomial-time computable rational parameter profile. Rational cycle-local capacities clip these intervals exactly and preserve rational endpoint profiles, including singleton intervals; infeasibility is decided exactly. When feasible, every rational linear flow objective has an exactly optimizing rational parameter scenario, with or without these capacities. Its scalar value is computable to any additive precision in polynomial bit time; exact scalar comparison of sums across cycles is a separate problem.

Proof. If $B = 0$, every physical flow and reference circulation is zero. The attainable intervals are all $[0, 0]$, and capacities reduce to constant tests at the zero flow. Reject any failed test; otherwise rational linear feasibility supplies an admissible point in each cycle and variable bridge parameter polytope. These rational profiles attain every endpoint and optimize every linear flow objective, whose value is rational. This also handles the connected edgeless graph. Henceforth $B > 0$, so $[-B, B]$ is a nontrivial root bracket.

The root map of (66) is continuous: by joint continuity in (q, θ_C) , strict signs at points just below and above a root trap nearby roots. This also gives one-sided continuity at the common bracket endpoints. The image of the compact connected P_C is therefore a compact interval, and theorem H.1 gives its endpoints and rational profiles. Cycle independence and consistency give every point of (68); bridges have fixed flows regardless of their law parameters.

We also need rational realization away from the endpoints. A rational target q is attainable exactly when the two rational LP values satisfy

$$v^- := \min_{\theta \in P_C} H_C(q, \theta) \leq 0 \leq v^+ := \max_{\theta \in P_C} H_C(q, \theta).$$

Let θ^-, θ^+ attain them. If $v^+ > v^-$, set

$$\lambda = \frac{-v^-}{v^+ - v^-}, \quad \theta = (1 - \lambda)\theta^- + \lambda\theta^+. \quad (69)$$

This rational profile belongs to P_C , has polynomial bit length, and satisfies $H_C(q, \theta) = 0$ exactly, because $H_C(q, \cdot)$ is affine and affine constant terms are preserved by convex combinations. If $v^+ = v^-$, both are zero and either profile works. No division by an edge flow is used.

Reduce the capacities to $[a_C, b_C]$ as in theorem H.2. The constrained interval is

$$[L_C, U_C] = [\max\{\ell_C, a_C\}, \min\{u_C, b_C\}].$$

Exact comparison of the separately encoded algebraic endpoints and rational clipping values decides emptiness in polynomial bit time. Every new endpoint is either an old endpoint with its stored rational profile or a rational value realized by (69); at ties either applicable witness works. A rational singleton is realized by interpolation; an irrational singleton must be an endpoint of the original interval and uses that endpoint's rational profile. Combining the independent witnesses satisfies every capacity exactly.

A rational linear objective is a rational constant plus $\sum_C \gamma_C q_C$ with rational γ_C . For maximization choose U_C when $\gamma_C > 0$, L_C when $\gamma_C < 0$, and either when $\gamma_C = 0$; minimization reverses the choices. The resulting rational profiles optimize exactly without comparing sums of algebraic values. For additive value output, enclose each selected endpoint q_C in a rational interval I_C and evaluate the objective at rational representatives $\hat{q}_C \in I_C$, for example their midpoints. The rational estimate $\hat{v}$ of the optimal value v satisfies

$$|\hat{v} - v| \leq \sum_C |\gamma_C| \text{width}(I_C).$$

Stop when this sum is at most the requested error $\varepsilon > 0$; it suffices to refine every interval to width at most $\varepsilon/(1 + \sum_C |\gamma_C|)$, which requires only polynomially many precision bits and rational arithmetic operations. The Square-Root Sum obstruction already present in the quadratic special case of theorem F.10 is why these two output contracts must be kept distinct. $\square$

In particular, $g_e(x; \beta) = \beta_e x |x|$ with explicit positive rational bounds and any rational polytope coupling the resistances within one cycle satisfies the hypotheses with $D = 2$; independent positive resistance intervals are the box special case. These special cases give exact capacitated continuous resistance design and local quadratic algebraic extrema without a bound on cycle size or parameter dimension.

H.3 Convex minimization with exact capacity recovery

Lemma H.4 (A rational Lipschitz extension from a cube). *Let a rational polynomial h be convex on $[-1, 1]^k$, $k \geq 1$, with a polynomial-time exact rational value and gradient oracle. Suppose rational bounds $V \geq |h|$ and $G \geq \max_j |\partial_j h|$, $G \geq 1$, are known on that cube. Put*

$$M = 1 + V + kG, \quad t(z) = \max\{1, \|z\|_\infty\},$$

$$\hat{h}(z) = t(z)h(z/t(z)) + M(t(z) - 1).$$

Then $\hat{h}$ is globally convex, agrees with h on the cube, and has an exact rational value and subgradient oracle. It is globally kC -Lipschitz in Euclidean norm for $C = G + 1 + 2(V + kG)$. Oracle and bound encoding lengths are polynomial in the input and query lengths whenever those of h, V, G are.

Proof. Extend h by $+\infty$ outside the cube. The perspective $sh(z/s) + M(s-1)$ is jointly convex on $s \geq 1$, $|z_j| \leq s$; perspective convexity and convexity under partial minimization are standard [14, Sections 3.2.5–3.2.6]. For fixed z and feasible s , set $w = z/s$. Differentiation gives

$$\frac{\partial}{\partial s} \{sh(z/s) + M(s-1)\} = h(w) - \nabla h(w)^T w + M \geq 1.$$

Thus the minimum over $s \geq t(z)$ occurs at $s = t(z)$ and equals $\hat{h}(z)$, which proves global convexity and cube agreement.

For the subgradient, including boundary points and ties, choose any $\tau \in \partial t(z)$ and put $w = z/t(z)$, $K = h(w) - \nabla h(w)^T w + M$. The gradient of h supports its restriction to the cube, also at the boundary. Multiplying that support inequality by the positive perspective variable gives

$$\hat{h}(z') \geq \hat{h}(z) + \nabla h(w)^T (z' - z) + K(t(z') - t(z)).$$

Here $1 \leq K \leq 1 + 2(V + kG)$, so the support inequality for t yields the subgradient $\nabla h(w) + K\tau$. One may choose $\tau = 0$ when $\|z\|_\infty \leq 1$, and otherwise a signed unit vector of a maximizing coordinate, any one in case of ties. Since $\|\tau\|_\infty \leq 1$, this subgradient has infinity norm at most C . Applying its support inequality at both z and z' proves

$$|\hat{h}(z) - \hat{h}(z')| \leq C\|z - z'\|_1 \leq kC\|z - z'\|_2.$$

At rational queries, t, w, K and the chosen subgradient use only rational evaluation and comparisons of polynomial bit complexity. $\square$

Theorem H.5 (Dense convex polynomial design). *In the independent-cycle model of theorem H.3, impose any of its permitted rational cycle-local capacities. Let $f \in \mathbb{Q}[x_1, \dots, x_m]$ be densely encoded and promised convex on $[-B-1, B+1]^m$, where $B = \sum_v |b_v|$. For every rational $\varepsilon > 0$, one can decide infeasibility or return a rational original parameter scenario satisfying every capacity exactly and having objective at most ε above the constrained minimum. Time is polynomial in the total dense input and requested accuracy bits, with no fixed bound on the number of cycles or the polynomial degrees. Convexity is a promise; sparse binary exponents, variable nominations, potential bounds, and flow constraints coupling distinct cycle coordinates are not covered.*

Proof. First use theorem H.3 to check feasibility and to store the exact constrained endpoints L_C, U_C with rational parameter profiles. If there are no edges, or $B = 0$, check the fixed flow and return any admissible rational profile. Otherwise put $S = B + 1$ and, writing $f = \sum_{\alpha} c_{\alpha} x^{\alpha}$, compute

$$V_f = \sum_{\alpha} |c_{\alpha}| S^{|\alpha|}, \quad G_f = 1 + \max_i \sum_{\alpha: \alpha_i > 0} |c_{\alpha}| \alpha_i S^{|\alpha|-1}.$$

These rational bounds have polynomial bit length under dense encoding. On the expanded flow box, $|f| \leq V_f$ and $\|\nabla f\|_{\infty} \leq G_f$, whence $|f(x) - f(y)| \leq G_f \|x - y\|_1$ on every segment in that box. Choose

$$\eta = \min\{1/2, \varepsilon/(16mG_f)\}.$$

Refine each constrained endpoint to rational enclosures $L_C^- \leq L_C \leq L_C^+$ and $U_C^- \leq U_C \leq U_C^+$ of widths at most η . If $L_C^+ \leq U_C^-$, retain the rational inner interval $[L_C^+, U_C^-]$; projection from the true interval moves a coordinate by at most η . Otherwise freeze the surrogate coordinate at L_C^+ and retain the stored profile of the exact endpoint L_C . The true interval then has width less than 2η , so every point in it lies within 2η of L_C^+ , while the stored endpoint is within η . This works for an irrational singleton even though it contains no rational target circulation.

Since cycle supports are disjoint, every feasible physical flow maps to this rational surrogate box within $2m\eta$ in one-norm. Conversely, every surrogate flow becomes feasible by replacing only its frozen coordinates with their stored exact endpoints, a change of at most $m\eta$. Every surrogate flow therefore lies in the expanded box, since it differs from a feasible flow by at most η per edge. The surrogate itself need not satisfy the physical capacities on frozen cycles.

To optimize on the surrogate, remove every fixed coordinate, including retained intervals of zero width, and affinely rescale the remaining rational intervals to $[-1, 1]^k$. If $k = 0$, evaluate the single surrogate point. Otherwise write its full flow as $x = Tz + a$ with rational T, a and set $h(z) = f(Tz + a)$, which is convex on the cube. Its exact rational value and gradient can be evaluated by composition without expanding a new multivariate coefficient list. Use $V = V_f$ and $G = 1 + G_f \max_j \sum_e |T_{ej}|$ in lemma H.4. Dense degrees bound rational powers and their bit growth polynomially, including for rational oracle queries outside the cube after the extension rescales them.

The resulting globally convex Lipschitz function has the explicit rational Lipschitz bound kC and a polynomial-time rational value oracle. The cube contains a Euclidean unit ball, lies in a ball of radius k , and has an exact membership oracle. The classical ellipsoid optimization theorem, in the encoding formulation of Dadush [20, Theorem 2.5.9 and Section 2.5.1], therefore returns a rational point *in* the cube within $\varepsilon/2$ of the minimum, in time polynomial in the lengths of the radii, the Lipschitz bound, and the accuracy. Tiny positive surrogate widths cause no exception, because their rational affine rescaling has polynomial bit length. The extension agrees with h on the cube, so this is the required surrogate optimizer.

Realize every retained rational circulation exactly by (69), use the stored endpoint profile on each frozen cycle, and choose any allowed rational bridge parameters. Every returned parameter is admissible, and its physical flow satisfies all capacities exactly. To bound the loss, let x^* minimize the true objective, project it to the surrogate for comparison with the surrogate optimizer, and recover the latter as just described. The total loss is at most

$$2mG_f\eta + \varepsilon/2 + mG_f\eta \leq 11\varepsilon/16 < \varepsilon.$$

All arithmetic across cycles is rational. Separate scalar root enclosures also approximate the returned objective with the same gradient bound; no common algebraic field or constitutive inverse condition number is required. $\square$

Corollary H.6 (Quadratic continuous resistance design). *For quadratic passive laws, independent positive rational resistance intervals, or independent positive rational resistance polytopes within cycles, admit the exact capacity and additive design guarantees above for every rational objective $f(x) = \frac{1}{2}x^T Qx + d^T x + c$ with $Q \succeq 0$. The conclusion also holds without capacities.*

Proof. The piecewise quadratic laws satisfy (65), and the objective is convex on the whole flow space. Apply theorems H.3 and H.5. $\square$

Rational output matters here because design takes place in the original model. An optimizer over rational approximations to an algebraic circulation box need not itself be attainable or satisfy a tight capacity; stored exact end-point profiles and rational interpolation resolve both issues. Finite resistance sets lack this interpolation property, and the one-cycle realization hardness in theorem E.11 precludes transferring this convex minimization conclusion to finite choices.

H.4 Convex maximization: coupling and measurement dimension

Theorem H.7 (Bounded-data convex performance hardness). *Maximizing a rational convex quadratic function of all flows is strongly NP-hard on simple cacti of maximum degree three with a fixed unit source and sink, all other nominations zero, and independent resistance intervals contained in $[1, 16]$. All fixed resistances also lie in that range. Additive value approximation with absolute error at most $1/4$ is NP-hard. On the explicit family below, existential integer-threshold attainment is NP-complete and robust satisfaction of a half-integer upper bound is coNP-complete.*

Proof. Given an unweighted Max-Cut graph J on n vertices, construct a chain of n triangles linked by bridges. Each triangle has distinct entry and exit vertices: the direct edge joins them and the two-edge alternate path uses its third vertex. Attach bridges at entry and exit, give every bridge resistance 4, and nominate $+1$ at the first entry and -1 at the last exit. This is a simple cactus of maximum degree three. Orient direct edges, alternate paths, and bridges from source to sink. Each triangle carries unit through-flow. Its alternate edge resistances are 2, 2 and its direct resistance is an independent $\beta_i \in [1, 16]$. If x_i is its direct flow,

$$\beta_i x_i^2 = 4(1 - x_i)^2, \quad x_i = \frac{2}{2 + \sqrt{\beta_i}} \in [1/3, 2/3].$$

These coordinates vary independently, so the vector z with $z_i = 3x_i - 1$ ranges over the entire cube $[0, 1]^n$, and all constructed physical flows are positive. The convex objective

$$F(x) = 9 \sum_{ij \in E(J)} (x_i - x_j)^2 = \sum_{ij \in E(J)} (z_i - z_j)^2$$

is a sum of squares of rational linear forms. A convex function on a cube has a maximizing vertex: fix coordinates successively, each at an endpoint

with no smaller value. At binary z , F is exactly the cut size, so its maximum is the maximum cut size of J , an integer. This is the classical quadratic Max-Cut encoding; see, for example, Del Pia et al. [21, Section 1.1] for the binary formulation.

For integer K , existence of $F \geq K$ is precisely the Max-Cut decision, and robust $F \leq K - 1/2$ is its complement. Endpoint choices $\beta_i = 16$ for $z_i = 0$ and $\beta_i = 1$ for $z_i = 1$ have rational physical flows and give polynomial-size cut certificates. This proves the stated memberships for the restricted family, without an NP membership claim for general algebraic performance values. All physical numerical data are constants, and expanded objective coefficients and thresholds have polynomial magnitude in $|J|$. The reduction therefore remains polynomial under unary numerical encoding, which proves strong hardness. Rounding a value estimate with error at most $1/4$ recovers the integer maximum, which gives the fixed-error claim. This is an absolute error for the displayed objective, without normalization to $[0, 1]$. $\square$

The physical network in this proof has only scalar cycle equations, but the comparison graph in the objective couples arbitrarily many cycles, so endpoint attainment does not make endpoint search efficient. A fixed number of measurements supplies the needed restriction, through classical zonotope enumeration [42, Lemmas 2.1–2.3 and Theorem 2.6].

Theorem H.8 (A fixed number of linear measurements). *Fix $k \geq 1$. In the independent polynomial-law model of theorem H.3, including its permitted local capacities, let $R \in \mathbb{Q}^{k \times m}$ and let g be a densely encoded rational polynomial convex on a box containing all attainable measurements Rx . One can decide infeasibility or return a rational admissible parameter scenario satisfying the capacities and maximizing $g(Rx)$ within additive $\varepsilon > 0$. This takes time $N^{O(k)}$ times a polynomial in the requested accuracy bits, where N is the dense input length.*

The conclusion also holds for the fixed-nomination quadratic cactus model with independent explicitly listed finite positive rational resistance sets, without capacity filters. No exact comparison of all candidate values, or exactly maximizing scenario, is promised.

Proof. For the continuous model, first reject infeasible capacities using theorem H.3; otherwise that theorem supplies the constrained affine box and a rational profile for each endpoint. For finite quadratic sets, theorem F.10 supplies the same representation for the convex hull of the attainable flows, and every box endpoint is attained by resistances from the original finite

sets. In either case denote the relevant bounds by L_C, U_C . The measurement region, or its convex hull, is the zonotope

$$y = Rx^0 + \sum_C a_C q_C, \quad a_C = RZ_C \in \mathbb{Q}^k, \quad L_C \leq q_C \leq U_C.$$

Its segment lengths can be algebraic, but its directions a_C are rational. In the finite case the convexity box also contains the whole zonotope, since it contains the finite set generating it.

Discard zero a_C and enumerate the full-dimensional strict sign cones of the central hyperplane arrangement $h^T a_C = 0$ in $\mathbb{R}^k$, explicitly and rationally as follows. Insert the hyperplanes one at a time. For each current sign pattern try both signs on the new hyperplane, retaining precisely those for which

$$\sigma_C h^T a_C \geq 1 \quad \text{for every inserted } C$$

is feasible. By homogeneity this is equivalent to strict sign feasibility, and rational LP gives a polynomial-bit representative. The number of full-dimensional regions of r hyperplanes in fixed dimension k is $r^{O(k)}$: adding a hyperplane creates at most as many new regions as its induced arrangement in dimension $k - 1$ has, which gives this bound by induction. Repeated or dependent hyperplanes cannot increase it. Thus every insertion and the entire enumeration use $N^{O(k)}$ bit operations up to polynomial rational LP factors.

For each retained sign pattern select $q_C = U_C$ when $h^T a_C > 0$ and $q_C = L_C$ otherwise, and combine the corresponding rational endpoint profiles; zero a_C can use either endpoint. Every vertex of the projected zonotope is obtained this way. A vertex has a full-dimensional set of exposing directions in the ambient space, even if the zonotope is lower dimensional, so we may choose such a direction off the finitely many nonzero-generator hyperplanes. It uniquely selects the appropriate endpoint of every positive-length segment. Zero-length generators merely refine this arrangement and do not change the selected point. The point zonotope is included. By convexity a maximum occurs at a zonotope vertex. In the finite case that vertex is an original admissible scenario, so the finite and convex-hull maxima agree. This convex-hull argument gives no conclusion for capacity-filtered finite sets.

It remains to compare candidates in bit complexity. Put

$$T_0 = B \max_j \sum_e |R_{je}|, \quad M_0 = \max_j \sum_C |a_{jC}|.$$

Every true measurement lies in $[-T_0, T_0]^k$. On $[-T_0 - 1, T_0 + 1]^k$, bound every partial derivative of g in absolute value by a rational $L \geq 1$ using the

absolute monomial coefficient sums, as in theorem H.5. Convexity is needed only on the attainable zonotope; this derivative bound uses only polynomial evaluation and holds on the enlarged box regardless of convexity there. All bound encodings are polynomial under dense degree encoding.

Refine every selected endpoint to rational error at most

$$\delta = \min \left\{ \frac{1}{1 + M_0}, \frac{\varepsilon}{8kL(1 + M_0)} \right\}.$$

Each measurement error is at most $M_0\delta$, so the estimated measurement stays in the enlarged box and the value changes by at most $kLM_0\delta \leq \varepsilon/8$. Dense univariate root refinement, rational measurement sums, and dense evaluation of g are polynomial in input and accuracy bits. Keep the candidate with the largest rational estimated value and return its stored original parameter scenario. Two estimation errors cost at most $\varepsilon/4$, so this scenario has the claimed additive guarantee and, in the continuous model, satisfies the capacities exactly. No common field of all cycle roots and no exact comparison of independent algebraic sums is used.

If $B = 0$, all a_C vanish, or g is constant, choose any of the stored feasible profiles; this avoids unnecessary enumeration and does not affect the bound. The same construction covers zero-width intervals and the absence of cycles. $\square$

Corollary H.9 (Fixed-rank quadratic maximization). *The preceding additive maximization algorithm applies to $\frac{1}{2}x^T Qx + d^T x + c$ when the rational positive-semidefinite matrix Q has fixed rank r , using at most $r + 1$ measurements. This holds in either scenario model of theorem H.8.*

Proof. Rational LDL^T elimination expresses the quadratic part as a sum of r nonnegative rational multiples of squares of rational linear forms, with polynomial bit lengths; a zero diagonal pivot in a positive-semidefinite Schur complement has zero row and column and can be omitted. Use these forms and $d^T x$ as measurements. The resulting polynomial in at most $r + 1$ variables is convex. $\square$

Fixed-rank convex binary quadratic maximization by zonotopes has the direct predecessor Ferrez et al. [26], and the general fixed-dimensional edge-direction framework is due to Onn and Rothblum [42]. The application here supplies rational directions, separately refined algebraic lengths, and admissible rational passive-network scenarios. These arithmetic and recovery steps also extend the measurement theorem from quadratic resistances to the independent polynomial-law polytopes treated above. The $N^{O(k)}$ bound is

not a fixed-parameter tractability claim in k . Together with theorem H.7, the theorem identifies measurement dimension as a meaningful restriction for coupled convex performance. It does not transfer the finite-set maximization result to convex minimization or exact target realization.

I Correlations, dissipation, and rational feasibility

The product structure in theorem H.3 permits arbitrary correlations within a cycle, but requires independence between cycles. Under global resistance correlations, three tasks behave differently: individual arc validation remains linear programming on a cactus (theorem H.2); maximizing total edge flow or minimizing dissipation becomes strongly hard; and maximizing dissipation admits an additive convex algorithm on every graph. Exact physical constraints also separate rational from algebraic parameter witnesses. Unless stated otherwise, this section uses the symmetric quadratic law and fixed rational balanced nominations.

I.1 A shared parameter and a curved attainable region

Example I.1 (Two correlated cycles). Join two source-to-sink triangles by a bridge of resistance one. Supply one unit at the first entrance and withdraw one at the last exit. The first triangle has a two-edge path with resistances $1/2, 1/2$; the second has a two-edge path with resistances $2, 2$. Both direct edges have the same resistance $\theta \in [1, 4]$. This is a simple degree-three cactus, and its uncertainty set is a rational polytope in the actual edge resistances. Equal path drops and conservation give the two direct flows

$$x_1 = \frac{1}{1 + \sqrt{\theta}}, \quad x_2 = \frac{2}{2 + \sqrt{\theta}}, \quad x_2 = \frac{2x_1}{1 + x_1}, \quad \frac{1}{3} \leq x_1 \leq \frac{1}{2}.$$

All edge flows are positive. The last graph is strictly concave and nonaffine, so the projected attainable region is nonconvex, and hence so is the full flow region. Explicitly, the midpoint of its endpoints is $(5/12, 7/12)$, whereas the graph at $5/12$ has ordinate $10/17$, a difference of $1/204$.

The linear objective $F = x_2 - x_1$ has value $1/6$ at both parameter endpoints. Writing $s = \sqrt{\theta}$ gives $F = s/((s+1)(s+2))$ and

$$\frac{dF}{ds} = \frac{2 - s^2}{(s^2 + 3s + 2)^2}.$$

Hence its unique maximum is at $\theta = 2$, where $F = 3 - 2\sqrt{2}$. The gain over either endpoint is

$$\frac{17}{6} - 2\sqrt{2} = \frac{1}{36(17/6 + 2\sqrt{2})} > \frac{1}{216};$$

the last inequality follows from $\sqrt{2} < 19/12$. Measuring one edge of the first alternate path and the second direct edge gives the positive unit-coefficient objective $1 + F$, with the same interior optimum. Multiplying all resistances by two makes the fixed resistances integral without changing any flow. The example disproves a shared-parameter endpoint rule; by itself it proves no hardness.

I.2 Bounded-data hardness for total flow and minimum dissipation

For a physical state define its constitutive dissipation by

$$D(\beta) = \sum_{e \in E} \beta_e |x_e(\beta)|^3 = b^\top \pi(\beta). \quad (70)$$

The equality follows by multiplying $A^\top \pi = g(x)$ by x and using $Ax = b$. Under unit source/sink nominations it is the prescribed terminal potential difference. If potentials represent squared gas pressures, this is a squared-pressure difference; the identity does not identify D with literal compressor power.

Theorem I.2 (Globally correlated cactus hardness). *On connected simple cacti of maximum degree three, with unit source/sink nominations and a global rational resistance polytope, each of the following problems is strongly NP-hard: maximizing $T = \sum_e x_e$, and minimizing D . This holds with resistances in $[1, 3]$, bounded integer coefficients in the polytope description, an acyclic orientation, and strictly positive flows on every edge. Thus T also sums all absolute edge flows, with coefficient one on every edge. Absolute-error $1/256$ value approximation is NP-hard for either problem, as is returning a rational admissible profile within $1/256$ of the appropriate optimum.*

On the explicit families constructed below, the weak existential tests $T \geq \tau$ and $D \leq \tau$ are NP-complete, and their strict universal complements are coNP-complete. The universal test $T \leq \tau$ is also coNP-complete on the total-flow family. No membership statement is asserted for arbitrary correlated cactus objectives.

Proof. Use an unweighted Max-Cut instance H with n_H vertices, $m_H \geq 1$ edges, and integer target $1 \leq K \leq m_H$. The binary cut encoding is classical; see, for example, Del Pia et al. [21, Section 1.1]. A trivial source instance can be replaced by a one-edge comparison graph with target one (yes case) or a comparison triangle with target three (no case). Introduce $\theta_i \in [0, 1]$. For every comparison edge ij , use two unit-through-flow triangles. Each triangle has a two-edge path and an alternate direct edge of resistance two. In the plus triangle, both path edges have resistance $2 + \theta_i - \theta_j$; in the minus triangle, both have resistance $2 - \theta_i + \theta_j$.

If the resistance on each path edge is r , its path flow $f(r)$ obeys $2rf(r)^2 = 2(1 - f(r))^2$. The positive solution and the triangle's terminal drop are

$$f(r) = \frac{1}{1 + \sqrt{r}}, \quad e(r) = \frac{2r}{(1 + \sqrt{r})^2}. \quad (71)$$

The three edge flows sum to $1 + f(r)$. Their dissipation is $e(r)$, the common path drop times the total through-flow one. Direct differentiation gives

$$f''(r) = \frac{1 + 3\sqrt{r}}{4r^{3/2}(1 + \sqrt{r})^3} > 0, \quad e'(r) = \frac{2}{(1 + \sqrt{r})^3}.$$

$$e''(r) = -\frac{3}{\sqrt{r}(1 + \sqrt{r})^4} < 0.$$

Consequently $f(2 + d) + f(2 - d)$ is even and convex on $[-1, 1]$, whereas $e(2 + d) + e(2 - d)$ is even and concave there.

For total flow, join all $2m_H$ triangles in a chain with $2m_H - 1$ bridges of resistance two. Place a path of n_H additional bridges before the first triangle, with actual resistance coordinates $r_i \in [1, 2]$, and set $\theta_i = r_i - 1$. Impose the equalities

$$\beta_{ij,+1} = \beta_{ij,+2} = 2 + r_i - r_j, \quad \beta_{ij,-1} = \beta_{ij,-2} = 2 - r_i + r_j.$$

All other resistances are fixed at two. These equations and interval bounds describe an injective affine image of the cube in the actual physical resistance vector, so no projection encoding is used. Their coefficients and constants belong to a fixed integer alphabet. Orient every route from the first source to the final sink and put nominations $+1, -1$ there. Every bridge carries one, each triangle carries one in total, and equation (71) makes every edge flow strictly positive. The graph is simple, is a cactus of maximum degree three, and has an acyclic orientation. Its vertex and edge counts are $6m_H + n_H$ and $8m_H - 1 + n_H$, so its cycle rank is $2m_H$.

At a binary cube vertex, one triangle pair contributes

$$A = 2f(2) = 2\sqrt{2} - 2 \quad \text{if uncut,} \quad B = f(1) + f(3) = \frac{\sqrt{3}}{2} \quad \text{if cut.}$$

Put $\kappa = B - A$ and $C = 4m_H - 1 + n_H + m_H A$. The full objective is

$$T(\theta) = 4m_H - 1 + n_H + \sum_{ij \in E(H)} [f(2 + \theta_i - \theta_j) + f(2 - \theta_i + \theta_j)].$$

It is convex on the cube. A convex function on a cube has a maximizing vertex: with all other coordinates fixed, at least one endpoint of an interval has value at least its interior value; apply this successively. At binary vertices $T = C + \kappa \text{cut}(\theta)$, whence

$$\max T = C + \kappa \text{MaxCut}(H). \quad (72)$$

The rational square comparisons $\sqrt{3} > 173/100$ and $\sqrt{2} < 283/200$ give $\kappa > 7/200 > 1/32$.

For minimum dissipation replace the prefix by $2n_H$ bridges, in pairs $r_i = 1 + \theta_i$ and $s_i = 2 - \theta_i$. Impose $r_i + s_i = 3$, and use the same triangle correlations in terms of r_i . This is again an injective affine cube in actual resistances, all in $[1, 3]$. The graph now has $6m_H + 2n_H$ vertices and $8m_H - 1 + 2n_H$ edges; its rank, degree, simplicity, orientation, and positivity properties are unchanged. Each prefix pair contributes constant dissipation three. Pairing is necessary: a single unit-flow bridge would contribute its variable resistance to the objective.

At binary parameters the pair energies are

$$A_E = 2e(2) = 24 - 16\sqrt{2}, \quad B_E = e(1) + e(3) = \frac{13}{2} - 3\sqrt{3}.$$

Put $\kappa_E = A_E - B_E$ and $C_E = 3n_H + 4m_H - 2 + m_H A_E$. The total energy is a sum of the concave pair responses and constant bridge energies, so it has a minimizing cube vertex by the reverse endpoint argument. Therefore

$$\min D = C_E - \kappa_E \text{MaxCut}(H). \quad (73)$$

The same rational square comparisons give $\kappa_E = 35/2 - 16\sqrt{2} + 3\sqrt{3} > 1/20 > 1/32$.

Thresholds. For total flow choose rational τ_T within $1/512$ of

$$Q_T = C + \kappa(K - 1/2) = 4m_H - 1 + n_H + (m_H - K + 1/2)A + (K - 1/2)B.$$

Dyadic enclosures of $\sqrt{2}, \sqrt{3}$ of width at most $1/[8192(m_H + 1)]$ suffice: substitute any enclosed rational values; the sum of the absolute sensitivity coefficients is at most $2m_H + m_H/2$. For energy choose rational τ_E within $1/512$ of

$$Q_E = C_E - \kappa_E(K - 1/2) = 3n_H + 4m_H - 2 + (m_H - K + 1/2)A_E + (K - 1/2)B_E.$$

Width $1/[65536(m_H + 1)]$ suffices, since the corresponding coefficient sum is at most $19m_H$. Dyadic bisection needs $O(\log(m_H + 1))$ bits; taking the first power of two satisfying either bound makes the common denominator $O(m_H + 1)$. Both thresholds then have polynomially bounded numerators in $m_H + n_H$ and polynomially bounded denominators. Thus all numerical input, including the thresholds, has polynomial unary encoding length, as strong hardness requires.

Gaps. In a yes case, equation (72) exceeds $\tau_T + 7/512$ and equation (73) is below $\tau_E - 7/512$. In a no case the respective optima are below $\tau_T - 7/512$ and above $\tau_E + 7/512$. Indeed half of either cut reward exceeds $1/64 = 8/512$, and the rational threshold incurs at most $1/512$ error. These strict gaps prove all hardness statements for the weak tests and their complementary universal tests, including robust $T \leq \tau_T$.

Membership. On these families an extremum occurs at binary θ , hence at integer resistances. Its value lies in the fixed field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and is exactly comparable with any input rational threshold in polynomial time. A binary cut is thus a polynomial certificate for existential attainment, or for strict violation of a universal bound, which proves the claimed memberships. The argument gives no such certificate for arbitrary sums of unrelated algebraic cycle responses.

Approximation. A value estimate within $1/256 = 2/512$ distinguishes the two cases. Suppose instead that an algorithm returns a rational admissible profile within that loss. In a yes total-flow case its value exceeds $\tau_T + 5/512$, whereas every no-case profile is below $\tau_T - 7/512$. For minimum energy the signs reverse. Evaluate the returned profile by enclosing the individual square roots in equation (71), adding the rational bridge contributions, and refining to total error at most $1/512$. All arguments of the roots belong to $[1, 3]$; the formulas have uniformly bounded derivatives there. Thus $O(\log(m_H + 1))$ additional accuracy bits beyond the output encoding suffice. A polynomial-time output algorithm has polynomial output length, so this evaluation is polynomial-time and separates the cases without exact comparison of a general radical sum. $\square$

The transported amount in this construction is fixed at one. Total edge

flow counts it again on every traversed edge; it is not end-to-end throughput. The cycle rank grows, so neither a fixed-rank lower bound nor constant relative-error or normalized-objective hardness follows. The reduction combines the classical convex/concave cube-vertex mechanism with Max-Cut; its specific content is the bounded passive-network realization under global resistance correlations. For comparison, Klimm et al. [32, Corollary 4 and Theorem 11] study conductance investment with installation costs, obtain convexity without fixed costs, and give a series-parallel approximation scheme when all variable conductance costs are positive. Linear correlations in resistance coordinates do not become linear correlations in conductance coordinates; those conclusions concern different feasible sets and objectives.

I.3 Maximum dissipation on arbitrary graphs

The opposite energy direction has a classical variational explanation. Let

$$V(\beta) = \min_{Ax=b} \sum_e \frac{\beta_e |x_e|^3}{3}, \quad D(\beta) = 3V(\beta). \quad (74)$$

The second equality is equation (70) and theorem A.1. As an infimum of affine functions of β , V is concave. For linear two-terminal networks, concavity in all edge resistances was proved by Shannon and Hagelbarger [48]; the present conclusion follows from the same variational structure with a cubic primitive. Our additions are an explicit conic formulation and its bit-complexity and exact-feasibility details. The energy principle and its homogeneous dual are classical (see appendix A and Groß et al. [29, Section 3, equation (7), and Lemma 3.4, equation (8)]); perspectives and the hyperbolic-cone construction are also established [14, Section 3.2.6] [36, Section 2.3].

Theorem I.3 (Additive maximum dissipation and rational profiles). *Let G be any connected graph with fixed rational balanced nominations. Let P be a nonempty bounded rational polytope of resistances, given by rational inequalities and explicit positive bounds $\beta_L \leq \beta_e \leq \beta_U$. Maximum physical dissipation over P has an exact formulation with two second-order cones per edge. For every rational $\varepsilon > 0$, a rational $\hat{\beta} \in P$ satisfying*

$$D(\hat{\beta}) \geq \max_{\beta \in P} D(\beta) - \varepsilon$$

is computable in time polynomial in input length and $\max\{0, \log(1/\varepsilon)\}$. The theorem imposes no additional physical capacity or potential filters.

Proof. For $\beta > 0$, the scalar maximum of $dx - \beta|x|^3/3$ occurs at $x = \text{sgn}(d)\sqrt{|d|/\beta}$ and equals the conjugate value $2|d|^{3/2}/(3\sqrt{\beta})$, also at $d = 0$. Fenchel's inequality gives

$$V(\beta) = \max_{\pi: \pi_{v_0}=0} \left[b^\top \pi - \frac{2}{3} \sum_e \sqrt{\frac{|(A^\top \pi)_e|^3}{\beta_e}} \right]. \quad (75)$$

Weak duality follows by summing Fenchel's inequality at any conserved flow. The physical potentials from theorem A.1 give equality on every edge, proving attained strong duality. Normalization changes neither edge drops nor the balanced pairing $b^\top \pi$. The conjugate is the convex perspective $\beta(2/3)|d/\beta|^{3/2}$, so the bracket is jointly concave in (π, β) . Combining its maximum over π with a maximum over $\beta \in P$ interchanges only two maxima.

For each edge introduce $z_e, u_e, t_e \geq 0$, put $d_e = (A^\top \pi)_e$, and impose

$$z_e \geq d_e, \quad z_e \geq -d_e, \quad z_e^2 \leq t_e u_e, \quad u_e^2 \leq \beta_e z_e. \quad (76)$$

These are two rotated second-order cones and linear inequalities; for example $z^2 \leq tu$ with $t, u \geq 0$ is $\|(2z, t - u)\|_2 \leq t + u$, with rational coefficients. If $z > 0$, multiplying gives $z^4 \leq \beta z t^2$, hence $z^3 \leq \beta t^2$. Conversely this last inequality with $z > 0$ forces $t > 0$, and $u = z^2/t$ satisfies both cones. At $z = 0$ choose $u = 0$ and any $t \geq 0$; conversely the second cone forces $u = 0$. If $t = 0$, the first cone forces $z = 0$. Thus eliminating z, u is exactly $t \geq \sqrt{|d|^3/\beta}$, including every boundary case. Maximize

$$J = b^\top \pi - \frac{2}{3} \sum_e t_e \quad \text{subject to } \beta \in P, \quad \pi_{v_0} = 0, \quad \text{equation (76)}. \quad (77)$$

Its optimum is $\max_P V$, whose triple is the requested dissipation.

It remains to obtain an exactly feasible rational output of polynomial bit length without assuming a general exact SOCP solver. If $b = 0$, every physical flow and dissipation is zero, and rational LP returns any profile in P . The connected edgeless case is included. Otherwise put $M = \frac{1}{2} \sum_v |b_v| > 0$ and $m = |E| \geq 1$. Orient each nonzero physical edge downhill in potential. This orientation is acyclic and its nonnegative flow decomposes into source-to-sink paths of total weight M ; one may add a supersource and supersink and repeatedly remove such a path. Hence $|x_e| \leq M$. A simple path from v_0 yields

$$|\pi_v| \leq m\beta_U M^2.$$

At a physical dual optimum, cone variables may be chosen as

$$z_e = \beta_e |x_e|^2, \quad u_e = \beta_e |x_e|, \quad t_e = \beta_e |x_e|^3. \quad (78)$$

Both cones are tight and these coordinates are uniformly bounded. Moreover, evaluating the primitive energy at the other profile's minimizer gives $|V(\beta) - V(\beta')| \leq (M^3/3)\|\beta - \beta'\|_1$. Thus V attains its maximum on compact P .

The following compact bounds preserve a global optimizer. Set $r = \min\{1, \beta_L\}/2$ and impose

$$\begin{aligned} |\pi_v| &\leq m\beta_U M^2 + 1, & 0 \leq z_e &\leq \beta_U M^2 + 2, \\ 0 \leq u_e &\leq \beta_U M + r + 1, & 0 \leq t_e &\leq \beta_U M^3 + 2/r + 1. \end{aligned}$$

Every bound has polynomial rational encoding length and contains equation (78). At a relative-interior point β^0 of P , use $\pi = 0, z_e = 1, u_e = r, t_e = 2/r$. Then $z_e > |d_e|$, $t_e u_e - z_e^2 = 1$, and $\beta_e z_e - u_e^2 \geq \beta_L - r^2 > 0$; all added upper and lower bounds are strict as well.

Next we construct β^0 and a quantitative interior ball. For each row $R_i \beta \leq c_i$ of P , maximize its slack by rational LP. A zero maximum identifies a universally tight row. For each other row retain a polynomial-bit rational maximizing vertex with positive slack. Average these witnesses (or take any feasible rational point if there are no such rows). All nonuniversal rows are then strict at the average. Basic-solution determinant bounds give polynomial encoding lengths for each vertex, their average, and every positive rational slack. The universally tight rows define the affine hull: within their solution space the strict nonuniversal rows leave a neighborhood of the average, so no further affine equality can hold throughout P .

Rational Gaussian elimination expresses this hull by $\beta = \beta^0 + Q\xi$, choosing free coordinates among original resistance coordinates. The entries of Q and the chosen center coordinates have polynomial bit length; the free coordinates have bounds inherited from P . Remove the fixed potential coordinate. Substitution leaves a compact convex region K in the remaining coordinates, with every nonconstant constraint a rational linear or quadratic inequality $p_j(y) \geq 0$ strict at a rational center y^0 . Universal equalities and redundant constant inequalities are removed. If P is a singleton, there are simply no free resistance coordinates.

On $\|y - y^0\|_\infty \leq 1$, bound $\|\nabla p_j(y)\|_1$ by a rational $G_j \geq 1$ using the absolute coefficients and the coordinate magnitudes $|y_i^0| + 1$. Since degrees are at most two and there are polynomially many coefficients, these bounds have polynomial bit length. The rational radius

$$\rho = \min \left\{ 1, \min_j \frac{p_j(y^0)}{2G_j} \right\} > 0$$

puts the Euclidean ball $B_2(y^0, \rho)$ inside K , by the mean-value inequality. The coordinate bounds give a rational outer radius about y^0 , for instance

the sum of bounds on the absolute coordinate differences. Both radii and the center have polynomial encoding length. Convexity here follows from the nonnegative rotated-cone constraints, not from an assumption that arbitrary quadratic inequalities are convex.

Membership of a rational point in K is decided exactly by rational linear and quadratic comparisons, supplying a weak membership oracle. The negative objective $-J$ is linear on the whole reduced coordinate space, hence is globally convex and Lipschitz with a rational bound obtained by summing the absolute coefficients; its rational value oracle is exact. Apply Dadush [20, Theorem 2.5.9, p. 48] to this centered convex body with error $\varepsilon/3$. This theorem returns a rational point *in* K , with an additive objective guarantee, in polynomial time in these encodings and accuracy bits. Its feasibility conclusion is essential here. Map its coordinates back to obtain rational $\hat{\beta} \in P$ exactly. Every feasible conic point satisfies $J \leq V(\hat{\beta})$ by equation (75); therefore

$$D(\hat{\beta}) = 3V(\hat{\beta}) \geq 3J \geq \max_{\beta \in P} D(\beta) - \varepsilon.$$

The auxiliary potentials need not be physical. Neither rounding a physical flow nor exactly evaluating a radical sum is required. $\square$

Corollary I.4 (Exact cactus capacities). *On a cactus, theorem I.3 remains valid with rational signed arc capacities: emptiness is decidable in polynomial time, and a nonempty instance admits the same additive algorithm with a rational profile satisfying all original physical capacities exactly.*

Proof. Apply theorem H.2 to obtain the exactly equivalent rational polytope $P_{\text{cap}} \subseteq P$, rejecting fixed bridge violations or emptiness. This is the cycle interval linearization of Aßmann et al. [5, Proposition 4.9, Lemma 4.10, Proposition 4.11], composed across blocks in theorem H.2. Apply theorem I.3 to P_{cap} . Its affine-hull reduction handles capacity equalities and singletons. Membership of the returned profile in P_{cap} guarantees its actual physical capacities; the conic auxiliary variables are used only for the objective guarantee. $\square$

An exact conic representation does not give an exact polynomial-time scalar comparison. For fixed resistances and unit terminal nominations on a cactus, D is the terminal potential difference, so the two reductions in theorem B.2 apply to its exact threshold comparison, already for a singleton parameter polytope. The theorem returns rational near-optimal resistance parameters; it promises neither a rational physical state nor an exact rational optimal parameter vector.

I.4 An exact certificate for the original global design problem

Proposition I.5 (Rational global design certificate). *Write the complete positive bounded resistance polytope as $P = \{\beta : R\beta \leq c\}$. Let rational $y, \lambda, \beta^0, \pi, t$ satisfy*

$$\begin{aligned} Ay = b, \quad \lambda \geq 0, \quad R^\top \lambda = w, \quad w_e = |y_e|^3/3, \\ R\beta^0 \leq c, \quad t_e \geq 0, \quad \beta_e^0 t_e^2 \geq |(A^\top \pi)_e|^3. \end{aligned}$$

Then with $U = c^\top \lambda$ and $L = b^\top \pi - (2/3) \sum_e t_e$,

$$\max_{\beta \in P} D(\beta) - D(\beta^0) \leq 3(U - L). \quad (79)$$

The certificate allows any additive potential gauge because b is balanced. All hypotheses and the gap are exactly checkable by rational arithmetic in time polynomial in their encoded lengths.

Proof. For every $\beta \in P$, the conserved trial flow yields $V(\beta) \leq \beta^\top w \leq c^\top \lambda = U$ by LP weak duality. The conjugate inequalities and equation (75) yield $L \leq V(\beta^0)$. Subtract and multiply by three. The checks use only rational equalities, inequalities, absolute cubes, and products. $\square$

For an exact zero-gap example, orient a triangle with two path edges and one direct edge from source to sink, and impose unit through-nominations. Let $P = \{\beta \in [1, 4]^3 : \sum_e \beta_e = 6\}$. Take

$$\begin{aligned} y = (1/2, 1/2, 1/2), \quad \beta^0 = (3/2, 3/2, 3), \\ (\pi_s, \pi_w, \pi_t) = (3/4, 3/8, 0), \quad t = (3/16, 3/16, 3/8). \end{aligned}$$

Each $w_e = 1/24$. Represent the sum equality by two inequalities and give its upper direction multiplier $1/24$, with all other multipliers zero. Then $U = 6/24 = 1/4$. The edge drops are $(3/8, 3/8, 3/4)$; each conjugate inequality is tight, and $L = 3/4 - (2/3)(3/4) = 1/4$. This proves the global maximum $D = 3/4$ and exact optimality of the displayed original profile. For the capacitated corollary, use the full rows of P_{cap} in the certificate. The trial flow y need only conserve nominations: it bounds the unconstrained physical primitive energy of every admissible profile, so it needs no capacity condition.

The certificate proves its stated gap, but an arbitrary conserved trial flow need not yield a small gap, and the proposition supplies no general numerical producer. The accompanying exact rational verifier checks the encoded graph, nominations, resistance bounds, all polytope rows, and the displayed

conditions. Its polynomial bound counts expanded rational numerators and denominators, as in appendix K. Its polytope input contains no independently interpreted operating constraints; deriving such rows from physical capacities is a separate equivalence check. A numerical conic producer for selected budget-constrained instances supplies candidate certificates; their guarantees come from rational verification, not solver status. This producer does not implement the convex-body algorithm of theorem I.3.

I.5 Irrational feasible parameters beyond cacti

Proposition I.6 (Capacities forcing irrational resistance). *There is a feasible rational-input quadratic design instance on a simple series-parallel graph of global cycle rank two, with one resistance interval and only nonnegative positive-width upper-capacity intervals, for which every feasible resistance profile is irrational.*

Proof. Use vertices u, v, r, w, z , a direct arc $a = (u, v)$, a connector (u, r) , and the two routes r, w, v and r, z, v , all oriented towards v . Set

$$\begin{aligned}\beta_a &= \theta \in [4/3, 11/8], & \beta_{ur} &= 1, \\ \beta_{rw} &= \beta_{wv} = 1/2, & \beta_{rz} &= \beta_{zv} = 1.\end{aligned}$$

Supply two at u and withdraw two at v . Impose capacities $[0, 1]$ on a and (u, r) , and $[0, 2]$ on the four remaining arcs. This simple graph is a parallel composition of a and a connector in series with two parallel paths, hence is series-parallel. Its rank is $6 - 5 + 1 = 2$. Conservation forces $x_a = x_{ur} = 1$. The two r - v path flows q_1, q_2 have sum one and the same sign because their drops agree and their resistance sums are positive. Thus both are positive. Their resistance sums are one and two, so

$$q_1^2 = 2q_2^2 = d, \quad q_1 = 2 - \sqrt{2}, \quad q_2 = \sqrt{2} - 1, \quad d = 6 - 4\sqrt{2}.$$

The connector adds drop one; the unit-flow direct arc therefore forces

$$\theta = 7 - 4\sqrt{2}, \quad \theta^2 - 14\theta + 17 = 0.$$

This is strictly between $4/3$ and $11/8$: the equivalent square-root bounds are $\sqrt{2} < 17/12$ and $\sqrt{2} > 45/32$, verified by squaring. Set $\pi_v = 0$, $\pi_r = d$, $\pi_u = 1 + d$, and $\pi_w = \pi_z = d/2$. These potentials and the displayed flows satisfy every original equation and capacity. Conversely the cut saturation forces all preceding values. The polynomial's discriminant is 128, which is not a rational square, so there is no rational feasible profile. $\square$

A smaller equality-constrained version identifies r with u and removes the connector. Set the direct interval to $[1/3, 3/8]$ and require $x_a = 1$. With the same nomination and branch resistances, the forced value is $\theta = 6 - 4\sqrt{2}$, satisfying $\theta^2 - 12\theta + 4 = 0$. The bounds are again the rational square comparisons just used, shifted by one. Potentials $\pi_u = \theta$, $\pi_v = 0$, and $\pi_w = \pi_z = \theta/2$ verify feasibility on this four-vertex, five-edge graph. Replacing its interval by the two-point set $\{1/3, 3/8\}$ makes $x_a = 1$ infeasible, while its interval hull remains feasible. Thus operating-constraint filtering can destroy an unfiltered hull identity. Discrete filtered feasibility is already hard on one cycle by theorem E.11; this smaller example illustrates the failure directly.

By theorem H.2, a nonempty rationally described capacity-filtered cactus domain has a rational original parameter witness. Every connected graph of global cycle rank at most one is a cactus, making rank two the first possible rank for the preceding obstruction. It is an output-representation obstruction, not an NP-hardness proof, and permits algebraic feasible output. Both cut capacities in proposition I.6 must be saturated; strict margins have a different conclusion below. General irrational arithmetic in potential-flow design is already discussed by Klimm et al. [32, Remark 8]; the point here is unavoidable irrationality of a feasible *parameter*, not merely an irrational physical state at rational input.

I.6 Uniform coefficient sensitivity, including zero flows

Theorem I.7 (Linear flow perturbation bound). *Let a connected graph have $m \geq 1$ edges and fixed balanced nonzero nominations, and put $M = \frac{1}{2} \sum_v |b_v|$. Consider two quadratic coefficient profiles whose coefficients lie in $[\beta_L, \beta_U]$, with $\beta_L > 0$. Either use one symmetric coefficient per edge, or two independent directional coefficients β_e^+, β_e^- as in appendix A.1. If every corresponding coefficient changes by at most δ , the physical flows satisfy*

$$\|x' - x\|_\infty \leq \frac{2mM\delta}{\beta_L}. \quad (80)$$

The bound holds when flows vanish or reverse sign. Its constant is not asserted to be optimal.

Proof. Both flows have magnitude at most M , by the downhill path decomposition used in theorem I.3; this also applies to positive directional coefficients. Their difference $h = x' - x$ is a circulation. Reorient every edge with negative h_e so that $h_e \geq 0$. Reorientation swaps the positive and negative coefficients of each directional law, for both old and new profiles. Thus the common bounds and the maximum coefficient change δ are preserved.

Let $H = \|h\|_\infty > 0$ and choose an arc $u \rightarrow v$ with $h_e = H$. There is a directed path from v to u using only arcs with $h_e \geq H/m$. Otherwise the vertices reachable from v on such arcs form a set containing v but not u . The incoming cut contains flow H , while every outgoing arc has flow strictly less than H/m . There are at most m outgoing arcs, so their total is strictly below H , contradicting circulation balance. Choose a simple return path. Together with $u \rightarrow v$, it forms a directed cycle C on which every $h_e \geq H/m$.

Write $f(x) = x|x|$. For $h > 0$,

$$f(x+h) - f(x) \geq \frac{h|x|}{2}, \quad f(x+h) - f(x) > 0. \quad (81)$$

For $x \geq 0$ or $x \leq -h$, expansion gives even the lower bound $h|x|$. In the crossing case $x = -th$, $0 < t < 1$, the difference between the two sides of the weak inequality, divided by h^2 , is $2t^2 - (5/2)t + 1$. Its minimum is $7/32 > 0$. Let g'_e denote the new law. In the symmetric case, or in either same-sign branch of a directional law,

$$g'_e(x+h) - g'_e(x) \geq \beta_L[f(x+h) - f(x)].$$

In the crossing case the left side is $\beta_e^{'+}(x+h)^2 + \beta_e^{-}x^2$, so the same inequality holds. Also, at any fixed x , $|g'_e(x) - g_e(x)| \leq \delta x^2$, including at zero.

Potential differences telescope around C . Hence

$$\sum_{e \in C} [g'_e(x_e + h_e) - g'_e(x_e)] = - \sum_{e \in C} [g'_e(x_e) - g_e(x_e)] \leq \delta \sum_{e \in C} |x_e|^2.$$

If $H > 2mM\delta/\beta_L$, every term with $x_e \neq 0$ on the left is at least $\beta_L h_e |x_e|/2$, strictly larger than $M\delta |x_e| \geq \delta |x_e|^2$. Each term with $x_e = 0$ is strictly positive, while its right-hand contribution is zero. Summation gives a contradiction. The case $H = 0$ is immediate. This proof uses no positive lower bound on differential resistance at zero flow. $\square$

The elementary circulation argument gives an explicit bound without differentiating the physical state. A regularized electrical sensitivity argument improves its constant and retains separate directional coefficients.

Proposition I.8 (A sharper coefficient perturbation bound). *Under the hypotheses of theorem I.7, define $q_e = |\beta'_e - \beta_e|$ in the symmetric case and $q_e = \max\{|\beta_e^{'+} - \beta_e^{+}|, |\beta_e^{-} - \beta_e^{-}|\}$ in the directional case. Then*

$$\|x' - x\|_\infty \leq \frac{M}{2\beta_L} \sum_e q_e \leq \frac{mM\delta}{2\beta_L}. \quad (82)$$

Proof. Interpolate the coefficients affinely by a parameter $s \in [0, 1]$, and add ρx to every law, for a fixed rational $\rho > 0$. The regularized primitive is the original primitive plus $\rho \|x\|_2^2/2$. Its physical flow $x^\rho(s)$ still has $|x_e^\rho(s)| \leq M$, because its law remains strictly increasing and positive at every positive flow.

To differentiate, choose a full-column-rank matrix C with image $\ker A$ and write a conserved flow as $x^0 + Cz$. The stationarity equations are $C^\top(g_s(x^0 + Cz) + \rho(x^0 + Cz)) = 0$. Directional quadratic laws are jointly continuously differentiable in x and their coefficients, including $x = 0$. The derivative in z is $C^\top \mathcal{R}C$, where $\mathcal{R}$ is diagonal with

$$r_e = 2\beta_e^{\text{active}}(s)|x_e^\rho(s)| + \rho > 0.$$

At zero either directional coefficient gives the same value $r_e = \rho$. The matrix is positive definite, so the implicit-function theorem and uniqueness make $x^\rho(s)$ continuously differentiable. If $\ker A = 0$, the flow is independent of all coefficients and the conclusion is immediate. The equations may be differentiated locally beyond either endpoint because all coefficients remain positive in a neighborhood.

At fixed s , let a_e be the incidence column of edge e , and let $j^{(e)}$ be the electrical flow with resistances r_a and nomination a_e . Thus $Aj^{(e)} = a_e$ and $\mathcal{R}j^{(e)} = A^\top v$ for some v . Existence and uniqueness follow by minimizing $\sum_a r_a j_a^2/2$ on that affine conservation space, exactly as in theorem A.1. Its nonzero flow is downhill, so unit source-to-sink path decomposition gives $|j_a^{(e)}| \leq 1$. Moreover

$$v_{\text{tail}(e)} - v_{\text{head}(e)} = \sum_a r_a (j_a^{(e)})^2 > 0,$$

so $0 \leq j_e^{(e)} \leq 1$. Consequently $|j_a^{(e)} - \mathbf{1}_{a=e}| \leq 1$ on every edge.

For a forcing d_e only on edge e , the differentiated state equation has $Ah = 0$ and $\mathcal{R}h + d_e \mathbf{1}_e \in \text{im } A^\top$. Its unique solution is

$$h = \frac{d_e}{r_e}(j^{(e)} - \mathbf{1}_e).$$

Indeed its divergence vanishes and its residual is $(d_e/r_e)\mathcal{R}j^{(e)}$, a potential gradient. Uniqueness follows by taking the scalar product of the homogeneous equation with its circulation solution. Summing these solutions handles simultaneous coefficient changes. The actual forcing on edge e is $\dot{\beta}_e x_e^\rho |x_e^\rho|$ in the symmetric case; in the directional case it is $\dot{\beta}_e^+(x_e^\rho)^2$ for positive flow,

$-\dot{\beta}_e^-(x_e^\rho)^2$ for negative flow, and zero at zero. Only one coefficient is active. In both cases

$$\frac{|d_e|}{r_e} \leq \frac{q_e|x_e^\rho|^2}{2\beta_L|x_e^\rho| + \rho} \leq \frac{q_e M}{2\beta_L}.$$

Thus $\|\dot{x}^\rho(s)\|_\infty \leq M \sum_e q_e/(2\beta_L)$, and integration gives equation (82) for the two regularized endpoint states.

Finally let $\rho \downarrow 0$. At each endpoint the regularized flows lie in the compact box $[-M, M]^E$. Every convergent subsequence has a conserved limit. For any fixed conserved trial flow y , regularized optimality gives

$$F_\beta(x^\rho) + \rho\|x^\rho\|_2^2/2 \leq F_\beta(y) + \rho\|y\|_2^2/2,$$

where F_β is the original primitive energy. Passing to the limit shows the limit minimizes F_β ; uniqueness makes it the original physical flow. Hence the full endpoint sequences converge. Passing to the limit in their uniform difference bound proves the assertion, including zero and reversing flows. Uniform convergence over all s is unnecessary for this endpoint argument. $\square$

Continuity in the resistances is known: Raber [45, Theorem 3.12, pp. 34–35] proves continuity of physical flows and normalized potentials in all positive resistances by parametric convex optimization. The explicit linear constant in equation (80), valid also for separate directional coefficients, provides the quantitative precision bound used here; qualitative continuity alone does not determine a bit budget.

Corollary I.9 (Strict margins and rational witness size). *For fixed rational nominations and independent rational positive coefficient intervals (symmetric or directional), suppose a real profile satisfies all rational signed arc bounds with common slack $\gamma > 0$. If $b \neq 0$, rounding each coefficient within its own interval by at most*

$$\delta \leq \frac{\beta_L \gamma}{4mM}$$

preserves at least $\gamma/2$ slack. For rational γ , a rational feasible profile exists with encoding length polynomial in the input length and $\max\{0, \log(1/\gamma)\}$. This is a conditional witness-size and safe-rounding statement, not an algorithm for finding the initial strictly feasible real profile.

Proof. Apply equation (80). For existence and encoding, round coordinates to a dyadic grid of spacing at most δ and clip to the rational interval endpoints. Clipping preserves the interval and cannot increase distance from the original point in that interval. The number of fractional bits and the integer parts are

polynomially bounded as claimed. The same construction applies separately to the two directional coordinates. With $b = 0$, every physical flow is zero and any rational allowed profile is feasible whenever the capacities contain zero. $\square$

The sharper estimate equation (82) preserves half the slack whenever $\delta \leq \beta_L \gamma / (mM)$; the radius in corollary I.9 also remains valid. The existence and size conclusion extends to any bounded rational coefficient polytope P within the same positive coefficient bounds, including affine equalities, although arbitrary coordinatewise rounding need not preserve its correlations. If $b = 0$, every physical flow is zero, and rational LP returns a polynomial-bit profile in P satisfying the same arc bounds. Otherwise choose $\delta = 2^{-k}$, where k is the least nonnegative integer such that

$$2^{-k} \leq \min\{1, \beta_L \gamma / (4mM)\}.$$

Then k is polynomially bounded in the input length and $\max\{0, \log(1/\gamma)\}$. Choose a vector a on the grid of spacing $\delta/4$ within $\delta/4$ of the strictly feasible real profile β , and form

$$P \cap \{z : \|z - a\|_\infty \leq \delta/2\}.$$

This is a nonempty bounded rational polytope because it contains β . The chosen grid and the bounds on β give a and the box inequalities polynomial encoding length in the original data and k . A rational LP vertex therefore has polynomial bit length by determinant bounds, and lies within $3\delta/4$ of β . The sensitivity estimate preserves the required physical slack. This proves existence of a short correlated witness; if a sufficiently accurate center a is supplied, LP recovers one. The argument does not locate the unknown strictly feasible profile or its center.

For pressure bounds, let a fixed path between terminals have k edges. On $[-M, M]$ every new directional law is $2\beta_U M$ -Lipschitz, since its one-sided derivatives have magnitude at most that number and the law is continuous at zero. Thus splitting the coefficient and flow changes on each edge and summing along the path gives

$$\begin{aligned} |(\pi'_s - \pi'_t) - (\pi_s - \pi_t)| &\leq k[\delta M^2 + 2\beta_U M \|x' - x\|_\infty] \\ &\leq kM^2\delta(1 + 4m\beta_U/\beta_L). \end{aligned} \tag{83}$$

Finitely many strict pressure-difference margins therefore survive sufficiently fine rational rounding, with the same polynomial bit bounds. Absolute potential bounds require fixing a reference potential, or explicitly choosing the additive gauge; with a fixed reference, take paths from that vertex in the preceding estimate.

I.7 Pressure filters and the conic boundary

Even on a triangle, arbitrary upper pressure filters cannot be appended as affine resistance restrictions to theorem H.2. Give its two-edge path total resistance A (each edge $A/2$) and its direct edge resistance B . Under unit terminal transfer, equal drops and conservation yield

$$D(A, B) = \frac{AB}{(\sqrt{A} + \sqrt{B})^2}.$$

On the rational resistance segment joining $(A, B) = (1, 9)$ and $(9, 1)$, the endpoint drops are $9/16$, while the midpoint $(5, 5)$ has drop $5/4$. Thus the upper restriction $D \leq 1$ accepts both endpoints and rejects their midpoint, even though every physical flow lies in $[0, 1]$. The feasible resistance set is nonconvex and cannot be a projection of a linear feasible set in these original coordinates.

For the opposite restriction under unit terminal transfer, $D \geq c$ is a convex resistance condition by equation (74). It has an exact extended conic representation: append $J \geq c/3$ to equations (76) and (77). If such auxiliary variables exist, weak duality gives $D \geq c$; conversely physical potentials and equation (78) attain $J = D/3$. Adding this inequality may remove the strictly feasible conic center constructed above, so this representation alone does not give an unconditional exact rational feasible-output guarantee at a degenerate boundary. It also concerns the single unit-transfer potential difference, not arbitrary collections of potential bounds. These distinctions reconcile the maximum-energy algorithm with the hard minimum-energy direction and the irrational capacity-witness example.

J Weighted blocks beyond cacti

The small-face theorem in theorem G.2 applies to every fixed maximum block rank. The remaining algorithmic issue is summation: many blocks can depend on the same few unfixed nominations. This section resolves that issue for one shared nomination parameter, with arbitrarily many noncactus blocks, and for any fixed number of shared nomination parameters when the total rank of the noncactus blocks is fixed. Both results include independent coefficient intervals and rational near-optimal original scenarios. The last subsection delimits the open case of several shared parameters and arbitrarily many higher-rank blocks.

Except in corollary J.5, laws in this section are quadratic and the allowed coefficient models are exactly those in theorem G.1: fixed positive asymmet-

ric coefficients, independent symmetric coefficient intervals, or independent directional coefficient intervals. No operating constraints filter the scenarios. Put $W = c^\top \pi$, where c is rational and balanced. All complexity statements refer to the Turing model and accuracy bits.

J.1 Exact local value graphs

Lemma J.1 (Conditional block values). *Fix d, r_0 . Let $b(z) = b^0 + Hz$ be balanced on a nonempty compact rational polytope $P \subseteq \mathbb{R}^d$, and suppose every block has rank at most r_0 . One can construct, for every block B , a rational quantifier-free formula $\Gamma_B(z, v)$ for the graph of its maximum weighted contribution $V_B(z)$, conditional on z and optimized over its allowed coefficients. Formula sizes and coefficient bit lengths are polynomial, and degrees are bounded in terms of d, r_0 . Moreover,*

$$\max_{z, \theta} W = \max_{z \in P} \sum_B V_B(z).$$

Each V_B has a computable rational absolute bound and a Lipschitz constant of polynomial bit length. At a rational $z \in P$, its value and an optimizing local coefficient profile can be computed as real algebraic data of polynomial representation size, without joining fields from different blocks.

Proof. Choose a spanning tree formed by a spanning tree in each block, and a rational tree flow w with $Aw = c$. Then $W = \sum_e w_e g_e(x_e)$ by (48). A block's effective nomination $b^B(z)$ is obtained by summing nominations over the components attached to each of its vertices. These components partition V , so b^B is rational affine and balanced. Its physical state depends only on b^B and its own coefficients. Block independence therefore proves the displayed identity, with each local contribution $\sum_{e \in B} w_e g_e(x_e)$.

Tree routing and a fundamental-cycle basis give $x_e = X_e(z, q_B)$, with $r_B \leq r_0$ circulation coordinates. A rational bound $B_0 \geq 1$ on total positive nominations bounds every physical flow; because the tree routing vanishes on chords, $|q_{B,i}| \leq B_0$. Enumerate the realizable flow-sign cells in (z, q_B) and use their closures. On such a cell put $Q_e = s_e X_e^2$. At fixed coefficients the cycle equations and local value are quadratic. For uncertain coefficients apply lemma C.9, with the r_B cycle equations as linking rows and $w_e Q_e$ as the single objective row. In the directional model use the interval of the selected sign; unused directional coordinates can take arbitrary allowed values. At zero flow both directional terms vanish, so closed sign-cell overlaps are exact. The symmetric model retains one original coefficient per edge.

This gives a polynomial-size fixed-degree formula for attainable (z, q_B, v) . Eliminating q_B gives $\Psi_B(z, v)$, and the desired graph is

$$\Psi_B(z, v) \wedge \neg \exists v' (\Psi_B(z, v') \wedge v' > v).$$

All projections have a fixed number of variables. The bounds in appendix A.5 prove the representation claims, including singular states and equality cases. Every fiber is nonempty and compact: all coefficients are in compact boxes, and the unique physical state depends continuously on them. At rational z , the same computation and lemma C.9 recover an exact local optimizer. A bridge needs no circulation variable and is included in this argument.

For the continuity bounds, let H^B be the linear part of b^B and put

$$T_B = \sum_{v,j} |H_{vj}^B|, \quad K_B = \beta_U B_0 T_B \sum_{e \in B} |w_e|, \quad M_B = \beta_U B_0^2 \sum_{e \in B} |w_e|.$$

Here β_U bounds every allowed directional coefficient. At fixed coefficients, (54), applied inside the block, and the edge-law Lipschitz constant $2\beta_U B_0$ give

$$|V_B(z) - V_B(z')| \leq K_B \|z - z'\|_\infty, \quad |V_B(z)| \leq M_B.$$

In the uncertain case, the first inequality follows by applying the fixed-profile inequality to a maximizer for z , evaluated at z' , and then reversing the roles; taking a maximum thus preserves the bound. All displayed constants have polynomial bit length, using a rational bounding box for P . They do not require lower bounds on nonzero flows or on derivatives. $\square$

J.2 A rational approximation lemma for one parameter

The next lemma concerns a scalar semialgebraic function, rather than a special radical formula. A quantifier-free graph description means an explicit Boolean formula in rational polynomial sign tests whose solutions over the stated interval are exactly $(t, f(t))$. Degrees may grow when the polynomials are densely encoded.

Lemma J.2 (Uniform rational approximation on an interval). *Let $f : [0, 1] \rightarrow \mathbb{R}$ have a rational quantifier-free semialgebraic graph description, and suppose rational $K, M_0 \geq 1$ satisfy $|f(t) - f(t')| \leq K|t - t'|$ and $|f(t)| \leq M_0$. Given rational $0 < \eta \leq 1$, one can construct rational closed intervals covering $[0, 1]$ and a rational polynomial on each interval whose error from f is at most η throughout that interval. The number of intervals, polynomial degrees, dense coefficient encodings, and construction time are polynomial in the graph description, the encodings of K, M_0 , and the accuracy bits. Adjacent polynomials need not agree at their shared endpoints.*

Proof. We give all approximation and precision bounds. Polynomial arithmetic, subresultants, root isolation, and real elimination in fixed dimension are the classical algorithms described in Basu et al. [8] and appendix A.5.

An annihilating polynomial and its exceptional parameters. Discard identically zero polynomials from the graph formula, replacing their tests by their truth values. Multiply its polynomials that depend on the value variable y , remove the content in $\mathbb{Q}[t]$, and take the squarefree part in $\mathbb{Q}(t)[y]$. Clear denominators and remove content again to obtain a nonzero primitive $F(t, y) \in \mathbb{Q}[t, y]$ with positive y -degree. This polynomial satisfies

$$F(t, f(t)) = 0 \quad (0 \leq t \leq 1). \quad (84)$$

Indeed, outside the finitely many roots of the nonzero t -only polynomials and the contents, some value-dependent polynomial must vanish at $(t, f(t))$. Otherwise all polynomial signs are locally constant, and the graph formula would contain a two-dimensional neighborhood, contrary to being a function graph. Taking a primitive squarefree part preserves these zeros away from finitely many additional specializations. Continuity extends the resulting identity to every remaining parameter. The same argument shows that a value-dependent polynomial exists in the first place.

The product has polynomial total degree and coefficient length. Gcd and squarefree computation over $\mathbb{Q}(t)$ have the same polynomial bounds: subresultant coefficients are determinants of polynomial-size matrices of univariate polynomials, with polynomial degree and bit length. Taking primitive parts and exact quotients preserves polynomial bounds. Thus the following construction does not require a bounded degree for F .

Write $F(t, y) = \sum_{j=0}^D a_j(t)y^j$, with $a_D \neq 0$. The polynomial

$$H_*(t) = a_D(t) \operatorname{Res}_y(F, \partial_y F)$$

is nonzero because F is squarefree over $\mathbb{Q}(t)$. Its degree E and coefficient lengths are polynomial. Its zeros include every finite parameter where a root can collide or the leading coefficient can vanish. Compute the distinct real parts of its complex roots. For example, eliminate v from $\Re H_*(u + iv) = \Im H_*(u + iv) = 0$ and isolate the resulting real u . This has two real variables and polynomial degree, so takes polynomial bit time. There are at most E such real parts; nonreal roots must be included. When H_* is constant the list is empty.

Short intervals around exceptional real parts. Choose $\rho = 2^{-s}$, where s is the least integer $s \geq 0$ such that

$$\rho \leq \min\{1/16, \eta/[64K(E+1)]\}.$$

If A denotes the right-hand side, then $A/2 < \rho \leq A$, so ρ has polynomial bit length. Keep every exceptional real part in $[-1, 2]$, enclose it in a rational interval $[l, u]$ of width at most ρ , and pad that interval to $[l - \rho, u + \rho]$. Include $[-\rho, \rho]$ and $[1 - \rho, 1 + \rho]$, intersect the union with $[0, 1]$, and merge overlaps. Call this union J . Its total length is at most $3E\rho + 4\rho \leq 4(E+1)\rho \leq \eta/(16K)$. On each component use a rational approximation, within $\eta/8$, to f at the rational midpoint. The Lipschitz bound makes its error throughout that component less than $\eta/4$. Values at rational arguments are computed by the graph formula and real-algebraic sampling, then isolated to the required rational accuracy. These operations have polynomial cost in the graph data and requested precision.

Analytic disks and rational panels. For a complementary gap $[a, b]$ put $d(t) = \min\{t - a + \rho, b - t + \rho\}$. Every exceptional real part is at distance at least $d(t)$ from t . For retained real parts this follows from the padding, also after merging; omitted ones are outside $[-1, 2]$, whereas $d(t) \leq 1/2 + \rho < 1$. Split the gap at its midpoint. Starting at $x = a$, make left-half panels

$$[x, y], \quad y = \min\{(a + b)/2, x + (x - a + \rho)/32\},$$

and reflect this rule for the right half. Before the last panel, $x - a + \rho$ grows by $33/32$. Hence there are $O(1 + \log(1/\rho))$ panels per gap, all with polynomial rational encoding. If τ and h are a panel's midpoint and half-width, then

$$h \leq d(\tau)/64, \quad R = d(\tau)/4, \quad h \leq R/16.$$

The closed complex disk $|t - \tau| \leq R$ stays at distance at least $3d(\tau)/4 \geq 3\rho/4$ from every zero of H_* .

All roots of $F(t, \cdot)$ are simple on a neighborhood of this disk, and $a_D(t)$ is nonzero. They form a finite unramified covering there: local implicit branches exist, and a uniform polynomial root bound prevents escape to infinity along a compact path. Since a disk is simply connected, the branch through $(\tau, f(\tau))$ is single valued and holomorphic on a neighborhood of the closed disk. On the real panel it equals f , because a continuous choice of distinct roots cannot change branches. This step allows nonsmooth branch switches inside J and does not assume that the original graph is irreducible.

Every such holomorphic branch is uniformly bounded. Write k_0 for the degree of a_D and $\lambda \neq 0$ for its leading coefficient. All its roots are zeros of H_* . Thus throughout these disks

$$|a_D(t)| \geq |\lambda|(3\rho/4)^{k_0}.$$

Also $|t| < 2$. Expanding the a_j gives a rational $T \geq 1$ bounding $|a_j(t)|$ there. The elementary Cauchy polynomial-root bound therefore gives

$$|f_{\text{hol}}(t)| \leq M := 1 + \frac{T}{|\lambda|(3\rho/4)^{k_0}}. \quad (85)$$

The logarithm and rational encoding length of M are polynomial, even if a complex singularity is extremely close to the real interval.

Rational interpolation with a certified error. Choose the least integer $q \geq 1$ with $M2^{-q} \leq \eta/8$; its numerical value is polynomially bounded. On a panel take the $q+1$ equally spaced rational nodes $t_j = \tau - h + 2hj/q$. Let I_q interpolate the exact values $f(t_j)$. The contour interpolation remainder on $|t - \tau| = R$ yields, for real x in the panel,

$$\begin{aligned} |f(x) - I_q(x)| &\leq \frac{MR}{R-h} \left(\frac{2h}{R-h} \right)^{q+1} \\ &\leq \frac{16M}{15} (2/15)^{q+1} \leq M2^{-q} \leq \eta/8. \end{aligned}$$

Here the remainder is the contour integral of $f_{\text{hol}}(t)\omega(x)/[(t-x)\omega(t)]$ divided by $2\pi i$, where $\omega(t) = \prod_j (t - t_j)$. Bound each numerator factor by $2h$, each denominator factor by $R - h$, and the contour length by $2\pi R$. At a node the error is zero directly.

No algebraic coefficient field is needed to construct the approximation. Replace each $f(t_j)$ by a rational value with error at most

$$\delta = \frac{\eta}{8(2q)^q}.$$

For the Lagrange basis polynomials L_j on these nodes,

$$\sum_{j=0}^q |L_j(x)| \leq \sum_{j=0}^q \frac{q^q}{j!(q-j)!} = \frac{(2q)^q}{q!} \leq (2q)^q.$$

The rational interpolant consequently differs from I_q by at most $\eta/8$, and from f by at most $\eta/4$. Its required node precision has $O(\log(1/\eta) + q \log(q+1))$ bits. Root isolation computes the node values independently. Expanding the rational Lagrange formula requires polynomially many arithmetic operations with polynomial-bit numbers: each product has q factors, each node difference has polynomial encoding, and summing $q+1$ such polynomials preserves that bound. Thus the dense rational coefficient representation also has polynomial size. The constant pieces on J and these analytic panels cover the whole interval and prove every assertion. $\square$

Corollary J.3 (Sums of scalar semialgebraic responses). *For a list of functions satisfying lemma J.2, both extrema of their sum on $[0, 1]$ admit rational enclosing intervals of width at most ε and rational ε -optimal arguments in time polynomial in the combined input and accuracy bits. The number of summands and their algebraic degrees may grow.*

Proof. For m summands apply the lemma with $\eta = \varepsilon/(16 \max\{1, m\})$, after replacing ε by $\min\{\varepsilon, 1\}$. Sort all rational panel endpoints. On each common closed subinterval select one valid polynomial per summand and add them. This gives a rational polynomial with uniform error at most $a = \varepsilon/16$ from the true sum. The number of subintervals, degrees, and coefficient lengths are polynomial; endpoints are covered by either adjoining valid choice.

Optimize each rational polynomial on its closed interval by univariate real-algebraic methods. These compute a rational interval $[L, U]$ of width at most $\varepsilon/8$ containing the largest surrogate maximum and an algebraic argument whose surrogate value is at least $L - \varepsilon/16$. Then $[L - a, U + a]$ encloses the true maximum and has width at most $\varepsilon/4$. The sample's true value is at least $L - \varepsilon/16 - a$, so its loss is at most $5\varepsilon/16$. Isolate the sample in an interval of width $\varepsilon/[4 \sum_i K_i]$ and choose a rational point in its intersection with $[0, 1]$. The true sum loses at most $\varepsilon/4$ by Lipschitz continuity. Every isolation precision has polynomial bit length. A constant sum can instead use any rational argument; the empty sum is zero. Negating every function proves the minimum statement. $\square$

J.3 One shared nomination parameter

Theorem J.4 (Weighted bounded blocks on an affine nomination line). *Fix r_0 . On a connected graph of maximum block rank at most r_0 , let $b(t) = b^0 + ht$ be a rational balanced nomination family on a nonempty compact rational interval. Objective support, the number of noncactus blocks, and total cycle rank are unrestricted. In all three coefficient models stated above, each joint extremum of W admits a rational enclosing interval of width at most ε and a rational ε -optimal original parameter scenario in accuracy-bit polynomial time.*

Proof. A singleton interval reduces to independent exact local computations and separately refined value intervals. Otherwise a rational affine change of variable makes the interval $[0, 1]$; it preserves polynomial encoding. Apply lemma J.1 with $d = 1$ and then corollary J.3, using tolerance $\varepsilon/2$. Increase any zero local bound to one to meet the approximation lemma's conventions.

This gives a certified interval and a rational parameter t' whose conditional joint value loses at most $\varepsilon/2$.

At this rational t' , recover a coefficient maximizer separately in each block by lemma J.1. These local choices form a global physical scenario because their nominations agree at the attachments and their cycle equations hold in every block. The output of this intermediate step may be algebraic. If $k \leq 2m$ original coefficients are uncertain, isolate each to width δ , intersect with its own original rational interval, and choose a rational value there. Choose

$$\|c\|_1 B_0^2 k \delta \leq \varepsilon/4.$$

If its left coefficient is zero there is no loss to control. The all-graph estimate (60) bounds the objective loss by $\varepsilon/4$. Shared symmetric coefficients are rounded once; directional coordinates are rounded separately, including inactive ones. Fixed and singleton coefficients are retained exactly. Isolation and rounding take polynomial time because local representations have bounded degree and polynomial length. There is no common field containing all block roots. The resulting rational nomination and coefficients satisfy their original unfiltered domains exactly, and their physical state has the required objective. Negate c for the minimum. $\square$

The theorem resolves the univariate summation issue even when the local functions are not expressed by radicals. It does not make exact comparison of their sum polynomial-time: it uses certified surrogates throughout, not separation bounds for exact sums.

Corollary J.5 (Fixed dense piecewise-polynomial laws). *The fixed-law conclusion of theorem J.4 remains true for the continuous, strictly increasing, densely encoded rational piecewise-polynomial edge laws vanishing at zero from appendix D.1. Degrees and piece counts may grow with the input. This extension concerns fixed laws and returns rational nominations.*

Proof. Use the same affine flow coordinates (t, q_B) and polynomial-bit bound $B_0 \geq 1$. Partition by every affine equation $X_e(t, q_B) = a$, where a is an edge-law breakpoint. There are polynomially many realizable cells in fixed dimension $1 + r_0$. On each closed cell the cycle equations and weighted contribution are rational polynomials whose degrees and dense encodings are polynomial in the input. Continuity makes the formulas agree at breakpoints. Conservation and cycle consistency still characterize the unique physical state by theorem A.1. Fixed-dimensional elimination therefore constructs a polynomial-size graph for each local value, now of polynomial rather than fixed degree.

Absolute coefficient sums over all edge pieces give polynomial-bit rational bounds G_0 and L_0 on their values and one-sided derivatives on $[-B_0, B_0]$. For a piece $\sum_j a_j x^j$ one can use $\sum_j |a_j| B_0^j$ and $\sum_{j \geq 1} j |a_j| B_0^{j-1}$, respectively. Continuity across finitely many breakpoints makes L_0 a Lipschitz constant for the complete edge law on that interval. The acyclic difference-flow proof of (54) uses only strict monotonicity, so remains valid. Consequently each block's value has absolute bound $G_0 \sum_{e \in B} |w_e|$ and nomination-parameter Lipschitz constant $(L_0/2) T_B \sum_{e \in B} |w_e|$. Lemma J.2 and corollary J.3 allow growing dense degree and supply the claimed value interval and rational argument. No law approximation or coefficient rounding is needed. Sparse binary-encoded large exponents are outside this statement. $\square$

J.4 Several parameters with finitely much noncactus rank

Define

$$\kappa(G) = \sum_{B: r_B \geq 2} r_B.$$

Thus $\kappa(G) = 0$ on cacti. A graph can have fixed $\kappa(G) > 0$ and arbitrarily large total rank, for example a subdivided theta block with arbitrarily many attached cycle blocks. Neither the lengths of the exceptional blocks nor the number of their uncertain coefficients are bounded by $\kappa(G)$.

Theorem J.6 (A fixed total rank in the noncactus blocks). *Fix d, κ_0 . Suppose $\kappa(G) \leq \kappa_0$ and nominations form a rational affine balanced family $b(z) = b^0 + Hz$ on a nonempty compact rational polytope $P \subseteq \mathbb{R}^d$. In all three coefficient models, with unrestricted objective support, both joint extrema of W admit rational intervals of width at most ε and rational ε -optimal original parameter scenarios in accuracy-bit polynomial time.*

Proof. Call the blocks of rank at least two exceptional. Choose the blockwise spanning tree and objective weights w as in lemma J.1. Retain the circulations q in all exceptional blocks; there are $\kappa = \kappa(G)$ of them. Every exceptional-edge flow $X_e(z, q)$ is rational affine, and $|q_i| \leq B_0$. Thus the exact exceptional core has fixed dimension $d + \kappa$, even if it contains arbitrarily many edges.

Enumerate the realizable exceptional flow-sign cells in these variables and use closed cells intersected with $P \times [-B_0, B_0]^\kappa$. For each cell use lemma C.9 with all κ exceptional cycle equations as linking rows and the *one* value row

$$v = \sum_{e \text{ exceptional}} w_e \beta_e s_e X_e(z, q)^2. \quad (86)$$

The linking matrix is block diagonal in its cycle rows. The box lemma eliminates every exceptional coefficient and gives a quantifier-free formula $\Psi(z, q, v)$ for exactly the attainable cores and total exceptional values. Its degree is bounded in terms of d, κ_0 , and its size and coefficient lengths are polynomial. Arbitrary objective support changes w , but does not add linking or value coordinates. Directional intervals are selected by the flow sign as in lemma J.1; fixed coefficients are singleton intervals. If there are no exceptional blocks, simply take $v = 0$.

The remaining blocks are cycles and bridges. Their through-nominations depend on z , and not on the exceptional circulations or coefficients: block aggregation determines them from the original nominations alone. Use the local candidate construction and rational approximations of appendix G for these blocks. That construction uses only their affine local nominations, their one cycle equation, and their weights w_e ; it does not require the other blocks of the graph to be cycles. In fixed dimension d it gives polynomially many common sign parts with rational surrogates $S(z)$ for their summed conditional maximum. Allocate local errors so that both the true conditional maximum and a recoverable selected local scenario differ from $S(z)$ by at most $a = \varepsilon/16$. These are the two guarantees proved after lemma G.7. Bridge endpoint choices are exact.

On a common part optimize

$$v + S(z) \quad \text{subject to} \quad \Psi(z, q, v).$$

The number of variables is $d + \kappa + 1$. Although the surrogate's degree grows with requested precision, its degree and dense encoding are polynomial. Denominators have known nonzero signs on each part; clear them exactly as in appendix G. The box lemma was used *before* this approximation, with degree-two coefficient data, so its fixed-degree hypothesis has not been changed.

The projected exceptional set is compact. Individual surrogate sign parts can be open, but the physical objective is bounded by (50) and the surrogate differs from it by at most a after conditional maximization. Hence all relevant suprema are finite. Bisection and a sample at a strictly smaller threshold, exactly as in (53), produce an enclosing interval of width at most $\varepsilon/4$ and an algebraic core (z_*, q_*, v_*) whose recoverable physical scenario loses at most $5\varepsilon/16$. The estimates apply because at fixed z coefficients in different blocks are independent: the exceptional and nonexceptional optimizers can be combined.

Recover the exceptional coefficients from this feasible core using the zonotope argument in the proof of lemma C.9. The recovery works for *any*

feasible algebraic core and value, not only for the box lemma's own maximizer: enumerate exposed vertices of the $(\kappa + 1)$ -dimensional image, find a convex combination of at most $\kappa + 2$ of them representing the prescribed image, and lift that combination to original box coordinates. Every operation uses the one ordered field of (z_*, q_*, v_*) , of polynomial degree and encoding. Recover each selected nonexceptional cycle profile in its own local extension of the field of z_* , as in appendix G. This gives consistent local physical states at the same nomination.

Finally isolate the coordinates of z_* to width h and solve rational linear feasibility in the intersection of that isolating box with P . It is nonempty because it contains z_* . Round all $k \leq 2m$ original coefficient coordinates inside their original intervals to width δ . With $H_1 = \sum_{v,j} |H_{vj}|$, choose positive rational widths such that

$$K_b H_1 h + \|c\|_1 B_0^2 k \delta \leq \varepsilon/4,$$

where K_b is from (55). Polynomial-bit choices suffice, including zero coefficients of either term. The all-graph continuity estimate (60) bounds the remaining loss. Rounding need not retain the selected algebraic sign part or its circulation: the original rational parameters have their own unique physical state. The total loss is below ε . All counts, representations, and precision demands are polynomial for fixed d, κ_0 . The minimum follows by changing c to $-c$. $\square$

Corollary J.7 (Balanced boxes with fixed support and exceptional rank). *Fix p, κ_0 . If $\kappa(G) \leq \kappa_0$, nominations range over a nonempty rational balanced box, and $|\text{supp } c| \leq p$, then the value and rational-scenario guarantees of theorem J.6 hold in each of the three coefficient models.*

Proof. Every block rank is at most $\max\{1, \kappa_0\}$. Apply theorem G.2; pruning and zero-objective block contraction cannot increase κ . Each of polynomially many faces has a fixed number of affine nomination parameters. Apply theorem J.6 with tolerance $\varepsilon/8$ on each nonempty face. Take the maximum of the certified lower endpoints and the maximum of the upper endpoints, and select the scenario associated with the largest lower endpoint. This loses at most one additional interval width. The resistance-independent face property and rational disaggregation in theorem G.2 complete the joint original-network guarantee. The zero objective is handled by rational feasibility alone. $\square$

J.5 The remaining several-parameter summation question

After the face reduction and lemma J.1, the case left open by the present algorithms is

$$\max_{z \in P} \sum_{i=1}^m V_i(z), \quad d \geq 2 \text{ fixed}, \quad (87)$$

where m is unrestricted, $P \subseteq \mathbb{R}^d$ is a compact rational polytope, and each V_i is a continuous bounded Lipschitz semialgebraic block-value function with a polynomial-size bounded-degree rational graph description and polynomial-bit bounds. A polynomial-size, uniformly accurate rational piecewise approximation in the shared z coordinates, constructible in accuracy-bit polynomial time, would suffice: take a common sign decomposition, add the rational surrogates, optimize in fixed dimension, and use the global continuity estimate for rational recovery. An optimization method that bypasses such representations could also suffice. Neither construction is asserted here for arbitrary m, d .

The difficulty no longer lies in univariate root isolation or the nomination face reduction, both handled above. A Lipschitz grid in d dimensions has a power of $1/\varepsilon$ points. Retaining one local value or circulation vector per block in a quantifier-elimination formula loses the fixed-variable bound. Counts of zeros of exact univariate sums alone supply neither effective near-cancellation control nor a several-variable optimizer. These observations explain only why those particular arguments do not give an accuracy-bit bound. They establish no impossibility result, no new hardness classification, and no need to solve the exact SRS comparison problem for additive optimization.

The approximation ingredients used above have substantial predecessors. Vigneron’s multiplicative approximation scheme for sums of nonnegative algebraic functions [52] is polynomial in $1/\varepsilon$ (appendix G). Our scalar lemma instead uses Lipschitz control near exceptional parameters and geometric analytic panels to obtain polynomial dependence on accuracy bits. It supplies rational polynomials in the *original common parameter*, so they can be added without introducing a new variable per block. No new root-isolation, complex inverse, or interpolation principle is claimed. Verified enclosure of piecewise analytic functions also has direct predecessors: Petras supplies such approximation algorithms under assumptions on the function representation [43]. Borcea, Bøgvad, and Shapiro give rational approximants converging exponentially to a dominant algebraic branch away from an exceptional locus [13, Theorems 2–3]; that statement does not by itself give uniform rational approximations across the exceptional parameters with the present

coefficient-bit and branch-selection guarantees.

Yomdin gives a bound of order $\log^3(1/\varepsilon)$ for the degree-based complexity of parametric polynomial approximations to planar semialgebraic sets [55, Definitions 5.1–5.2 and Theorem 5.6]. This approximates a parametrization of a set, rather than the value of a function in a prescribed original coordinate. In several variables, Binyamini and Novikov give polynomial bounds for the number and semialgebraic complexity of smooth parameterizing charts [9, Theorem 1]. Such charts parameterize each set separately; their stated geometric bounds do not by themselves supply a rational-coefficient bit algorithm for adding many value functions in their original shared coordinates. These are distinctions between the inspected statements and the needed interface, not claims that approximation theory cannot supply it.

K Certified computation from rational witnesses

The additive algorithms above fix a tolerance before computation. A numerical solver raises a different question: what can be proved about its actual output? This section gives rational witnesses for energy, individual flows, linear flow goals, and recovered resistance scenarios. They are verified in exact arithmetic, without assuming that a numerical method has reached its stopping tolerance. The global resistance-design certificate in proposition I.5 already treats maximum dissipation over a polytope; the present certificates concern the physical state of a fixed law and its use in the arc-envelope reduction.

The underlying principles are classical. Conjugates and Slater duality are used in their standard sense [14, Sections 3.3, 5.1.6, and 5.2.3]. The divergence below is precisely the convex-function divergence of Bregman [16, p. 206, equation (1.4)], with its second argument equal to the unknown physical state. Primal-dual gaps as computable bounds on uniformly convex errors are established a posteriori methods; see Bartels and Milicevic [6, Section 3, Proposition 3.1]. The conservation projection is the electrical cut/cycle projection [37, Chapter 2, Section 4]. Our purpose is to give their complete asymmetric-quadratic specialization, exact rational checks, degeneracy conditions, and original-instance interpretation. We make no novelty claim for these general principles.

The accompanying Lean formalization (topics 03 and 16) proves the deterministic energy, Bregman, support, and curvature guarantees below, including conditional scenario recovery and the two-path comparison. They do not cover the original uncertainty-to-envelope mapping, the numerical producers,

the Python verifiers and parsers, conservation repair, the bit-complexity claims, or the effective-resistance reformulation. In the formalization, the rational goal-acceptance predicate checks a supplied factor $C(v)$; optimality of C_* is proved by separate theorems.

K.1 Energy witnesses and rational construction

Let A have $+1$ at the tail and -1 at the head of each oriented edge. In this section the graph may be a finite loopless multigraph, possibly disconnected, with $m \geq 1$; nominations must be feasible on each component. Define

$$\begin{aligned} E(x) &= \sum_e E_e(x_e), & E_e(t) &= \frac{1}{3} \begin{cases} c_e^+ t^3 & t \geq 0, \\ c_e^- (-t)^3 & t < 0, \end{cases} \\ G_e(t) &= E'_e(t) = c_e^{\text{sgn}(t)} t|t|, & \beta_L &= \min_{e, \sigma \in \{+, -\}} c_e^\sigma > 0. \end{aligned} \quad (88)$$

All coefficients and nominations are rational. The notation $c_e^{\text{sgn}(t)}$ selects c_e^+ at $t = 0$; every formula at zero is independent of this convention. Strict convexity and coercivity imply that E has a unique minimizer x^* on $Ax = b$. Stationarity on this affine space gives $G(x^*) = A^\top \pi^*$, with one arbitrary potential gauge per component. Thus x^* is the passive physical flow.

Proposition K.1 (Rational energy and uniform flow certificate). *Let $y \in \mathbb{Q}^m$ satisfy $Ay = b$ and let $p \in \mathbb{Q}^V$ be arbitrary. For $d = A^\top p$, suppose rational $\zeta_e \geq 0$ satisfy*

$$\zeta_e^2 c_e^{\text{sgn}(d_e)} \geq |d_e|^3.$$

Set

$$L = b^\top p - \frac{2}{3} \sum_e \zeta_e, \quad \delta = E(y) - L. \quad (89)$$

Then $0 \leq E(y) - E(x^) \leq \delta$. Any rational $\eta \geq 0$ with $\beta_L \eta^3 \geq 6\delta$ gives $|y_e - x_e^*| \leq \eta$ for every edge. All hypotheses, the claimed gap, and the radius condition are verifiable by rational arithmetic in polynomial time.*

Proof. For $d > 0$ the maximizer of $dt - E_e(t)$ is positive, since negative t gives a nonpositive value whereas small positive t gives a positive value. Differentiation gives $t = \sqrt{d/c_e^+}$ and value $(2/3)\sqrt{d^3/c_e^+}$. The negative case uses c_e^- and the maximizer $-\sqrt{|d|/c_e^-}$; at zero the maximizer and value are zero. Hence

$$E_e^*(d) = \frac{2}{3} \sqrt{\frac{|d|^3}{c_e^{\text{sgn}(d)}}}.$$

The defining inequality $E_e(t) + E_e^*(d) \geq dt$, summed over the physical flow, proves $E(x^*) \geq L$. Minimality gives $E(y) \geq E(x^*)$.

The scalar modulus used in equation (38) also holds for all laws (88). The function $G_e(t) - \beta_L t|t|$ is nondecreasing, and

$$(u|u| - v|v|)(u - v) \geq \frac{1}{2}|u - v|^3.$$

On either same-sign half-line this follows by factoring; for opposite signs it is $|u|^2 + |v|^2 \geq (|u| + |v|)^2/2$. Writing $a = y_e - z_e$ and integrating with $u = z_e + ta$, $v = z_e$, $0 \leq t \leq 1$, yields

$$D_e(y_e, z_e) := E_e(y_e) - E_e(z_e) - G_e(z_e)(y_e - z_e) \geq \frac{\beta_L}{6}|y_e - z_e|^3. \quad (90)$$

Indeed the integrand $[G_e(z_e + ta) - G_e(z_e)]a$ is at least $\beta_L t^2|a|^3/2$. At $z = x^*$ the linear terms sum to zero by $A(y - x^*) = 0$. Summation proves the asserted radius. The checks require only rational additions, products, comparisons, and cubes, with polynomially bounded intermediate bit lengths. $\square$

Here and below, verification complexity counts the expanded binary lengths of rational numerators and denominators. A parser may accept other exact string forms, including scientific exponent notation; the bound applies after their expansion, not to their raw JSON length. Neither y nor p needs to satisfy the constitutive equations; an inaccurate candidate simply has a larger certified gap.

A practical construction repairs conservation before evaluating the energy. Choose a spanning tree in each component, round the chord flows to dyadics, subtract their incidence contributions from b , and recover tree flows by leaf elimination. At a leaf the unique remaining tree flow is its residual nomination, with its orientation sign; transfer that residual to its parent and delete the leaf. The last residual vanishes because the component sum of b is zero. This proves exact conservation. If a conserved reference flow was approximated on every chord to accuracy ρ , each tree correction differs from that reference by at most $m\rho$: removing a tree edge expresses its flow as a cut sum, with at most m chord errors. Thus the repair preserves convergence of increasingly accurate approximations, and it runs in time polynomial in the input and rounding precision. The numerical helper implements only the connected simple loopless case and rejects other graphs; this implementation restriction does not narrow the mathematical certificate.

For a nonnegative rational a/b , a dyadic upper enclosure of its k th root, $k \in \{2, 3\}$, is $j/2^q$, where j is the least nonnegative integer satisfying

$$j^k b \geq a 2^{kq}. \quad (91)$$

An integer square root, or binary search for an integer cube root, followed by one exact upward comparison computes j in polynomial time in the expanded input length and q . Its excess over the root is less than 2^{-q} . This constructs the conjugate enclosures and η without algebraic-number arithmetic; potentials can be rounded independently. These constructions certify achieved accuracy; they give no polynomial stopping bound for an arbitrary numerical solver.

K.2 Bregman intervals, pressure objectives, and sign recovery

Proposition K.2 (Separate-edge rational intervals). *Assume the certificate in proposition K.1 has been verified. If $\delta > 0$, rational endpoints $l_e \leq y_e \leq u_e$ satisfying*

$$D_e(y_e, l_e) \geq \delta, \quad D_e(y_e, u_e) \geq \delta \quad (92)$$

certify $x_e^ \in [l_e, u_e]$. If $\delta = 0$, $x^* = y$ and the singleton intervals suffice. Starting from $y_e \pm \eta$, rational bisection constructs the endpoints in (92) to any specified additive boundary accuracy in polynomial bit time.*

Proof. Conservation and stationarity give the exact identity

$$E(y) - E(x^*) = \sum_e D_e(y_e, x_e^*) \leq \delta. \quad (93)$$

Every summand is nonnegative. For fixed y_e , differentiation with respect to z gives

$$\frac{d}{dz} D_e(y_e, z) = 2c_e^{\text{sgn}(z)} |z| (z - y_e).$$

The function is continuously differentiable at zero as well. Its derivative is strictly negative on $z < y_e$ except possibly at the single point zero, and strictly positive on $z > y_e$ with the same exception. Consequently it is strictly decreasing to zero at y_e and strictly increasing thereafter. It tends to infinity at both ends by (90). Thus its sublevel set at δ is one interval. The endpoint inequalities, in the stated outward directions, enclose this sublevel set and hence x_e^* . The initial endpoints $y_e \pm \eta$ are outside or on the sublevel boundary because $\beta_L \eta^3 / 6 \geq \delta$. On each side keep y_e as an inside endpoint and the current outer endpoint with $D_e \geq \delta$. Bisect and replace the appropriate endpoint using exact cubic evaluation. After k steps the boundary bracket length is $\eta 2^{-k}$. All rational sizes grow polynomially in k . A zero gap gives $E(y) = E(x^*)$, so uniqueness proves the singleton claim without a curvature assumption. $\square$

Increasing boundary accuracy at a fixed positive gap does not make these intervals arbitrarily narrow: it approaches the two roots of $D_e(y_e, z) = \delta$. Away from zero the divergence grows quadratically in displacement; near zero its cubic behavior persists. This is the source of its improvement over a common cube-root radius, and also its limitation.

The physical edge drop lies in the rational interval $[G_e(l_e), G_e(u_e)]$. Along a fixed path, sum these intervals with orientation signs to enclose the terminal potential difference. For any rational zero-sum weighted potential objective $a^\top \pi$ on a connected graph, choose a spanning-tree flow q with $Aq = a$. Then $a^\top \pi = q^\top G(x^*)$; signed interval arithmetic on the tree edges gives its enclosure. Componentwise zero sums give the same statement for disconnected graphs. These are deterministic-state certificates on arbitrary graphs. An uncertainty interpretation for that objective still requires its own graph-dependent reduction. In particular, an envelope target-drop interval concerns the envelope law, whose target coefficient may differ from the selected original resistance scenario.

Proposition K.3 (Posterior endpoint-scenario recovery). *Suppose (88) is a correctly constructed envelope for an original target-flow extremum, with allowed resistance endpoints $\underline{\beta}_e, \bar{\beta}_e$ and fixed nominations. Select $\hat{\beta}_e = c_e^+$ if $y_e \geq 0$ and $\hat{\beta}_e = c_e^-$ otherwise. Let $[l_e, u_e]$ be verified intervals containing y_e and x_e^* , and put*

$$a_e = \begin{cases} \max\{0, -l_e\} & y_e \geq 0, \\ \max\{0, u_e\} & y_e < 0, \end{cases} \quad r_e = (\bar{\beta}_e - \underline{\beta}_e) a_e^2.$$

If $\hat{x}$ is the physical flow at the selected scenario, then

$$\|\hat{x} - x^*\|_\infty^2 \leq \frac{2}{\beta_L} \sum_e r_e. \quad (94)$$

In particular, resolved signs give an exactly optimal rational endpoint scenario even when the exact flow is irrational. More generally, exact optimality holds whenever every edge satisfies at least one of

$$\begin{aligned} c_e^+ = c_e^- = \hat{\beta}_e, \quad l_e \geq 0 \text{ and } \hat{\beta}_e = c_e^+, \\ u_e \leq 0 \text{ and } \hat{\beta}_e = c_e^-, \quad l_e = u_e = 0. \end{aligned} \quad (95)$$

Proof. A disagreement between the selected coefficient and the envelope coefficient at the true sign can occur only on the opposite side of zero from y_e . In that event $|x_e^*| \leq a_e$, whence $|\hat{\beta}_e x_e^*| - G_e(x_e^*) \leq r_e$. Write $d = \hat{x} - x^*$

and $D = \|d\|_\infty$. The two physical potential gradients both annihilate d . Pairing with d and using the scalar monotonicity inequality gives

$$\frac{\beta_L}{2} \sum_e |d_e|^3 \leq \sum_e (G_e(x_e^*) - \hat{\beta}_e x_e^* |x_e^*|) d_e \leq D \sum_e r_e.$$

If $D = 0$ there is nothing to prove; otherwise division by D proves (94). Under (95), the selected law and envelope law agree at x^* on every edge. The envelope state therefore satisfies the selected scenario's equations, and uniqueness gives $\hat{x} = x^*$. Non-strict signs suffice because both laws vanish at zero. The envelope theorem supplies the extremum interpretation. $\square$

An upward rational square root of the right side of (94) certifies scenario loss. Using only $|y_e - x_e^*| \leq \eta$ recovers the uniform bound from theorem E.6; in particular $(1 + 2m\beta_U/\beta_L)\eta$ is a valid, possibly conservative, loss bound when $\beta_U = \max_e \bar{\beta}_e$. There is no universal finite-precision sign-identification guarantee here: a zero or very small physical flow may keep an interval across zero. Exact optimality is a sufficient posterior conclusion when (95) succeeds.

K.3 Conserved-energy support certificates for linear goals

Separate-edge intervals omit the coupling $Ax = b$. For a rational linear goal $w^\top x^*$, retain that coupling in the compact convex set

$$S(y) = \{x : Ax = b, E(x) \leq E(y)\}.$$

This set contains x^* , but its other points need not be physical states of the fixed law.

Theorem K.4 (Rational support certificate and completeness). *Let $Ay = b$. For rational $\lambda > 0$, $v \in \mathbb{Q}^V$, set $s = w - A^\top v$. If rational $R_e \geq 0$ satisfy*

$$R_e^2 \lambda c_e^{\text{sgn}(s_e)} \geq |s_e|^3, \quad U_w = \lambda E(y) + v^\top b + \frac{2}{3} \sum_e R_e, \quad (96)$$

then $w^\top x \leq U_w$ for every $x \in S(y)$. Applying the construction to $-w$ gives the interval $[-U_{-w}, U_w]$ for $w^\top x^$. For $\lambda = 0$, a finite dual bound exists exactly when $w = A^\top v$; then $w^\top x = v^\top b$ on $Ax = b$ and no roots are needed.*

If $E(y) > E(x^)$, the infimum of these rational upper bounds equals $\max_{x \in S(y)} w^\top x$. Hence rational witnesses can approach the exact support value arbitrarily closely. As conserved trial energies decrease to $E(x^*)$, the exact support intervals converge to $w^\top x^*$ for every w .*

Proof. The scalar calculation in proposition K.1, with coefficient $\lambda c_e^{\text{sgn}(s_e)}$, gives

$$(\lambda E_e)^*(s_e) = \frac{2}{3} \sqrt{\frac{|s_e|^3}{\lambda c_e^{\text{sgn}(s_e)}}}.$$

Fenchel's inequality gives $s^\top x \leq \lambda E(x) + \sum_e (\lambda E_e)^*(s_e)$. Add $v^\top Ax = v^\top b$ and use $E(x) \leq E(y)$. When $\lambda = 0$, the conjugate of the zero function is finite only at $s = 0$, proving the stated condition. These are exactly the goals constant on the conservation affine space, because $(\ker A)^\perp = \text{im } A^\top$. They include cut flows and each bridge flow; rational linear elimination finds v from rational A, w whenever it exists.

To prove completeness of the unrounded dual, minimize $-w^\top x$ subject to $Ax = b$, $E(x) \leq E(y)$. Its value is finite and attained by coercivity. If $E(y) > E(x^*)$, the point x^* is strictly feasible for the energy inequality relative to $Ax = b$. Slater duality therefore gives equality of the support value and the dual infimum, and a dual minimizer. Redundant conservation rows may first be removed, so their presence causes no qualification issue. If w is nonconstant on $Ax = b$, a finite dual minimizer cannot have $\lambda = 0$ by the preceding conjugate calculation. For $\lambda > 0$ the displayed unrounded objective is continuous in (λ, v) , including at $s_e = 0$: the two asymmetric terms tend to zero there. Approximate a dual minimizer by rational data keeping $\lambda > 0$, and then take increasingly accurate upward rational square roots. The resulting valid rational upper bounds tend to the support value. Constant goals already have an exact rational witness at $\lambda = 0$.

For convergence, the cubic modulus gives the stronger quantitative statement

$$|z_e - x_e^*|^3 \leq \frac{6(E(y) - E(x^*))}{\beta_L} \quad (z \in S(y), e \in E).$$

Indeed, apply (90) at x^* to the feasible flow z . Conservation cancels the summed linear terms, and $E(z) \leq E(y)$ gives the displayed bound. Thus, if $Ay_k = b$ and $E(y_k) \rightarrow E(x^*)$, the whole set $S(y_k)$ eventually lies in any fixed positive-radius supremum-norm ball about x^* . Continuity of each fixed linear goal proves convergence of its exact support interval. This estimate is in terms of energy error and gives no rate in producer runtime or certificate size. $\square$

The support problem has the cubic second-order-cone lift of theorem E.6, with a linear objective and an energy sublevel constraint, so numerical conic solvers can propose its dual variables. The accompanying support module supplies exact verification, conversion of supplied rational dual data to root

witnesses, and analytic demonstrations; it does not implement a general conic search for these dual data. The completeness statement gives no polynomial certificate-size or producer-runtime bound at a specified support tolerance, and it does not assert that a support interval always improves a Bregman interval. In particular, (93) uses physical stationarity at x^* ; one may not assume the same separate-edge Bregman constraints for all $x \in S(y)$.

K.4 Curvature bounds and the constrained Laplacian

Theorem K.5 (Conservation-aware quadratic goal certificate). *Let $y \in \mathbb{Q}^m$ satisfy $Ay = b$. Suppose $0 \leq E(y) - E(x^*) \leq \delta$ is verified, and rational intervals $[l_e, u_e]$ contain both y_e and x_e^* . Define*

$$h_e = \begin{cases} 2c_e^+ l_e & l_e > 0, \\ -2c_e^- u_e & u_e < 0, \\ 0 & l_e \leq 0 \leq u_e. \end{cases} \quad (97)$$

Let $P = \{e : h_e > 0\}$ and $Z = \{e : h_e = 0\}$. For any rational v with $s = w - A^\top v$ satisfying $s_Z = 0$, put

$$C(v) = \sum_{e \in P} \frac{s_e^2}{h_e}. \quad (98)$$

Every rational $r \geq 0$ with $r^2 \geq 2\delta C(v)$ proves

$$|w^\top(x^* - y)| \leq r. \quad (99)$$

If $\delta = 0$, every goal is exact regardless of h or s_Z .

Proof. The second derivative is $E_e''(t) = 2c_e^{\text{sgn}(t)}|t|$, including its continuous zero value. Formula (97) is its minimum on the interval. Taylor's integral formula along the segment between x_e^* and y_e gives $D_e(y_e, x_e^*) \geq h_e(y_e - x_e^*)^2/2$. Therefore

$$\frac{1}{2} \sum_e h_e (x_e^* - y_e)^2 \leq \delta. \quad (100)$$

Since $A(x^* - y) = 0$, the goal difference equals $s^\top(x^* - y)$. The identity $s_Z = 0$ and weighted Cauchy-Schwarz on P bound its square by $C(v) \sum_e h_e (x_e^* - y_e)^2 \leq 2\delta C(v)$. Empty sums are zero, so the proof also covers $P = \emptyset$. For a zero gap use strict convexity, which gives $x^* = y$ directly. $\square$

Proposition K.6 (Optimal factor and zero-curvature obstruction). *Write $D = \text{diag}(1/h_e : e \in P)$. The zero-edge condition is feasible exactly when*

$$w_Z^\top z = 0 \quad \text{for every } z \in \ker A_Z. \quad (101)$$

When feasible, the minimum C_ of (98) is attained by a rational solution of*

$$\begin{bmatrix} A_P D A_P^\top & A_Z \\ A_Z^\top & 0 \end{bmatrix} \begin{bmatrix} v \\ \xi \end{bmatrix} = \begin{bmatrix} A_P D w_P \\ w_Z \end{bmatrix}. \quad (102)$$

Singular potential gauges and redundant zero-edge constraints are allowed. The value $\sqrt{2\delta C_}$ is the exact supremum of $|w^\top d|$ over*

$$Ad = 0, \quad \frac{1}{2} \sum_e h_e d_e^2 \leq \delta \quad (103)$$

for every $\delta > 0$. If (101) fails, that supremum is infinite. Thus optimizing the Laplacian constant is sharp for this quadratic information, without asserting sharpness for the physical error or for the additional interval constraints.

Proof. The equation $A_Z^\top v = w_Z$ is solvable exactly when w_Z is orthogonal to $\ker A_Z$, by the fundamental subspace identity. If it is inconsistent there is $z \in \ker A_Z$ with $w_Z^\top z \neq 0$. Extending z by zeros on P gives a zero-curvature circulation; arbitrary multiples satisfy (103) and have unbounded goal.

For the feasible case, minimize the squared norm $\|D^{1/2}(w_P - A_P^\top v)\|^2$ on the affine space $A_Z^\top v = w_Z$. Its affine image in Euclidean space is closed, so a least-norm point exists. First-order optimality on that affine space says its gradient lies in $\text{im } A_Z$. Writing a multiplier ξ gives exactly (102); conversely these equations imply global optimality of the convex quadratic. The system is rational and consistent. Exact elimination with free coordinates set to zero returns rational v, ξ , even with redundant rows. Determinant bounds after clearing denominators give polynomial output bit length and polynomial bit complexity; a singular system need not be perturbed.

For sharpness, take an optimal solution and put $d_P = D(w_P - A_P^\top v)$ and $d_Z = -\xi$. The first row of (102) gives $Ad = 0$, and the second gives $s_Z = 0$. Consequently

$$w^\top d = s^\top d = C_*, \quad \sum_e h_e d_e^2 = C_*.$$

If $C_* > 0$, multiply d by $\sqrt{2\delta/C_*}$ to attain the upper bound from weighted Cauchy–Schwarz. If $C_* = 0$, $s = 0$ on all edges, so $w = A^\top v$ and every circulation goal is zero. This also handles $P = \emptyset$: consistency makes w a cut goal, and the optimal factor is the empty sum zero. $\square$

When every $h_e > 0$, let $H = \text{diag}(h)$ and $L_H = AH^{-1}A^\top$. The minimum can equivalently be written

$$C_* = w^\top H^{-1}w - (AH^{-1}w)^\top L_H^+(AH^{-1}w), \quad (104)$$

where L_H^+ is the Moore–Penrose inverse. To verify this expression, expand $C(v)$ as a quadratic in v , solve $L_H v = AH^{-1}w$, and substitute. The right side is in $\text{im } L_H = \text{im } A$, so the substitution is unaffected by component gauges. For a single-edge selector $w = \mathbf{1}_e$, writing a_e for that incidence column gives

$$C_* = \frac{1}{h_e} - \frac{R_{\text{eff}}(e)}{h_e^2}, \quad R_{\text{eff}}(e) = a_e^\top L_H^+ a_e.$$

Here the auxiliary network has conductances $1/h_e$. For a bridge the unit terminal current has no alternative route, so $R_{\text{eff}}(e) = h_e$ and the factor vanishes, as conservation also proves. This is the usual electrical projection and effective resistance identity, now used as a goal-error constant.

The supplied producer solves (102) by dense exact rational elimination. It is intended for modest certificate sizes; no sparse scalability claim follows. Certificate validity does not require optimal potentials: with positive curvatures, even a rounded numerical Laplacian solution gives a valid, perhaps larger, factor. With zero curvatures, all equalities $A_Z^\top v = w_Z$ must instead hold exactly, and regularizing away an inconsistent constraint cannot produce a valid witness. The implementation first handles zero energy gap, then rejects the zero-curvature obstruction. The full pipeline may omit this optional goal witness and retain its independently verified Bregman interval.

K.5 An exact comparison on two long paths

Example K.7 (Information lost by separate-edge bounds). Consider two internally disjoint source–sink paths, each of length $L \geq 1$, with every edge oriented toward the sink, all coefficients one, source nomination two, sink nomination minus two, and zero internal nominations. The physical flow is one on every edge: its path drops are both L , and strict convexity gives uniqueness. For rational $0 < \varepsilon < 1$, choose $y = 1 + \varepsilon$ on the first path and $y = 1 - \varepsilon$ on the second. Its exact energy gap is

$$\delta = \frac{L}{3}((1 + \varepsilon)^3 + (1 - \varepsilon)^3 - 2) = 2L\varepsilon^2. \quad (105)$$

Integral physical potentials, decreasing by one on each path edge, give this exact dual energy value, with rational conjugate roots.

Every conserved flow has path values $t, 2 - t$. Its energy is strictly convex, symmetric about $t = 1$, and strictly increasing with $|t - 1|$. Consequently $S(y)$ is exactly $1 - \varepsilon \leq t \leq 1 + \varepsilon$, even though the energy formula uses absolute cubes outside $[0, 2]$. The exact support interval of a first-path edge is therefore of width 2ε . Its upper dual can also be attained rationally. Choose

$$\lambda = \frac{1}{4L\varepsilon}, \quad (A^\top v)_e = w_e - \lambda t_e^2,$$

where $t_e = 1 + \varepsilon$ on the first path and $1 - \varepsilon$ on the second, and w selects one first-path edge. The two sums of these prescribed potential drops agree because $1 - \lambda L(1 + \varepsilon)^2 = -\lambda L(1 - \varepsilon)^2$. Thus rational node potentials exist. The residual is $s_e = \lambda t_e^2$, with exact root $R_e = \lambda t_e^3$; equality in Fenchel and in the energy constraint gives $U_w = 1 + \varepsilon$. Exchanging the path values and using $-w$ gives the lower support bound as well.

For fixed L and rational $\varepsilon \downarrow 0$, let $z = y_e + a$ remain positive. Direct expansion gives

$$D_e(y_e, y_e + a) = y_e a^2 + \frac{2}{3} a^3.$$

The two boundary displacements satisfying $D_e = 2L\varepsilon^2$ therefore obey $a/\varepsilon \rightarrow \pm\sqrt{2L}$. The separate-edge Bregman width is

$$2\sqrt{2L}\varepsilon + o(\varepsilon). \tag{106}$$

All interval endpoints tend to one, so their curvatures tend to two. The circulation space is one-dimensional, generated by d equal to $+1$ on one path and -1 on the other. Directly optimizing (103), or using proposition K.6, gives $C_* = 1/\sum_e h_e \rightarrow 1/(4L)$ for the selected edge. Thus the optimized Laplacian width is

$$2\sqrt{2\delta C_*} = 2\varepsilon + o(\varepsilon). \tag{107}$$

These asymptotics hold for exact boundaries. They also hold for rational outer endpoints and roots with errors $o(\varepsilon)$. They are pointwise in L . Choosing L large and then ε sufficiently small makes the improvement factor arbitrarily large; no uniform-in- L asymptotic or computational lower bound is implied. The physical solution of this family is elementary.

The centers of these intervals differ. The Laplacian interval is centered at $w^\top y$, whereas the exact support interval need not be. Equal asymptotic widths therefore do not imply equal endpoints. For a concrete warning against combining their hypotheses, take $L = 1$ and $\varepsilon = 1/10$. The opposite support endpoint has $D_1(11/10, 9/10) = 29/750 > \delta = 1/50$. It belongs to $S(y)$ but

fails a separate-edge physical Bregman constraint. Intersecting independently verified enclosures of x^* is valid; silently restricting the support problem using those physical constraints is a different optimization problem.

K.6 Verification of the original uncertainty instance

The preceding certificates validate a deterministic energy model. Its JSON data alone cannot establish that it is the right envelope for a specified uncertainty problem. The complete pipeline additionally verifies a connected simple loopless graph with no K_4 minor, fixed balanced nominations, independent positive rational resistance intervals or explicit finite sets, a target edge, and maximum or minimum direction. It accepts no operating side constraints or correlated uncertainty. Its simple-graph restriction is narrower than the multigraph scope of the mathematics.

The verifier checks indices, simplicity, connectivity, dimensions, and exact balance. Allowed sets must be nonempty and strictly positive. It recognizes its graph class by repeatedly eliminating a vertex of degree at most two and joining its two neighbors when needed. The K_4 -minor-free characterization underlying lemma E.3 implies that such a vertex always exists in the remaining graph. Suppressing a degree-two vertex is a minor operation; degrees zero or one also preserve the class. Conversely a completed elimination order reverses to adding vertices whose existing neighbors form cliques of size at most two. Starting with bags of at most three vertices and attaching a new bag at its neighbor clique gives a tree decomposition of width at most two; therefore the original graph has no K_4 minor. This also explains why the test accepts dangling cyclic blocks.

Next solve the grounded all-unit-conductance Laplacian for a unit injection at the target tail and withdrawal at its head. Exact rational elimination computes every adjoint drop and its sign, including exact zeros. Let these signs be σ_e , and let $d = 1$ for maximization and $d = -1$ for minimization. Set

$$t_e = d\sigma_e \quad (e \neq a), \quad t_a = -d.$$

The directional coefficients are

$$c_e^+ = \begin{cases} \bar{\beta}_e & t_e > 0, \\ \underline{\beta}_e & t_e \leq 0, \end{cases} \quad c_e^- = \begin{cases} \bar{\beta}_e & t_e < 0, \\ \underline{\beta}_e & t_e \geq 0. \end{cases}$$

For finite sets, $\underline{\beta}_e, \bar{\beta}_e$ are their minimum and maximum. The target's reversed rule and the zero-sign choice are those of theorem E.4. The confluence and finite-secant proofs there justify this exact mapping for all positive secant

resistances. The verifier recomputes it from the original instance, rather than trusting stored coefficients or floating-point signs.

It then checks an energy witness for this envelope and a second energy witness for the selected original endpoint scenario. Each yields Bregman intervals, optionally intersected with independently verified linear-goal intervals for the reconstructed target selector. Empty intersections are rejected. Let the resulting target intervals be $I^* = [l^*, u^*]$ and $I^\beta = [l^\beta, u^\beta]$. The original-scenario loss is bounded directly by

$$\begin{cases} 0 \leq \text{OPT} - x_a(\beta) \leq u^* - l^\beta & \text{for maximization,} \\ 0 \leq x_a(\beta) - \text{OPT} \leq u^\beta - l^* & \text{for minimization.} \end{cases} \quad (108)$$

Indeed the original envelope theorem identifies OPT with x_a^* , the scenario belongs to the allowed product, and interval subtraction bounds their difference. If (95) holds, the checker instead reports zero loss by exact endpoint optimality. A zero loss certifies the selected scenario, not an exact numerical value for its possibly irrational flow.

The second witness requires no second physical-flow optimization. The producer may reuse the conserved envelope trial flow, round it again if necessary, and recompute a dual potential proposal under the selected symmetric law. Its independent primal-dual check certifies the resulting scenario interval; poor reuse only widens it. This is a posterior alternative to the uniform recovery factor.

K.7 Scaling, implementation scope, and reproducible evidence

Let $B = \sum_{b_v > 0} b_v$ and $C = \max_{e, \sigma} c_e^\sigma$. For $B > 0$ the producer solves and certifies normalized data $\tilde{b} = b/B$, $\tilde{c} = c/C$. The exact lift is

$$\begin{aligned} y &= B\tilde{y}, & p &= CB^2\tilde{p}, & \zeta &= CB^3\tilde{\zeta}, & \delta &= CB^3\tilde{\delta}, & \eta &= B\tilde{\eta}, \\ [l_e, u_e] &= B[\tilde{l}_e, \tilde{u}_e], & h_e &= CB\tilde{h}_e, & v &= \tilde{v}, & C(v) &= \tilde{C}(v)/(CB), & r &= B\tilde{r}. \end{aligned} \quad (109)$$

Here ζ denotes the vector of conjugate-root bounds, and l_e, u_e are interval endpoints. Cubic homogeneity proves the energy and root identities: $|A^\top p|^3/c$ scales by C^2B^6 . The second-derivative formula proves the curvature identity, and $r^2 \geq 2\delta C(v)$ then scales by B^2 . Thus every rational inequality is preserved. Certifying in normalized units before this exact lift makes the rounding error covariant under common rational changes of flow and resistance units. Normalizing only the floating-point solve would not have that property.

When $B = 0$, balance implies $b = 0$ and the producer uses the exact zero physical state. A positive-gap zero-load trial circulation is still a legitimate input to the general verifier and may exhibit the zero-curvature obstruction.

The numerical energy producer combines the explicit cubic cone lift (solved with CVXPY and Clarabel), conservation repair, rounded least-squares potential proposals, and exact root bounds. It is a practical proposal mechanism, not an implementation of the rational convex-body algorithm of theorem E.6, and it offers no guarantee for a requested accuracy. Extreme coefficient ratios can cause large certified intervals or a failed numerical solve. Verification uses only Python’s standard library and explicit exceptions, so its mathematical checks remain active under `python -O`. The original-instance reader also rejects duplicate JSON keys, unknown fields, nonstring rational fields, and Boolean indices. The simpler base-energy reader certifies the deterministic instance encoded in its own file and does not enforce this complete original-instance schema. The Bregman helper requires prior base verification, and the integrated goal verifier repeats that check.

The implementation files, in `code/potential_flow_mpd/`, have distinct roles:

File	Verified role
<code>envelope_rational_certificates.py</code>	Deterministic energy, conjugate roots, conserved flow, and uniform radius.
<code>envelope_bregman_bounds.py</code>	Outward scalar interval sharpening after base verification.
<code>goal_flow_certificate.py</code>	Support verification and exact constrained-Laplacian production and verification.
<code>certified_envelope.py</code>	Original data, exact envelope mapping, two state certificates, optional target witnesses, and scenario-loss interpretation.

The saved six-edge deterministic witness gives a uniform interval width below $3.405 \cdot 10^{-4}$. Rechecking it and sharpening by exact Bregman arithmetic gives maximum edge width below $2.894 \cdot 10^{-6}$ and maximum edge-drop width below $4.0 \cdot 10^{-6}$; all six edge intervals exclude zero. Applying the exact Laplacian producer without another physical solve gives maximum goal width below $9.447 \cdot 10^{-7}$. These are widths of verified enclosures, not measurements of the true numerical error. Exact optimality of an original scenario would additionally require an independently checked envelope mapping for this

saved base witness.

The two-path demonstration at $\varepsilon = 10^{-3}$ gives the following rounded displays of exactly checked bounds. The support column is the analytic width from example K.7.

L	Separate-edge width	Laplacian width	Exact support width
2	0.00399800593	0.00200200435	0.002
8	0.00799603843	0.00200401743	0.002
32	0.0159922895	0.00200807006	0.002

A replay of the original-instance benchmark produced valid certificates in ten cases, including maximum and minimum directions, mixed finite and interval sets, reversed orientations, an 80-edge graph of rank 39, a full coefficient ratio of $3 \cdot 10^6$, supply 10^{-8} , a dangling zero-adjoint block, zero nominations, and an ambiguous-sign case. Nine cases certified exact endpoint optimality. Some numerical solves reported `optimal_inaccurate`; the reported guarantees came from the exact witnesses. The ambiguous-sign instance retained a nonzero certified loss: the goal witness reduced its target enclosure width from about $4.99 \cdot 10^{-4}$ to $4.37 \cdot 10^{-6}$ and its loss upper bound from about $6.92 \cdot 10^{-4}$ to $5.28 \cdot 10^{-6}$, approximately 114-fold and 131-fold reductions in verified bounds. This says nothing about the actual scenario error or exact sign resolution.

One case adapts the topology and pipe geometry of the open water-network dataset accompanying Vrachimis et al. [53]. It uses 11 nodes and 13 pipes, synthetic unit demands at the nine nonreservoir nodes numbered 2 through 10, zero demand at the leak subdivision node, and supply nine at node 1. Its quadratic proxy coefficient is $(\text{length}/1000)(300/\text{diameter})^5$, with a $\pm 10\%$ interval. The target is original pipe 11, from node 4 to node 9. This transformed instance tests an external topology; it does not reproduce the source’s hydraulic law, measurements, demand scenarios, or leakage estimates, and it establishes neither hydraulic prediction accuracy nor large-network scalability.

The retained verification record includes ordinary and optimized Python runs of independently written deterministic, Bregman, goal, and original-instance checks. Their controls include exact signed states, zero gaps, zero-curvature cut and cycle goals, parallel edges, disconnected gauges, malformed witnesses, and all 771 connected labeled graphs on two through five vertices tested against the branch-set definition of a K_4 minor. The two small numerical cases were additionally compared with all 128 endpoint scenarios combined, using a separate numerical physical solver; this latter comparison

is a regression check, not a proof. Exact scale tests at 10^{400} and 10^{-400} verify the covariance in (109). A further diagnostic checks the sharp quadratic constant directly in circulation coordinates, including singular zero-edge cases, and the rational two-path support equalities. These finite checks supplement the proofs and do not establish their universal statements.

From the repository root, the saved complete certificate can be checked without scientific packages by

```
cd code/potential_flow_mpd
python -S -O certified_envelope.py verify \
    certified_envelope_example.json
```

The deterministic saved witness is checked with the base module's `-verify` option; the goal module runs the analytic demonstration with no arguments and accepts `-certificate` for the saved six-edge comparison. The numerical benchmark entry point is `certified_envelope_benchmarks.py`; it requires the scientific Python environment and writes example and result JSON files beside itself. The retained replay uses an isolated copy so the distributed saved witnesses remain unchanged. Source hashes, full commands, interpreter and package versions, exact rational comparison data, and return codes accompany the verification record. No solver status, rounded display, or historical benchmark result substitutes for rechecking the exact witness against its original instance.